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Company Name _____

Fuji Scalable Placement Platform

NXT/NXT II**Training Text for Service Engineers**

(Edition 2.0)

- Also refer to the latest technical report when using this manual for servicing.
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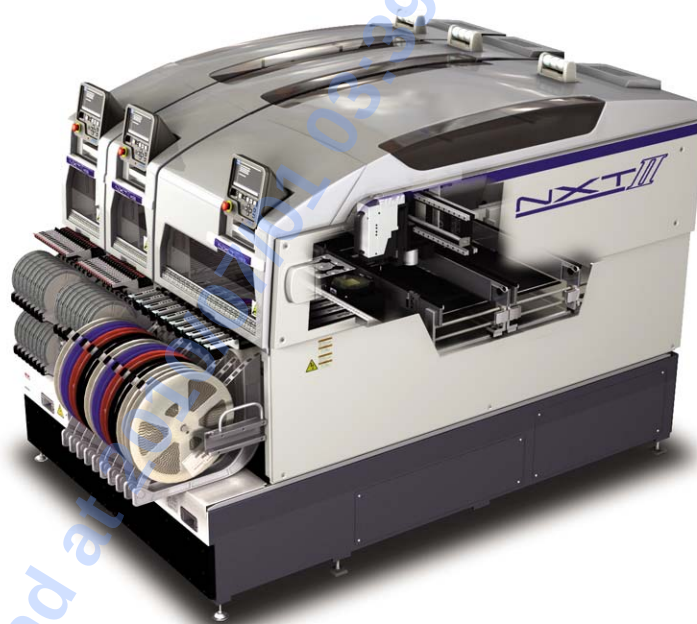
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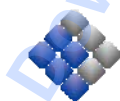
NXT/NXTII

Fuji Scalable Placement Platform

Training Text for Service Engineers



**FK-9F98-77-0E
Edition 2.0**



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Contents

Section	Page
Chapter 1 Accessory Software Upgrading	
1.1. Install Screen Explanation	1-2
1.2. Upgrading the various PCs.....	1-6
Chapter 2 Upgrading the Machine software	
2.1. Upgrading the machine application software	2-1
2.2. Various Unit Software.....	2-4
2.3. Servo Amplifier Software Update	2-14
Chapter 3 Emergency Install	
3.1. Pulling out the module	3-1
3.2. About Emergency Install.....	3-2
3.3. Performing Emergency Install for the base CPU Box.....	3-3
3.4. Performing Emergency Install for the module CPU Box.....	3-17
3.5. Replacing the base CPU Box	3-25
3.6. Replacing the boards in the base CPU BOX(NXTII only)	3-31
3.7. Replacing the module CPU Box.....	3-35
3.8. Replacing the boards in the module CPU BOX(NXTII only)	3-37
Chapter 4 Motor MT Reset & Origin Adjustment	
4.1. Preparation.....	4-1
4.2. Starting I/O Monitor	4-1
4.3. Starting Motion Tool Mini.....	4-4
4.4. XY axis MT Reset.....	4-5
4.5. If the origin setting is required	4-8
Chapter 5 Feeder Pallet Levelness Measurement	
5.1. Required Jigs	5-1
5.2. Preparing the jig	5-1
5.3. Preparing the job	5-2

5.4. Starting the device levelness measurement application	5-2
5.5. Preparation prior to measurement	5-7
5.6. Installing the feeder pallet levelness measurement software	5-20
 Chapter 6 Matrix Measurement for NXTII	
6.1. Preparation for the Matrix Measurement	6-1
6.2. Performing the Directload of the Matrix Measurement	6-8
6.3. Performing Matrix Measurement (V4.30 or later)	6-14
6.4. Performing Matrix Measurement (V4.30 or earlier)	6-27
 Chapter 7 Matrix Measurement for NXT	
7.1. Preparation for the Matrix Measurement	7-1
7.2. Performing the Directload of the Matrix Measurement	7-10
7.3. Performing Matrix Measurement (V4.30 or later)	7-17
 Chapter 8 Placing Quality Problems and countermeasures	
8.1. Examples of placing quality problems	8-1
8.2. Causes of these problems	8-3
8.3. Mechanism of placing quality problems	8-4
8.4. Countermeasures for placement quality problems	8-7
 Chapter9 Conveyor operation during unit breakdown	
9.1. Conveyor Operation Function Availability	9-1
9.2. Procedure flow	9-2
9.3. Work Procedure	9-3
 Chapter 10 Acquiring Trace Data for Different Types of Error	
10.1. Overview of the various trace data acquisition methods	10-1
10.2. Selecting the target module(s) for trace data acquisition	10-2
10.3. Trace data acquisition procedures	10-3
10.4. Trace data acquisition methods for various problems	10-10
 Chapter 11 Countermeasures for Various Error Types	
11.1. The standby lamp does not switch from flashing to lit during startup	11-1

11.2. “Communication Wating……timeout” error during stratup	
→ “PO 1394 Connection is not insert” displays	11-3
11.3. Module control box fuse error (Error code: **** 1001)	11-6
11.4. Module air pressure error	11-8
11.5. Vacuum pressure error (**** 6501)	11-11
11.6. Module vacuum error (error code: :::1203, ****1204)	11-13
11.7. Y-axis linear motor temperature error (Error code:**** 1B14)	11-16
11.8. Nozzle lower lever error (Error code: **** 1910)	11-19
11.9 .Work Procedure	11-22

Supplement

Supplement 1	S-1
Supplement 2	S-5
Supplement 3	S-9
Supplement 4	S-12
Supplement 5	S-14

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Chapter 1

Accessory Software Upgrading

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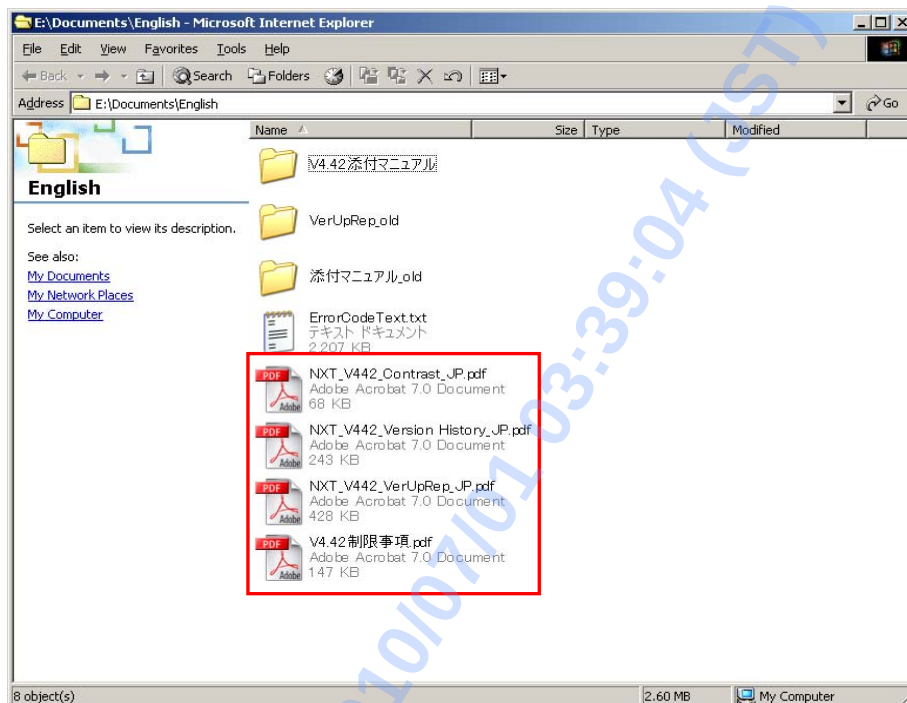
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1. Accessory Software Upgrading

This chapter describes the version upgrade procedure for the various types of NXT software. If you would like to make a new installation (rather than a version upgrade) please refer to the procedure described in the "Fuji Accessory Software Installation" section of the NXT II Setup Manual.

Precautions when upgrading

Note: First refer to the upgrade report located in the "Documents" folder of the NXT Application CD to check new functions, corrections, and limitations before performing the upgrade.



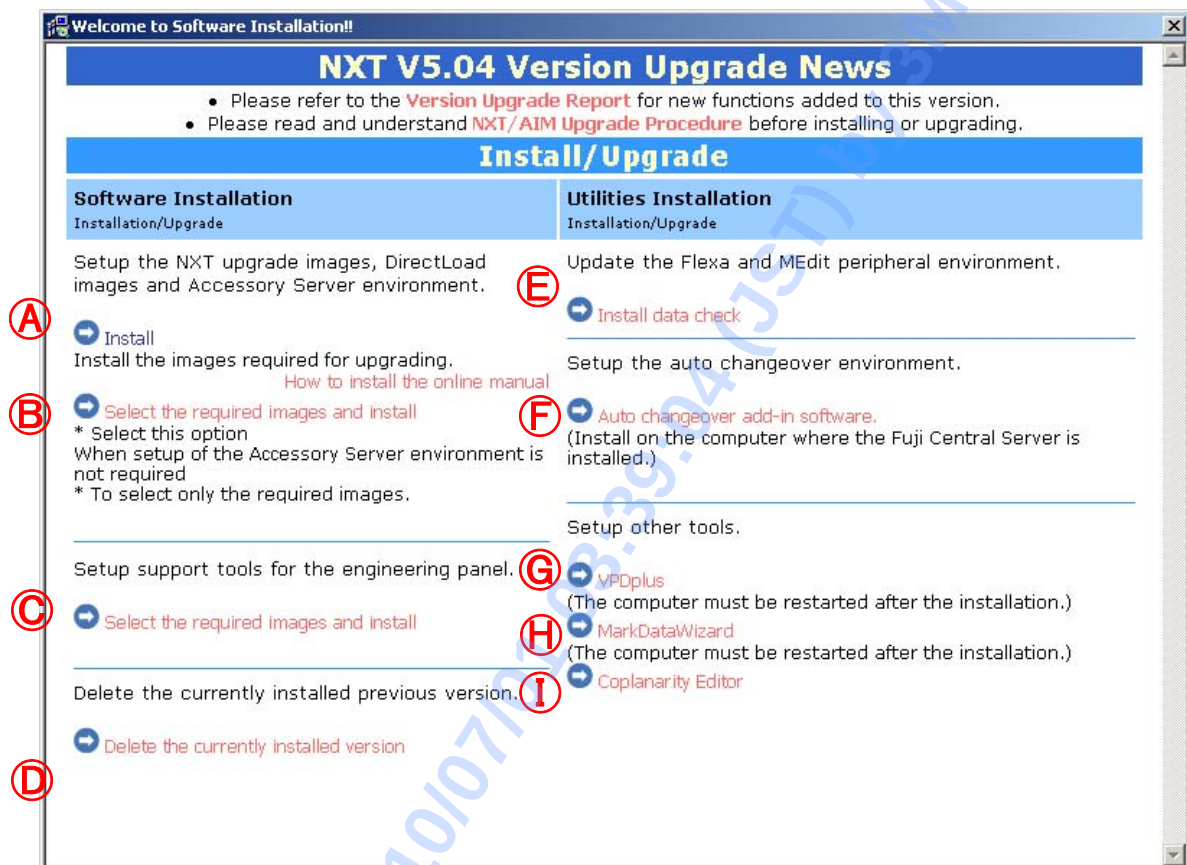
Note: When upgrading NXT, do not turn OFF the base power supply until the upgrade has completed for all modules on the base. The module may not operate properly if the power supply is turned OFF during the upgrade.

Note: Perform Emergency Install if the machine does not function properly following problems with the upgrade. Refer to "1. Emergency Install" in this training text for a detailed description of the Emergency Installation procedure.

1.1 Install Screen Explanation

Insert the NXT Application CD in the PC on which NXT Accessory Software has been installed (Accessory Software Server). The following screen is displayed automatically.

Note: The NXT SOFTWARE is distributed on 2 CDs from V4.50. Disk 1 is the application installation disc and disk 2 is the installation disc for the online manual.

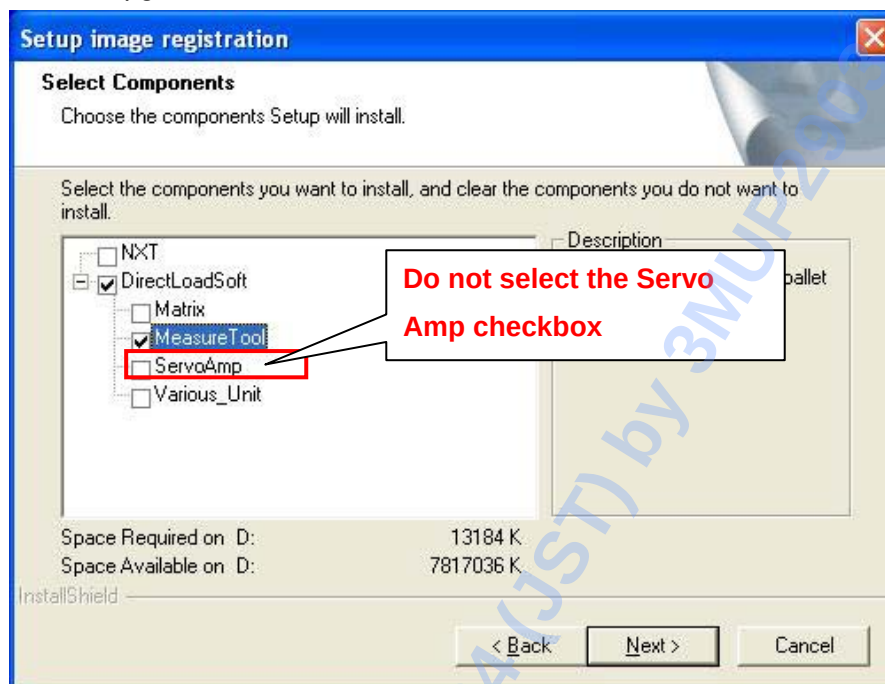


Explanation of each item

- (A) Install all the images required for the NXT application upgrade in the Accessory Software Server. This method is usually used for the installation procedure. The following software will be installed in the accessory software server.

- Accessory software
- NXT Application Software
- Direct Load Software (Matrix, Measure Tool, Various Units)
- Fuji Realtime Workingrate Buffer Service (RWB)

- ⑥ Choose and install only the required images for installation. Usually installation is not performed using this method because it is normally necessary to install all the images for an upgrade.

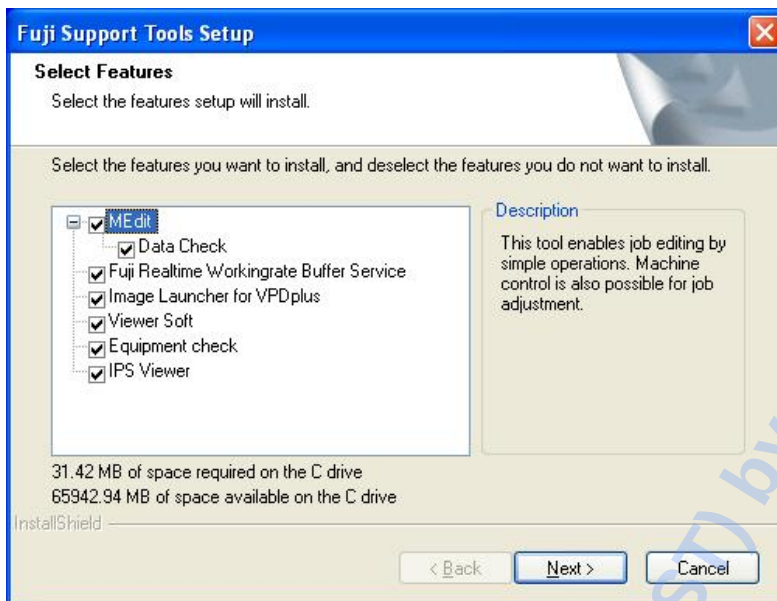


The screen above displays if ⑥ is selected. Check the required checkboxes to install the software.

NXT:	NXT Application
Matrix:	Software used for Matrix Measurement
Measure Tool:	Software used for Feeder Pallet Levelness Measurement
ServoAmp:	<u>Do not select this checkbox. See warning below.</u>
Various_Unit:	The software required to upgrade the various units (placing head, feeder, feeder pallet, MTU-L etc.)

Warning: The servo amplifier software has been removed from the NXT application CD in order to protect the intellectual property rights of the servo amplifier manufacturer and prevent possible malfunction of the motor control function due to overwriting of the software. From now onwards the servo amplifier version upgrade will be performed by Fuji or agent engineers when determined to be necessary. For further details, please refer to Rep News №10039. If installation is performed with the ServoAmp checkbox selected, a "ServoAmp Install Error" will occur. In this case, click "OK" to continue the installation.

- © Install the required software for the engineering panel on the client PC.



MEEdit: Software used for editing and testing a job.

Data Check: Software used to check the job correction. Usually installed to the PC using MEEdit. (Selecting this will also install MEEdit).

Fuji Realtime Workingrate Buffer Service: Services used for acquiring production information.

Image Launcher for VPDplus: Function to launch VPDplus based on the information taken out from the vision image.

Viewer Soft: Tool used for checking picture image.

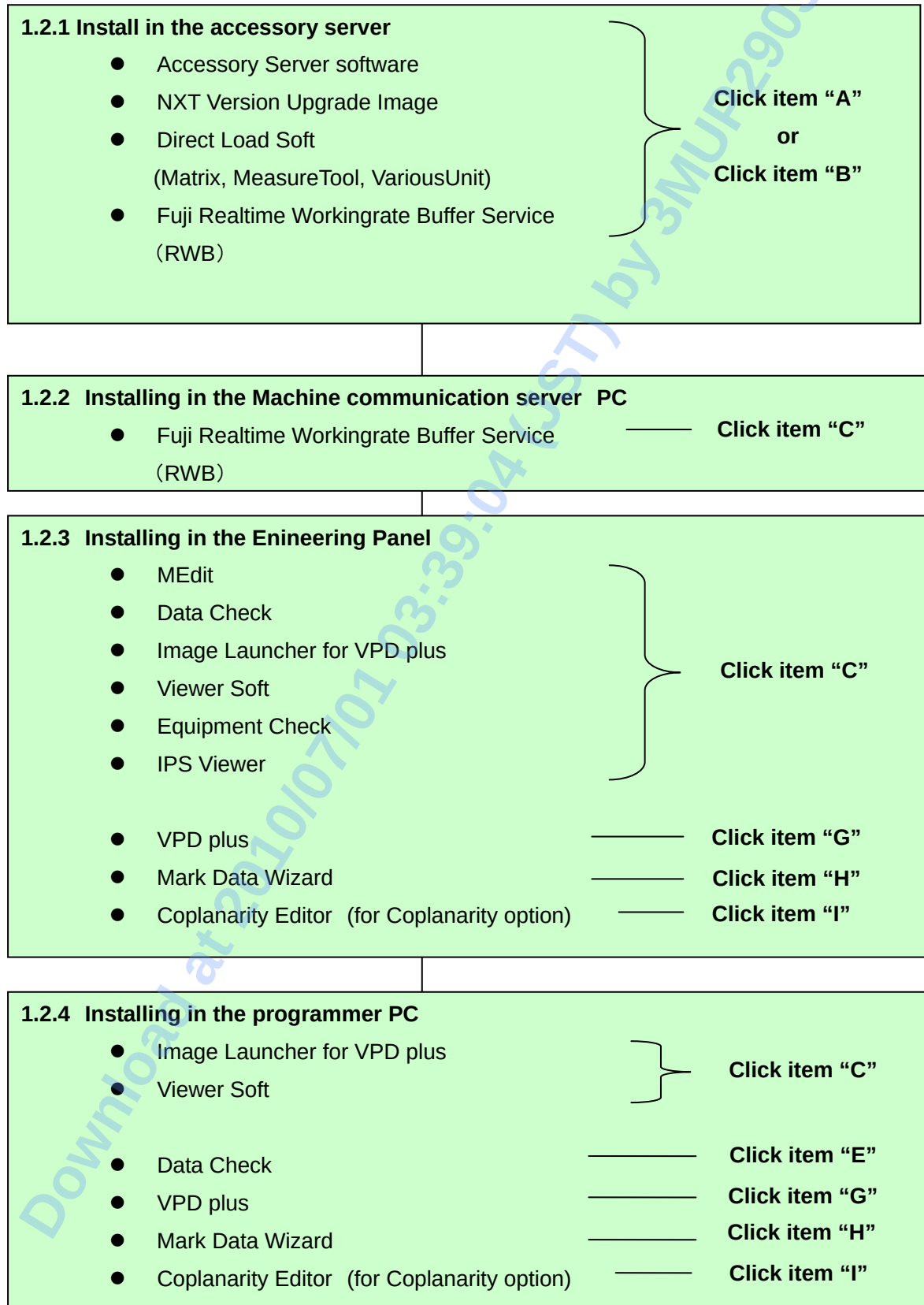
Equipment Check: Software used to check the I/O check, functions of base, module, conveyor and MTU-L, LT

- ① Delete the currently installed previous version. It will be saved on the server as long as the old version is not deleted.
- ② Install Data Check software. (If you want to install only Data check)
- ③ Install the [Automatic Changeover] Add-In software. Install on the computer where Fuji Central Server is installed.

- ⑥ Install the VPDplus software used to create part data information.
- ⑦ Install MarkDataWizard. This software is used to edit the irregular shaped fiducial marks. Editing procedure is started by selecting MarkDataWizard from MEdit.
- ⑧ Install Coplanarity Editor. This is used for coplanarity unit procedure of the option unit.

1.2 Upgrading the various PCs

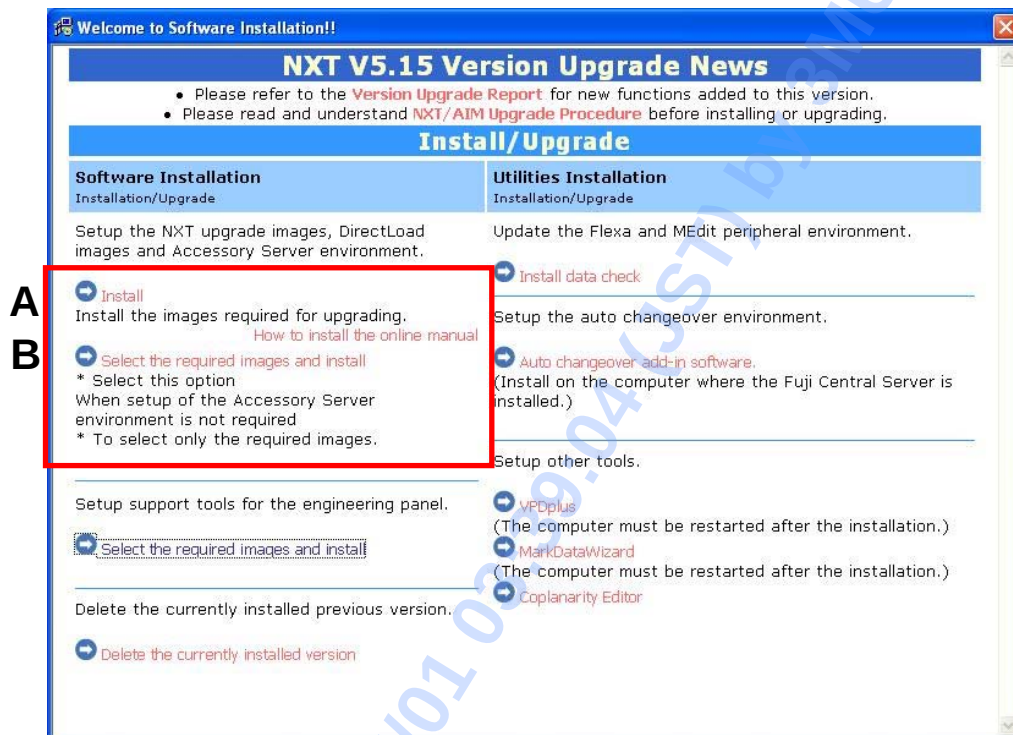
Upgrade Flow



1.2.1 Installing in the Accessory Server

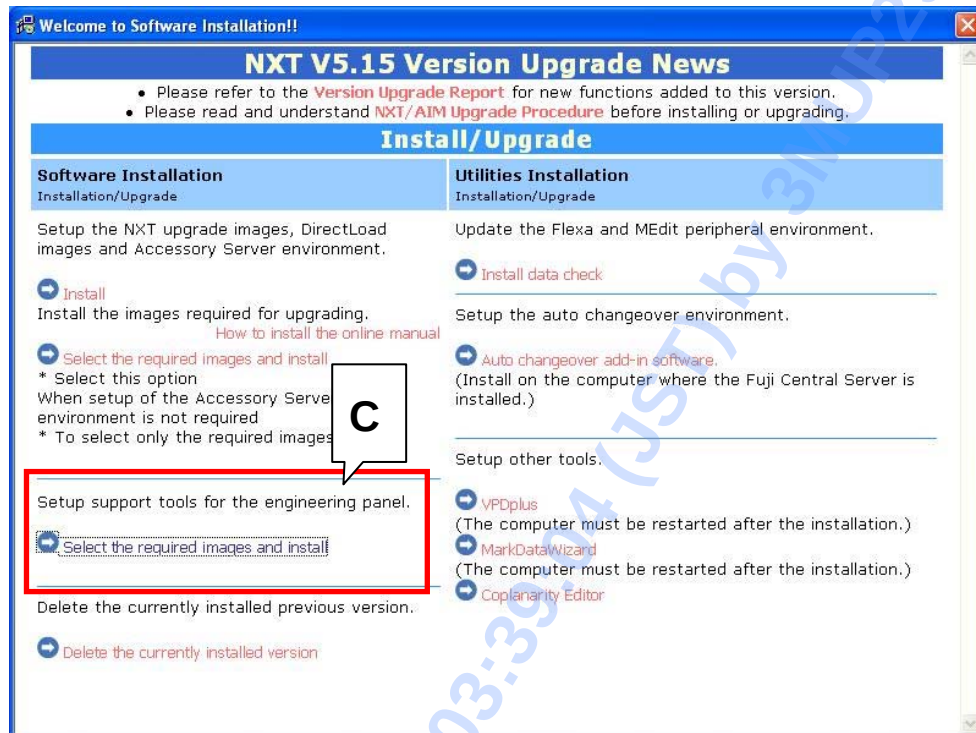
1. Set the NXT application software CD No.1 in the drive of the PC on which NXT Accessory Software has been installed (the Accessory Server). After a moment, the menu screen displays.

To install all the images required for upgrading, click "A". To select only the required images, select "B". Follow the wizard instructions to complete the installation.



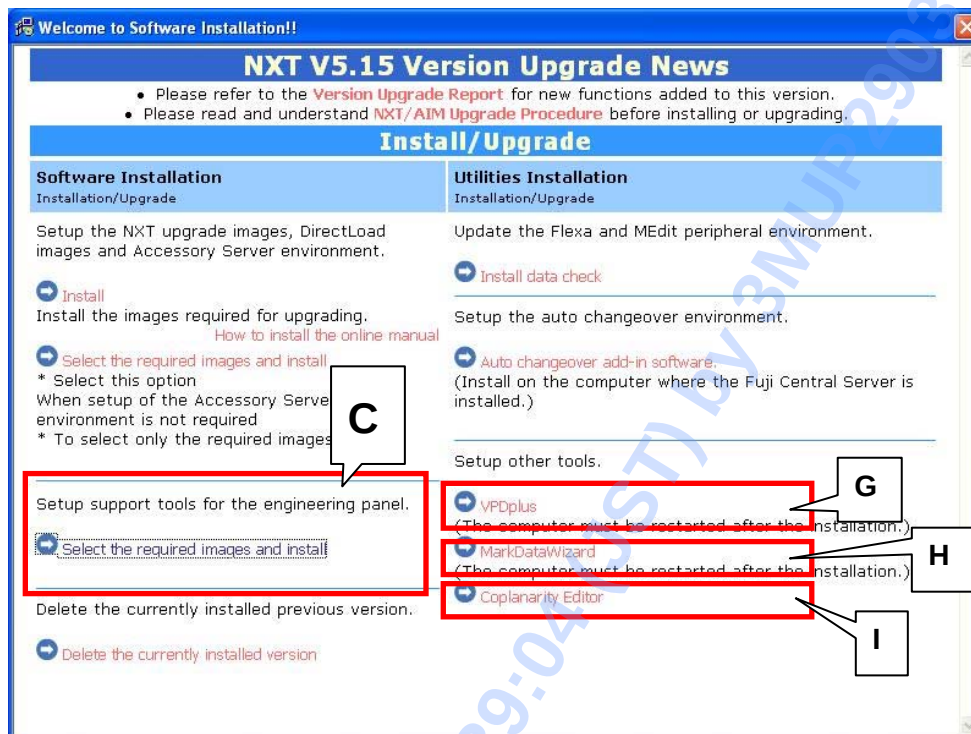
1.2.2 Installing in the Machine Communication Server PC

1. Set the NXT application software CD No.1 in the drive. When the menu screen displays, click "C". Check the Fuji Realtime Workingrate Buffer Service checkbox, and then follow the wizard instructions to complete the installation.



1.2.3 Installing in the Engineering Panel

1. Set the NXT Application Software CD No.1 in the Engineering Panel drive. And then the following menu screen will be displayed.



2. Click “C”, select the required items, and then follow the instructions in the wizard to complete the installation.

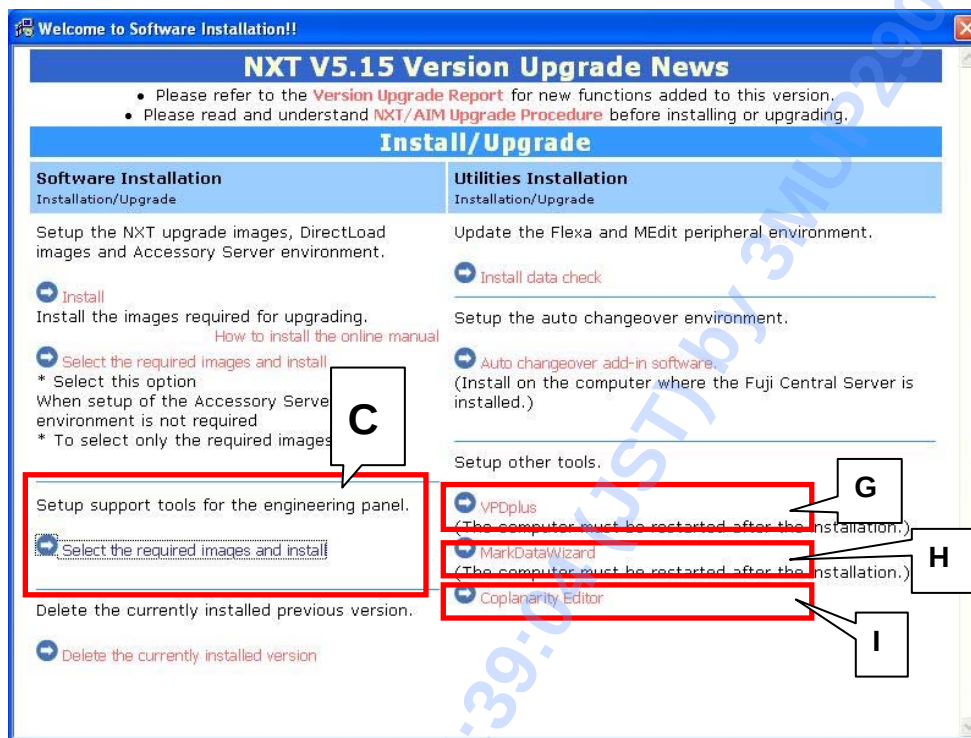
- MEdit
 - Data Check
 - Image Launcher for VPD plus
 - Viewer Soft
 - Equipment Check
 - IPS Viewer
- } **C**

3. To install VPD plus, Mark Data Wizard and Coplanarity Editor, click “G”, “H” and “I” respectively.

- VPD plus (license required) ----- **G**
- Mark Data Wizard ----- **H**
- Coplanarity Editor (for Coplanarity Option) ----- **I**

1.2.4 Installing in the programming PC

1. Set the NXT Application Software CD No.1 in the Engineering Panel drive. And then the following menu screen will be displayed.



2. Click “C”, select the required items, and then follow the instructions in the wizard to complete the installation.

- MEdit
 - Data Check
 - Image Launcher for VPD plus
 - Viewer Soft
- } **C**

3. To install VPD plus, Mark Data Wizard and Coplanarity Editor, click “G”, “H” and “I” respectively.

- VPD plus (license required) ----- **G**
- Mark Data Wizard ----- **H**
- Coplanarity Editor (for coplanarity option) ----- **I**

Chapter 2

Upgrading the Machine Software

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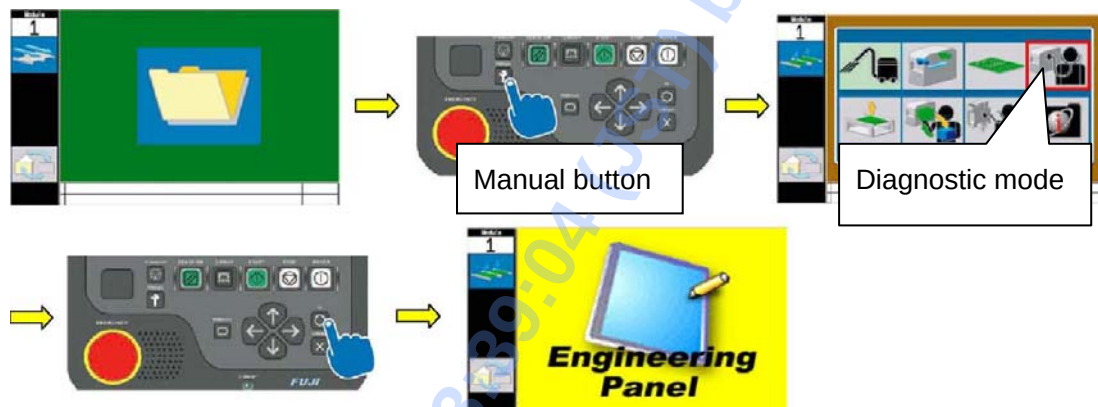
2. Upgrading the Machine software

2.1 Upgrading the machine application software

Installing the application software to the base and modules.

2.1.1 Preparation on the NXT

1. Remove the board if there is any during production using the board removal command.
2. Return the nozzle using the Nozzle Exchange command.
3. Set all the module to Diagnostic mode.

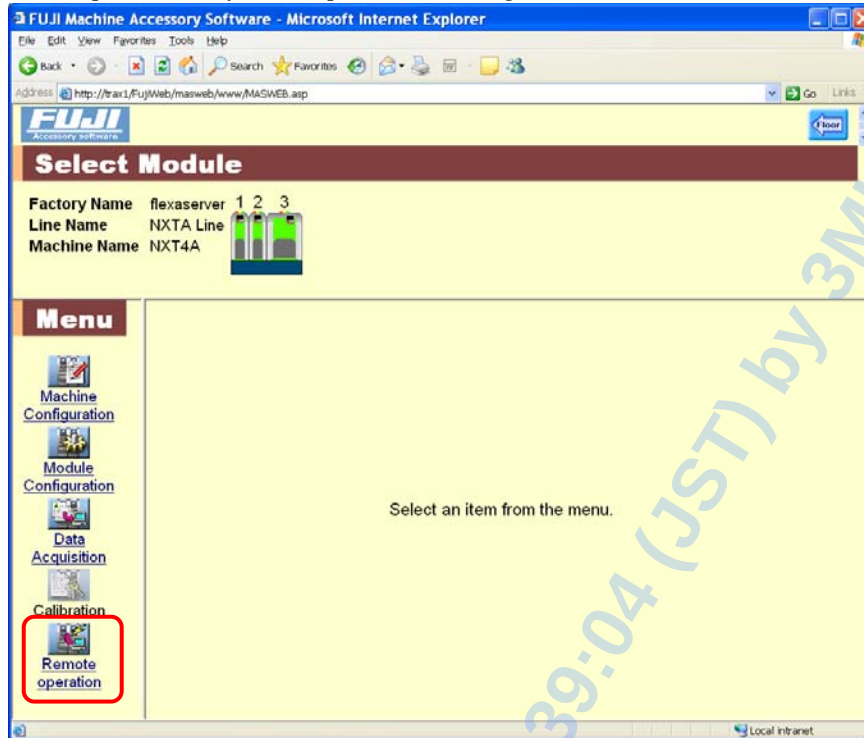


Note: Machine upgrade can still be performed even when the module is not in sleep mode or self-diagnosis mode. Be cautious that the following data will be cleared depending on setup made in Accessory Software.

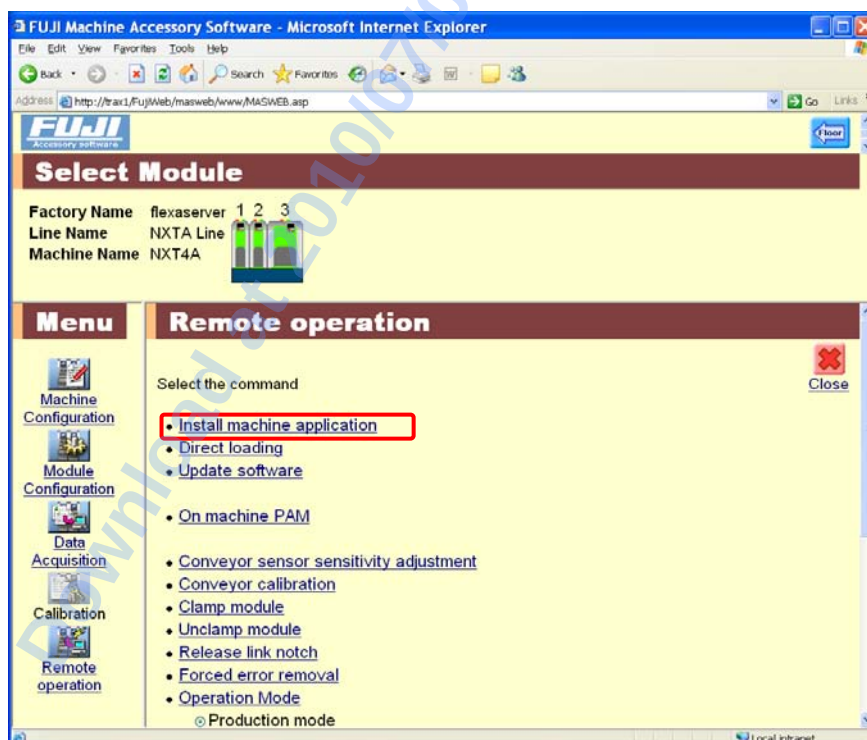
1. Machine Configuration Settings
 2. Production job currently being transmitted
 3. Automatic Calibration
-

2.1.2 Accessory Software Procedure

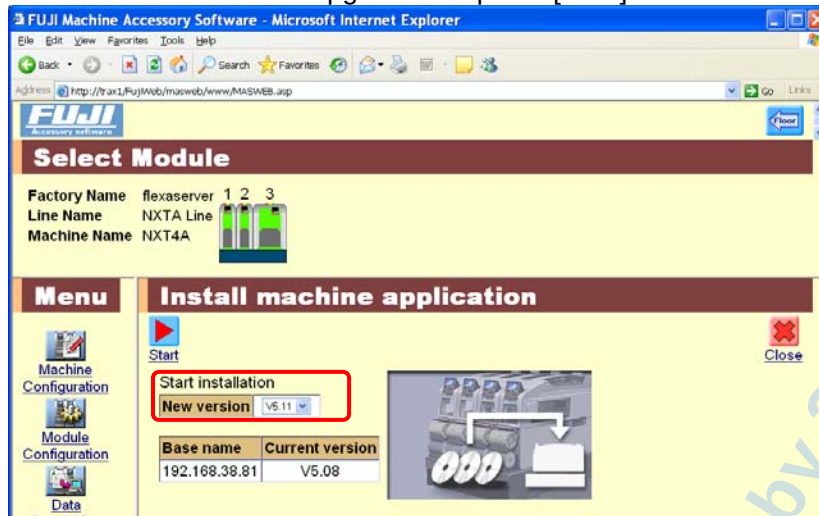
1. Start Accessory Software.
2. Select [Remote Operation] after selecting the whole line.



3. Select [Install Machine Application].



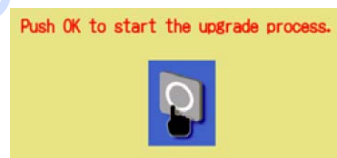
4. Select the version to be upgrade and press [Start].



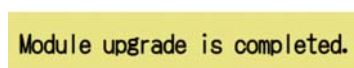
5. [OK] is displayed when the installation is completed.
6. Reboot the machine.

2.1.3 Upgrading the module version

1. Press the module POWER button to wake the module and display the “Push OK to start the upgrade process” dialog.



2. Press the [OK] button to start the upgrade. Wait until “Module upgrade is completed” displays on all the modules



3. Reboot the NXT.
4. Refer to “Chapter 2 – Upgrading the various unit software” to upgrade the software for the various units.

2.2 Various Unit Software

The following peripheral units of NXT/AIM operates using software. Unit software needs to be updated when certain functions addition or defect correction is made. The update method of various unit software is explained here.

2.2.1 Peripheral units which can be updated

The following peripheral units can be update.

1. Conveyor Unit (Loader Program)
2. Head Unit
3. Device Pallet
4. Tray Unit-L
5. Tray Unit-M
6. Feeder

Note: The change of DIP switch is required for device pallet upgrade. Refer to the "Device Pallet & Tape Feeder Upgrade Manual" in NXT Software CD-ROM for detailed information.

Perform various unit software updates using Accessory Software to update all the units mentioned above.

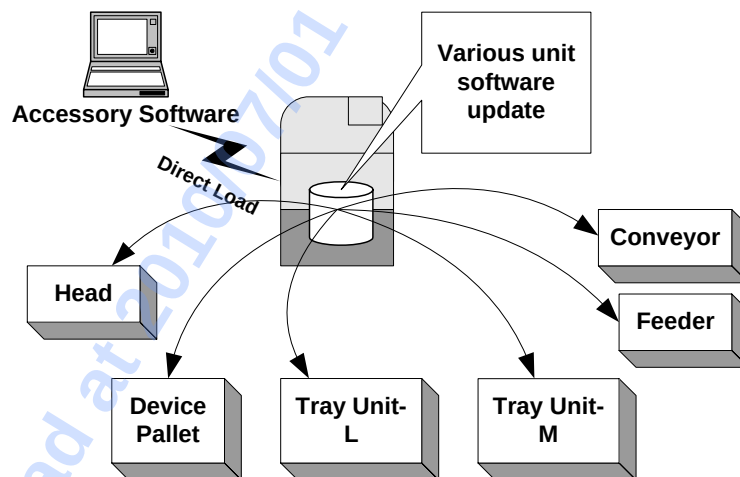
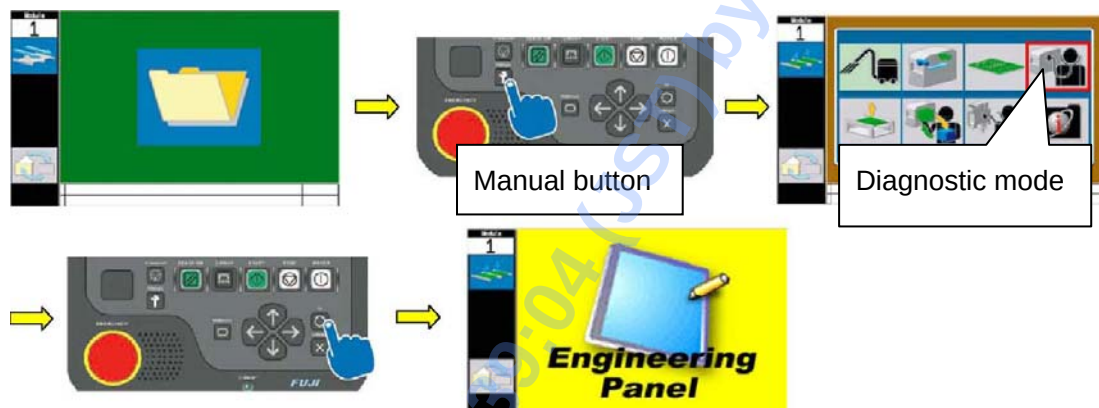


Fig. 2.1 Various unit software update

2.2.2 Preparation for Various Unit software update

The various unit software update is performed using [Remote Procedure] in Accessory Software (Engineering Panel). The procedure is shown below.

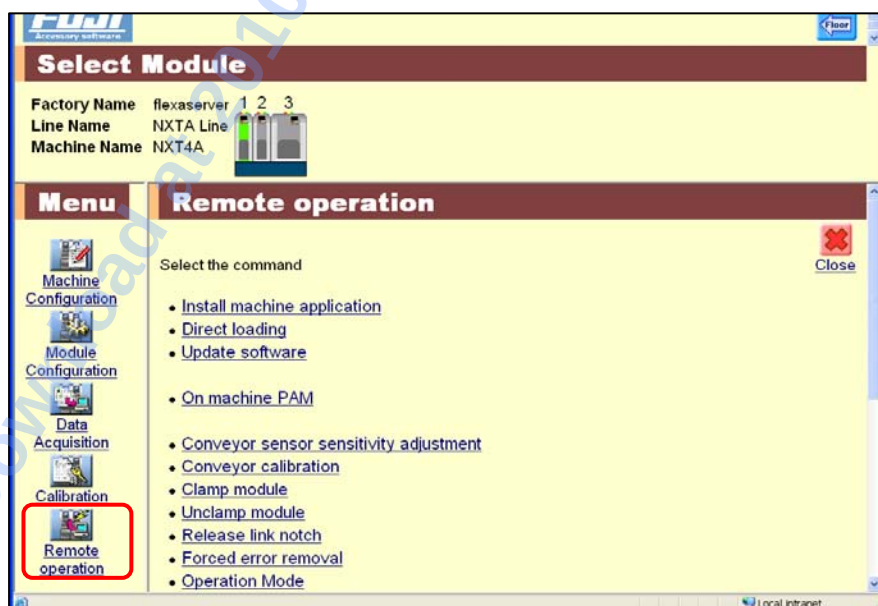
1. Remove the board if there is any during production using the board removal command.
2. Return the nozzle using the Nozzle Exchange command.
3. Prepare and load the units to be software updated into the module.
4. Refer to the "Device Palette & Tape Feeder Upgrade Manual" when updating the device pallet. Do not load other units especially feeder or tray unit-M.
5. Set the target module to Diagnostic mode.



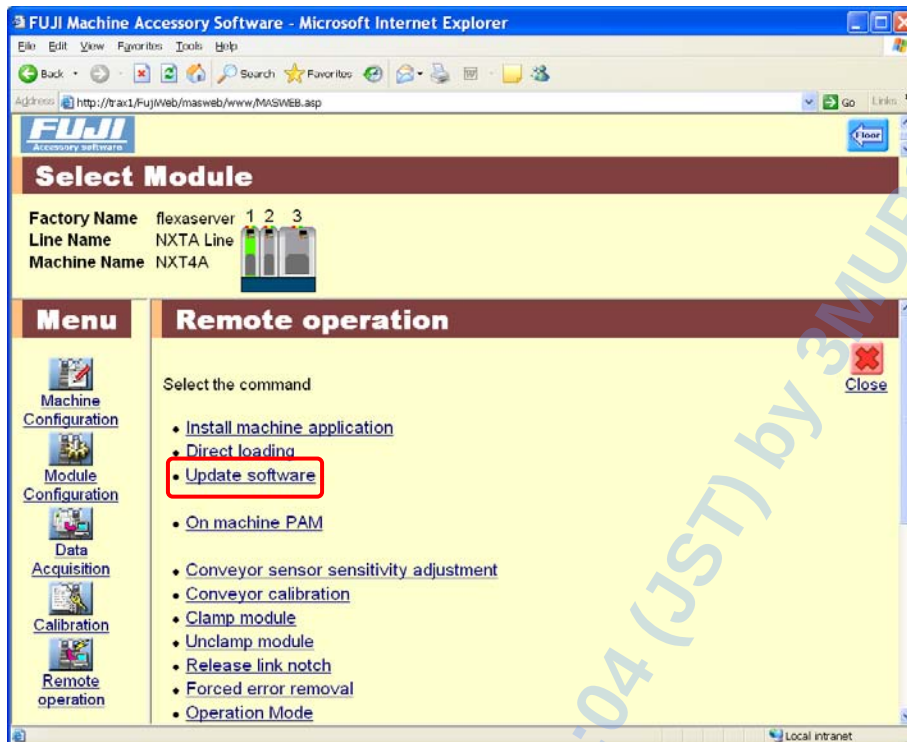
2.2.3 Accessory Software Procedure

Various units software update is performed in Accessory Software as follows.

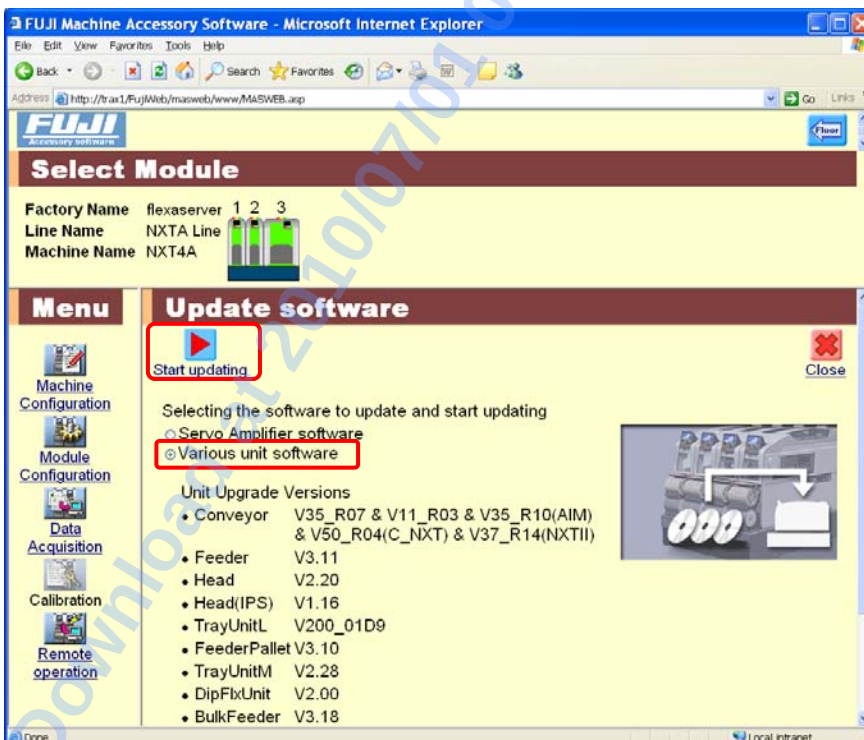
1. Start Accessory Software.
2. Select [Remote Operation] after selecting the module.



3. Select [Update Software].



4. Select [Various units Software] – [Start Updating].



5. The module various unit software update software starts after a while.

2.2.4 Screen Composition Explanation & Procedure

The screen composition of various units software update is shown and explained below.

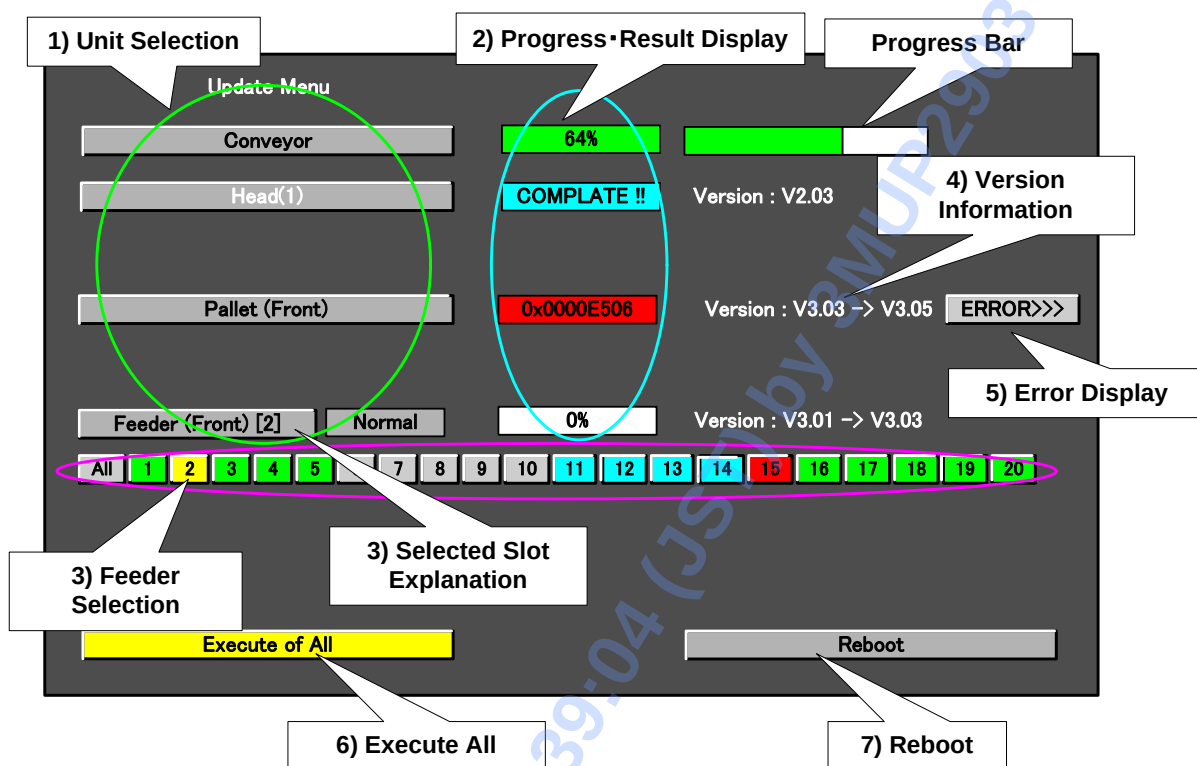


Fig. 4.1 Menu Screen

<Screen Detailed Explanation>

1) Unit Selection

Perform units selection for update. The selected items turns to yellow.

The units that can be upgrade is shown in black letter. If there is nothing white letter is displayed.

[Pallet (Front) is the tray unit-L when tray unit-L is loaded.]

2) Progress•Result Display

The progress state and result is displayed for each unit.

[Not Updated]	: White "0%" displayed
[Same Version]	: Blue "SAME" displayed
[Finish Update]	: Blue "COMPLETE!!" displayed
[Error]	: Red Error Code displayed
[Updating]	: Green Progress% displayed
[Not loaded]	: No display

3) Feeder Selection

Select the slot for update.

Each feeder update condition, result etc. are displayed by color classification.

[Not updated]	: Green
[Same Version or Finish Update]	: Blue
[Error]	: Red
[Selecting]	: Yellow
[Not loaded]	: Grey

The explanation of the unit which exists in a selected slot is displayed above.

4) Version Information

The units current version and version to be upgrade are displayed.

The left sides shows the current version and the right sides shows the version to be upgrade.

5) Error Display

Error occurred during update procedure is displayed.

The error detailed information is displayed when selected.

6) Execute All

Select this to perform an update on all units.

7) Reboot

Select this to end the update procedure.

End the various units software update software and reboot. The normal application starts.

2.2.5 Operation Method Explanation

1) Unit Selection

Use the operation panel arrow keys   to select unit.

Update starts when OK switch is pressed when unit is selected.

However, the following units condition cannot be selected.

- Finish Update
- Unit is not loaded
- Same Version (Can be selected only if the conveyor version is the same)

2) Slot Selection

All feeders or separate slot can be selected for feeder.

Separate slot can also be selected for Tray Unit-M

After selecting the feeder items with the   arrows keys, select the slot using   arrows keys. The feeder conditions and version is displayed only on the slot currently being selected.

Press OK switch to start the update.

[All] : All the loaded feeder and tray unit-M is updated in order.

[Slot] : Update the feeder or tray unit-M currently loaded into the reserved slot.

Same as the unit selection, the following units condition cannot be selected

- Finish Update
- Unit is not loaded
- Same version

2.2.6 All units batch update

Figure 4.1 6) [Execute of All] button can be selected when various units software update software is started.

All the loaded units starts update in order when the OK switch is pressed.

<Bundle/Batch Update Characteristics/Features>

- Update all the loaded units.
- No update if the version is the same except for conveyor.
- The unit which failed the update stops processing and moves to the next units update.
- The batch update ends automatically when processing of all the units is carried out normally and normal application starts afterwards .
- The unit which had error occurred can be checked if any failed units exists during the batch update. Select [Reboot] to starts the normal application.

Warnings

- ***Do not remove the unit during update procedure.***
 - ***Do not turn OFF the machine breaker during update procedure.***
 - ***The unit cannot be used when it is removed during update procedure.***
 - ***Refer to the "Device Palette & Tape Feeder Upgrade Manual" when performing device pallet upgrade.***
-

2.2.7 Only Conveyor Controller Update

The conveyor unit loader programme can be updated when [Conveyor] is selected in the unit selection.

Only conveyor unit can be updated even when the same version is used. Select [Reboot] after the update ends to starts the normal application.

Note: Perform conveyor unit update procedure once again if the machine breaker has been turned OFF during loader program update.

2.2.8 Only Head Update

The head unit can be updated when [Head(1)] is selected in the unit selection.

Select [Reboot] after the update ends to start the normal applications.

Warnings

- *Do not remove the head during update procedure.*
 - *Do not turn OFF the machine breaker during update procedure.*
 - *The unit cannot be used when it is removed during update procedure.*
-

2.2.9 Only Device Pallet, Tray Unit-L Update

The device pallet or tray unit-L can be updated when [Pallet(Front)] is selected in the unit selection.

Select [Reboot] after the update ends to start the normal applications.

Notes:

- *Refer to the "Device Palette & Tape Feeder Upgrade Manual" when performing device pallet upgrade.*
- *Perform Device Pallet upgrade before performing the feeder and tray unit-M update.*
- *There is no problem if [Execute of All] is selected because the device pallet will be update first.*
- *If the tray unit-L is attached to replace the device pallet, tray unit-L update will starts.*

Warnings

- *Do not remove the device pallet or tray unit-L during the update procedure.*
 - *Do not turn OFF the machine breaker during the update procedure.*
 - *The device pallet and tray unit-L cannot be used if they are removed during the update procedure.*
-

2.2.10 Only Feeder, Tray Unit-M Update

The feeder or tray unit-M can be updated when [Feeder(Front)] is selected in the unit selection.

Select [Reboot] after the update ends to start the normal applications.

Feeder Update Characteristics/Features

- The update can be performed on slot unit by selecting arrow keys after selecting the [Feeder(Front)].
- All the loaded feeders and tray unit-M updates can be performed when [All] is selected in the slot selection.
- The loaded feeders and tray unit-M in the reserved slots updates can be performed when slot selection is performed.
- Since the feeder insertion is being monitored, remove the feeder which had finished the update. Then insert the new feeder and perform the update again.
- Since the tray unit-M insertion is being monitored, remove the tray unit-M which had finished the update. Then insert the new tray unit-M and perform the update again.

Note: Sometimes the device pallet update is required before updating the feeder and tray unit-M.

Feeder : Require Device Pallet V3.04 and above

Tray Unit-M : Require Device Pallet V3.06 and above

Refer to the [Device Pallet & Tape Feeder Version Up Manual] in the NXT Software CD-ROM for detailed information.

Warnings

- **Do not remove the tray unit-M during the update procedure.**
 - **Do not turn OFF the machine breaker during the update procedure.**
 - **The feeder and tray unit-M cannot be used if they are removed during the update procedure.**
-

2.2.11 When error occurred during unit update

Perform the following measures when error occurred or unit upgrade procedure went wrong.

<Coping with error occurred>

1. The figure 4.1 5) button can be selected when an error occurred. Select the arrow keys and press OK button.
2. Note down the error code displayed when the page changed.
 - Error Code
 - Sub Error Code
3. Check the unit connection conditions.
 - The RS-232C connector between the conveyor controller and CPU box
 - Head unit connector fitting and extraction pin
 - Device pallet, tray unit-L seating condition
 - The feeder, tray unit-M power check
4. Repair is needed when there is a unit connection problem and when the unit cannot be used. Contact our company call center and attach the error code to the unit which failed in update.

Note: The update procedure can be performed once again for conveyor unit.

2.3. Servo Amplifier Software Update

The servo amplifier update software has been removed from the NXT application CD from version V5.17 onwards and it is now necessary to sign a licensing agreement to obtain a copy of the software. Please refer to Rep News No.10039 for further details. Only Fuji staff and agent engineers are allowed to have copies of the software.

2.3.1 Required items

- PC (WindowsXP SP2 and higher)
- Servo amplifier update software CD(direct load).

2.3.2 Cautions

Caution: *If the servo amplifier is one of the models shown below, do not perform the version upgrade unless instructed to do so in the version upgrade report. In certain cases the version upgrade fails and the servo amplifier cannot be restored.*

Model: CACR-0401IS04-FK (For M3 & M6 modules)

Model: CACR-0401IS03-FK (For M3 & M6 modules)

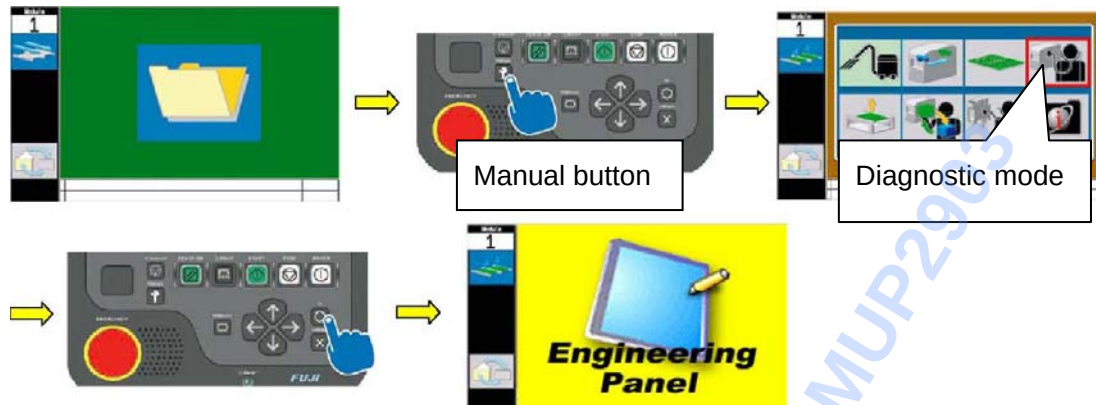
For further details please refer to RepNews No.05140.

Caution: *The update cannot be performed without the servo amplifier update software supplied by Fuji. Nor can it be performed using data copied from the CD to other storage media such as hard disks. Therefore, be sure to use the CD supplied by Fuji.*

2.3.2 Servo Amplifier Software Update

1. Remove the board if there is any during production using the board removal command.
2. Return the nozzle using the Nozzle Exchange command.

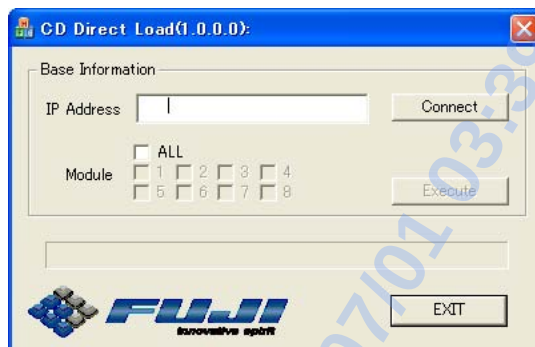
3. Set the target module to Diagnostic mode.



4. Set the servo amplifier update software in the PC.

5. After a while the software starts up and the following screen displays. **(Note: the software may take longer to start up than other applications).**

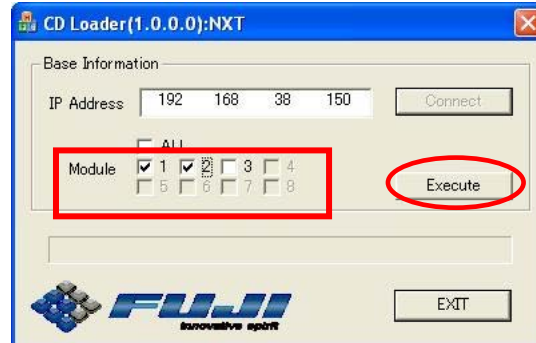
If it fails to start up, double-click the "CDLoader.exe" file.



6. Enter the IP Address of the target base in the [IP Address] field, and then click [Connect]. If the following error message displays, check the network connection between the PC and base.



7. Once connected to the base it becomes possible to select the module checkboxes which were previously grayed out. Select the number of the module you wish to update or [ALL] to select all the modules, and then click [Execute].

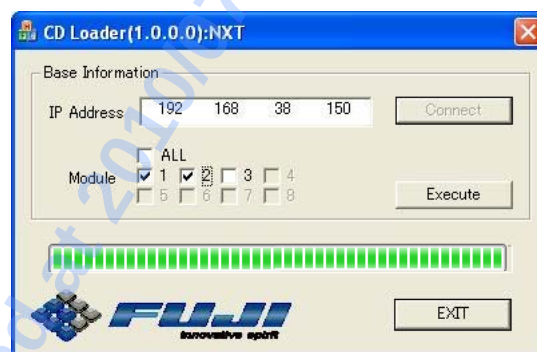


8. Click [Yes] in the confirmation dialog.



9. Direct load starts and after a while the module operation panel monitor turns yellow and the update proceeds automatically. The update is complete once "DONE" displays on the module operation panel monitor.

10. Click [EXIT] to close the application.



11. Turn the base main switch OFF, restart the machine in normal mode and check that it operates normally after the upgrade.
12. The servo amplifier update is now complete.

Chapter 3

Emergency Install & CPU Box Replacement

Download at 2010/07/01 03:35:04 (JST) b12MUP2903

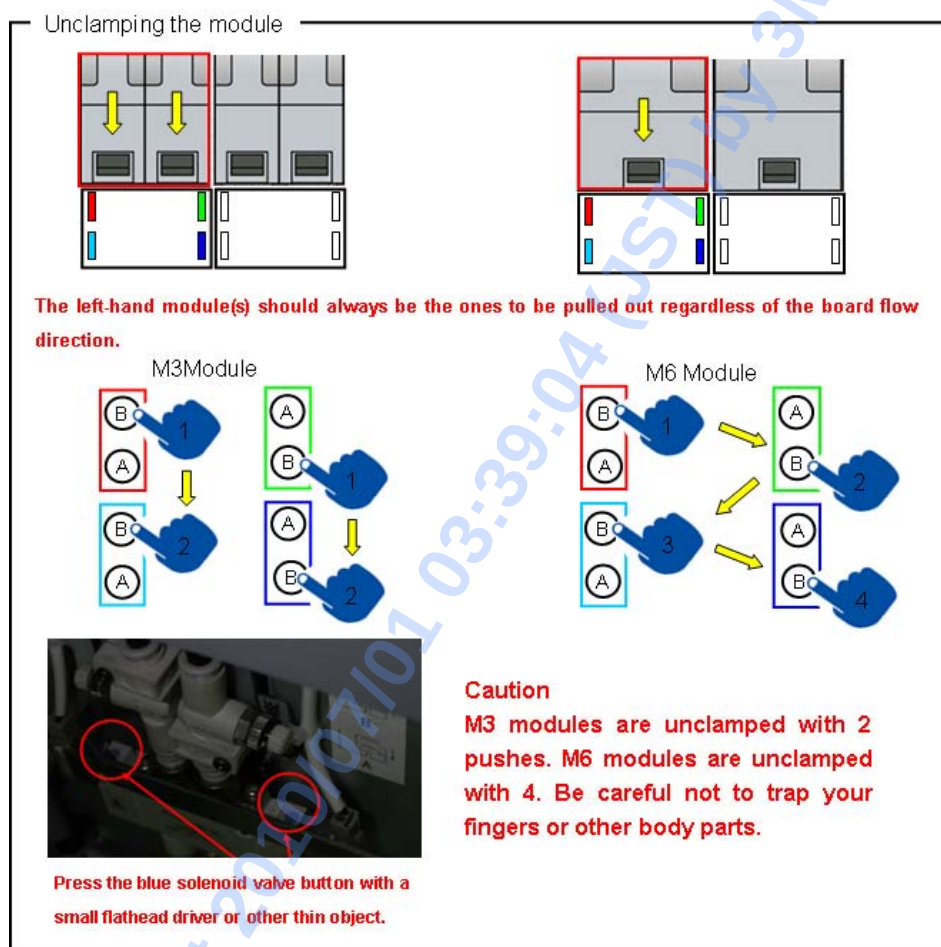
CONFIDENTIAL

Download at 2010/07/01 03:39:04 (JST) by 3MUP2903

3. Emergency Install & CPUBox Replacement Procedure

3.1. Pulling out the module

Neither the manual command nor Accessory Software can be used to unclamp the module when the CPU BOX cannot startup. In such cases it is necessary to turn the base main switch OFF and then follow the procedure below to manually unclamp the module with the solenoids.



Caution: the interlock function is disabled when using the solenoids to unclamp the module. Pulling out more modules than is necessary may result in the machine tipping over.

Caution: take care to avoid trapping your fingers or hitting your head while unclamping the modules.

3.2. About Emergency Install

Emergency Install is used to recover the system when the machine software fails to startup for some reason. If the NXT application installation is interrupted by a power outage or network error, the NXT will not be able to start up. In such cases it is necessary to perform Emergency Install. If the BASE CPU BOX cannot start up correctly, all the modules on the base will not be able to start up. If a particular MODULE CPU BOX cannot start up correctly, the problem will be limited to that module. In this way you can tell if the problem is a result of trouble with the BASE CPU BOX or the MODULE CPU BOX. During Emergency Install a PC is connected directly to the CPU BOX so it is necessary to prepare a PC for this purpose.

Preparation

The following items are required to perform Emergency Install:

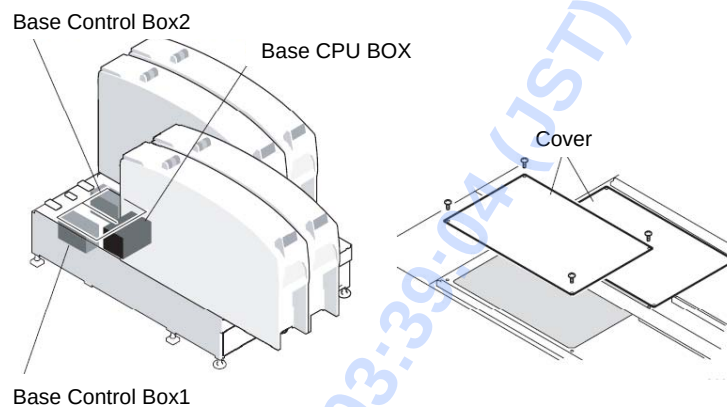
- Laptop PC (required)
- Cross Ethernet cable (required)
- NXT Application software CD (the same version as that installed on the machine) (required)
- Final Offset Data (required for base).
- Matrix measurement data (required for module).
- Device levelness measurement data (required for module).
- 4 nozzle head parts camera measurement data (required for module CPU Box replacement if the application version is lower than V4.10).

3.3. Performing Emergency Install for the Base CPU Box

3.3.1 Formatting the PC Card

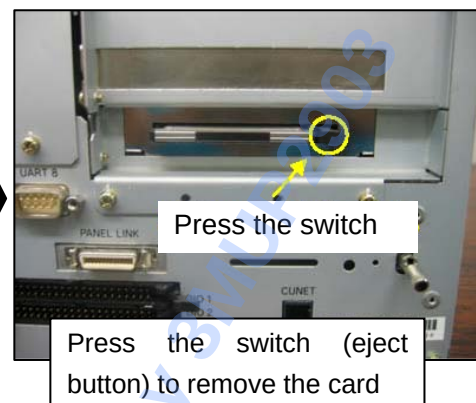
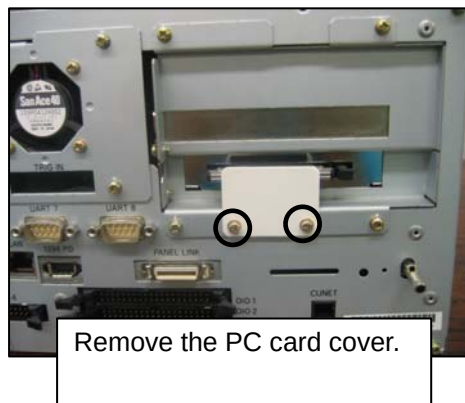
When the CPUBOX is damaged, the PC card data may also be damaged, therefore it is necessary to format the PC card. When formatting the PC card, the final offset data and machine function settings are erased. Be sure to backup the final offset data in advance.

1. Unclamp and pull out the left-hand module(s) (2 in the case of M3 and 1 in the case of M6). If the CPU BOX cannot boot up and the module cannot unclamp, follow the procedure in "3.1 Pulling Out the Module" to unclamp and pull out the module.

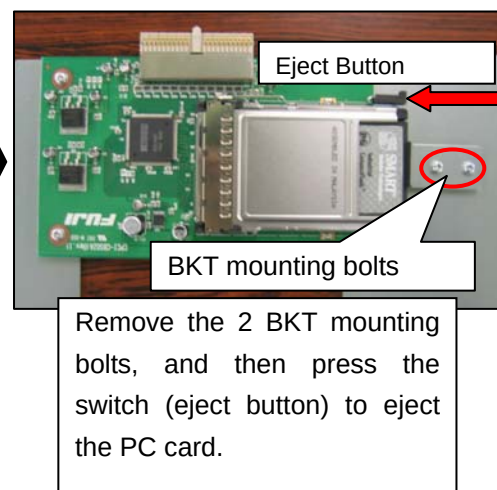
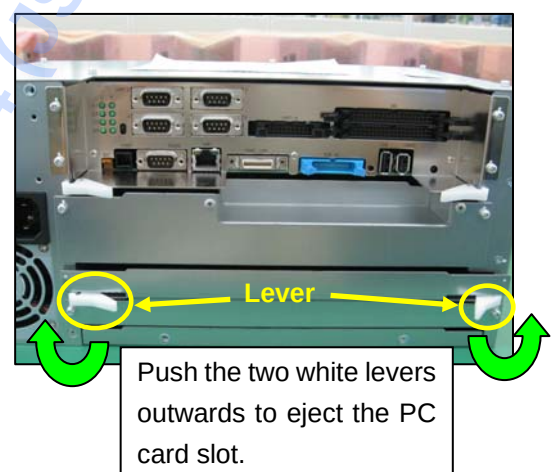
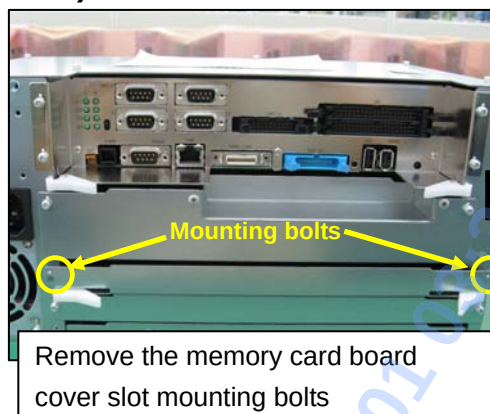


2. If the base power is ON, turn it OFF at the main switch.

3. Remove the PC card from the CPUBOX.
(NXT)



(NXT II)

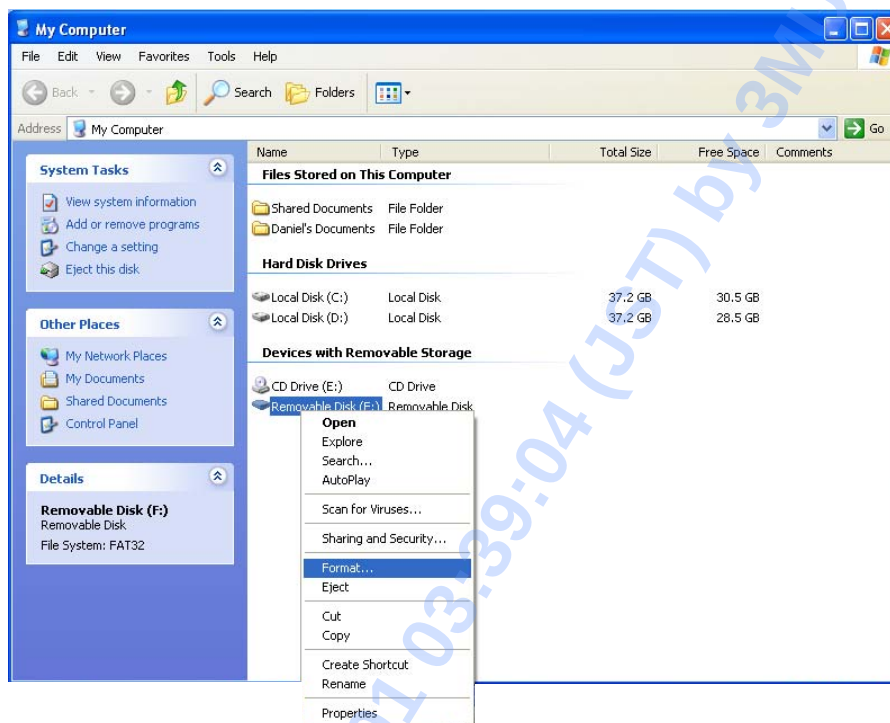


4. If you cannot find the backup disk, insert the PC card in the PC card slot of your PC and double-click [My computer]. Select [Removable Disk] to view the contents of the PC card. The final offset file is required after system reinstallation, so copy and save the final offset file to your desktop.

Location of final offset data:

Fuji/System/Data/FinalOfs

5. After saving the "FinalOfs" data to the desktop, format the PC card. As shown in the screen capture below, right-click the PC card drive and select Format.



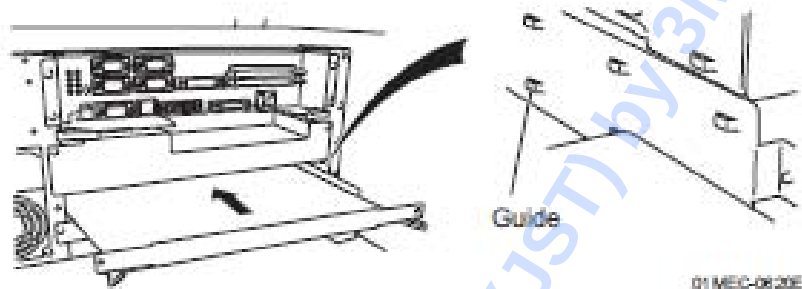
6. When the screen below displays, make the following settings.



- File System: Select FAT
(Do not select FAT32)
- Do not select the Quick Format checkbox
(Do not perform quick format).

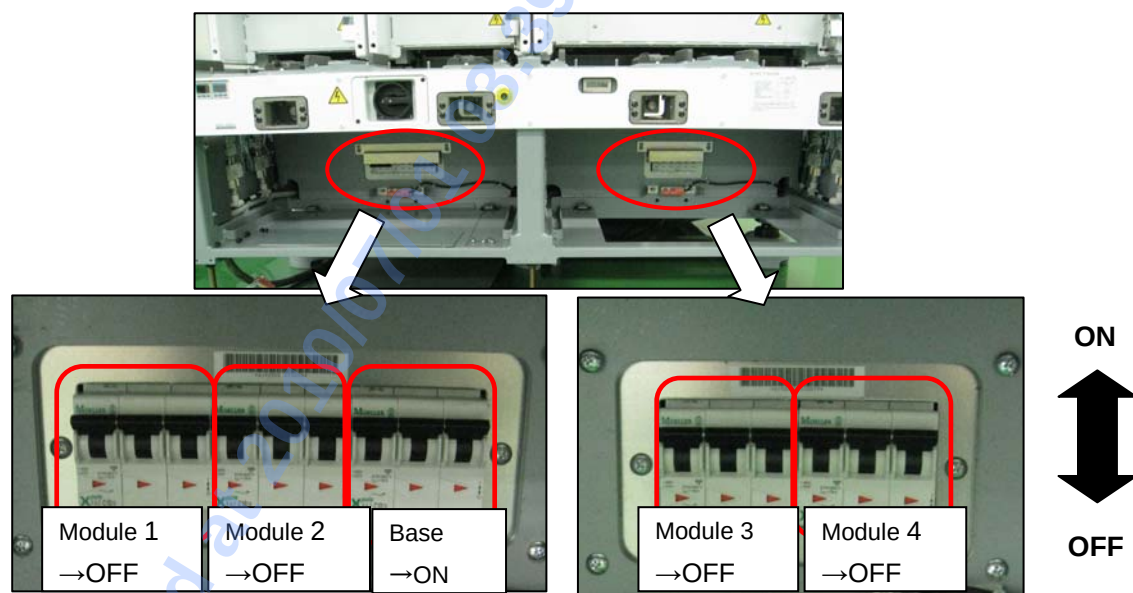
7. Once the format is complete, click the taskbar icon  and stop the PC card.
8. Insert the PC card to the memory card board and put the BKT.
9. Using the guides, insert the memory card board tray into the CPU box and tighten the screws.

Caution: Forcing the board into the CPU box without using the guides may result in damage. Be sure to run the board along the guides when inserting it.



3.3.2 Base CPU BOX Emergency Install

1. Turn OFF all the circuit protectors except the one for the base.



Circuit Protector Label

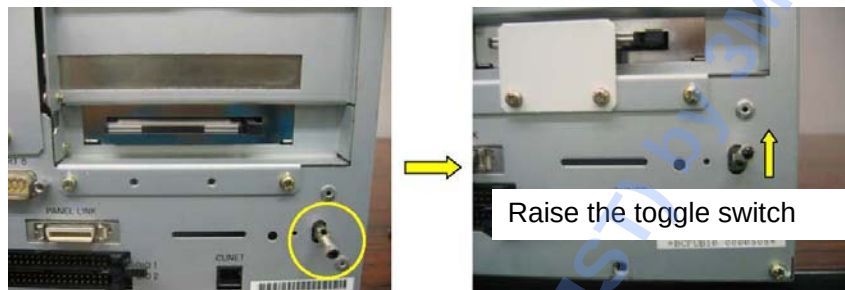
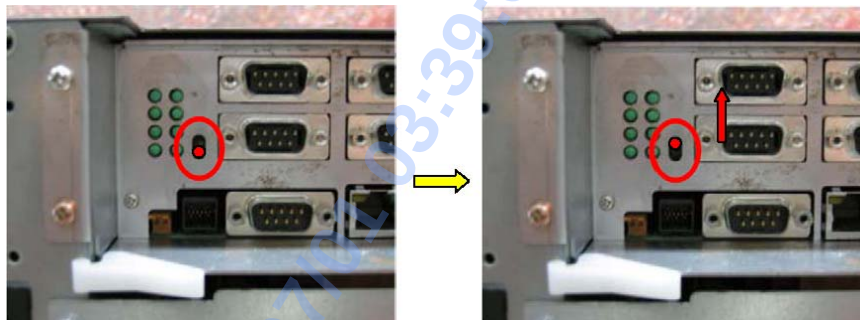
NXT

M1CP1: Module 1
M1CP2: Module 2
M1CP11: Base
CPA: Module 3
CPB: Module 4

NXT II

M1CP1: Module 1
M1CP2: Module 2
M1CP11: Base
M1CP3: Module 3
M1CP4: Module 4

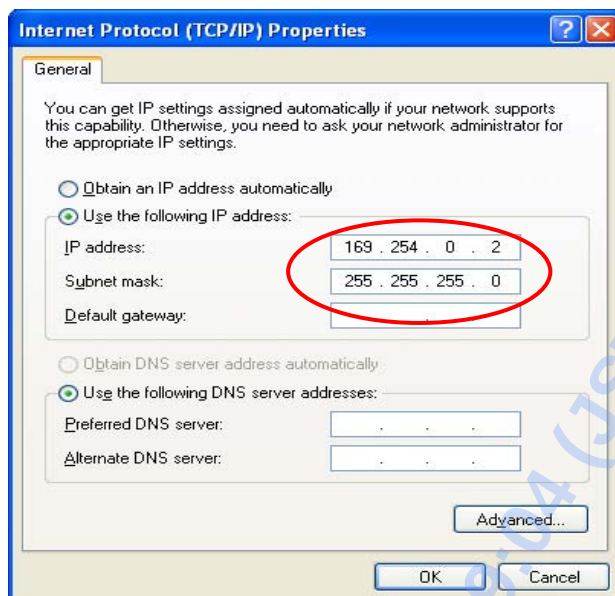
2. Remove the CPUBOX LAN cable and connect the cross Ethernet cable to the LAN connector.
3. Connect the other end of the cross cable to the LAN connector on the PC.
4. Raise the CPUBOX toggle switch.
Raising the CPUBOX toggle switch lever sets the CPUBOX IP address to 169.254.0.1.

(NXT)**(NXT II)**

5. Right-click [My Network] on the computer and select [Properties] – [Local Area Connection] – [Properties].

6. Select the [Internet Protocol (TCP/IP)] – [Properties] tab. Set the IP address of the PC to 169.254.0.2 and the subnet mask to 255.255.255.0 (the final digit of the IP address can be set to any value other than 1 or 0).

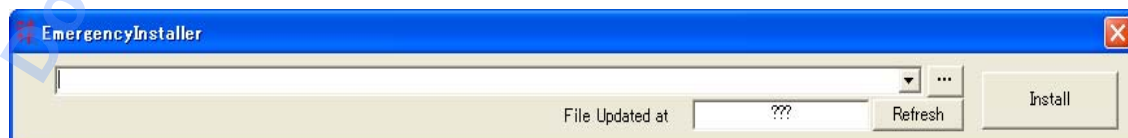
Note : The final digit of this IP address can be set to any value other than 1 or 0.
The IP address should not be set to 169.254.0.0 or 169.254.0.1.



7. Turn the main switch ON. The CPU boots up in safe mode with a fixed IP address of 169.254.0.1.
8. At the PC select [Start] – [Program] – [Accessories] – [Command Prompt] and enter “ping 169.254.0.1” to check the connection to the base CPU box. If you receive a reply from 169.254.0.1, then the connection has been established. If a “time out” message displays, recheck the LAN cable connection and network settings.
9. Insert the NXT application version CD in the PC CD drive. It should be the same version as the version installed on the machine.
10. Select CD: /NXT/Tool and double click “EmergencyInstaller.exe”.



11. The following screen displays.



12. Click [...] and select the bin file for the type of base you wish to restore (the file differs depending on the base type. Please refer to the table below). Select the file and click [Open] to display the path and file name in the text box.

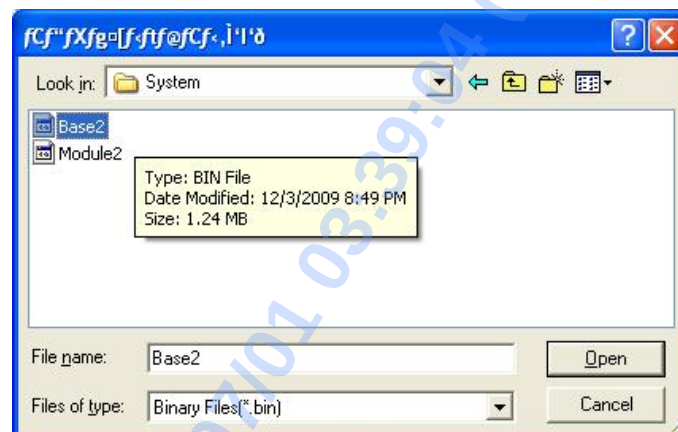
Caution: If you perform the installation with the wrong file, the CPU will not boot up.

<NXT>

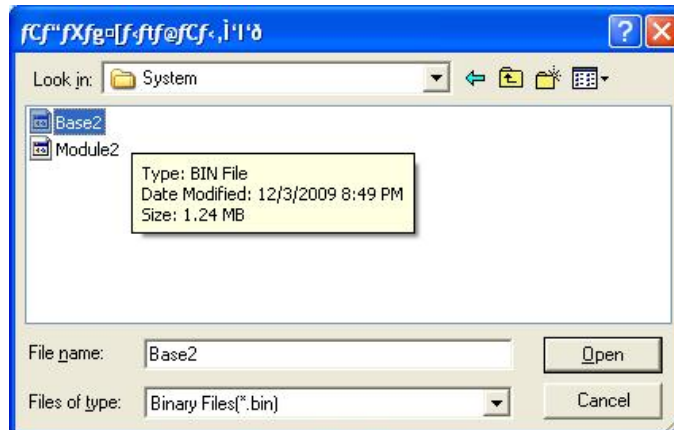
Install target		bin file
Base	2M, 4M, 8M	NXT\FujiInstSoft\Fuji\System\ Program\System\Base.bin

<NXT II>

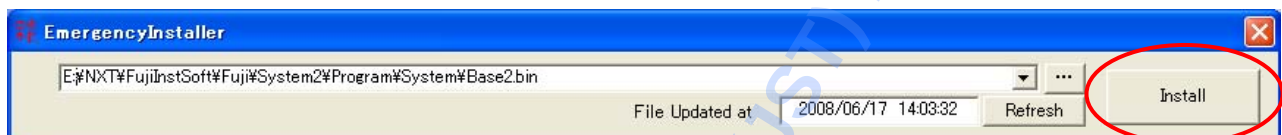
Install target		bin file
Base	2M II , 4M II	NXT\FujiInstSoft\Fuji\System2\ Program\System\Base2.bin



13. After selecting the bin file, click [Open].



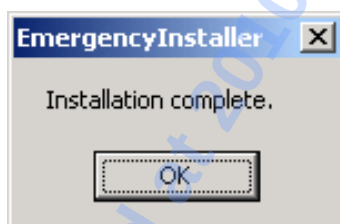
14. Click [Install] to install the selected file in the target CPU.



15. Installation progress displays in the "Installation Progress" dialog.



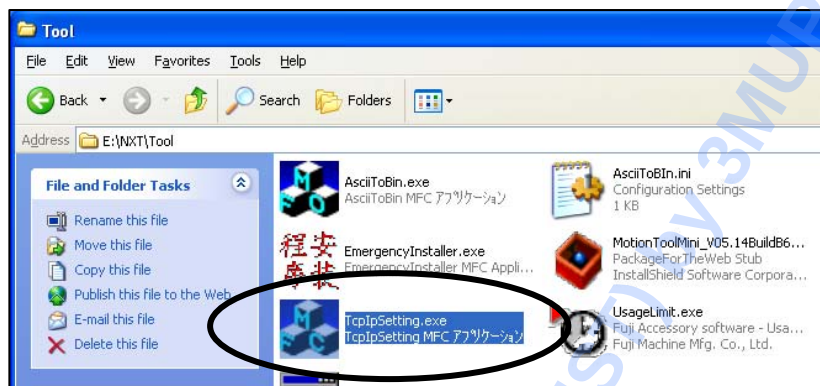
16. Once the installation is complete, the following message displays. Click [OK] to close the message box.



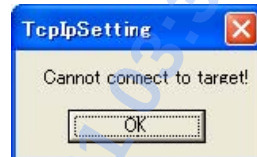
3.3.3 Setting the IP Address

TcpIpSetting can be used to change the base network settings when the base is set to safe mode (the toggle switch lever is raised).

1. If the CPU BOX is set to normal mode, set it to safe mode.
2. Turn the main switch ON.
3. Insert the NXT application software CD in the PC CD drive. Double-click Explorer\CD drive\NXT\Tool\TcpIpSetting.exe to open the TcpIp setting.

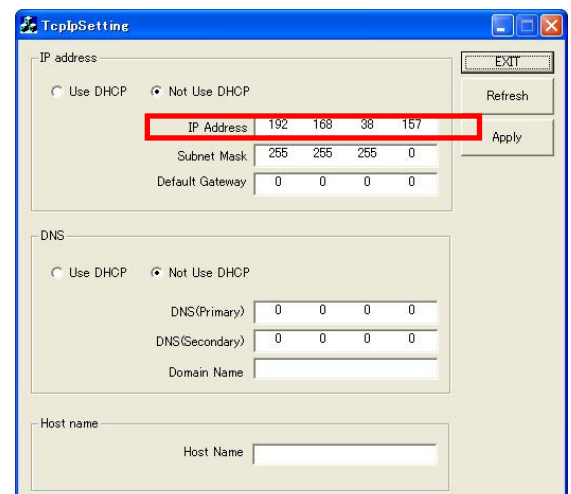
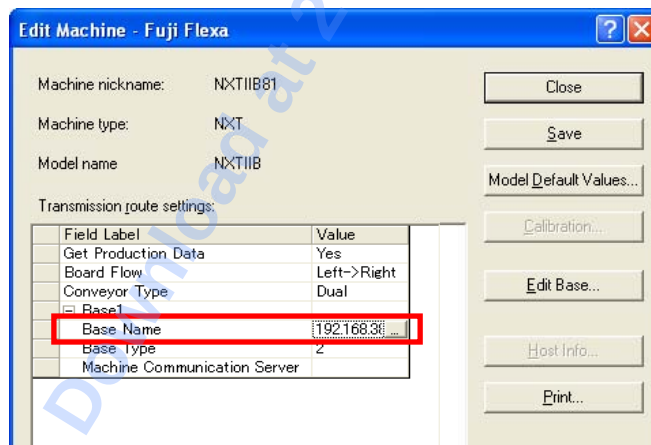


Caution: “Cannot connect to target” displays if the PC cannot connect to the CPUBOX. If this message displays, check the connection with the CPUBOX.

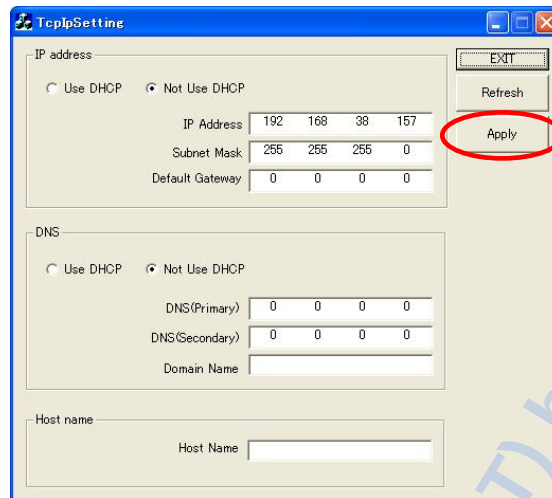


4. In the TcpIpSetting screen IP Address, enter the BaseName IP Address registered in FujiFlexa. Select FujiFlexa – Director – line (main) – and double-click the target machine to open the Machine Edit window.

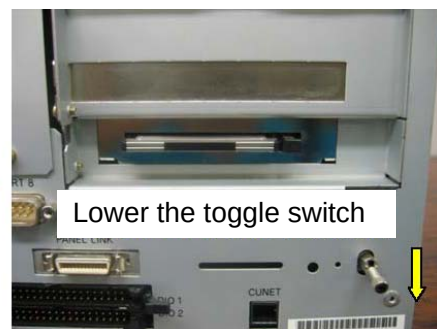
Enter the target machine's BaseName IP address in the IP Address field of the TcpIpSetting screen.



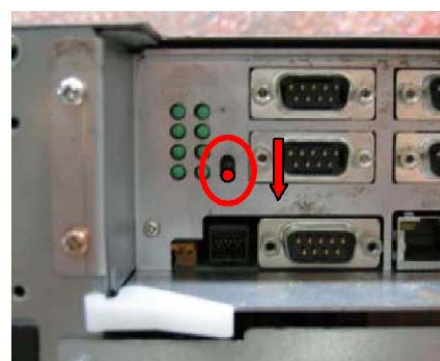
- Click [Apply] in the TcpIpSetting screen (it takes approximately 1 second for the setting to be applied after clicking [Apply]). Other items may need to be set depending on the user's network environment. A completion screen does not display after the settings are made.



- Turn the NXT base main power OFF.
- Switch the base CPU BOX back to normal mode.
(NXT)



(NXT II)

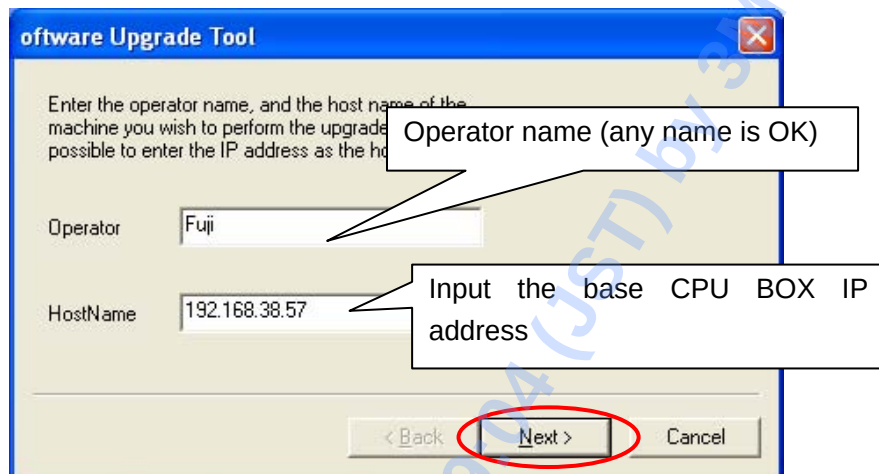


- Remove the cross Ethernet cable from the CPU BOX and connect the machine LAN cable in its place.

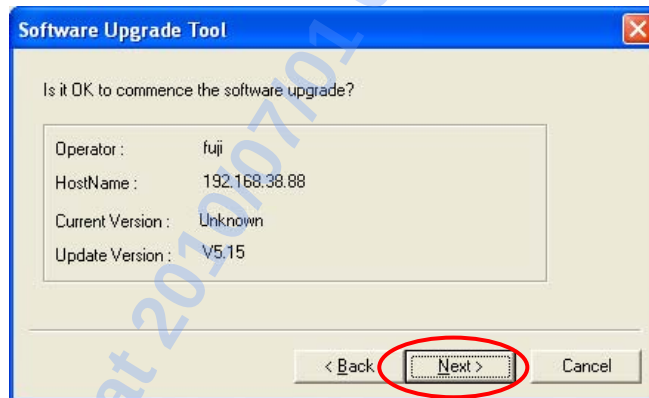
3.3.4 Installing the base CPU BOX application (VersionUp.exe)

VersionUp.exe does not use the Accessory Software, so it is possible to install the application software directly from the PC.

1. Turn the NXT base main power ON.
2. Return the PC IP address to the original address.
3. From the application CD select CD: /NXT and double-click "VersionUp.exe". Input an operator name (any name is OK) and the base IP address, then click [Next].



4. When the dialog shown below displays, click [Next].

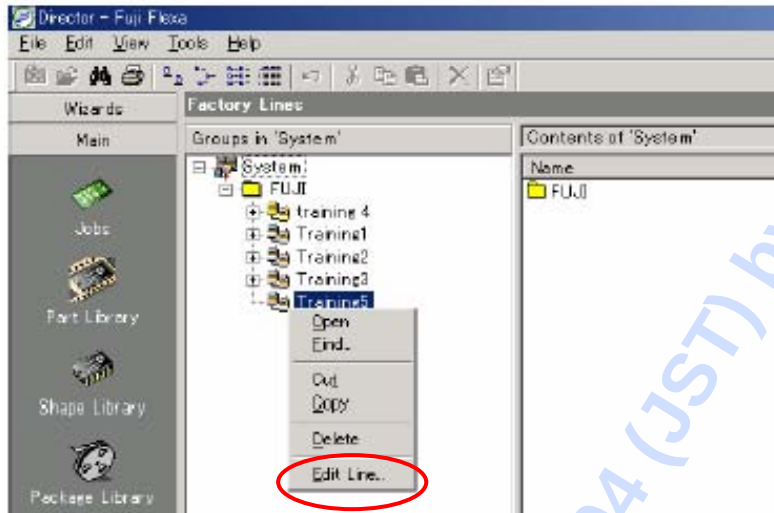


5. The installation takes around 15 to 30 minutes. Once it is complete, turn the base main switch OFF.
6. Turn ON the module circuit breakers, and then turn the base main switch ON.

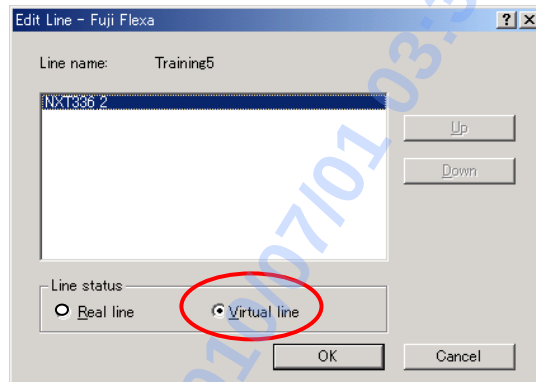
3.3.5 Checking the communication between the NXT and the HOST

Report line changes from Flexa to establish communication between NXT and the host.

1. Open director and select the line containing the NXT on which emergency install was performed. Right-click the line and select [Edit line].

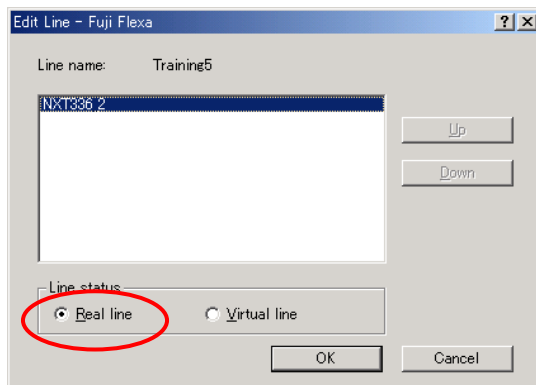


2. Select "virtual line" and click [OK].

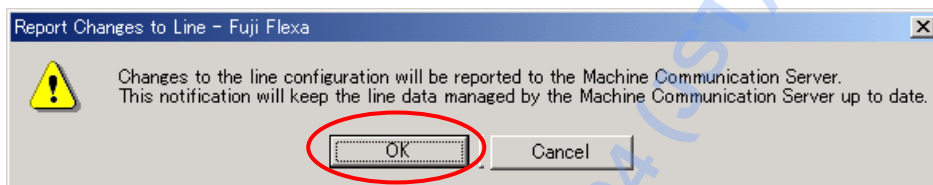


3. As in step 1, select the line containing the NXT on which emergency install was performed. Right-click the line and select [Edit line].

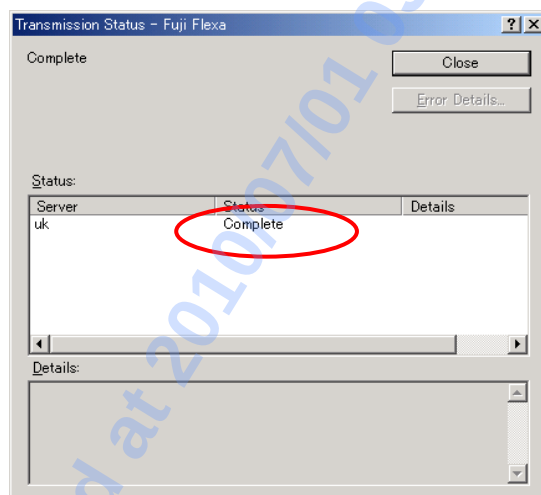
4. Select "Real line" and click [OK].



5. If the following message displays, click [OK].



6. Check that transmission was completed. If an error occurred, check the IP address and LAN cable.



3.3.6 Registering the Base CPU BOX data

Transmit the Final Offset file (bin file) to the base CPUBOX. Also check the machine configuration settings.

1. Turn the base main switch ON.
2. Press the module POWER button on the target base to enter wake mode.



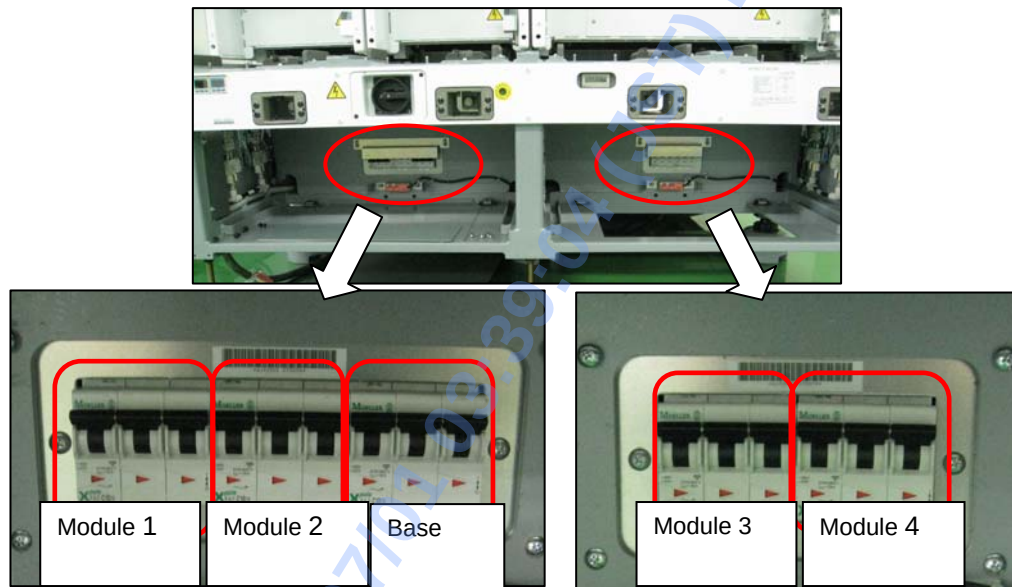
Caution: if the module application software is different to that of the base, “Push OK to start the upgrade process” displays on the screen. In this case, temporarily turn OFF the main switch and refer to “3.3.4 Installing the CPU BOX application (VersionUp.exe)” to install the same application version as the module to the base CPU BOX.

3. Please refer to “Supplement 2. Registering the Base CPU BOX data”.

3.4. Performing Emergency Install for the Module CPU Box

3.4.1 Module CPU BOX emergency installation

1. Unclamp and pull out the module on which you wish to perform Emergency Install. If you cannot unclamp the module because the base CPU BOX cannot boot up, follow the procedure in “3.1 Pulling out the module” to unclamp the module manually.
2. If the base power is ON, turn it OFF at the main switch.
3. Turn ON the circuit breaker for the base and for the module on which Emergency Install will be performed. Turn all the other circuit breakers OFF.



Circuit Protector Labels

NXT

M1CP1: Module 1
M1CP2: Module 2
M1CP11: Base
CPA: Module 3
CPB: Module 4

NXT II

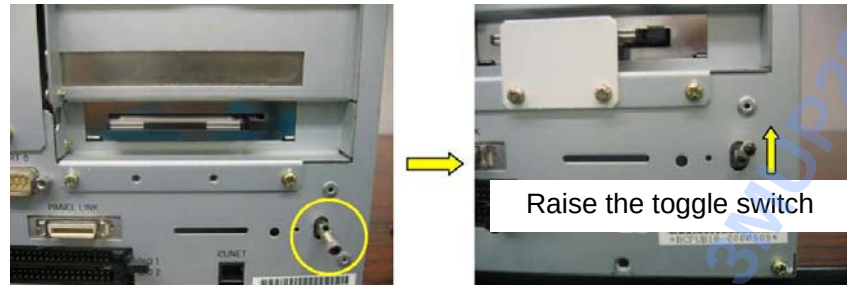
M1CP1: Module 1
M1CP2: Module 2
M1CP11: Base
M1CP3: Module 3
M1CP4: Module 4

4. Connect a cross Ethernet cable to the module CPU BOX.
5. Connect the other end of the cross cable to the LAN connector on the PC.

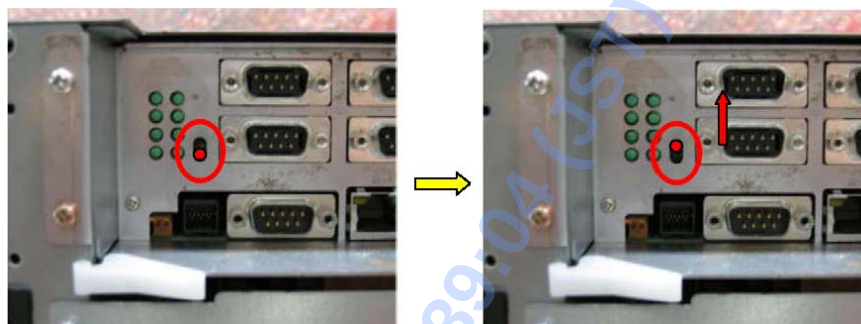
6. Raise the CPUBOX toggle switch.

Raising the CPUBOX toggle switch sets the CPUBOX IP address to 169.254.0.1.

(NXT)

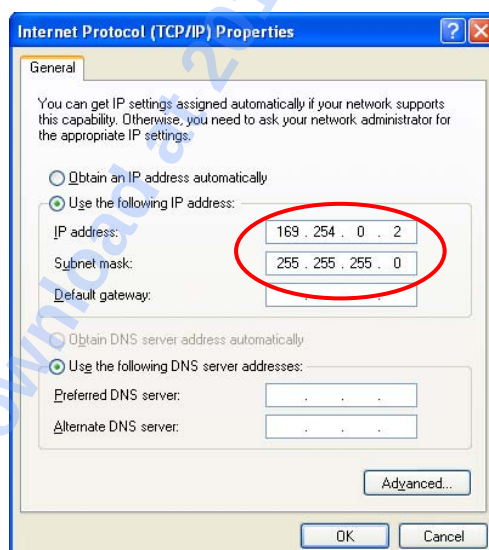


(NXT II)



7. At the PC, right-click [My Network] and select [Properties] - [Local area connection] - [Properties].
8. Select the [Internet Protocol (TCP/IP)] - [Properties] tab. Set the IP address of the PC to 169.254.0.2 and the subnet mask to 255.255.255.0 (the final digit of the IP address can be set to any value other than 1 or 0).

Note: The final digit of this IP address can be set to any value other than 1 or 0. The IP address should not be set to 169.254.0.0 or 169.254.0.1.



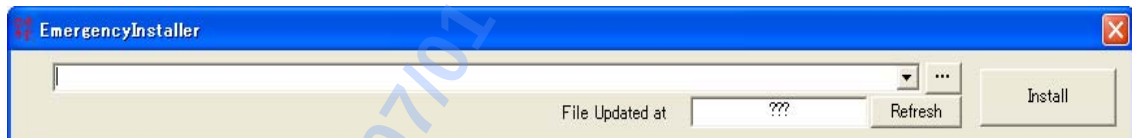
9. Turn the base main switch ON.



10. Booting up the CPU in safe mode sets the CPU BOX IP address to 169.254.0.1.
11. At the PC select [Start] – [Program] – [Accessories] – [Command Prompt] and enter “ping 169.254.0.1” to check the connection to the base CPU box. If you receive a reply from 169.254.0.1, then the connection has been established. If a “time out” message displays, recheck the LAN cable connection and network settings.
12. Insert the NXT application version CD in the PC CD drive. It should be the same version as the version installed on the machine.
13. Double-click the CD drive and select [NXT] – [Tool] – [Emergency Installer].



14. Click “...” when the following screen displays.



15. Select the appropriate bin file for the module you wish to restore. The bin files are different for NXT and NXTII (see table below). Select the file and click [Open] to display the path and file in the text box.

Caution: If you perform the installation with the wrong file, the CPU will not boot up.

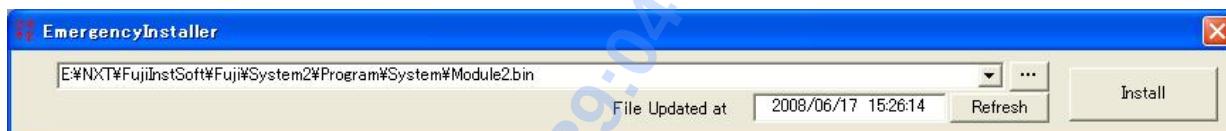
NXT

Install target		bin file
Module	M3,M6 M3S,M6S	NXT\FujiInstSoft\Fuji\System\ Program\System\Module.bin

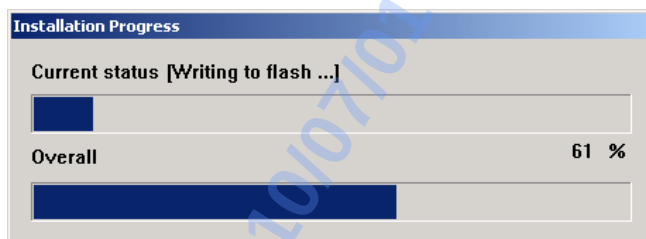
NXT II

Install target		bin file
Module	M3II, M6II	NXT\FujiInstSoft\Fuji\System2\ Program\System\Module2.bin

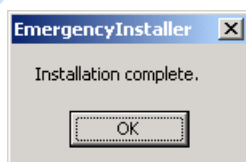
16. After selecting the bin file, click [Open].
17. Click [Install] to install the selected file in the target CPU.



18. Installation progress displays in the "Installation Progress" dialog.



19. Once the installation is complete, the following message displays. Click [OK] to close the message box.



20. Turn OFF the NXT base main switch.
21. Lower the toggle switch on the module CPU BOX to deactivate safe mode.

3.4.2 Registering module data

Transmit the matrix measurement, feeder pallet levelness data, and 4 nozzle head parts camera measurement data (not required for NXT V4.10 and higher) to the target module CPUBOX. Also check the module configuration settings.

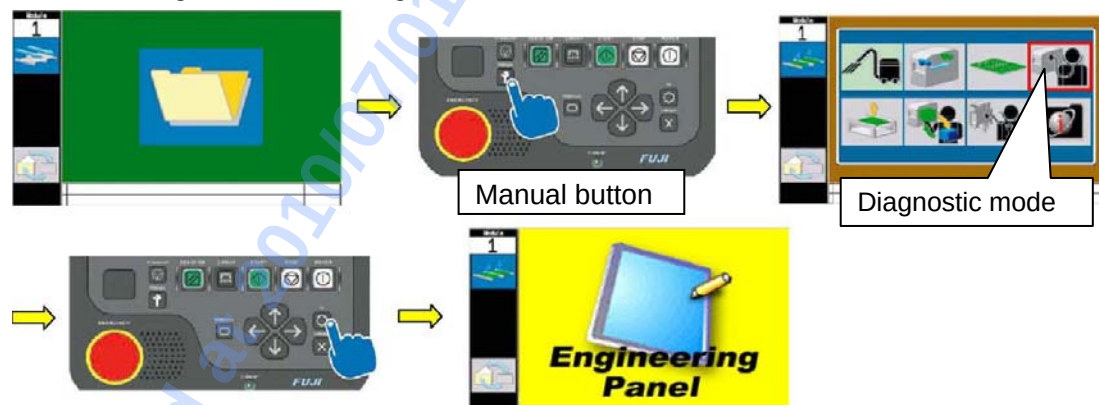
1. Turn the main switch ON.
2. Press the POWER button on the target module on the base to wake the module.



Caution: If the module application software version is different to that of the base, “Push OK to start the upgrade process” displays on the screen. In this case, press the OK button to upgrade the module software. Restart the machine once the “Module upgrade is completed” message displays.



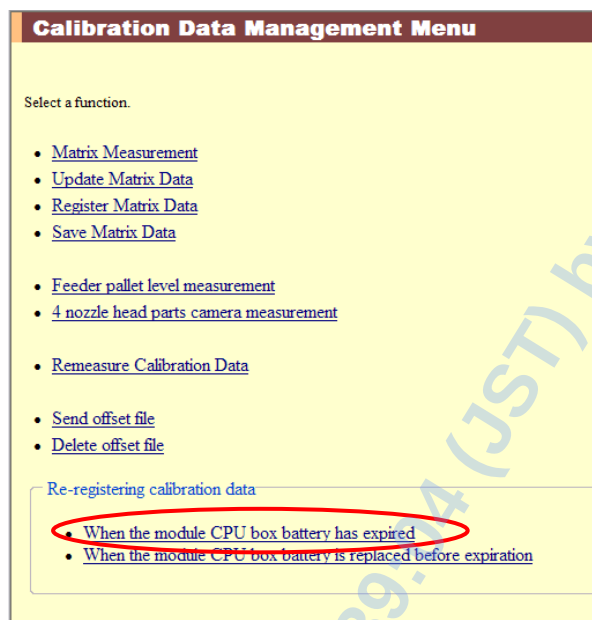
3. Set the target module to Diagnostic mode.



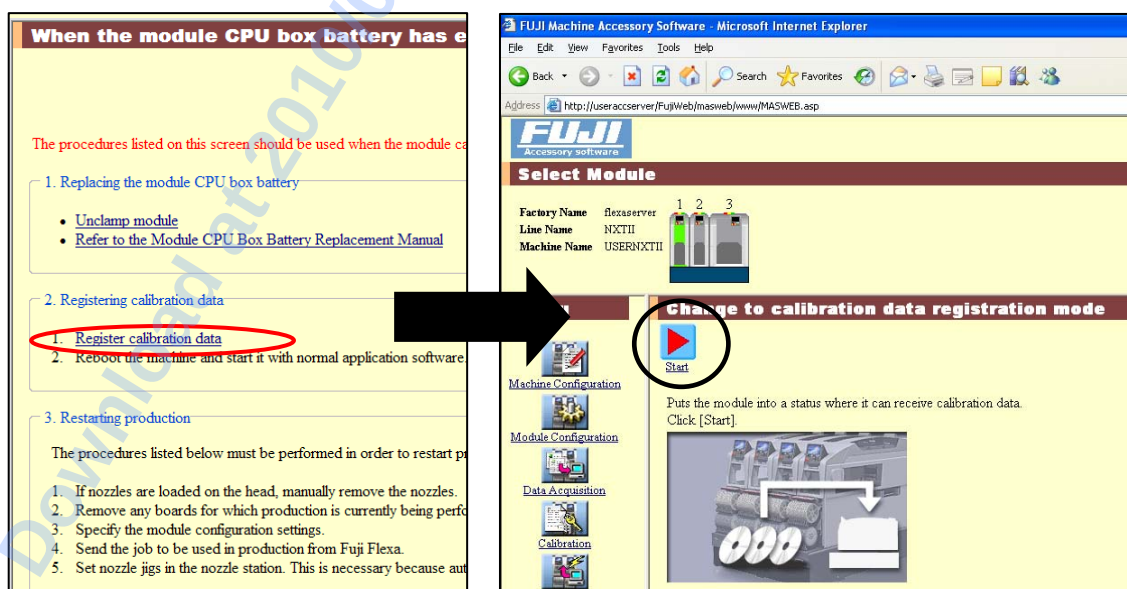
4. Open Accessory Software.
5. Select the line, machine and module, and then click [Calibration].

6. Select [When the module CPU box battery has expired]. This function is used to register the matrix and feeder pallet levelness measurement data.

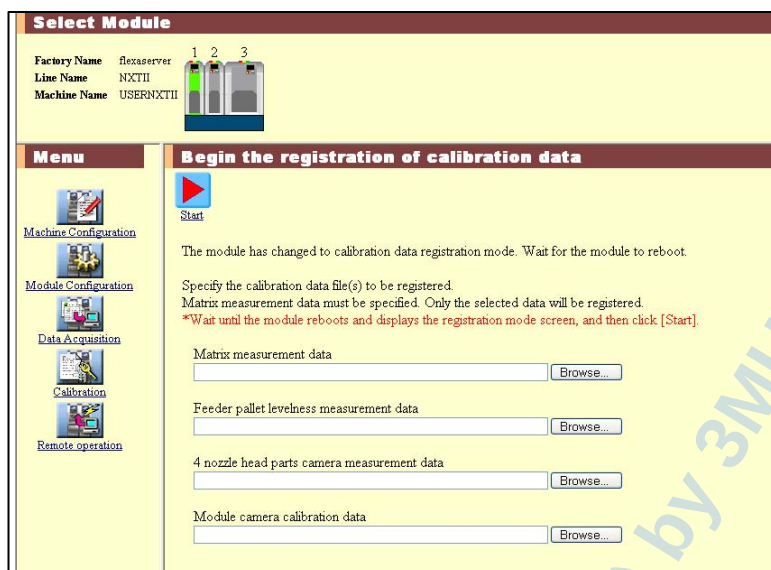
Caution: In application versions lower than V4.10, this data must be registered separately using the [Register matrix data] and [Register feeder pallet levelness data] commands in [Calibration data management].



7. Click [Register calibration data] and then click [Start] after displaying the “Change to calibration data registration mode” screen.



8. Wait until the accessory software changes to the screen below.

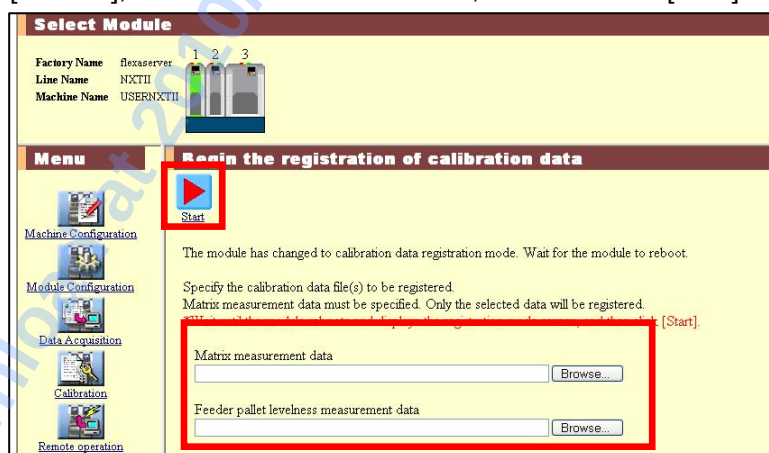


9. Once the target module has rebooted and entered sleep mode, press the module [POWER] button to prepare the machine to receive data.



This screen shows that the module can receive calibration data.

10. Click [Browse], select the various data files, and then click [Start].



11. Once the data registration is complete, reboot the machine.

12. Select [Module Configuration] in Accessory Software. Check the settings and then click [Send to machine].

3.4.3 Calibration

If the module CPU BOX has been replaced, it is necessary to perform calibration.

1. Transmit a job to the target machine.
2. Prepare the following jigs.

(NXT)

- Nozzle jig

(NXTII)

- Nozzle jig
- Hybrid calibration nozzle
- Hybrid calibration glass part, chip.

3. In Accessory Software select the module on which the CPU BOX was replaced, and then click [Calibration].
4. Select [Remeasure calibration data], and then click [Start].
5. Start the machine operation to perform the calibration. Emergency Install is complete once the calibration is finished.

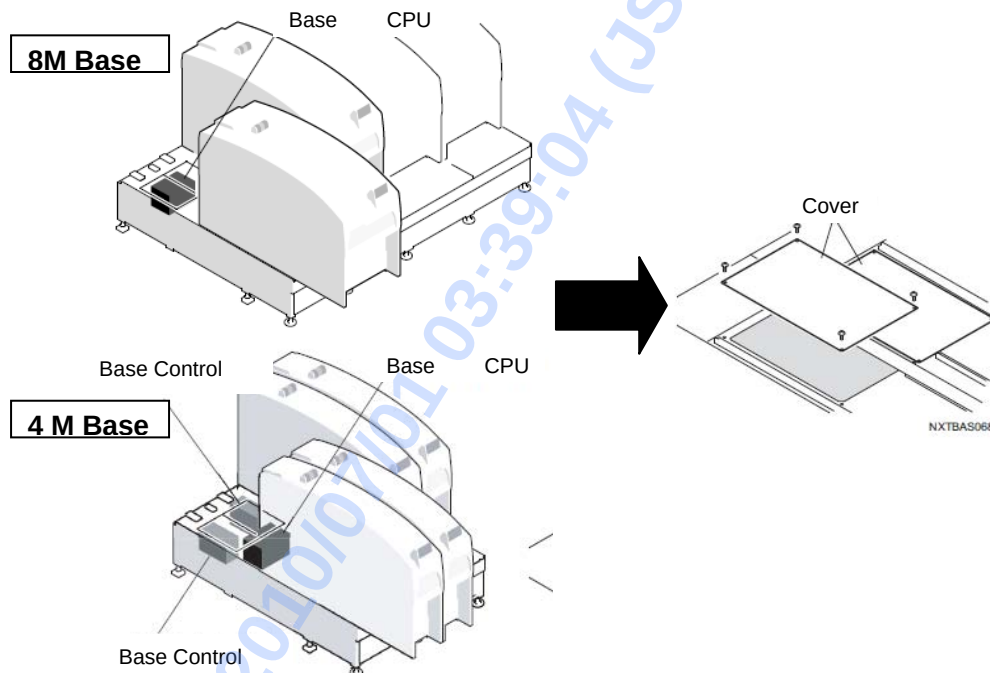
3.5. Replacing the base CPU BOX

When replacing the memory card in the base CPU BOX, be sure to prepare a backup of the final offset data in advance.

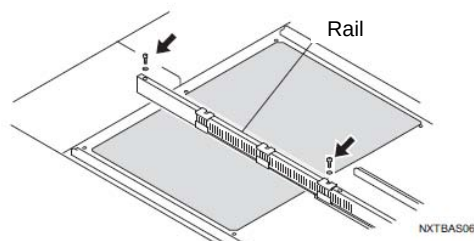
3.5.1 Removing the base CPU BOX

For NXT 8M and 4M base

1. Unclamp and pull out the left-hand module(s) (2 in the case of M3 and 1 in the case of M6). If the CPU BOX cannot boot up and the module cannot unclamp, follow the procedure in "3.1 Pulling Out the Module" to unclamp and pull out the module.
2. If the base power is ON, turn it OFF at the main switch.
3. Remove the screws, then remove the cover.

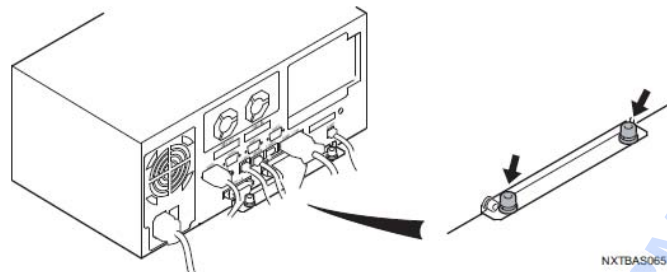


4. Remove the two bolts which fix a section of the rail. Keep this parts of the rail away from the opening when working in the base.

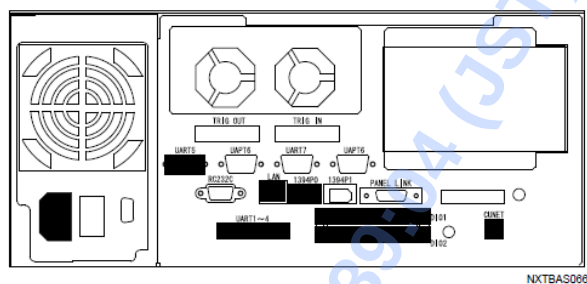


For NXT 8M and 4M base

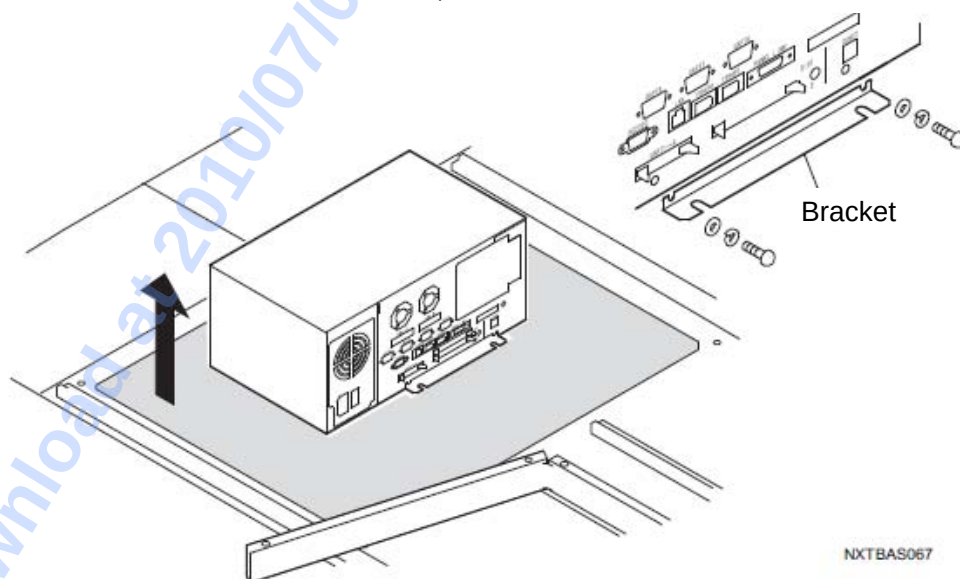
5. Remove the bolts which fix the CPU BOX.



6. Disconnect the connectors in the black marked positions in the illustration below.

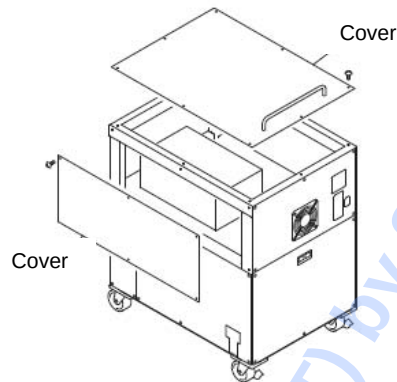


7. Take out the CPU box from the base, and detach the bracket.

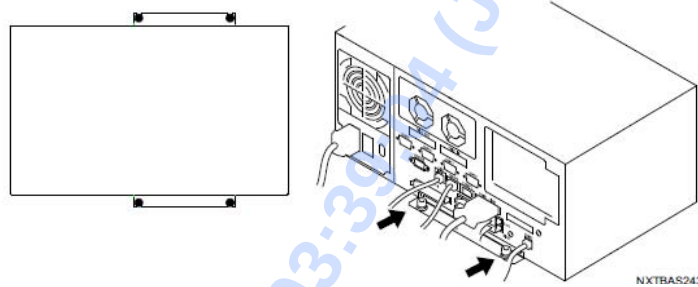


For NXT 2M base (External control box)

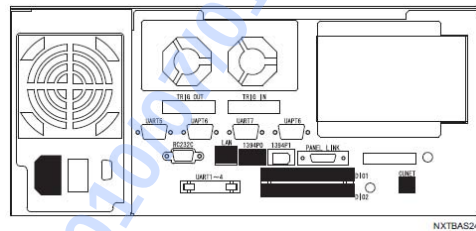
1. Turn off the power to the machine by turning the main switch off.
2. Remove the 4 screws and then remove the covers.



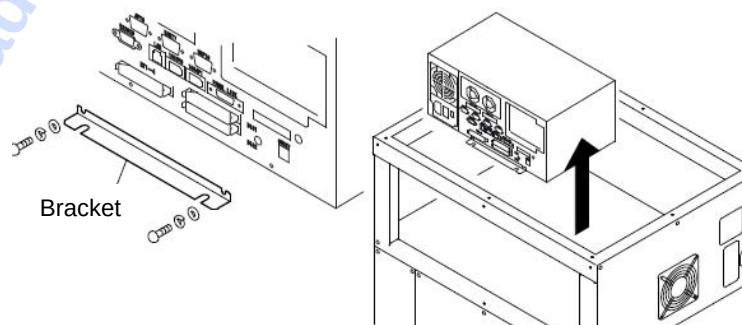
3. Remove the 4 bolts which fix the CPU BOX.



4. Disconnect the connectors in the black marked positions in the illustration below.

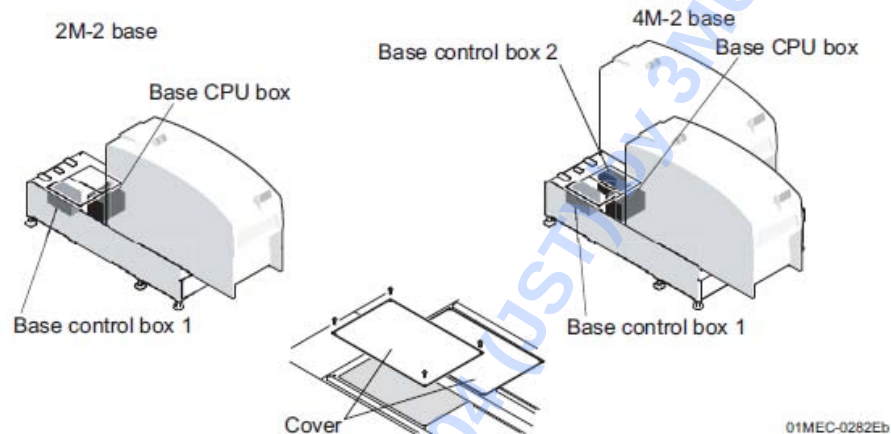


5. Take out the CPU BOX from the control box and detach the bracket.

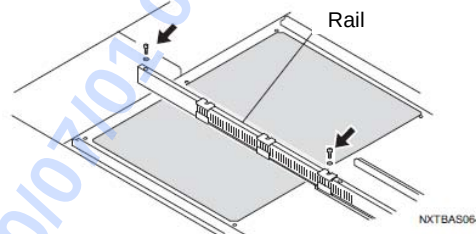


For NXTII 4M-II and 2M II base

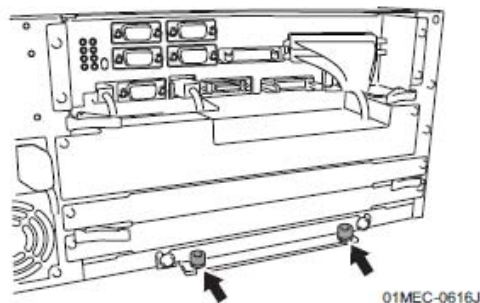
1. Unclamp and pull out the left-hand module(s) (2 in the case of M3 and 1 in the case of M6). If the CPU BOX cannot boot up and the module cannot unclamp, follow the procedure in "3.1 Pulling Out the Module" to unclamp and pull out the module.
2. If the base power is ON, turn it OFF at the main switch.
3. Remove the 4 screws and then remove the covers.



4. Remove the two bolts which fix a section of the rail. Keep this part of the rail away from the opening when working in the base



5. Take out the CPU box from the base, and detach the bracket.

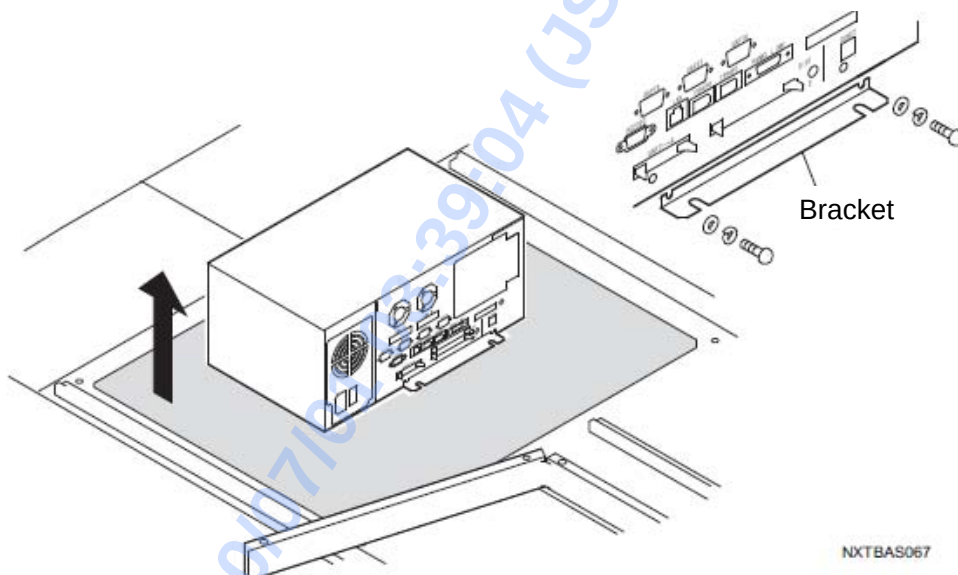


For NXTII 4M-II and 2M II base

6. Disconnect the connectors in the black marked positions in the illustration below.



7. Take out the CPU box from the base, and detach the bracket.



NXTBAS067

3.5.2 Installing the base CPU BOX

1. Remove the memory card from the old CPU BOX and set it in the new CPU BOX. Refer to step 3 of “3.3.1 Formatting the PC card” for details on how to remove the memory card.

Caution: replace the memory card if you suspect it may be damaged.

2. Install the new CPU BOX using the reverse of the removal procedure.

3.5.3 Setting the IP address

Follow the procedure in “3.3.3 Setting the IP Address” to set the IP address.

3.5.4 Installing the application version in the base CPU BOX (VersionUp.exe)

Follow the procedure in “3.3.4 Installing the base CPU BOX application” to install the application.

3.5.5 Checking the communication between the NXT and HOST. Follow the procedure in “3.3.5 Checking the communication between the NXT and the HOST”.

3.5.6 Registering base data

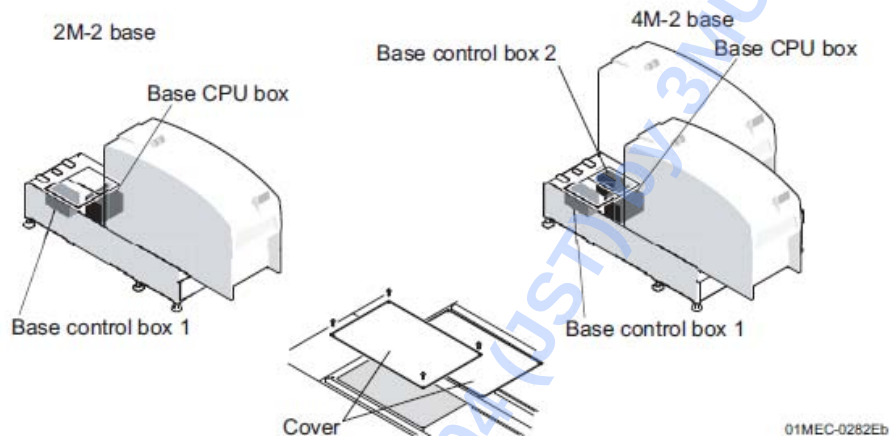
Follow the procedure in “3.3.6 Registering base data” to register the base data.

The base CPU BOX replacement procedure is now complete.

3.6. Replacing boards in the base CPU BOX (NXT II only)

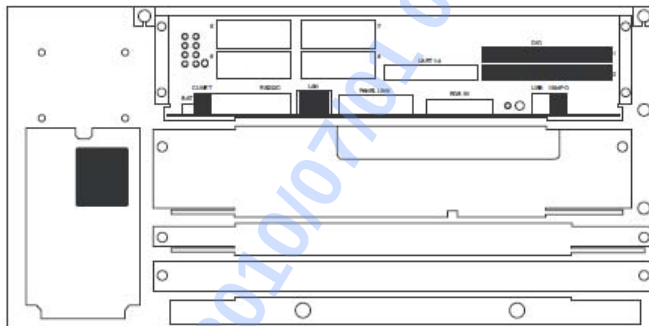
3.6.1 Removing covers and cables

1. Unclamp and pull out the left-hand module(s) (2 in the case of M3 and 1 in the case of M6). If the CPU BOX cannot boot up and the module cannot unclamp, follow the procedure in "3.1 Pulling Out the Module" to unclamp and pull out the module.
2. If the base power is ON, turn it OFF at the main switch.
3. Remove the screws and then remove the covers.

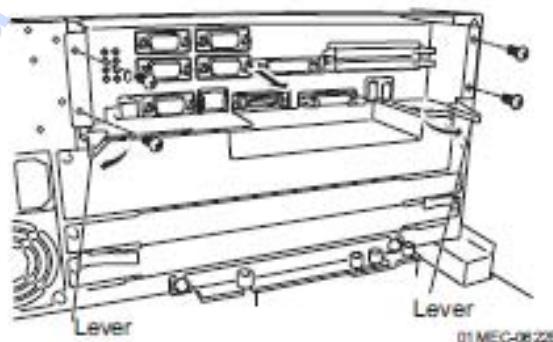


3.6.2 Replacing the CPU board

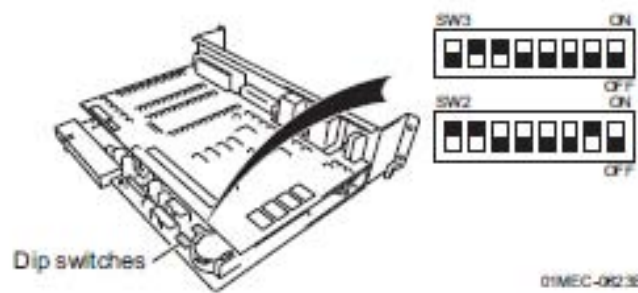
1. Disconnect the connectors in the black marked positions in the illustration below.



2. Remove the 4 screws, open the levers and remove the CPU board.

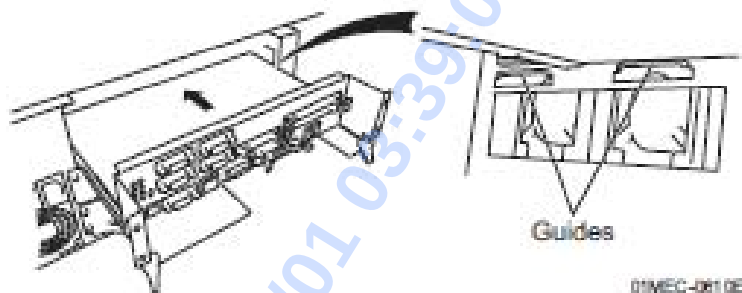


3. Confirm that the dip switch settings on the new CPU board are correct.



4. Using the guides, insert the new CPU board into the CPU box and tighten the screws.

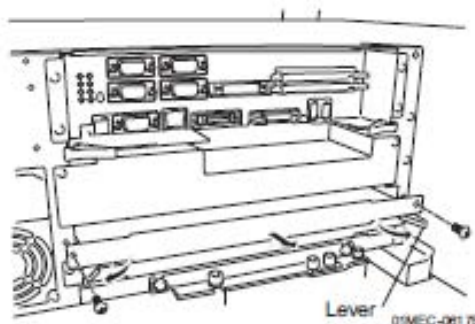
Caution: Forcing the board into the CPU box without using the guides may result in damage. Be sure to run the board along the guides when inserting it.



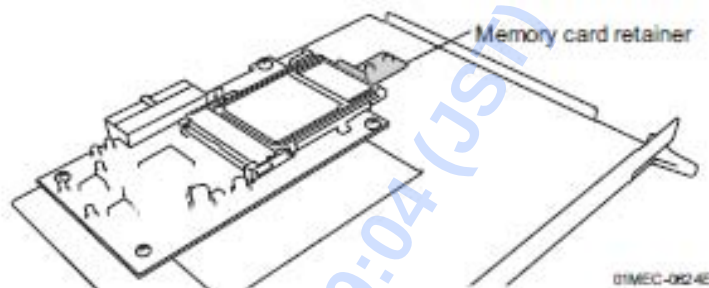
5. Connect the cables.
6. Turn the base main switch ON, and confirm that it boots up correctly.
7. Follow the procedure in "3.3.3 setting the IP address" to set the IP address.
8. Follow the procedure in "3.3.4 Installing the base CPU Box application" to install the application.
9. Follow the procedure in "3.3.5 Checking the communication between the NXT and the HOST" to check the communication.

3.6.3 Replacing the memory card board

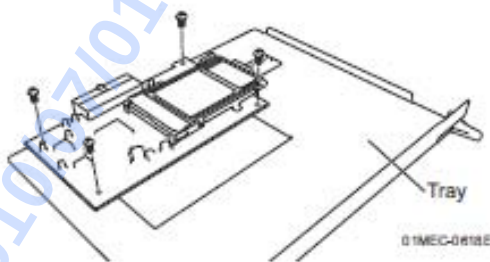
1. Remove the 2 screws, open the levers and pull out the memory card board tray.



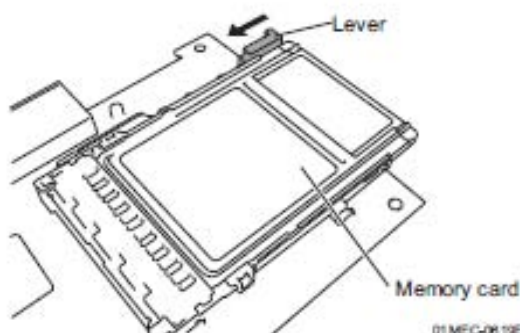
2. Loosen the 2 screws on the memory card retainer,



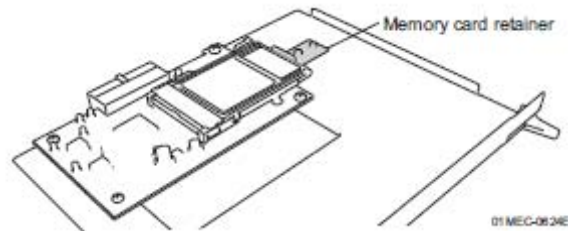
3. Remove the 4 screws and remove the memory card board from the tray.



4. Release the lever and remove the memory card in the new board.

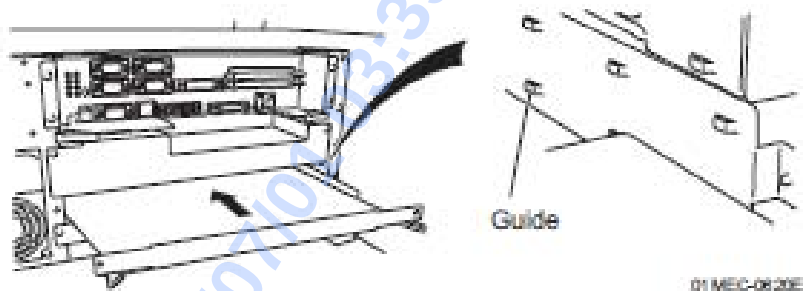


5. Attach the memory card board to the tray align the position of the memory card retainer and tighten the screws.



6. Using the guides, insert the memory card board tray into the CPU box and tighten the screws.

Caution: Forcing the board into the CPU box without using the guides may result in damage. Be sure to run the board along the guides when inserting it.



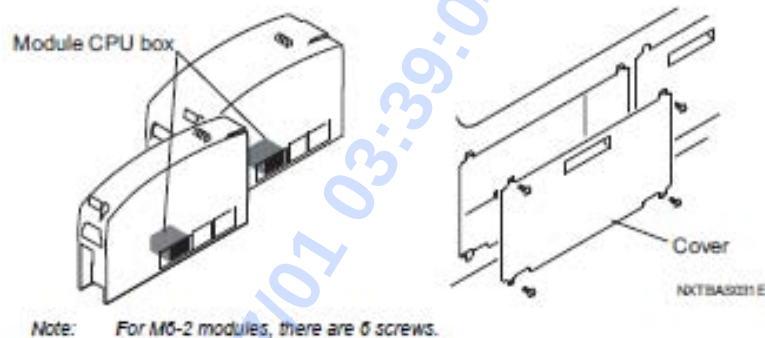
3.7. Replacing the module CPU BOX

Prepare the following backup data (FD or CD) before replacing the module CPU BOX.
This data will be registered in the CPU BOX following replacement.

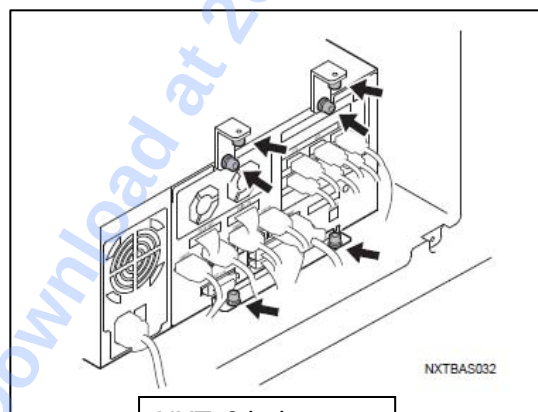
- Matrix measurement data
- Feeder pallet levelness measurement data
- 4 nozzle head parts camera measurement data (not required for NXT V4.10 and higher)

3.7.1 Replacing the module CPU BOX

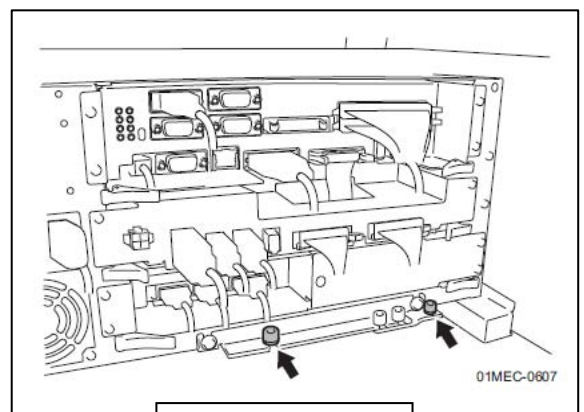
1. Unclamp and pull out the module on which you wish to replace the CPU BOX. If you cannot unclamp the module because the base CPU BOX cannot boot up, follow the procedure in "3.1 Pulling out the module" to unclamp the module manually.
2. If the base power is ON, turn it OFF at the main switch.
3. Remove the bolts and the cover.



4. Remove the CPU BOX mounting bolts.

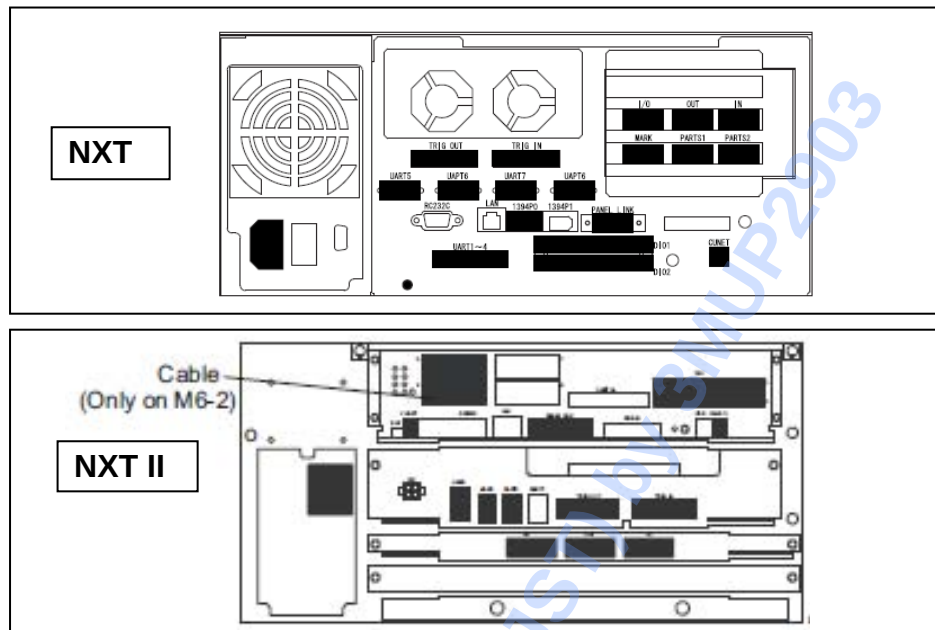


NXT: 6 bolts

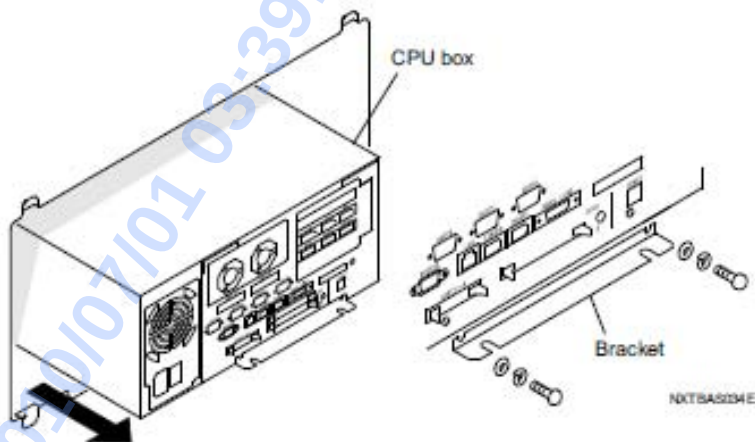


NXT II: 2 bolts

5. Disconnect the connectors in the black marked positions in the illustration below.



6. Take out the CPU BOX from the module body, and detach the bracket from the CPU BOX.



7. Use the reverse procedure when installing the new CPU BOX.

3.7.2 Registering module data

Follow the procedure in "3.4.2 Registering module data" to register the data.

3.7.3 Calibration

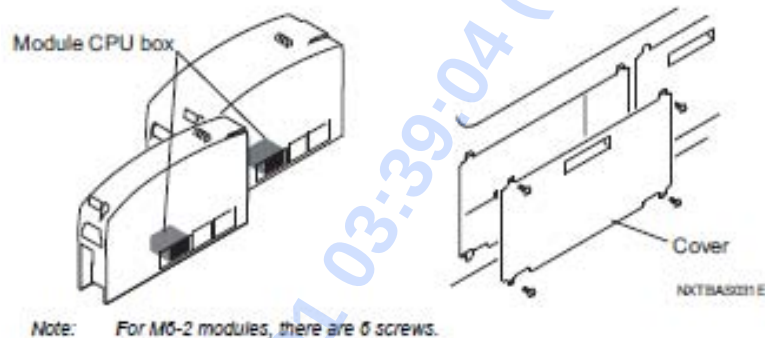
Follow the procedure in "3.4.3 Calibration" to perform the calibration.

The module CPU BOX replacement is now complete.

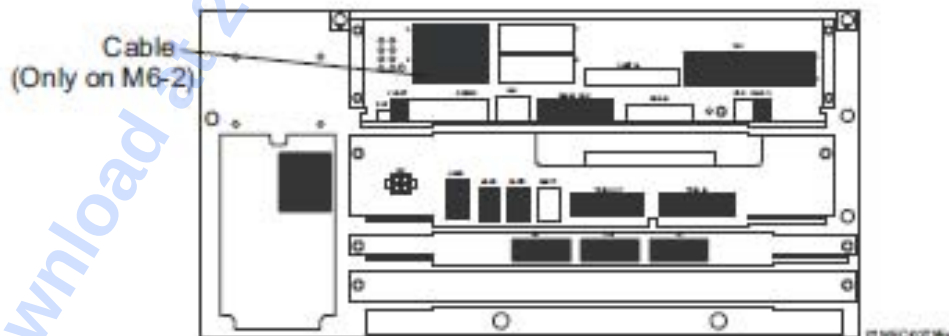
3.8. Replacing boards in the module CPU BOX

Prepare the following backup data (FD or CD) in advance when replacing the module CPU board. This data will be registered after replacing the CPU board.

- Matrix measurement data
 - Feeder pallet levelness measurement data
1. Unclamp and pull out the module on which you wish to perform the replacement. If you cannot unclamp the module because the base CPU BOX cannot boot up, follow the procedure in “3.1 Pulling out the module” to unclamp the module manually.
 2. If the base power is ON, turn it OFF at the main switch.
 3. Remove the bolts and cover.

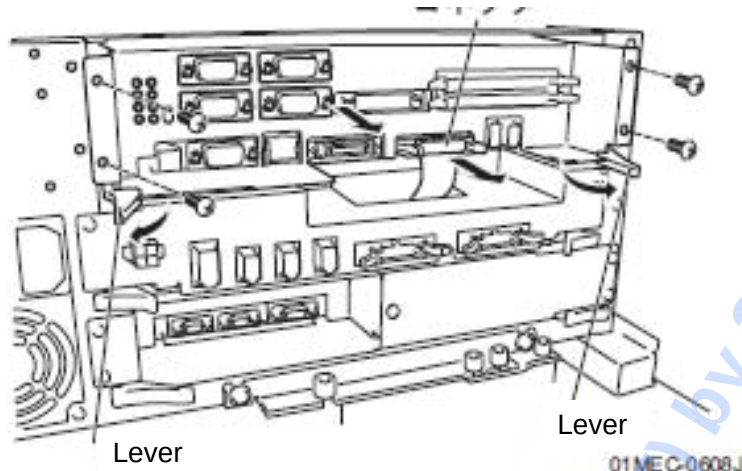


4. Remove the connectors indicated by the black rectangles in the following diagram.

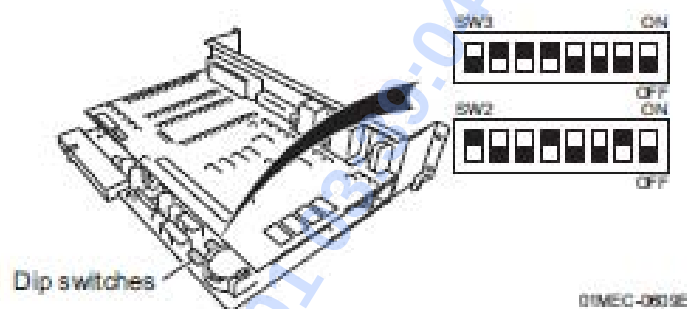


3.8.1 Replacing the CPU board

1. Remove the connector for the cable connecting the CPU and vision processing board. Remove the 4 bolts, and then push the levers outward to eject the CPU board.

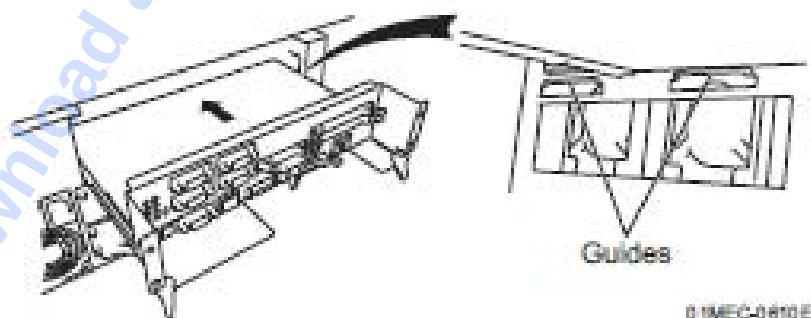


2. Check the dip switch settings on the new CPU board.



3. Insert the new CPU board making sure it runs along the guides. Once correctly inserted, fix it in place with the mounting bolts.

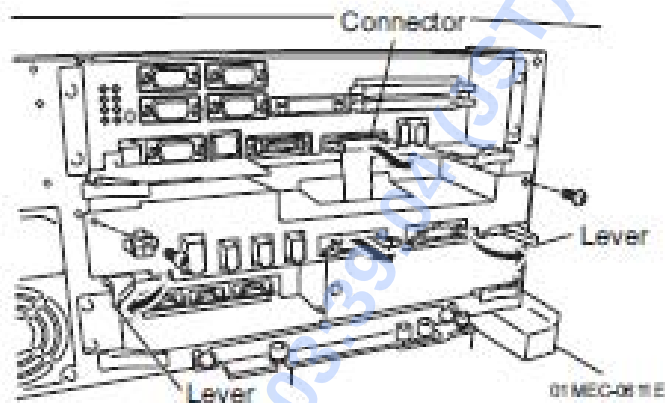
Caution: Forcing the board into the CPU box without using the guides may result in damage. Be sure to run the board along the guides when inserting it.



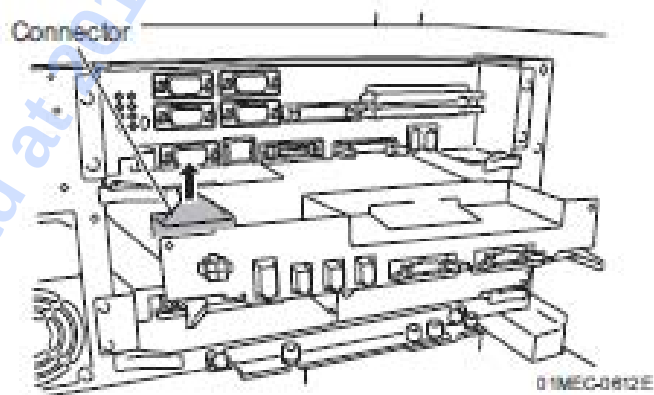
4. Connect the cables.
5. Turn ON the base main switch, and confirm that it boots up correctly.
6. Follow the procedure in "3.4.2 Registering the module data" to register the data.
7. Follow the procedure in "3.4.3 Calibration" to perform the calibration.

3.8.2 Replacing the vision process board

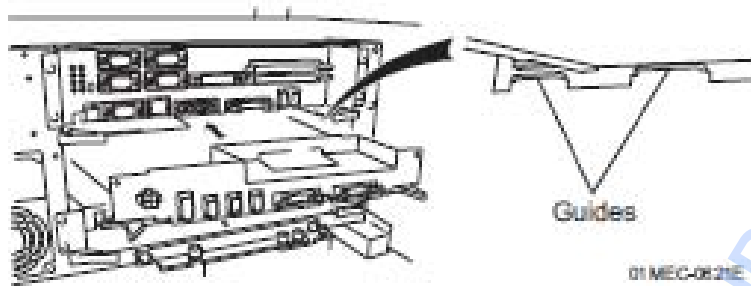
1. Disconnect the cable connecting the CPU board to the vision board. Remove the 2 bolts and push the levers outwards so that the vision board is approximately halfway out.



2. Disconnect the cable connecting the ATX power supply to the vision board, and then remove the vision board.



3. Insert the new vision board about halfway, making sure it runs along the guides.

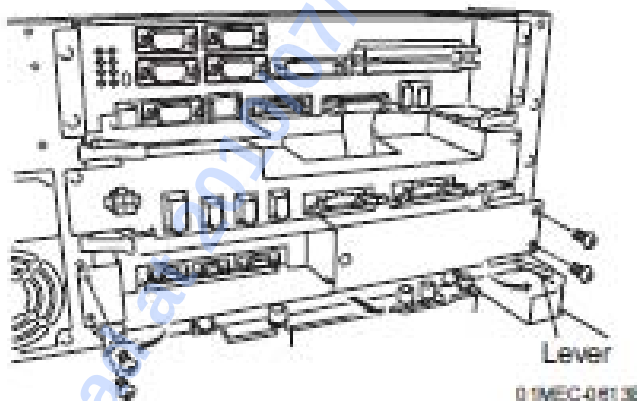


Caution: Forcing the board into the CPU box without using the guides may result in damage. Be sure to run the board along the guides when inserting it.

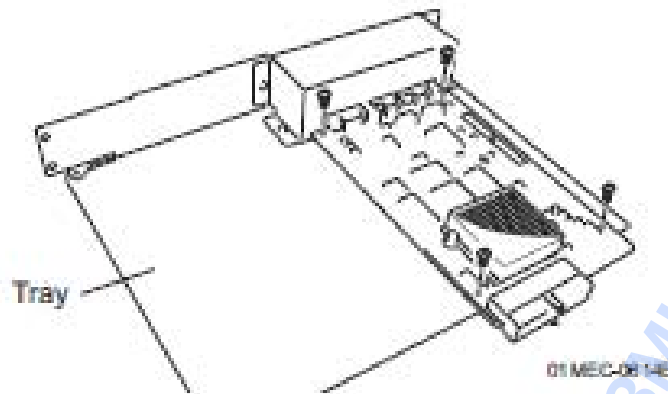
4. Connect the cable connecting the ATX power supply to the vision board, and then fully insert the vision board.
5. Connect the cable connecting the CPU board to the vision board and tighten the mounting bolts.

3.8.3 Replacing the servo board

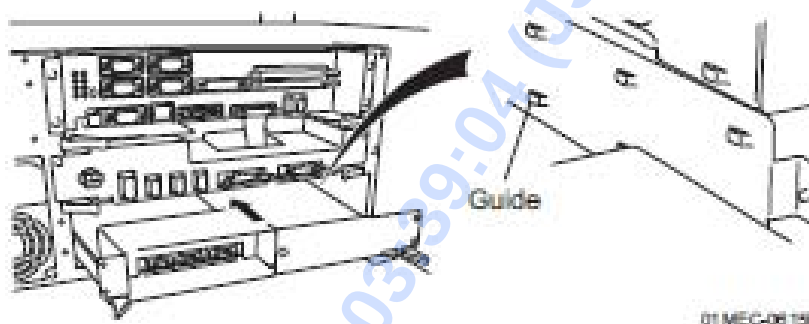
1. Remove the 4 bolts, and then use the lever to pull out the servo board.



2. Remove the 4 bolts, and then replace the servo board on top of the tray.



3. Insert the servo board tray into the CPU BOX, making sure it runs along the guides. Finally, tighten the mounting bolts.



Caution: Forcing the board into the CPU box without using the guides may result in damage. Be sure to run the board along the guides when inserting it.

MEMO:

Download at 2010/07/01 03:39:04 (JST) by 3MUP2903

Chapter 4

MT Reset & Origin Adjustment

CONFIDENTIAL

Download at 2010/07/01 03:39:04 (JST) by 3MUP2903

4. Motor MT Reset & Origin Adjustment

4.1 Preparation

The I/O Monitor (Equipment check) and Motion Tool Mini applications are both required for this procedure. They should be installed in advance from the NXT application CD-ROM.

I/O Monitor

Open the NXT application CD and select [Support Tools] – [Equipment check] from the menu to install the application.

Motion Tool Mini

Double-click CD\NXT\Tool\MotionToolMini.exe to start the installation and follow the

MT reset will be performed using Motion Tool mini. However, to use Motion Tool mini it is necessary to forcibly turn ON the module servo power. I/O monitor is used to perform this operation.

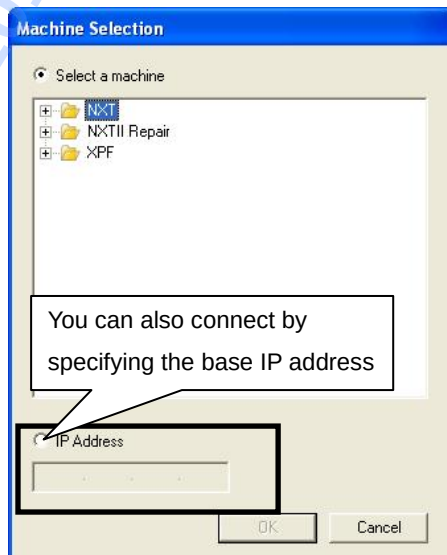
4.2 Starting I/O Monitor

1. Follow the procedure below to turn the I/O output signals ON, clear the multi-turn memory and reset the motor origins.
2. Select [Start] – [Program] – [Fuji Accessory Software] – [Equipment Check].

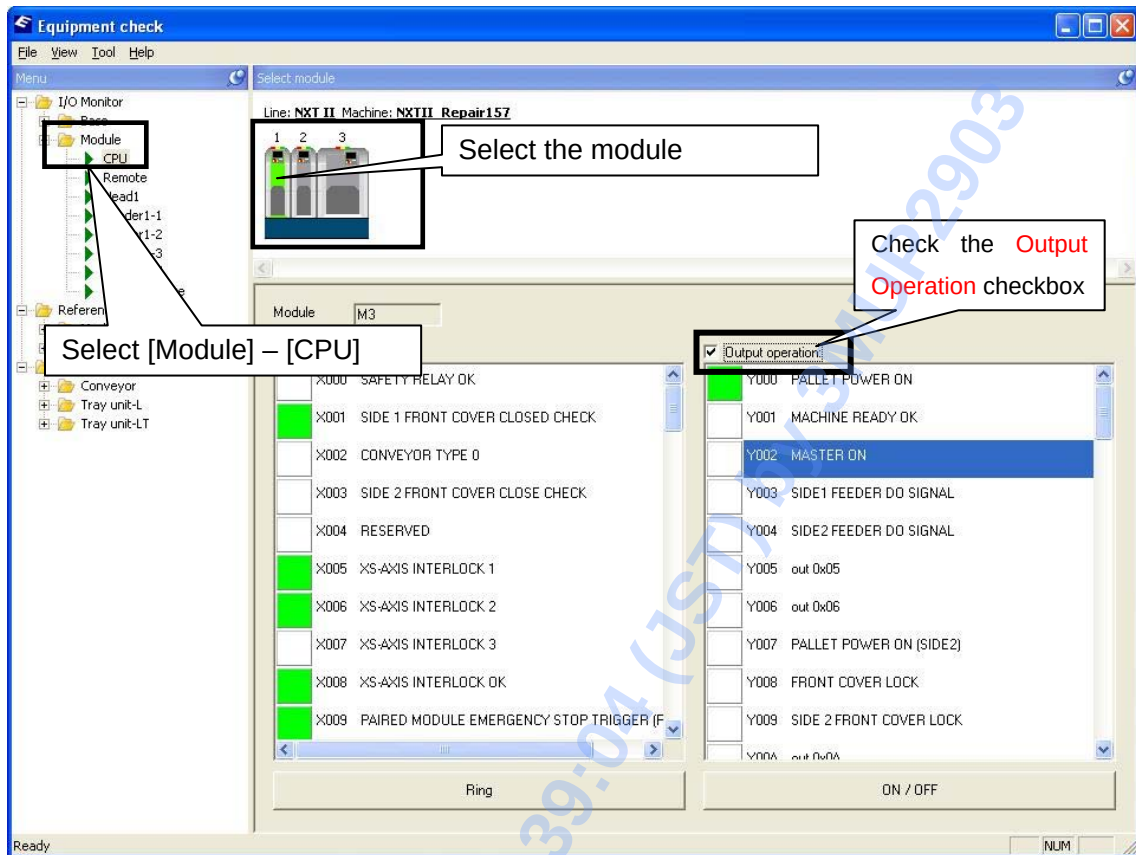


Equipment Check.Ink

3. In the **equipment selection** dialog, select the target equipment and click [OK] to connect. Alternatively, you can connect by specifying the IP address of the base you wish to connect to.



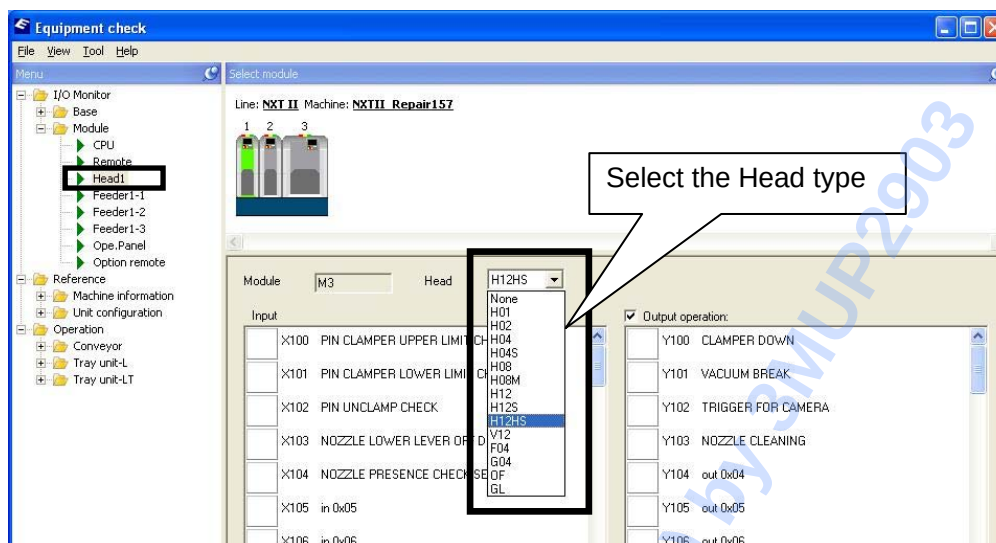
4. Select the target module, and then select [Module] – [CPU] in the left-hand window menu. Check the [Output operation] checkbox.



5. Set the following signals. (■ = ON)

- Y00D Servo control power, Head power: select ON.
- Y002 Master ON: select **OFF**.

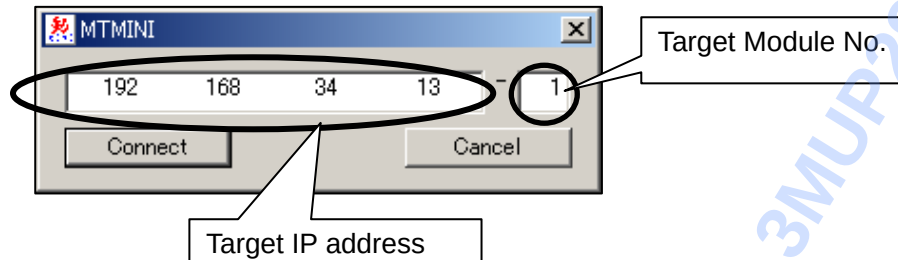
6. From the [Head Type] dropdown list, select the type of head loaded on the target module.



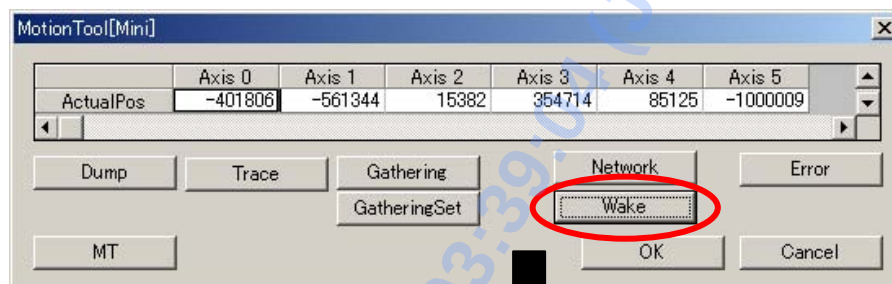
7. From the left-hand menu, select [Module] – [CPU]. This is to wake the head shell.
8. Remove the output signal checkbox.
9. To check the I/Os on another module, first clear the checkbox, and then select the target module.

4.3 Starting Motion Tool Mini

1. Start Motion Tool Mini.exe.
2. Enter target base IP address and Module number and select [Connect].

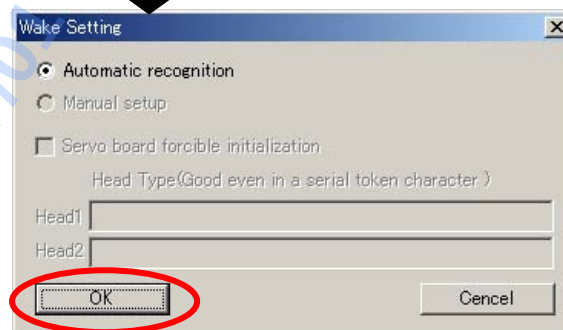


3. To start the servo amp, select [Wake] → [Automatic recognition]. Wait for approximately 10 seconds. Now the current servo counter can be referred.



Axis map

Axis 0	Y
Axis 1	X
Axis 2	Z
Axis 3	R
Axis 4	Q
Axis 5	XS



4.4 XY Axis MT Reset

1. To return to normal condition it is necessary to perform MT reset at the position shown in table 2. Move the target axis to the position shown in table 2 and check that the pulse count exceeds the number of pulses/rotation indicated in table 1. MT Reset is only required if the pulse count exceeds these values.

Warning: the machine may run out of control if MT Reset is performed at the wrong position.

Table 1: number of pulses/rotation for each motor

Module	Axis	Pulse
M3,M3S, M3 II	X	60000
	XS	50000
M3,M3S	Y	64000
M6S, M6 II	X	50000
M6	Y	64000

Table 2: XY axis MT reset positions

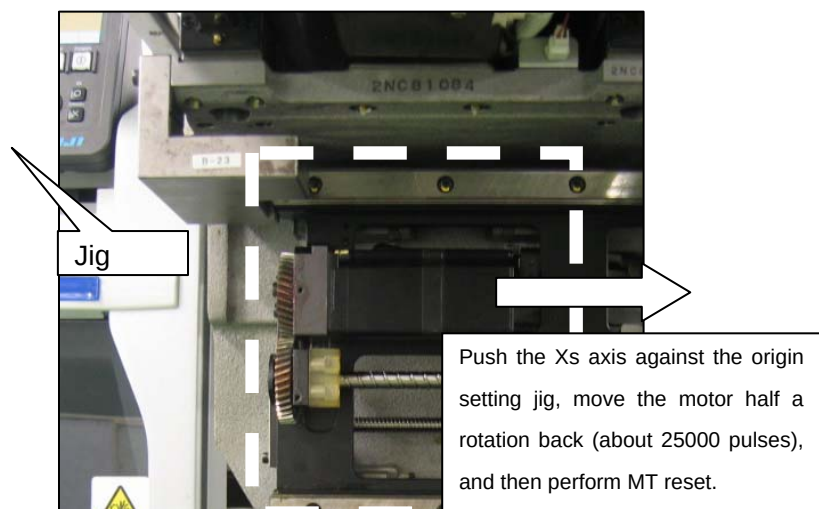
Module	Axis	MT reset position
M3,M3S, M3 II	X	Motor side plus mechanical stopper
	XS	25000 pulses from the Xs-axis origin setting jig
M3,M3S	Y	Motor side plus mechanical stopper
M6S, M6 II	X	Motor side plus mechanical stopper
M6	Y	Motor side plus mechanical stopper
M3II,M6S, M6 II	Y	not necessary

Caution: The position of the X-axis mechanical stopper on the M3(S/II) is opposite to that on the M6(S/II).

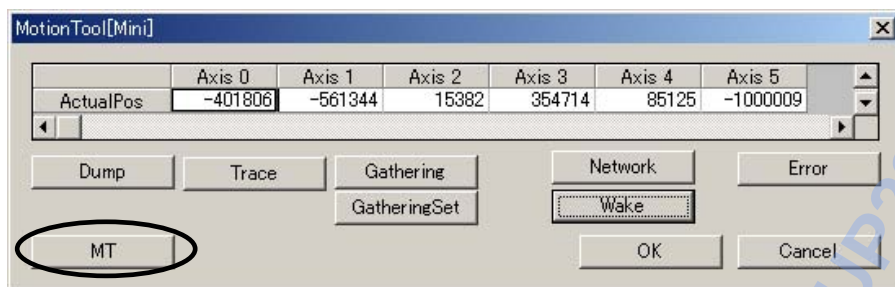
M3/M3S/M3II: on the left-side as viewed from the machine front.

M6/M6S/M6II: on the right-side as viewed from the machine front.

Supplemental information: an origin setting jig is set on the Xs-axis, and MT reset performed half a rotation (25000 pulses) from the position the axis contacts the jig. This is only necessary for the Xs-axis.



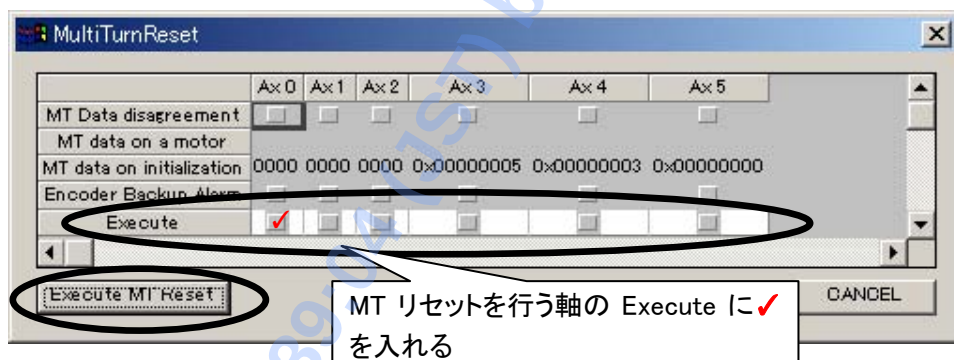
- If the target axis exceeds the number of pulses per 1 rotation of the motor, click [MT] in Motion Tool Mini.



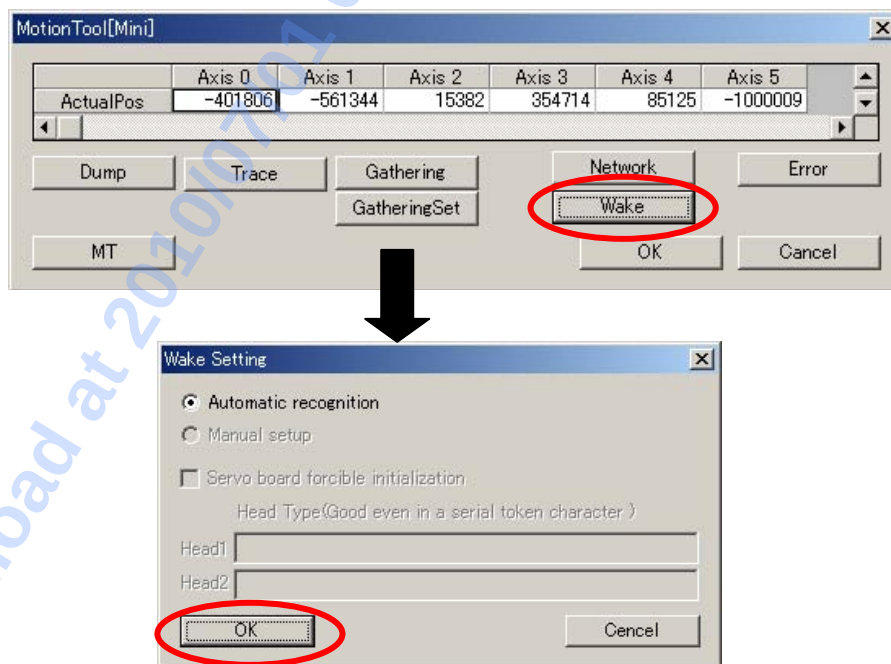
- Check the checkbox for the target axis, and then click [Execute MT reset] to clear the multi-turn memory for that axis. Refer to the Axis map for the number of the target axis.

Axis map

Axis 0	Y
Axis 1	X
Axis 2	Z
Axis 3	R (C)
Axis 4	Q
Axis 5	XS



- Select [Wake] – [Automatic recognition] again to display the servo counter.



Caution: the correct servo count will not display unless Wake is performed.

5. After performing MT reset, confirm that the servo count at the “Confirmation position” is within the tolerance shown in “Servo counter after performing MT reset” (see table below). If the count still exceeds the number of pulses per 1 rotation of the motor, MT reset may not have been performed correctly and should be performed again. If the pulse count is within the number of pulses per 1 rotation of the motor, but outside the tolerance specified in the table, a mechanical adjustment is required. In this case refer to “4.5 If the origin setting is required” to perform the adjustment.

Table 3: Servo counter after MT reset

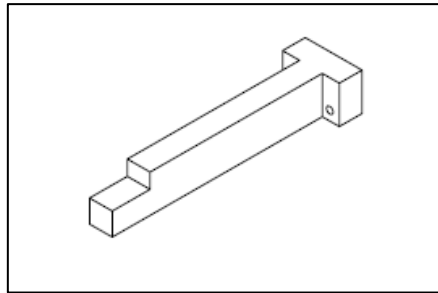
Module	Axis	Confirmation position	Servo counter after performing MT reset (± Tolerance)
M3,M3S, M3 II	X	Motor side plus mechanical stopper	+25000 (±2500)
	XS	X-axis origin setting jig	+0 (±1000)
M3,M3S	Y	Motor side plus mechanical stopper	+10000 (±2000)
M6S, M6 II	X	Motor side plus mechanical stopper	+13750 (±1250)
M6	Y	Motor side plus mechanical stopper	+10000 (±2000)

6. Check that there are no errors and the machine operates correctly during automatic operation. If an Xs interlock error occurs after MT Reset, follow the procedure in “4.5.5 Xs-axis interlock sensor check” to check the sensor and adjust it if necessary.
7. The X and Y-axis MT reset procedure is now complete.

4.5 If the origin setting is required

The mechanical origin adjustment for each axis is explained here. These adjustments are necessary if the servo count exceeds the tolerance after performing MT reset. They are not required if the machine can be restored by performing MT reset alone.

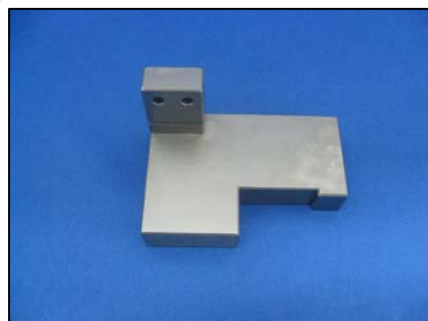
Required jig:



M3/M3S/M6 Y-axis zero set positioning jig (Y-axis ball screw type)
PM23180



M3/M3S X & Xs-axis
PM57630



M3II X & Xs-axis
PM070Y0



M6(S) X-axis
PM18620



M6II X-axis zero set positioning jig
PM090R*

4.5.1 Y-axis origin adjustment (M3/M3S & M6 Module)

The following explains the adjustment procedure for the M3/M3S/M6 modules which use ball screws for the Y-axis. For the M3II, M6S and M6II modules, which use linear motors, please refer to the “Linear scale and motor replacement manual”.

Table 4: Y-axis servo counter table**M3/M3S** Y-axis : 1 pulse = 0.5 μ m (0.0005 mm)

Position	Tolerance (Pulses)
Motor side plus mechanical stopper (5 mm / 0.0005 = 10000 pulses)	+10000 (+/- 2000)
5 mm from the plus mechanical stopper	0
Y-axis zero setting position (motor side + jig (25 mm))	0 (+/- 500)
Minus mechanical stopper	$\leq$ -2097000
Full stroke (1055.5mm / 0.0005 = 2111000 pulse) +/- 2 mm	2111000 (+/- 4000)

M6 Y-axis: 1 pulse = 0.5 μ m (0.0005 mm)

Position	Tolerance (Pulses)
Motor side plus mechanical stopper (5 mm / 0.0005 = 10000 pulses)	+10000 (+/- 2000)
5 mm from the plus mechanical stopper	0
Y-axis zero setting position (motor side + jig (25 mm))	0 (+/- 500)
Minus mechanical stopper	$\leq$ -2097000
Full stroke (1055.5mm / 0.0005 = 2111000 pulse) +/- 2 mm	2111000 (+/- 4000)

1. Remove the mechanical stopper (rubber stopper) and attach the Y-axis jig.
2. Turn on the power, wake the module, and then perform the procedures in “4.2 Starting I/O Monitor” and “4.3 Starting Motion Tool Mini” to display the servo counter. The Y-axis servo counter value is “Axis 0”.

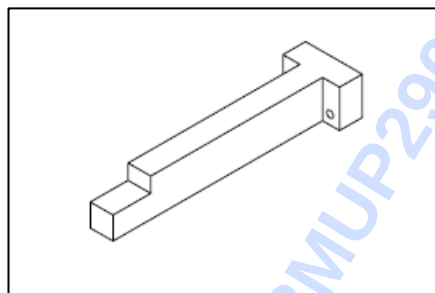
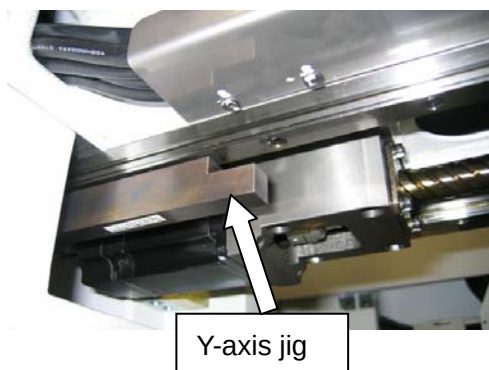
Axis map

Axis 0	Y
Axis 1	X
Axis 2	Z
Axis 3	R (C)
Axis 4	Q
Axis 5	XS



Y-axis = Axis 0

3. Push the Y-axis against the jig and check the servo counter (Axis 0). If the servo count is outside the Y-axis zero setting position tolerance of 0 (+/- 500) specified in the above table, it is necessary to loosen the two coupling bolts and turn the motor by hand to adjust the counter.



PM23180: M3/M3S/M6 Y-axis origin setting jig
(For ball screw type Y-axis)

4. Once the counter is within tolerance, tighten the 2 bolts at the Y-axis ball screw side.
5. Lock the coupling bolts with a torque wrench. (14.7 Ncm)
6. Remove the jig and reattach the mechanical stopper.
7. Move the Y-axis by hand and check that the servo count at the plus and minus mechanical stoppers are within the tolerance specified in table 4. Also calculate the full stroke from these servo counter values and confirm that it is within tolerance.
8. Turn I/O Y00D, Y002, Y001 OFF and select [Stop] – [Close] to close I/O monitor.

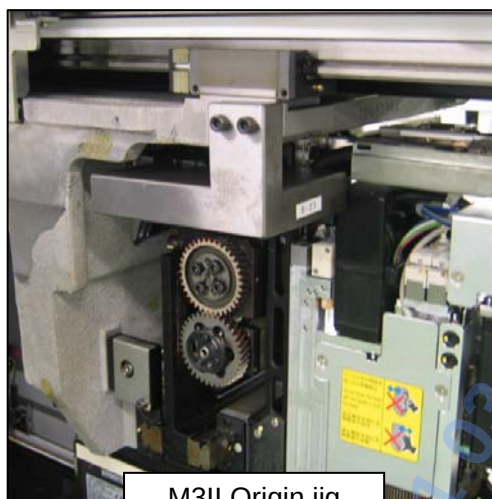
4.5.2 X-axis origin setting (M3/M3S/M3II module)

Table 5: M3/M3S/M3II X-axis servo counter table

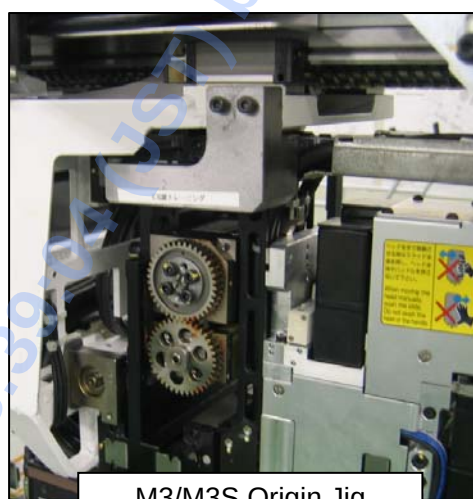
M3/M3S/M3II X-axis 1 pulse = 0.2 μ m (0.0002 mm)

Position	Tolerance (Pulses)
Motor side plus mechanical stopper (5 mm / 0.0002 = 25000 pulses)	+25000 (+/- 2500)
5 mm from plus mechanical stopper	0
X-axis zero setting position (Motor side + jig (20 mm))	-75000 (+/- 500)
Minus mechanical stopper	$\leq$ -1080000
Full stroke (222 mm / 0.0002 = 1110000 pulses) +/- 0.8 mm	1110000 (+/- 4000)

1. Attach the X and Xs-axis origin setting jigs as shown below.



M3II Origin jig
PM070Y0



M3/M3S Origin Jig
PM57630

2. Slide the Xs-axis against the jig, and then slide the X-axis against the jig. (The photograph shows NXT II).

Caution: accurate origin settings cannot be made unless the Xs-axis is correctly positioned against the jig before the X-axis. Failure to do so may result in a crash or error.



Position Xs-axis against jig first



Next position X-axis against jig

3. Turn the machine power ON, wake the module, and then follow the procedures in “4.2 Starting I/O Monitor” and “4.3 Starting Motion Tool Mini” to display the servo counter.

Axis map

Axis 0	Y
Axis 1	X
Axis 2	Z
Axis 3	R (C)
Axis 4	Q
Axis 5	XS



X = Axis 1

4. Perform the adjustment below if the servo count is outside the M3/M3S/M3II X-axis zero setting position range of -75000 (+/- 500).
5. Set a spanner on the flange as shown in the photo below.



6. Insert two Allen bolts in the spare holes on the mechanical lock and slowly tighten them to release the lock.
7. Use an Allen wrench to remove the 4 Allen bolts, and then remove the mechanical lock.
8. Turn the motor by hand so that the counter is within the table 5 X-axis zero setting position tolerance of -75000 (+/- 500).
9. Once the counter is within tolerance, make sure the motor and ball screw gears are flush, and then evenly tighten the 4 bolts to lock the mechanical lock.
10. Check that the gear backlash is within the range **5 ~ 40 μ m**. Adjust it if it is outside the range.
11. Tighten the mechanical lock bolts with **3.92N.m** torque force.
12. Repeat step 2 to push the X-axis against the jig and confirm that the M3/M3S/M3II X-axis zero setting position is within the tolerance of -75000 (+/- 500).

13. Move the X-axis by hand and check that the servo counts at the plus and minus mechanical stoppers are within the tolerance specified in table 5. Also calculate the full stroke from the servo counts and confirm that it is within tolerance.
14. Turn I/Os Y00D, Y002, and Y001 OFF, and then select [Stop] – [Close] in I/O monitor.

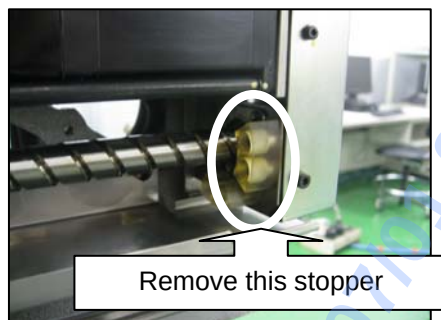
4.5.3 X-axis origin setting (M6/M6S/M6II modules)

Table 6: M6/M6S/M6II X-axis servo counter table

M6/M6S/M6II X-axis 1 pulse = 0.4 μ m (0.0004 mm)

Position	Tolerance (Pulses)
Motor side plus mechanical stopper (5.5 mm / 0.0004 = 13750 pulses)	+13750 (+/- 1250)
5.5 mm from the plus mechanical stopper	0
X-axis zero setting position (motor side + jig (20 mm))	-65000 (+/- 500)
Minus mechanical stopper	$\leq$ -1346250
Full stroke (545 mm / 0.0004 = 1362500 pulses) +/- 1.5 mm	1362500 (+/- 3750)

1. Remove the X-axis motor side stopper shown in the photo below.

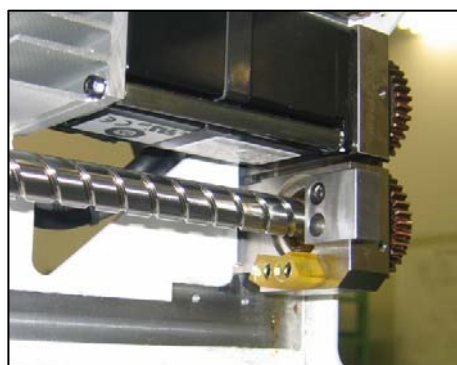


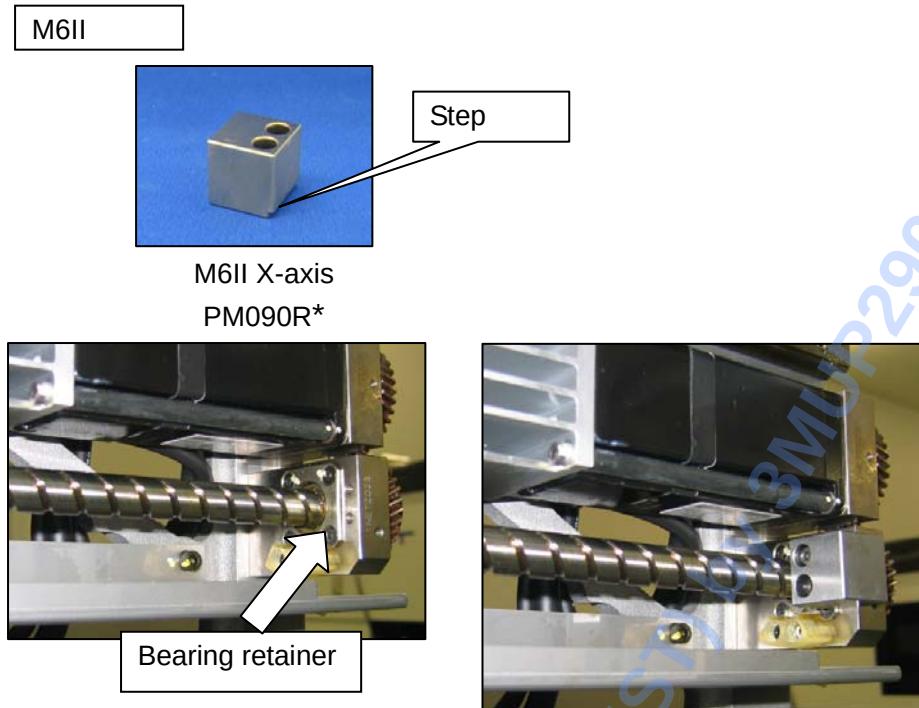
2. Attach the origin setting jig in place of the mechanical stopper.

M6/M6S



M6/M6S X-axis
PM18620





Caution: There is a step on one side of the M6II jig to prevent it interfering with the bearing retainer. Take care to ensure it is attached in the correct direction.

3. Turn the machine power ON, wake the module, and then follow the procedures in “4.2 Starting I/O Monitor” and “4.3 Starting Motion Tool Mini” to display the servo counter.

Axis map

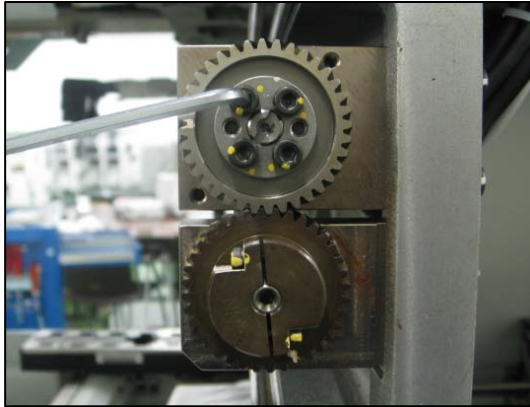
Axis 0	Y
Axis 1	X
Axis 2	Z
Axis 3	R (C)
Axis 4	Q
Axis 5	XS



X = Axis 1

4. Perform the adjustment below if the servo count is outside the table 6 X-axis zero setting position tolerance of -65000 (+/- 500).

5. Use an Allen wrench to loosen the 4 Allen bolts on the X-axis motor gear mechanical lock.

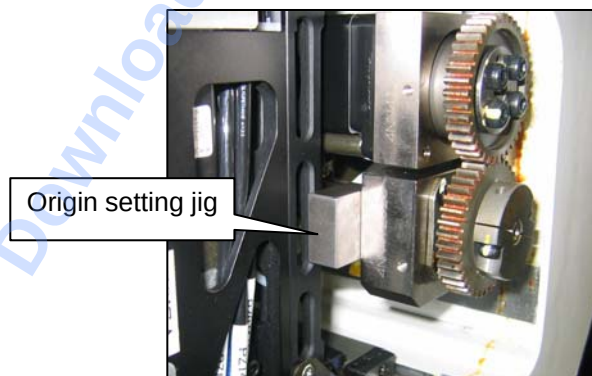


Caution: do not insert an Allen wrench in the X-axis ball screw nut when securing the gear. This will result in damage to the ball screw grooves.

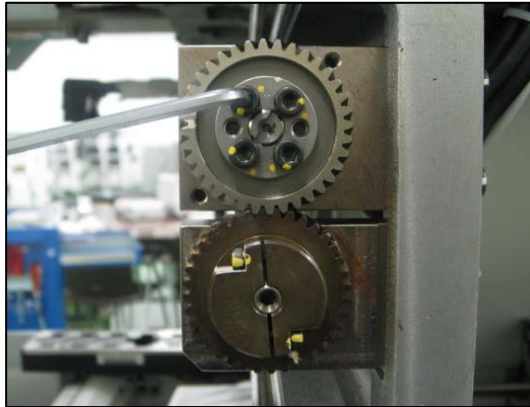
6. Insert two Allen bolts in the spare holes on the mechanical lock and slowly tighten them to release the lock.



7. Use an Allen wrench to remove the 4 bolts from the mechanical lock, and then remove the mechanical lock.
8. Rotate the motor by hand so that the servo count with the X-axis against the origin setting jig is within the zero setting position tolerance of -65000 (+/- 500) specified in table 6.



9. Once the counter is within tolerance, make sure the motor and ball screw gears are flush, and then evenly tighten the 4 bolts to lock the mechanical lock.

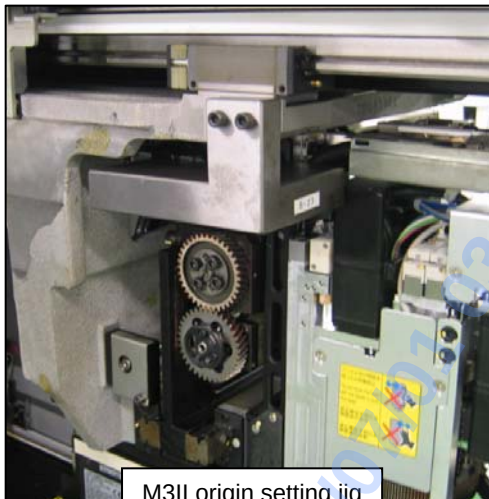


10. Check that the gear backlash is within the tolerance of **5 ~ 40 μm** . Adjust it if it is outside the tolerance
11. Lock the mechanical lock with **3.92 N.m** torque force.
12. Repeat step 2 to confirm that the servo counter with the X-axis against the jig is within the table 6 X-axis origin setting position tolerance of -65000 (+/- 500).
13. Move the X-axis by hand and check that the servo counts at the plus and minus mechanical stoppers are within the tolerance specified in table 6. Also calculate the full stroke from the servo counts and confirm that it is within tolerance.
14. Turn I/Os Y00D, Y002, and Y001 OFF, and then select [Stop] – [Close] in I/O monitor.

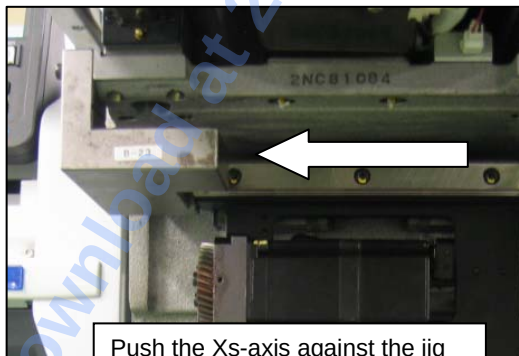
4.5.4 Xs-axis origin setting (M3/M3S/M3II Module)**Table 7: M3/M3S/M3II Xs-axis servo counter****M3/M3S/M3II** Xs-axis 1 pulse = 0.04 μ m (0.00004 mm)

Position	Tolerance (Pulses)
Motor side plus mechanical stopper (2 mm / 0.00004 = 50000 pulses)	+50000 (+/-25000)
2 mm from plus mechanical stopper	0
Xs-axis zero setting position (X motor side + jig (20 mm))	0 (+/-1000)
Position 2 (-1000000)	Adjacent LEDs are ON -995000 ~ -1005000
Position 3 (-1875000)	Adjacent LEDs are ON -1880000 ~ -1870000
Minus mechanical stopper	$\leq$ -2900000
Full stroke (119 mm / 0.000047 = 2975000 pulses) +/- 1 mm	2975000 (+/-25000)

1. Attach the X and Xs-axis setting jigs as shown in the photo below.

M3II origin setting jig
PM070Y0M3/M3S origin setting jig
PM57630

2. Push the Xs-axis against the jig.



Push the Xs-axis against the jig

3. Turn the machine power ON, wake the module, and then follow the procedures in “4.2 Starting I/O Monitor” and “4.3 Starting Motion Tool Mini” to display the servo counter.

Axis map

Axis 0	Y
Axis 1	X
Axis 2	Z
Axis 3	R (C)
Axis 4	Q
Axis 5	XS

**XS = Axis 5**

4. Perform the following adjustment if the servo counter is outside the Xs-axis zero setting position tolerance of 0 (+/- 1000).
5. Use an Allen wrench and spanner to loosen the mechanical lock (4 bolts) attaching the Xs-axis motor gear to the motor rod.

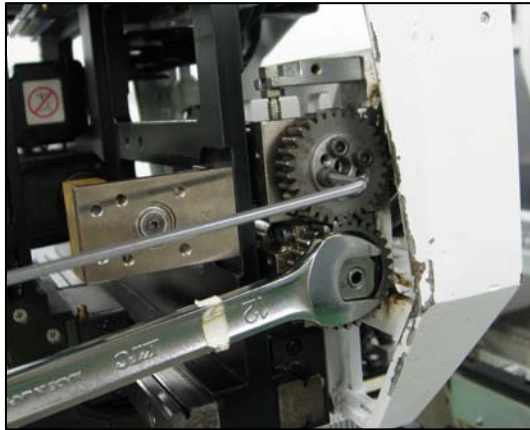


6. Insert two Allen bolts (M03X010) in the spare holes on the mechanical lock and slowly tighten them to release the lock.



7. Use an Allen wrench to remove the 4 bolts from the mechanical lock, and then remove the mechanical lock itself.
8. Rotate the motor by hand so that the servo count (when the Xs-axis is against the origin setting jig) is within the tolerance of 0 (+/- 1000) specified in table 7.

9. Once the counter is within tolerance, make sure the motor and ball screw gears are flush, and then evenly tighten the 4 bolts to lock the mechanical lock.



15. Check that the gear backlash is within the range **5 ~ 40 μ m**. Adjust the backlash if it is outside the tolerance.
16. Lock the mechanical lock with 3.92 N.m torque force.
17. Repeat step 2 to confirm that the servo counter with the Xs-axis against the jig is within the table 7 Xs-axis origin setting position tolerance of 0 (+/- 1000).
18. Move the X-axis by hand and check that the servo counts at the plus and minus mechanical stoppers are within the tolerance specified in table 7. Also calculate the full stroke from the servo counts and confirm that it is within tolerance.
19. Turn I/Os Y00D, Y002, and Y001 OFF, and then select [Stop] – [Close] in I/O monitor.
20. Follow the procedure in “4.4.5 Checking the Xs-axis sensor position” to adjust the position of the Xs-axis interlock sensor dog.

4.5.5 Checking the Xs-axis sensor position

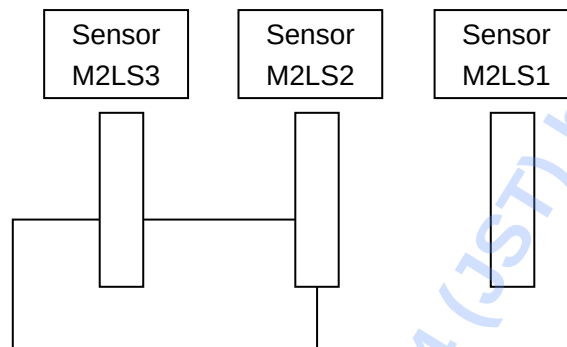
1. Check that the sensor responds correctly at the following positions. If the position is wrong, follow steps 2~6 to adjust it.

Position where sensors M2LS2 and M2LS turn ON⇔OFF

Xs-axis servo counter: within -995000 ~ -1005000 pulses

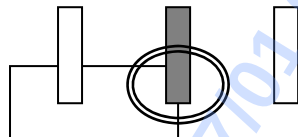
Position where sensors M2LS1 and M2LS2 turn ON⇔OFF

Xs-axis servo counter: within -1880000 ~ -1870000 pulses

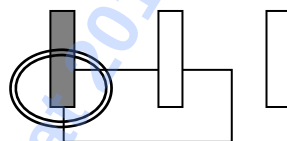


Viewed from module rear

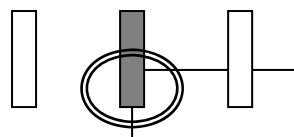
2. Adjust the dog position so that the M2LS2 turns ON and OFF between -990000 ~ -992500. (Apply glue to the two bolts fixing the dog).



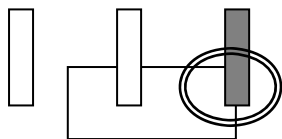
3. Adjust the M2LS3 (sensor) so that it turns ON and OFF between -1006250 to -1011250.



4. Verify that the M2LS2 turns ON and OFF between -1878750 to -1888750.



5. Adjust the M2LS1 (sensor) so that it turns ON and OFF between -1863750 to -1868750.



6. Check that the sensor is ON at both ends of the Xs-axis.
7. Return to step 1 and confirm that the sensor responds at the correct position.
8. Check that the machine operates normally in automatic operation.

MEMO:

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Chapter 5

Feeder Pallet Levelness Measurement

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CONFIDENTIAL

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5. Feeder Pallet Levelness Measurement

Warning:

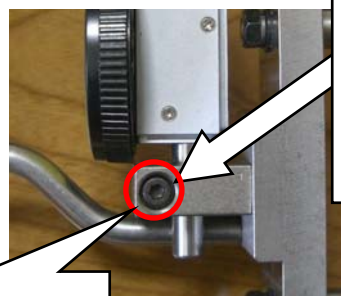
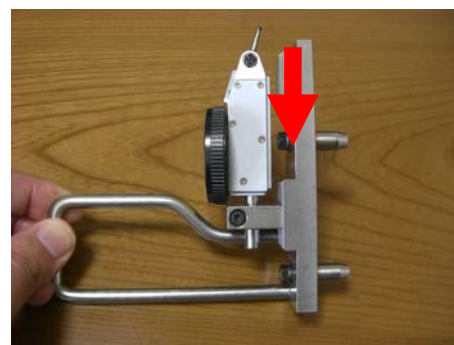
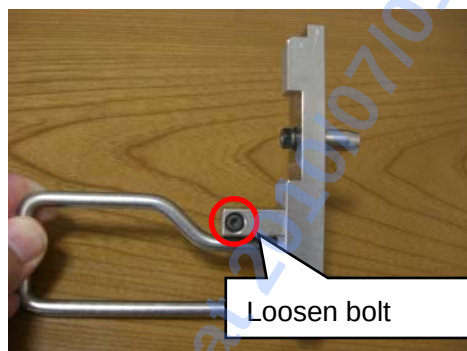
Never insert fingers or other body parts inside a moving machine. Always press the EMERGENCY STOP button and confirm that the machine has stopped before inserting your hands inside the machine to set or remove the jig.

5.1. Required Jigs

- Inspection jig (AA15S04): 3 sets
- Custom nozzle jig(AB22200): 1
- H12(HS,S), H08 head: 1
- H12(HS,S), H08 head nozzle station: 1
- Jig nozzle (11 for the H12HS,S head), (7 for the H08 head)
- Machine application CD (must be same version as the machine). This is only required if the feeder pallet levelness measurement software is not installed on the accessory software server PC.

5.2. Preparing the jig

Loosen the bolt shown in the photo below, and then insert the dial gauge in the installation hole from above.



Insert to about the middle point of the dial gauge shaft and then lock the bolt.

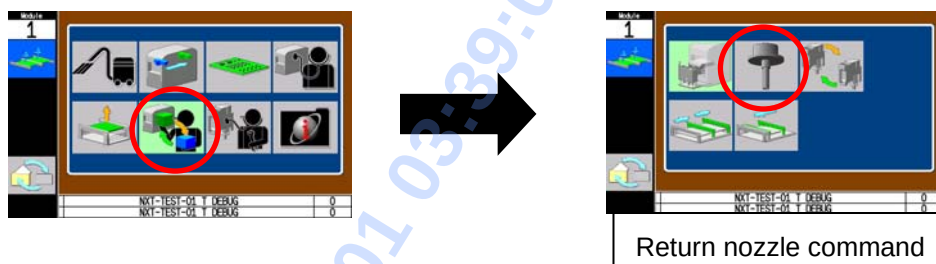
5.3. Preparing the job

Prepare a job with the placing sequences assigned to the module on which the feeder pallet levelness will be measured. There should be at least 1 placing sequence. Set the placing head to H12 (HS, S) or H08. Transmitting a job without a placing sequence will result in an "8F5A****" error.

5.4. Starting the device levelness measurement application

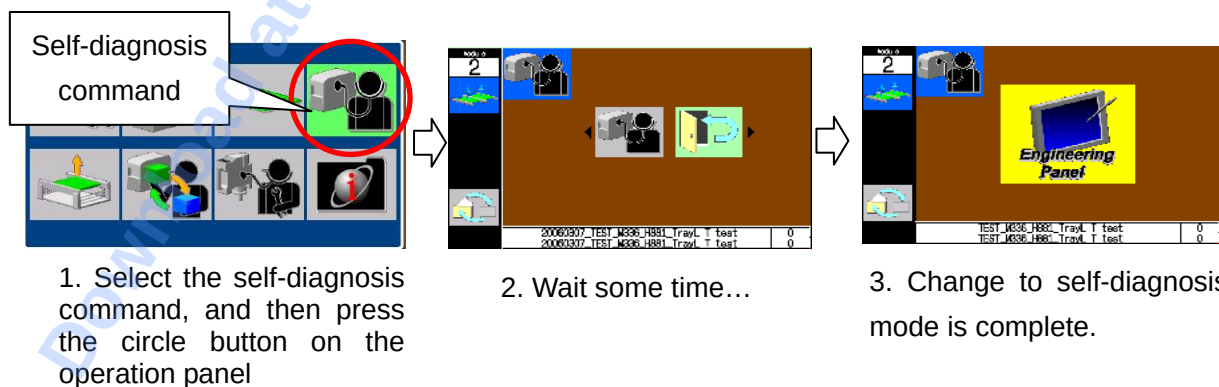
Loading the device levelness measurement software on the module.

1. In the case of M3/M3S/M3II, press the [Manual] button on the operation panel of the measurement module and the pair module next to it. In the case of M6/M6S/M6II, press the [Manual] button on that module only.
2. Use the return nozzle command in the [MANUAL] screen to return any nozzles on the head to the nozzle station.



Failure to return the nozzles here will result in an "8F55 ****" error when starting the feeder pallet levelness measurement.

3. Select the self-diagnosis command as shown in the diagram below. (In the case of M3/M3S/M3II, the pair module should also be set to self-diagnosis mode).



4. Start the accessory software on the accessory software PC. Startup Internet Explorer.



5. Enter the address (URL) of the accessory software in the address bar. Enter the following address.

http://accessory software server PC computer name/fujiweb/index.asp

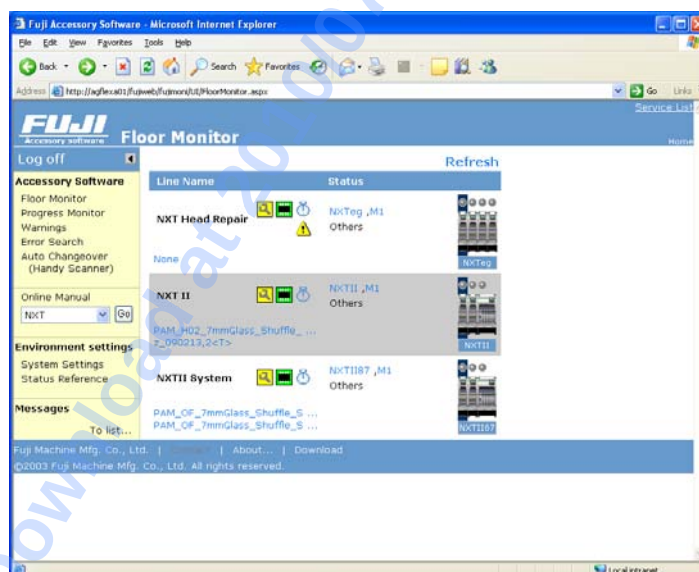


6. Startup screen display

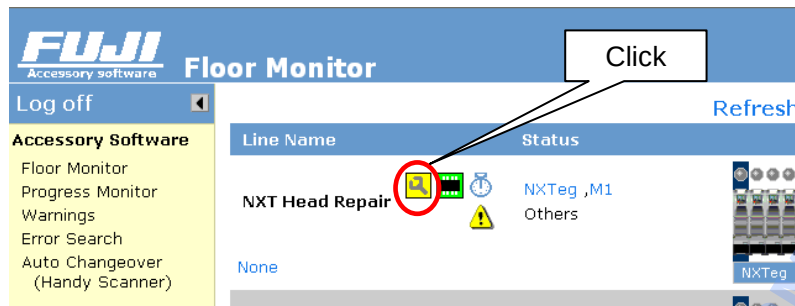
The application startup screen displays. Once startup is complete the accessory software homepage will display automatically. Please wait a little while.



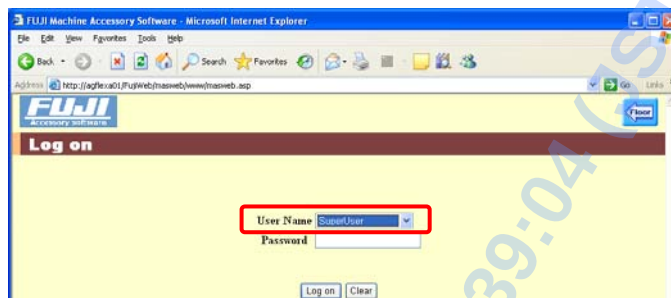
7. The Fuji accessory software homepage displays automatically once the application is ready.



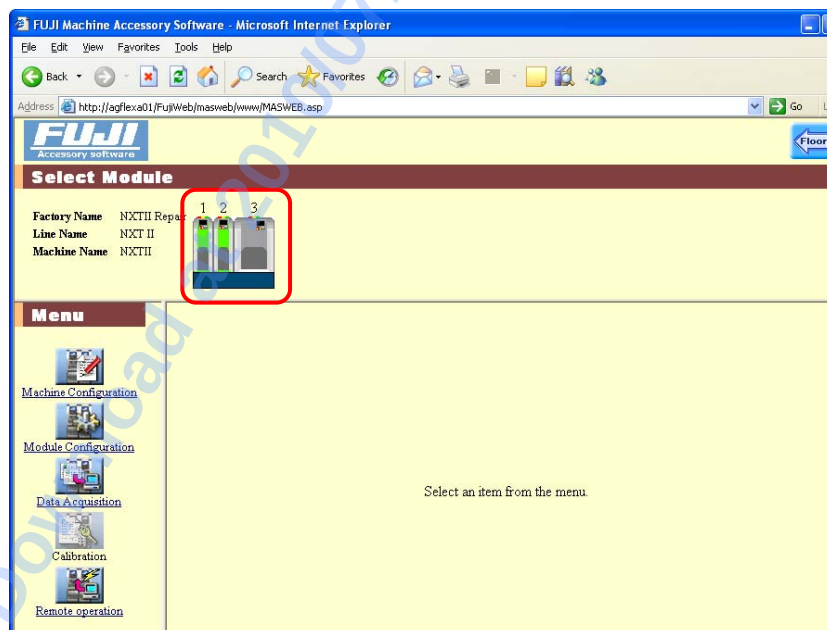
8. Click the  icon for the line on which the feeder pallet levelness will be measured. The **machine settings** screen displays.



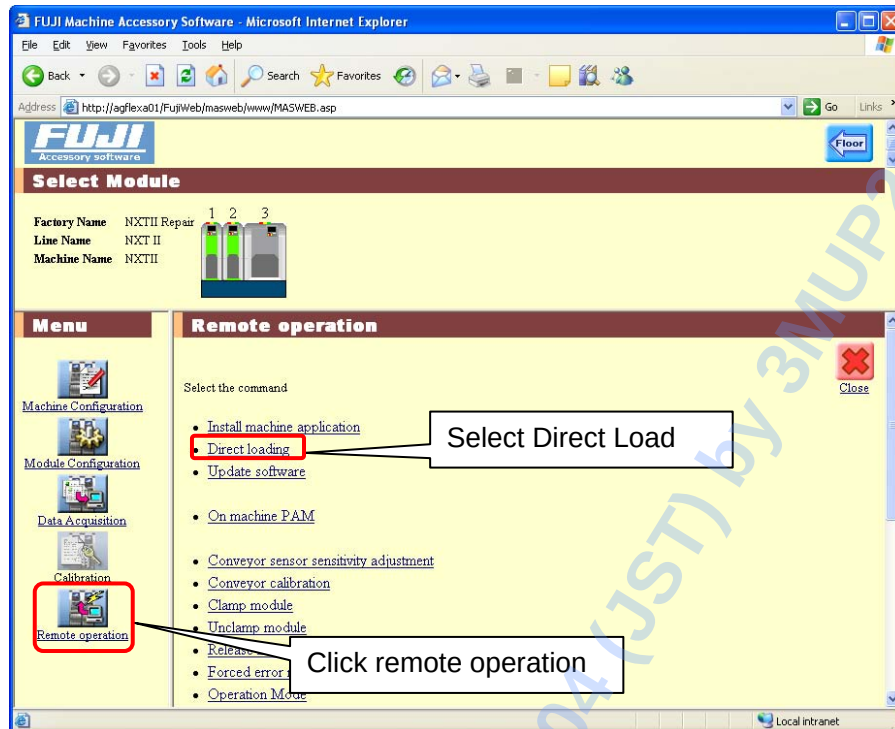
9. Enter your user name, and then click [Log On].



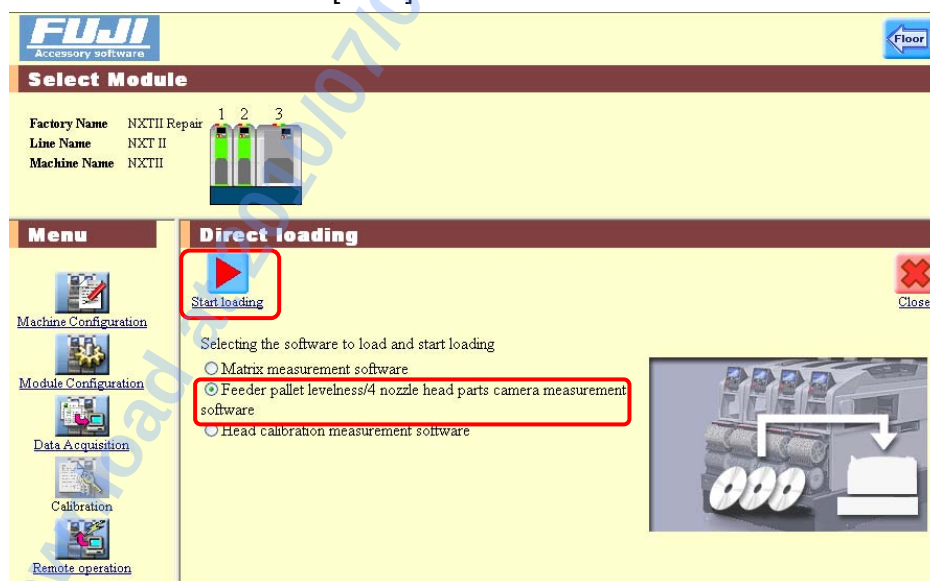
10. In the case of M3/M3S/M3II, select the measurement module and the pair module next to it. In the case of M6/M6S/M6II, select the measurement module only.



11. Click [Remote Operation] – [Direct Load].

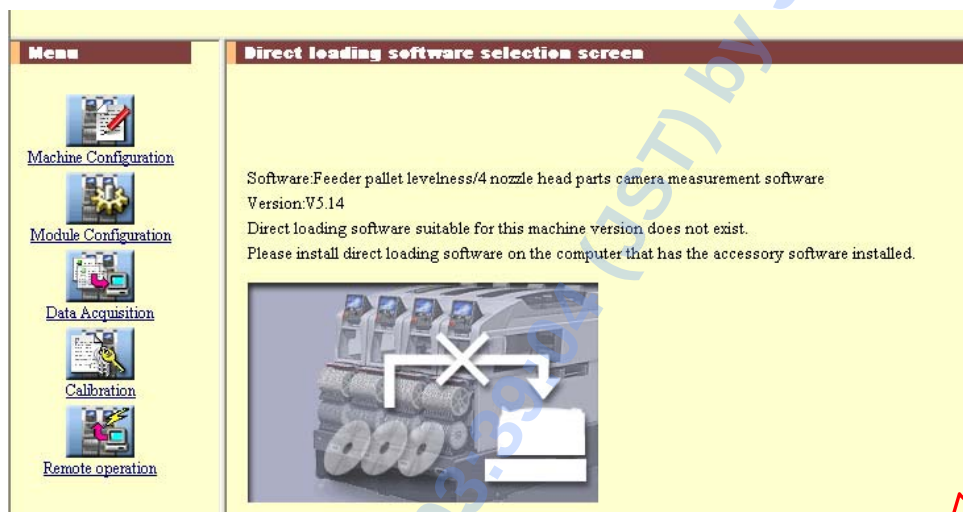


12. Select “Feeder pallet levelness/4 nozzle head parts camera measurement software” and then click [Load].



If you are requested to install the Direct loading software

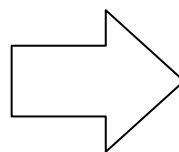
If the following request screen displays, the feeder pallet levelness measurement software version and the machine software version do not match. In this case, it is necessary to install the version of the feeder pallet measurement software that matches the machine version displayed on the screen. To install the software, follow the procedure in “5.6 Installing the feeder pallet levelness measurement software on the accessory software PC”.



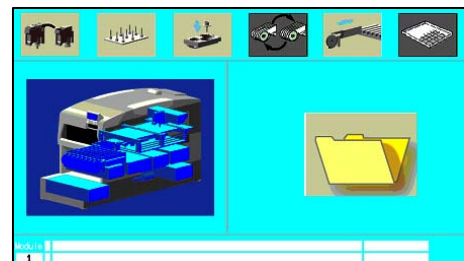
13. The feeder pallet levelness measurement software is loaded from the PC to the machine. After a while the module will restart in sleep mode (in this mode nothing is displayed on the screen). Press the “POWER” button on the module. The device levelness measurement software boots up.



The module is in sleep mode after restarting.



Press the
POWER button

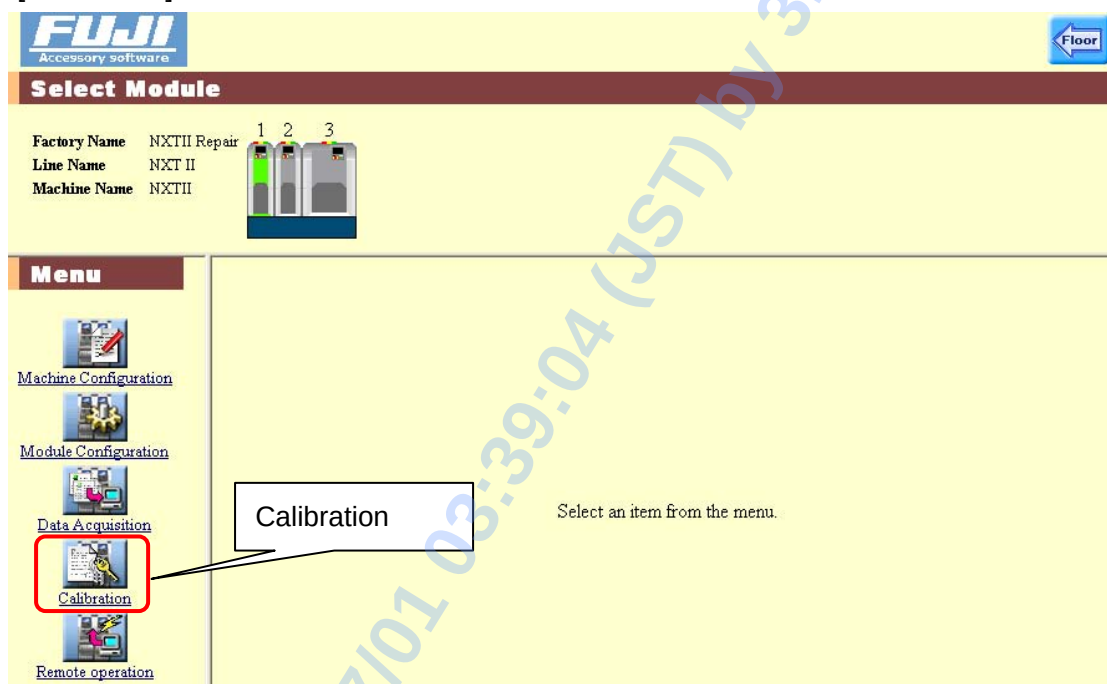


The device levelness measurement software boots up. (Job wait screen).

5.5. Preparation prior to measurement

5.5.1. Clearing the existing device levelness measurement data

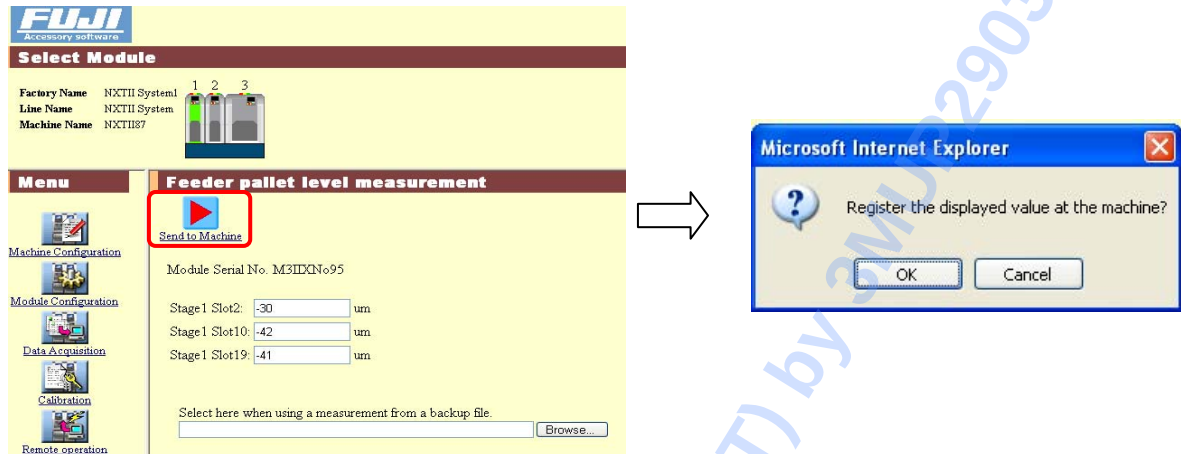
1. Start up Accessory Software.
2. In floor monitor, select the machine on which the device levelness will be measured. Click , and then enter your user name and log in.
3. Select the module on which the device levelness will be measured and click **[Calibration]**.



4. Click **[Feeder Pallet Level Measurement]**.



5. To backup the existing data, click [**Send to Machine**], and then click [OK] in the confirmation dialog. If you do not need to backup the data, proceed directly to step 8.



6. Click [Save] in the “**Save to PC**” confirmation dialog.



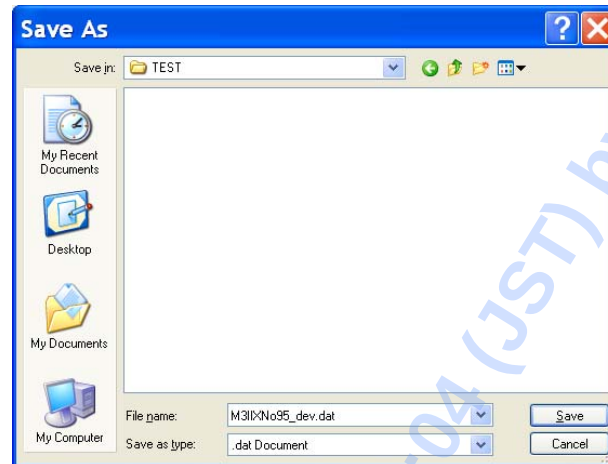
7. When the following window displays, specify the file name and save destination, and then click [Save]. The default file name format is shown below.

Backup file name: M*XNo****_dev.dat

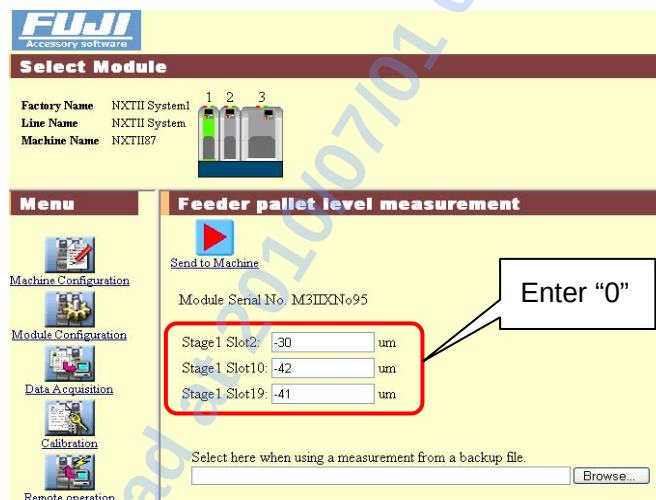
(Module serial number + _dev. Dat)

It is okay to change the default file name, but be sure to choose a name that makes it easy to recognize which module the data is for.

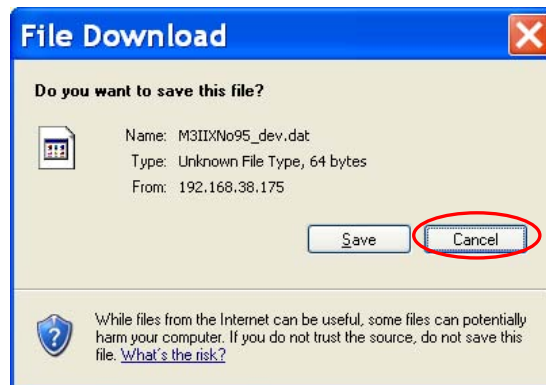
Example: M3IIXNo****_dev_org.dat



8. Set the data for slots 2, 10 and 19 (slots 2, 23, and 44 in the case of M6/M6S/M6II) to "0". Click [Send to machine] – [OK].

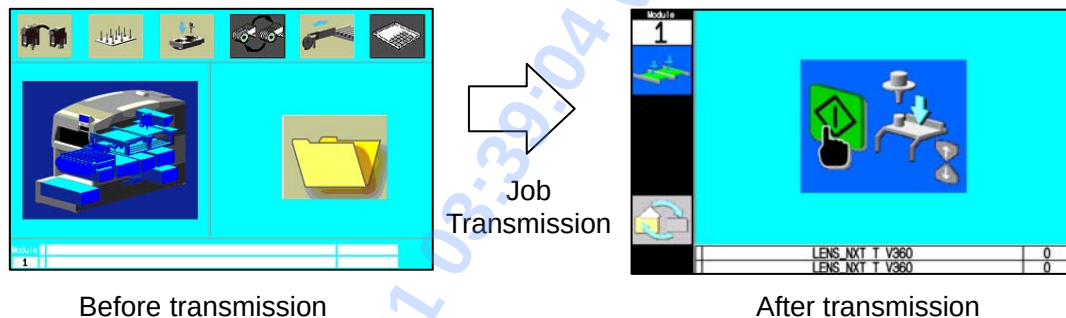


9. The “Save to PC” confirmation dialog displays, but it is not necessary to save the data here so click [Cancel].

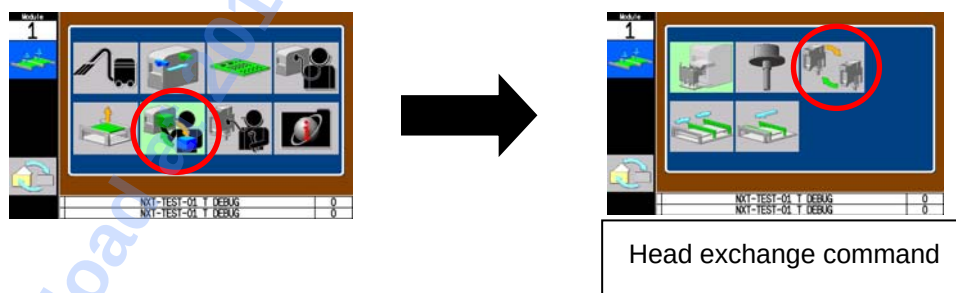


5.5.2. Performing the calibration

1. Transmit the job to the machine.



2. If an H12 (HS, S) or H08 head is not set on the module, use the head exchange command in the [MANUAL] screen to exchange the head.



Note: An “8F55 ****” error occurs if the nozzles on the head have not been returned to the nozzle station. If this error occurs, remove the nozzles by hand and return them to the nozzle station.

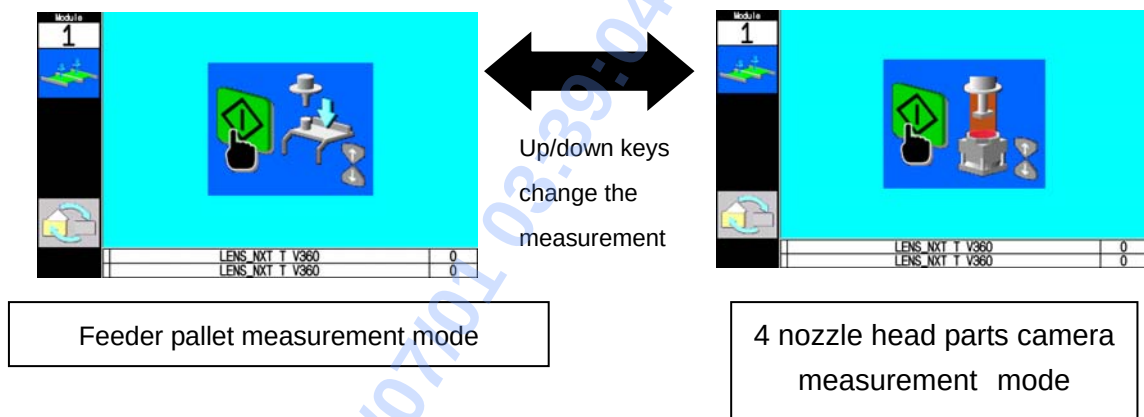
3. Press [START] to begin operation.



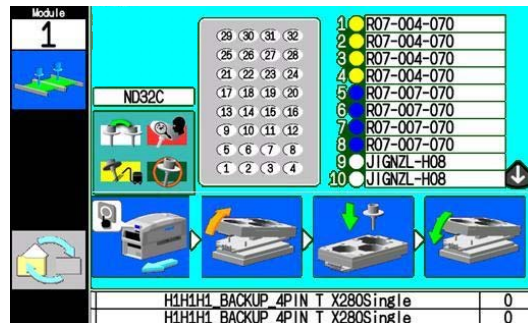
Press [START]

Note 1: If a Job without a placement sequence is transmitted, an “8F5A ****” error (“There is no nozzle on the nozzle station that matches the placement conditions.”) occurs. In this case, check the Job.

Note 2: Pressing the up and down arrow keys when the following screen is displayed will change the measurement mode. Be sure to set the mode to “Feeder pallet level measurement mode”.



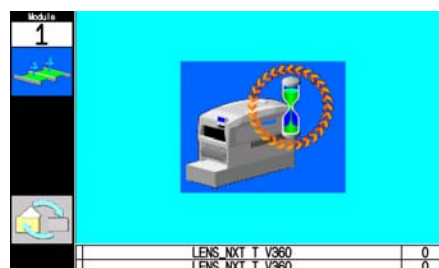
4. The following screen displays, indicating that the required nozzles and nozzle jig should be set in the nozzle station. Set the height measurement nozzle jig (shown in the photo below) in position 1 in the nozzle station. Set the other standard nozzle jigs and nozzles in any open slot in the nozzle station.



Height measurement nozzle jig

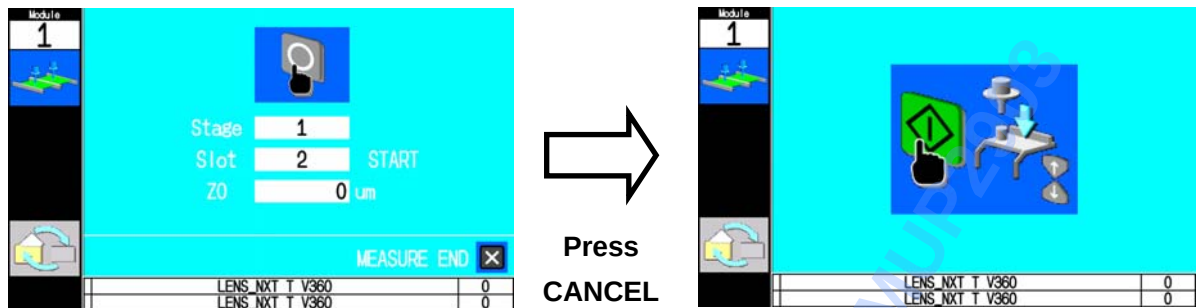


5. After setting the nozzles and nozzle jigs, press [START] to start the calibration. Wait until the calibration is finished and the start measurement screen displays.



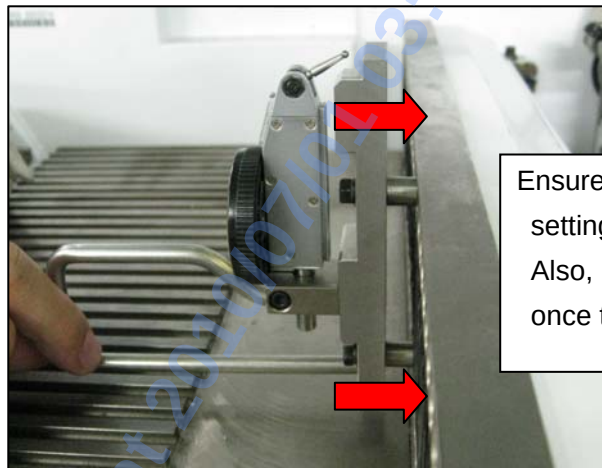
Performing calibration...

6. If the following measurement screen displays, press the "X CANCEL" button.

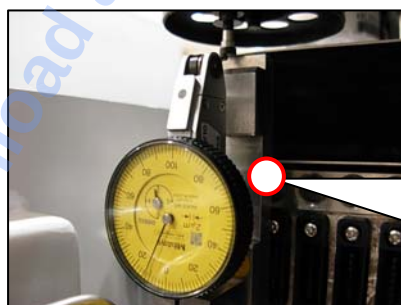


5.5.3. Attaching the pickup height measurement jigs (AA15S04)

1. Press the **EMERGENCY STOP button** and open the module front cover.
2. Set the pickup height measurement jigs (AA15S04) in the following slots on the device pallet.
 - ◆ M3/M3S/M3II module: slots 2, 10, 19
 - ◆ M6/M6S/M6II module: slots 2, 23, 44



Ensure the jigs are straight when setting them in the device pallet. Also, check they are still straight once they have been set.



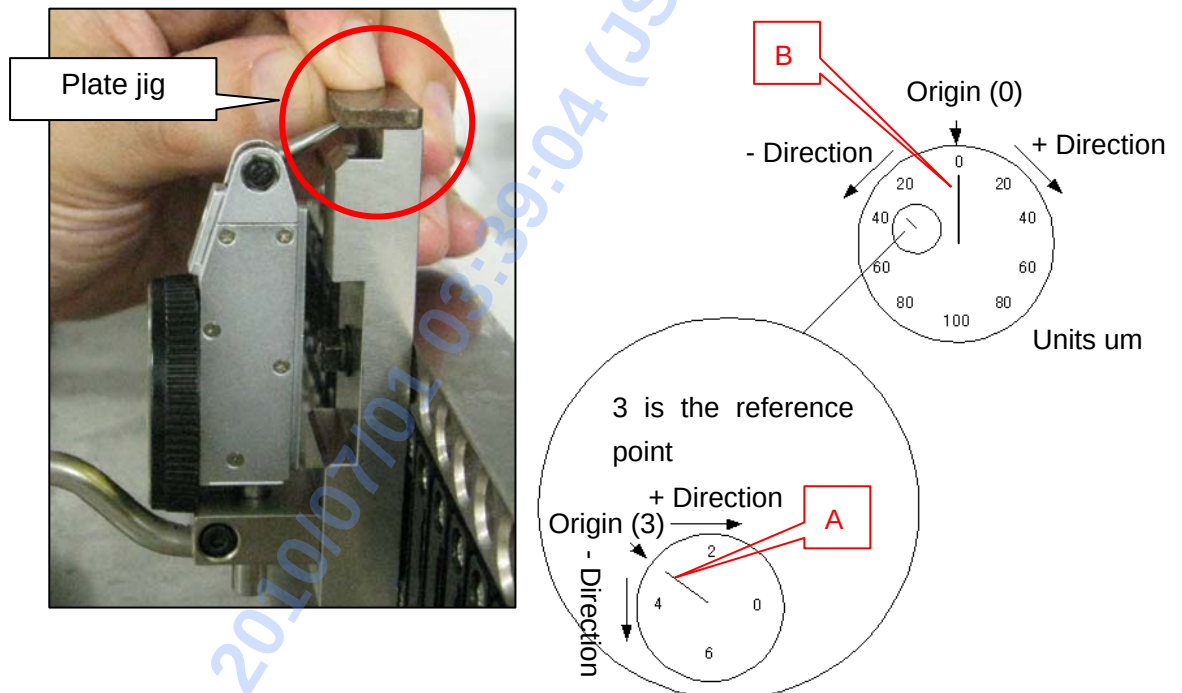
There is a possibility that the jig may be tilted, so lightly press here to ensure an accurate reading.

3. Prepare the plate jig.



4. Use the plate jig to set the dial gauge on the height measurement jig in slot 2 to zero (see photo below).

Set the dial gauge so that the short dial (A) is close to 3, and the long dial (B) is at zero. One unit on the dial gauge is equivalent to 2 μm . Therefore, 1 rotation is equivalent to 200 μm . After 1 rotation the small dial will read 2, indicating 200 μm .



Set the dial gauge so that dial B is close to "3" and dial A is at zero.

5. Repeat the procedure to set the dial gauge to zero on the pickup height measurement jigs in slots 10 and 19 (M3/M3S/M3II)/23 and 44 (M6/M6S/M6II).

5.5.4. Measurement

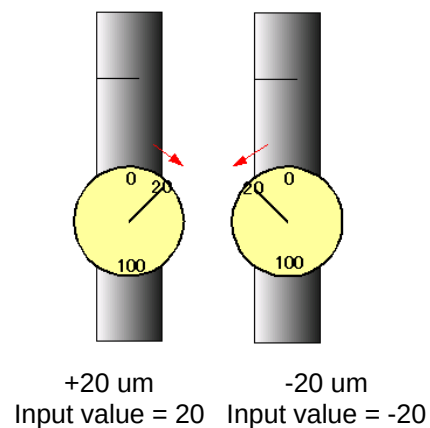
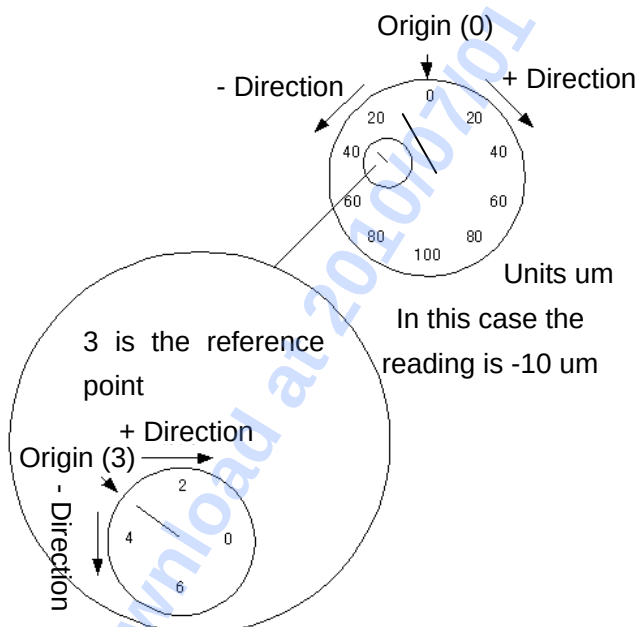
Caution: The machine starts moving once [START] is pressed. Therefore, never insert your hands inside the machine or touch the jigs once [START] has been pressed.

1. Press [START] to start the machine operation.
2. When the following screen displays, press the circle (OK) button. The head moves to the measurement point and the Z-axis slowly descends so that the jig nozzle pushes down the dial gauge on the pickup height measurement jig. Wait until the dial gauge needle stops moving.

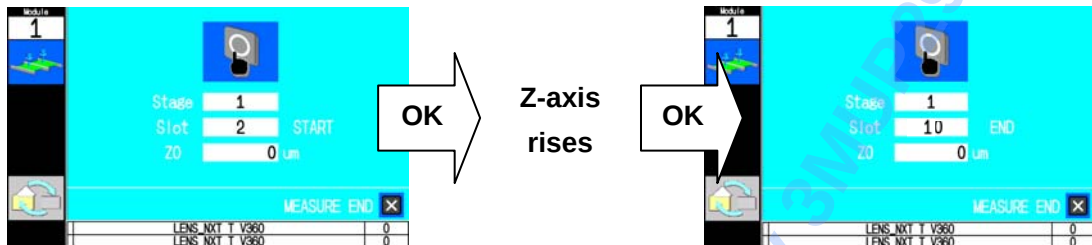


Caution! The head moves to the D2 slot measurement point.

3. Once the needle has completely stopped, note down the value shown on the slot 2 dial gauge

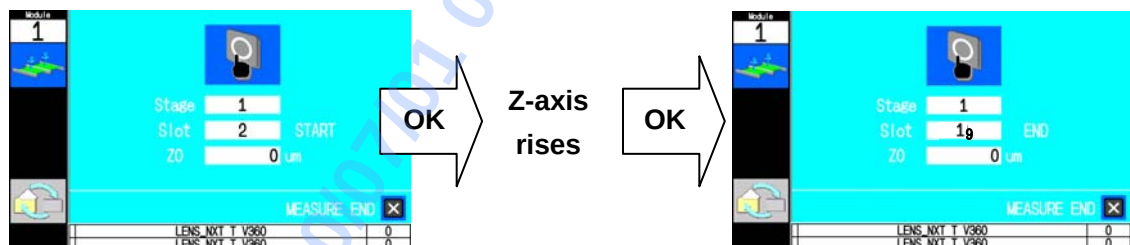


4. Once you have noted down the measurement value, press the circle (OK) button to raise the Z-axis. Press the circle (OK) button again. The head moves to the slot 10 measurement point, and the Z-axis slowly descends so that the jig nozzle pushes down the dial gauge on the pickup height measurement jig. Wait until the dial gauge needle stops moving.



Caution! The head moves to the D10 slot measurement point.

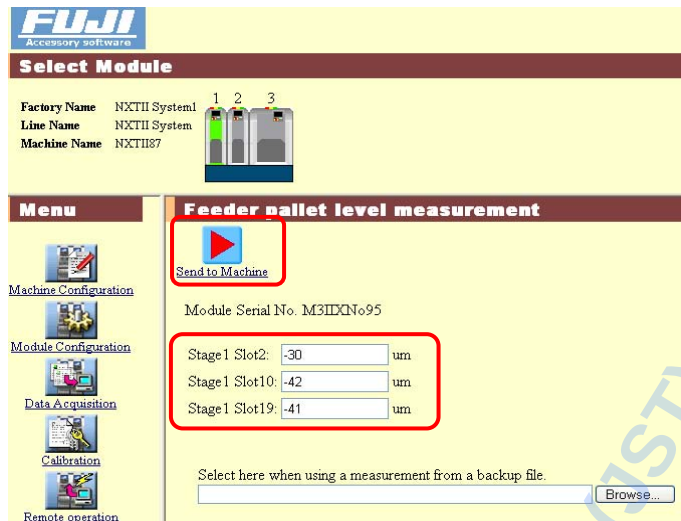
5. In the same way as step 3, note down the slot 10 dial gauge measurement value.
6. Once you have noted down the measurement value, press the circle (OK) button to raise the Z-axis. Press the circle (OK) button again. The head moves to the slot 19 measurement point, and the Z-axis slowly descends so that the jig nozzle pushes down the dial gauge on the pickup height measurement jig. Wait until the dial gauge needle stops moving.



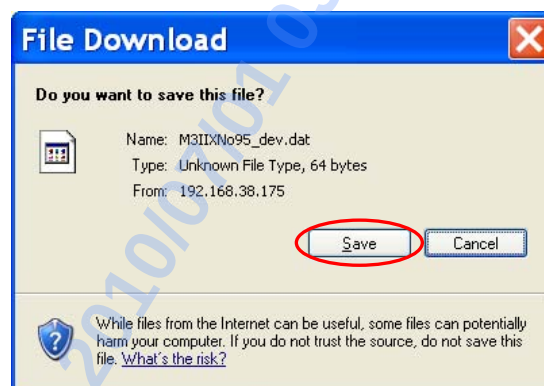
Caution! The head moves to the D19 slot measurement point.

7. In the same way as step 3, note down the slot 19 dial gauge measurement value.
8. Press the circle (OK) button to raise the Z-axis. Press the X (cancel) button to finish the measurement. (If you would like to repeat the measurement from slot 1, press the circle (OK) button).

9. At the PC, select the target module in accessory software, and then select **[Calibration] – [Device Levelness Measurement]**. Enter the measurement values for the 3 slots. (The values should be entered as they are. Do not reverse the + and – signs).



10. In accessory software, select [Send to machine] – [OK] to send the values to the machine.
11. When the save confirmation dialog displays on the PC, click [Save].

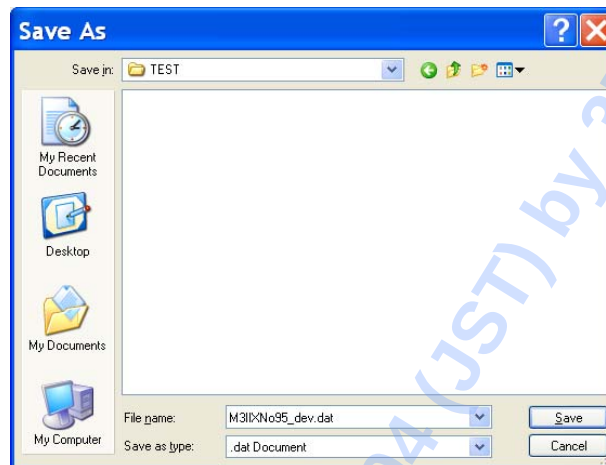


12. When the screen below displays, specify the file name and save location, and then click [Save]. By default the file is saved in the following format.

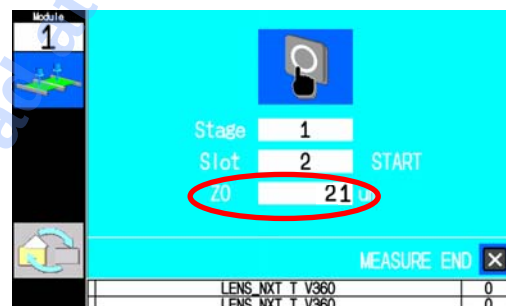
Backup file name: M*XNo****_dev.dat

(Module serial number + _dev. Dat)

It is okay to change the default file name, but be sure to choose a name that makes it easy to recognize which module the data is for.



13. Check if the offset values have been saved. Press [START] to start the machine operation.
14. When the following screen displays, click the circle (OK) button. The head moves to the measurement point, and the Z-axis slowly descends so that the jig nozzle pushes down the dial gauge on the pickup height measurement jig. The device levelness offset displays in the "Z0" field. Repeat the procedure for slots 10 and 19. Note that the offset values displayed will not be exactly the same as the values input in step 9.

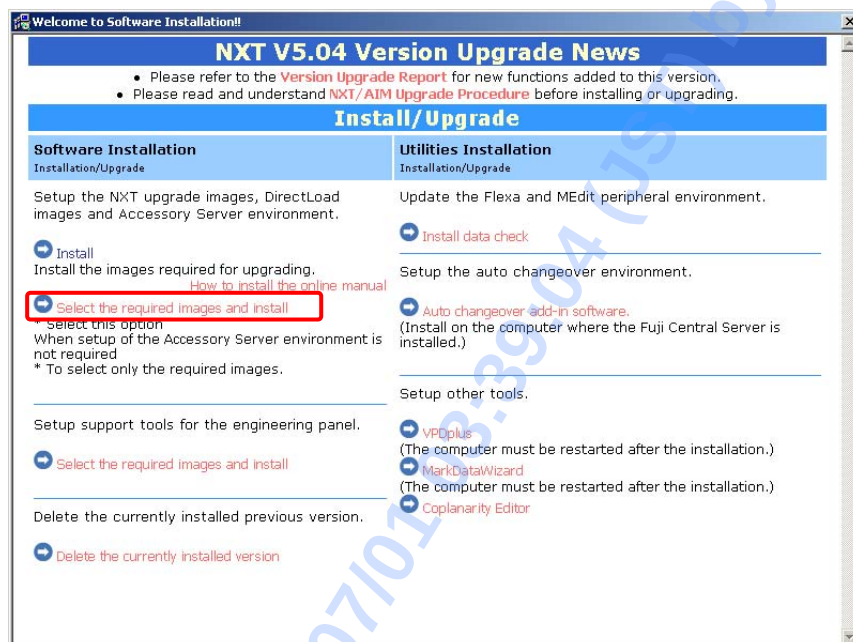


15. Repeat steps 3 to 8 to measure the height of slots 2, 10 and 19 and confirm that the offset values are reflected in the results.
16. Once the measurement is complete, turn the machine power OFF.
17. Remove all the jigs, and then turn the machine power ON again to start the machine in the normal application mode.
18. The feeder pallet measurement is now completed.

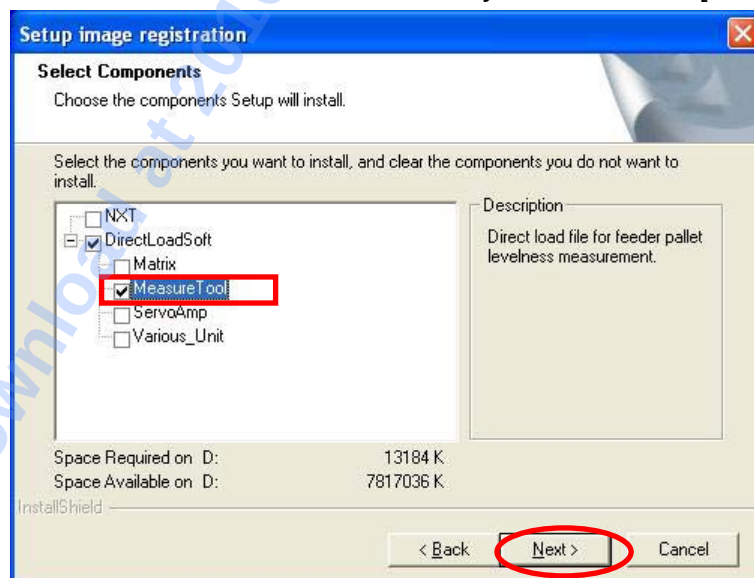
5.6. Installing the feeder pallet levelness measurement software

This section is only required if you were requested to update the direct load software in “5.5.

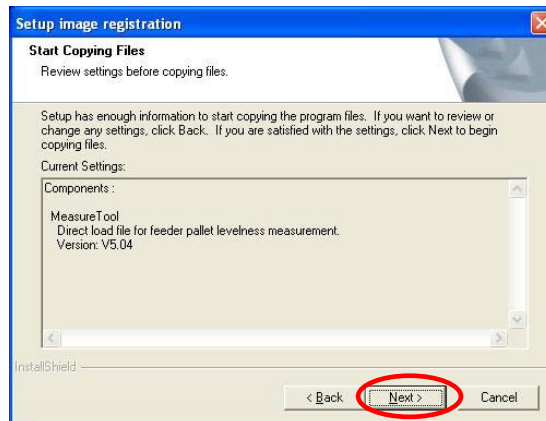
1. You will need an application CD that is the same version as the software application installed on the machine.
2. Insert the CD disk No.1 in the Accessory Software server PC disk drive.
3. The following window opens. Click “Select the required images and install”.



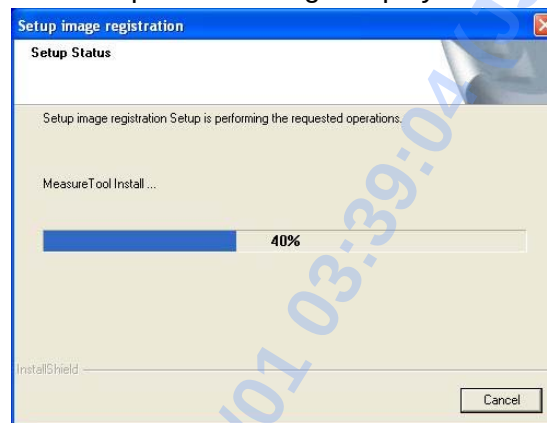
4. Select the “Measure Tool” checkbox only, and then click [Next].



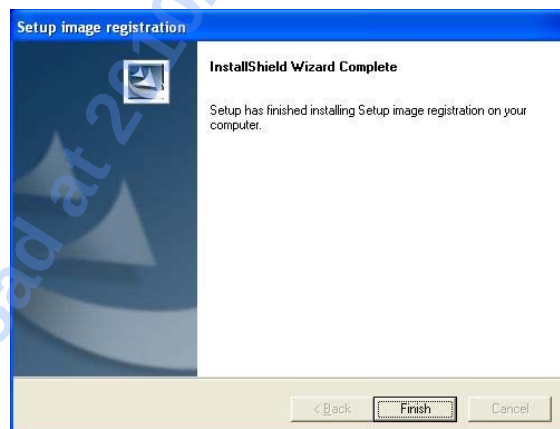
5. Click [Next].



6. The installation starts. A command prompt window opens during the installation, but do not close this window. Do not operate the computer until the “Install Shield Wizard Complete” message displays.



7. When “Install Shield Wizard Complete” displays, click [Finish] and restart the PC to complete the installation.



8. Return and continue the procedure from section “5.4. Starting the device levelness measurement application”

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Chapter 6

Matrix Measurement for NXT

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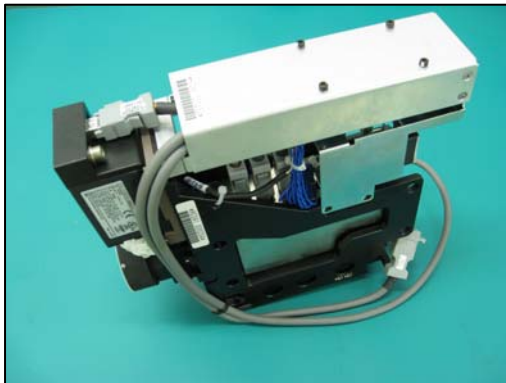
Download at 2010/07/01 03:39:04 (JST) by 3MUP2903

6. Matrix Measurement for NXT

This chapter explains the matrix measurement procedure for NXT. For the NXTII procedure, refer to “Chapter 7 Matrix Measurement for NXTII”.

6.1 Preparation for the Matrix Measurement

Required Jigs:



Matrix measurement head
(AA02S05)



Matrix glass panel
(AA1RW00)

Caution:

Note: It is possible to use the matrix glass board (AA1RW00) used in M3/M3S for matrix measurement of M3II.

Note: The matrix glass board (AA1RW00) used in NXT cannot be used when performing matrix measurement for M6 II/M6 IISP because the fiducial mark position on a conveyor differs from M6/M6S conveyor.

Note: Turn ON the module (keeping-warm state) for 3 hours or more before performing Matrix Measurement.

Note: Be sure to push the emergency stop switch when you put in the body in a module when the module power supply is ON.

Note: Perform measurement where there is no module next to a measuring object module. (Check that there are no other peripheral equipment around the shuttle conveyor, double conveyor etc. next to a measuring object module)

Note: Matrix measurement cannot be performed on machines configured to use MTUL/LT. Attach a feeder pallet to perform matrix measurement on these machines.

Note: The matrix board will interfere with the auto backup pin stacker on machines equipped with this function. To avoid this, remove the bracket on which the 2D code is attached.

Note: Matrix measuring head (AA02S0*) for NXT and NXTII are common.

1. To perform the matrix measurement for the M3/M3S modules, it is recommended that the modules next to the target M3/M3S modules be removed.
2. To perform the matrix measurement for the M6/M6S modules, it is not necessary to remove the modules next to the target M6/M6S modules.
3. Matrix measurement cannot be performed on M6/M6S modules equipped with MTU-L/MTU-LT units. A feeder pallet should be attached to perform matrix measurement on these machines.
4. To perform the matrix measurement for the M3/M3S modules, it is necessary to turn OFF the Xs-axis interlock. First, turn OFF the base power.

Caution: When the module power is OFF but the base power is ON, the power is still supplied to the control box. Ensure that the base power is OFF.

Note: For M3/M3S module, the Xs-axis moves over the maximum range during the matrix measurement. To allow this, it is necessary to turn OFF the Xs-axis interlock.

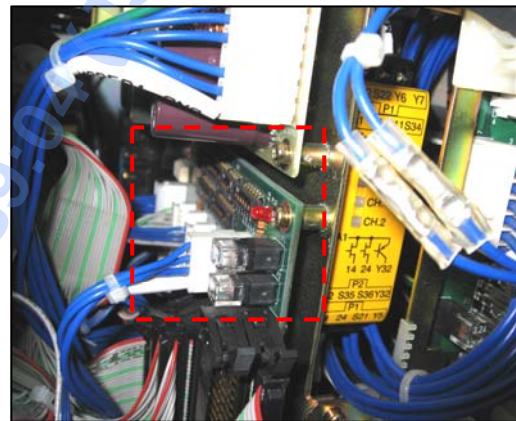
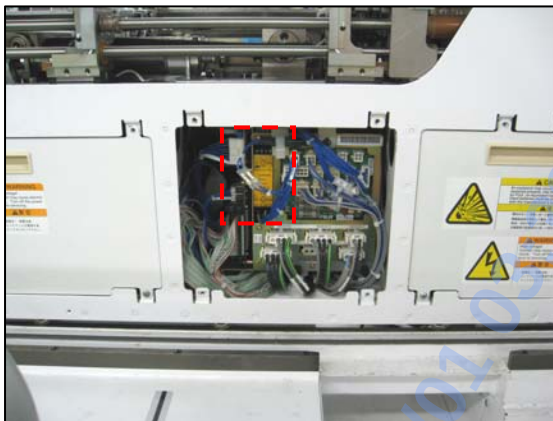
5. Open the control box cover of the M3/M3S modules.



Note: It is also possible to use the Module Change Unit (MCU) to make it easy to access to the control box.



6. Locate the interface board of the control box.

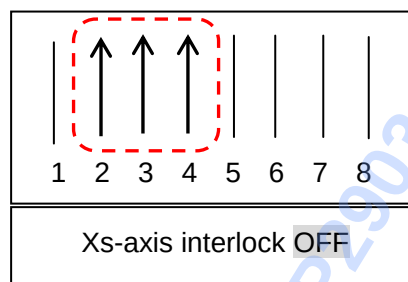
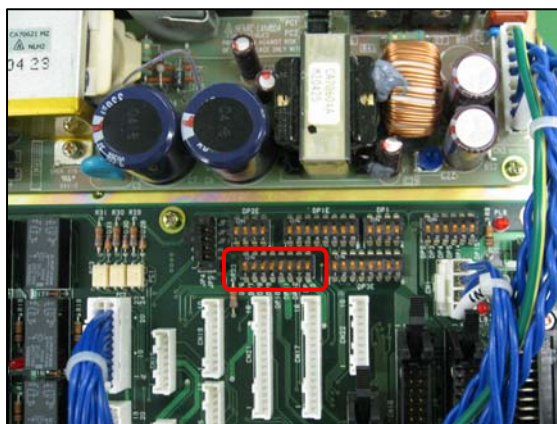


7. First, record the dip switch settings before changing their settings.
8. Turn the 2, 3, 4 dip switches UP by using a tool such as tweezers or small standard screwdriver. (When the dip switches are already turned UP, it is not necessary to change the settings.)

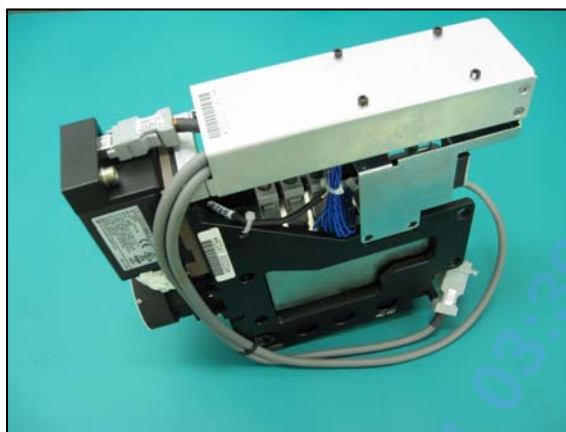
Caution: Do not change any other dip switches. Only change the 2, 3, 4 dip switch settings.

Caution: The original dip switch settings vary with the module type and the period of manufacture.

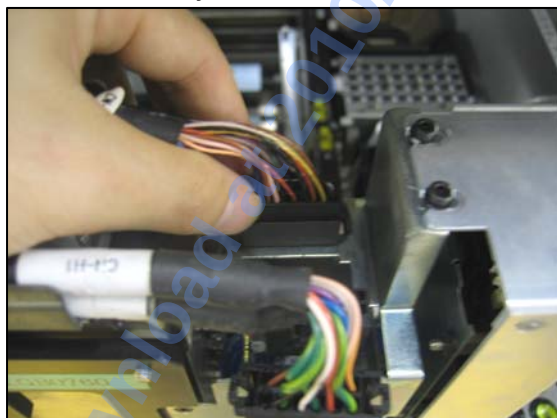
Caution: It is NOT necessary to remove the control box from the module to change the dip switch settings.



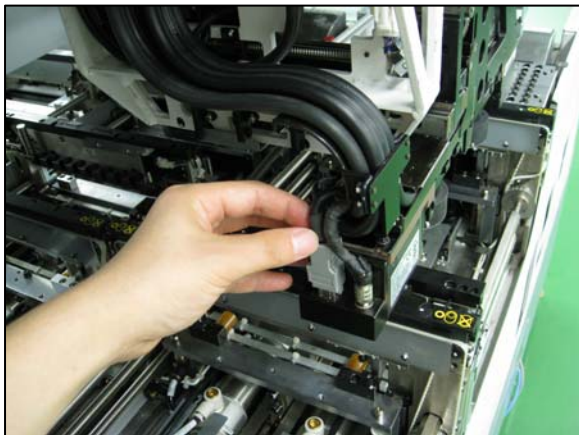
9. Remove the placement head from the module.
10. Set the matrix measurement head (AA02S05) to the module.



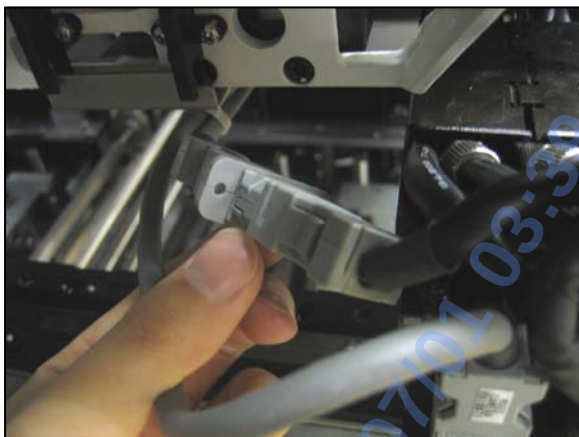
11. Connect the connector to the matrix measurement head. (Connect just one connector. It is not necessary to connect the rest two connectors.)



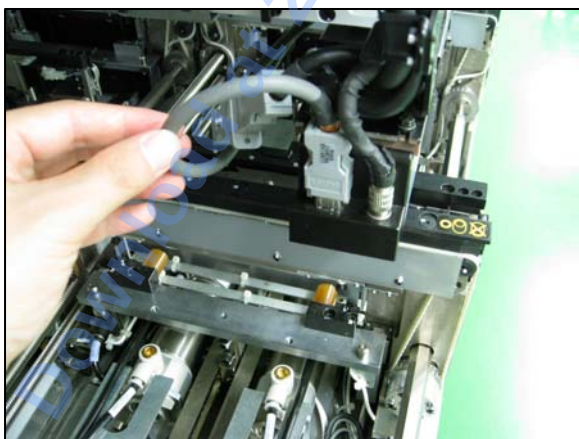
12. Remove the IEEE 1394 cable connector from the mark camera unit.



13. Reconnect the IEEE 1394 removed in the previous step to the matrix measurement head.



14. Connect the IEEE 1394 cable connector of the matrix measurement head to the mark camera unit.



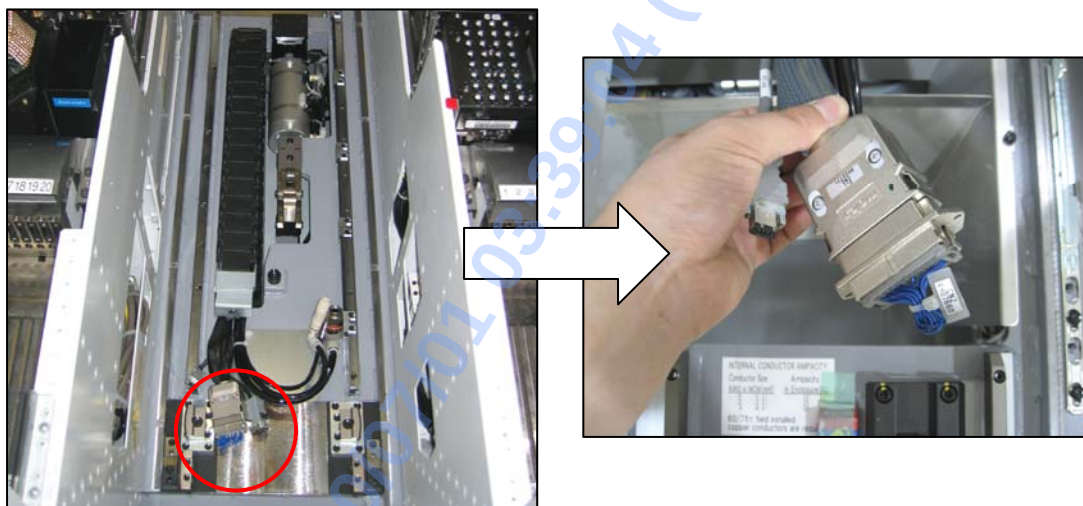
15. Ensure that the IEEE1394 cable does not interfere with the XY-robot assembly. Also, ensure that the IEEE 1394 cable does not interfere with any part of the conveyor unit and the nozzle station.

Caution: After attaching the matrix measurement head, leave the head with the module power ON condition for more than 2 hours to warm up the mark camera and matrix head camera.

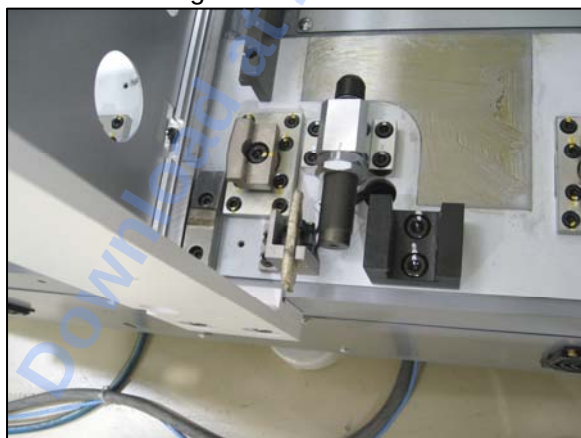
16. When performing the matrix measurement for the M3/M3S modules and having removed the modules next to the target M3/M3S modules, complete the steps 17, 18, and 19. When you perform the matrix measurement for the M6/M6S modules, go to step 20.
17. Attach the 50-pin jumper connector to the 50-pin connector cable that was connected to the module removed in step 1.



50-pin jumper connector



18. Set a material such as feeler gauge or ruler to turn ON the clamp position check sensor and the docking check sensor.



19. Put a magnet to turn ON the clamp check sensor.

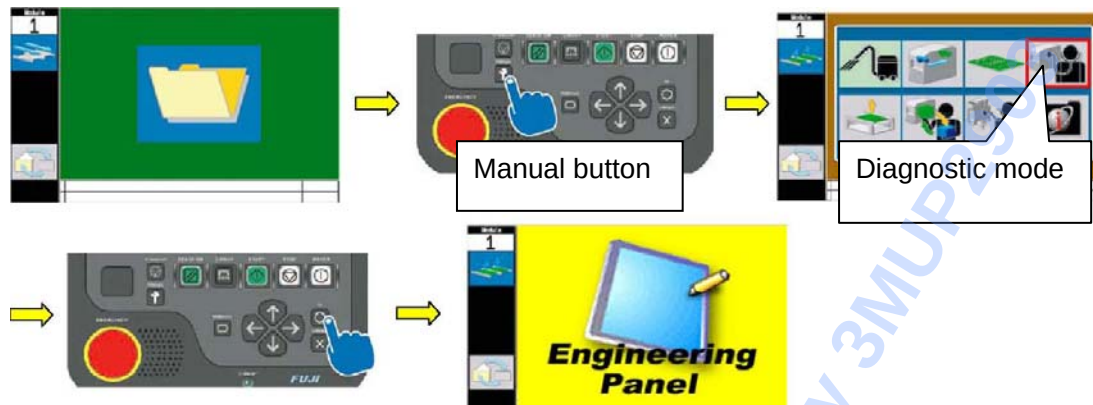


20. Turn ON the base power and the module power.

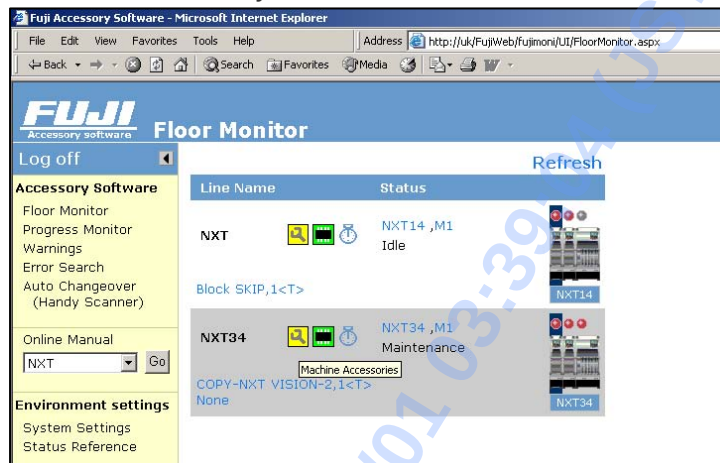
Note: After attaching the matrix measurement head, the power should be supplied in wake mode for at least two hours to prevent changes in temperature affecting the measurement results.

6.2 Performing the Directload of the Matrix Measurement

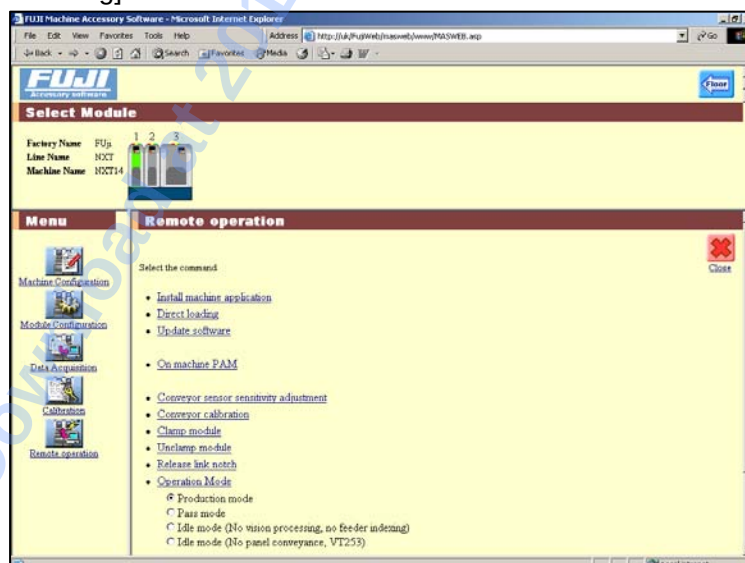
1. Change the module to the Diagnostic mode



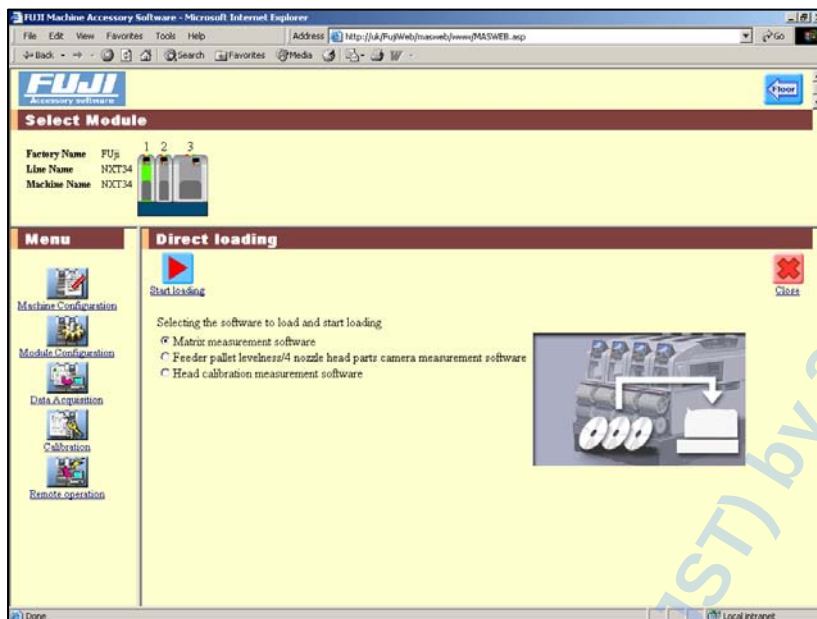
2. Start the NXT Accessory Software.



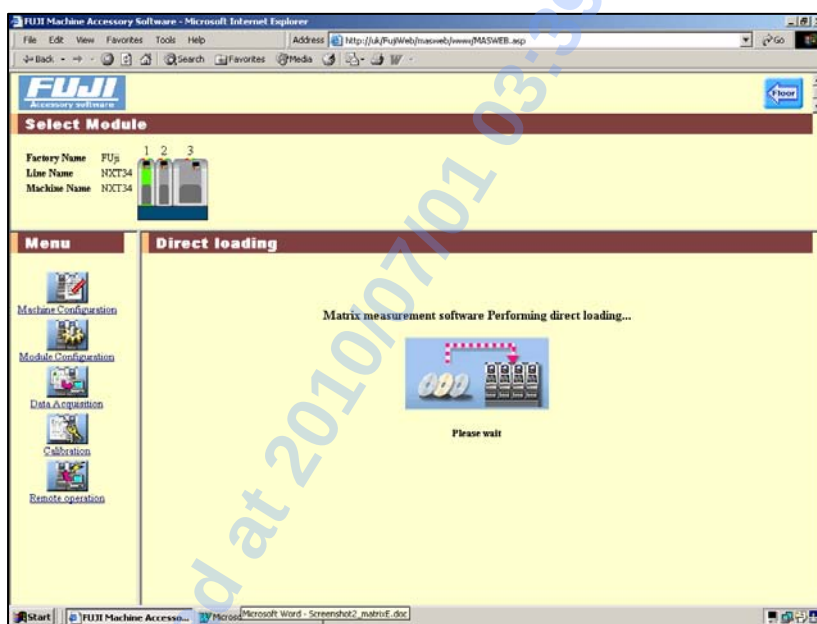
3. Select the module to perform the matrix measurement. Select [Remote Operation] – [Directloading].



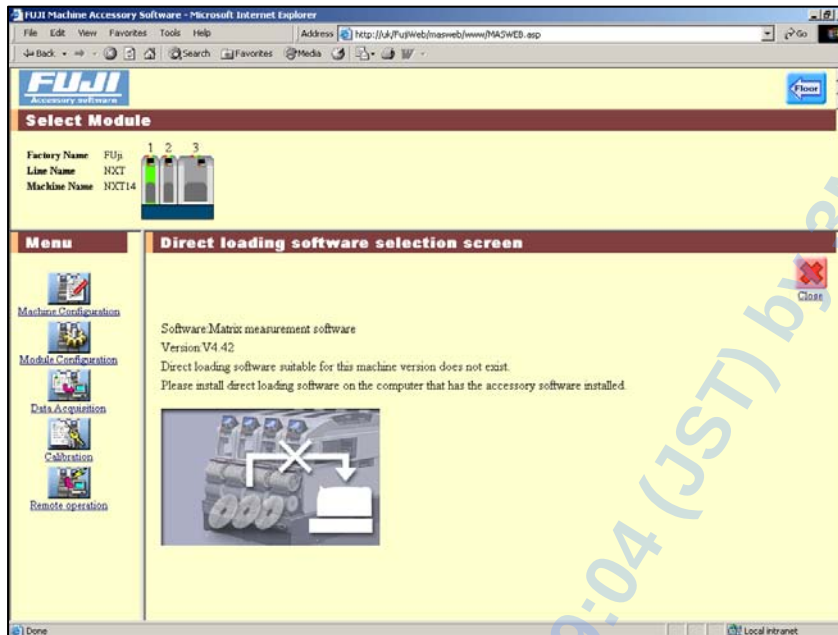
4. Select [Matrix measurement software] and then click [Start loading].



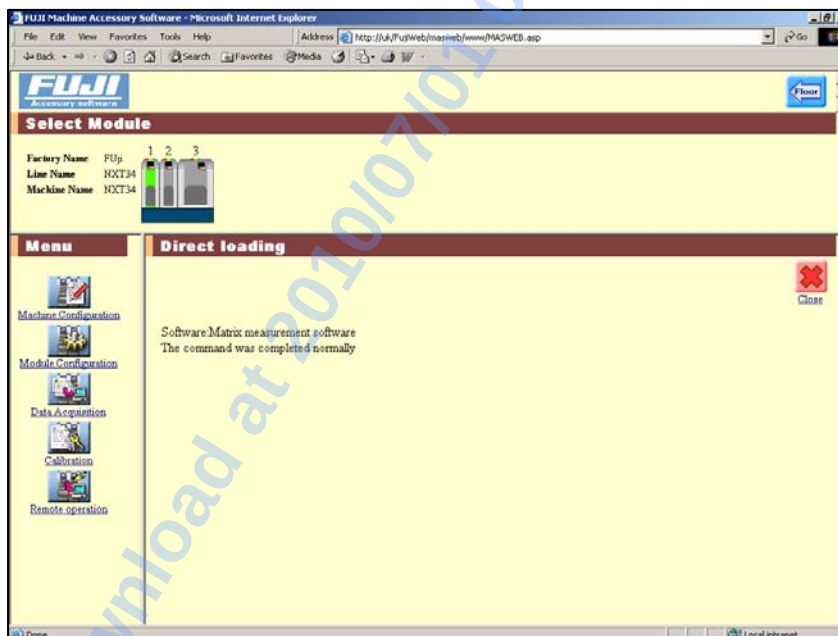
5. The following screen appears.



Note: If the following screen appears, check the application version of the module and install the Directload files to the PC from the NXT software CD-ROM which is the same version of the module.

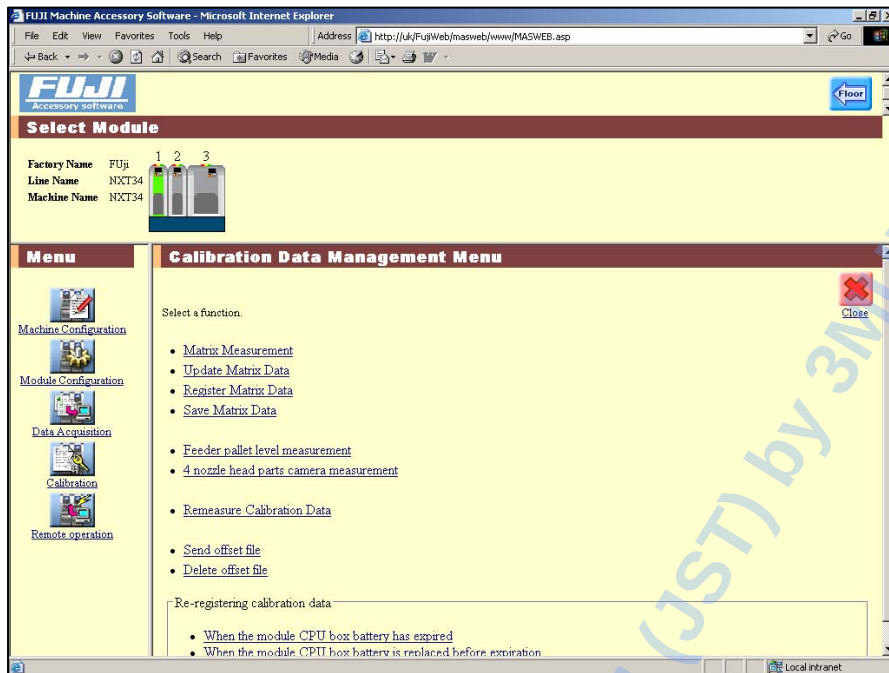


6. When the directloading is complete, the following screen appears.

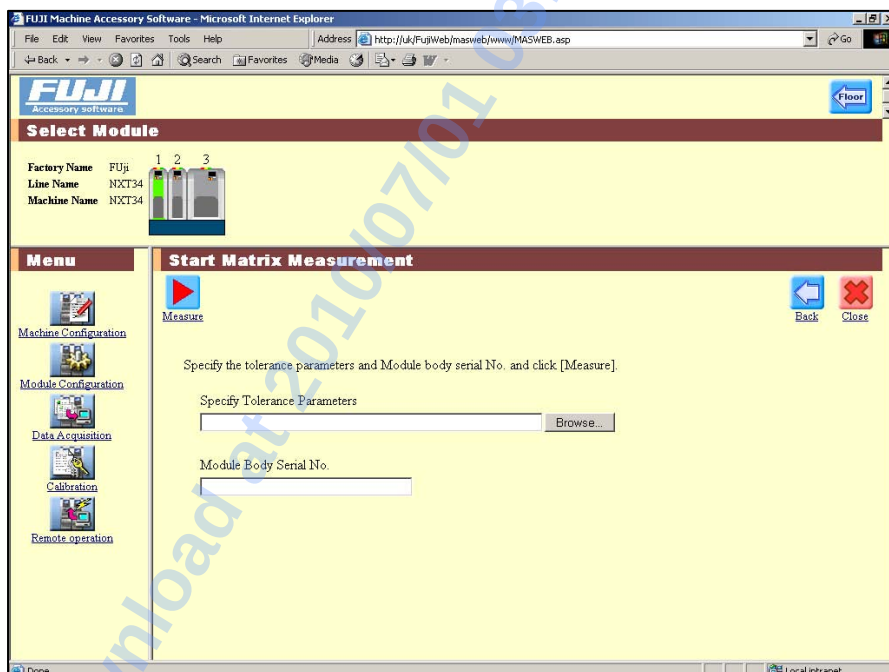


7. When the directloading is complete, the module begins to reboot. Then, press the [POWER] button to turn ON the module.

8. Select [Calibration] – [Matrix Measurement] in the accessory software.



9. The following screen appears.

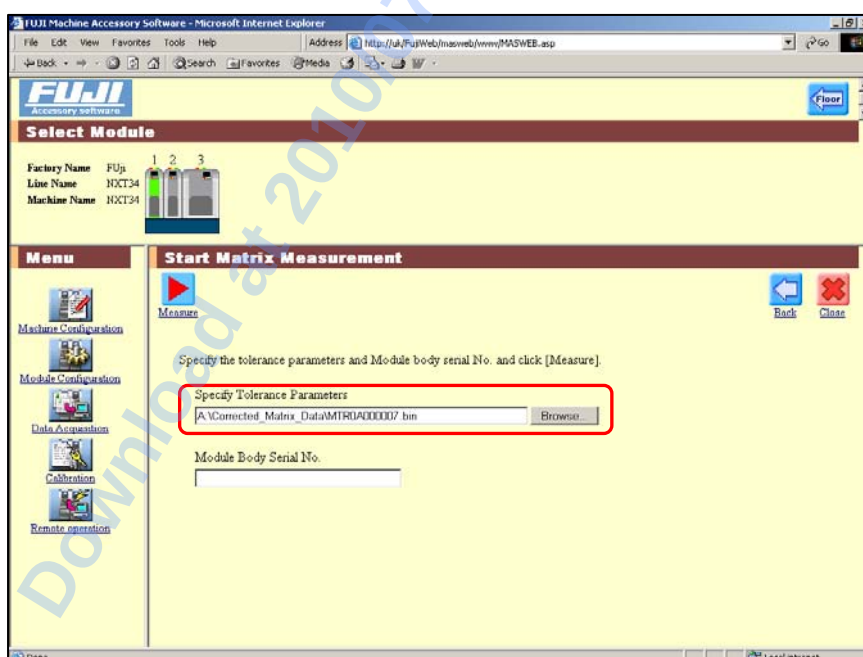


10. Install the floppy disc attached with the matrix glass panel to the floppy disc drive of the PC. This floppy disc includes the corrected matrix data file.

Note: Each matrix glass panel has its own corrected matrix data. Ensure that the serial number of the matrix glass panel matches the one of the floppy disc.

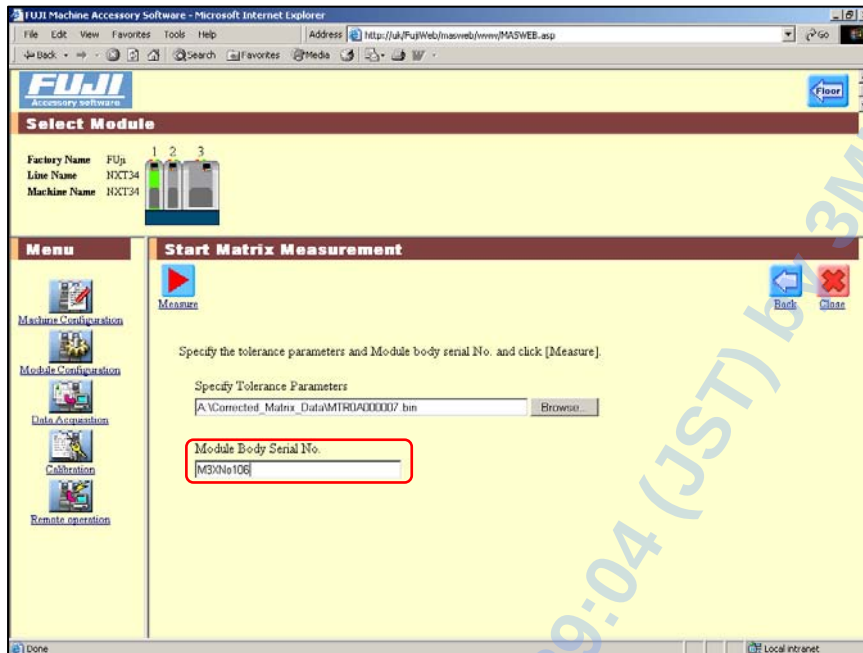


11. Click [Browse] and open the corrected matrix data file.

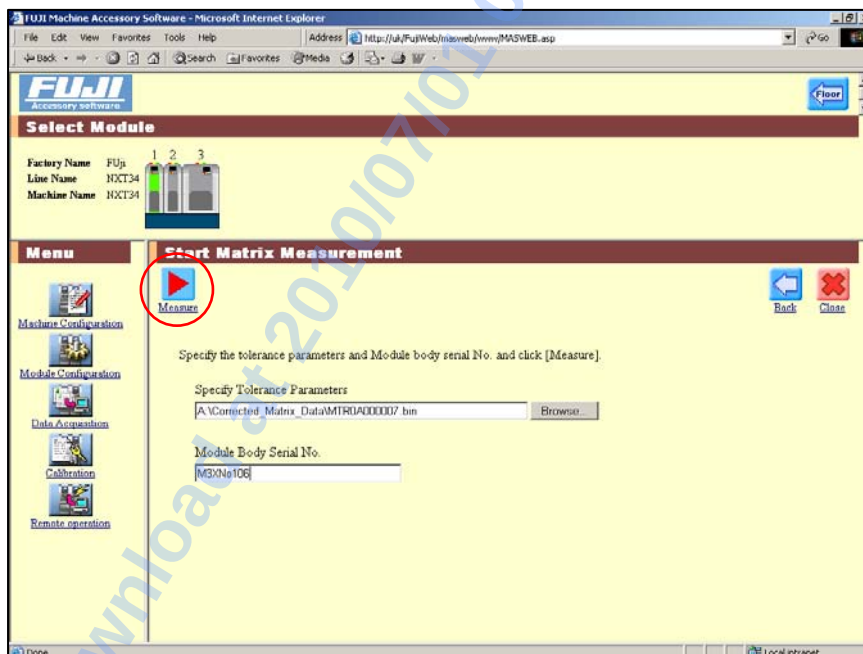


12. Type the serial number of the module. For example, when the M3S module serial number is 106, type M3XNo106.

Note: If the module type is either M3S or M6S, do NOT type M3SXNo or M6SXNo but type M3XNo or M6XNo.



13. Click [Measure].



6.3 Performing the Matrix Measurement (V4.30 or later)

Caution: The measurement procedure varies with the application software version. When the application software installed in the module is V4.30 or later, refer to the following procedures. When the application software version is V4.20 or earlier, refer to "6.4 Performing the Matrix Measurement (V4.20 or earlier)."

1. The operation panel displays the following screen and the [START] button flashes. Then, press the [START] button.



2. The matrix measurement head begins to move and read three fiducial marks on the conveyor. Then, the conveyor rail moves to the matrix measurement position.



3. Ensure that the matrix measurement head and conveyor stop. Then, press the [EMERGENCY] button.



4. Set the matrix glass panel (AA1RW00) on the module. Ensure that the serial number of the panel comes to the module front side.

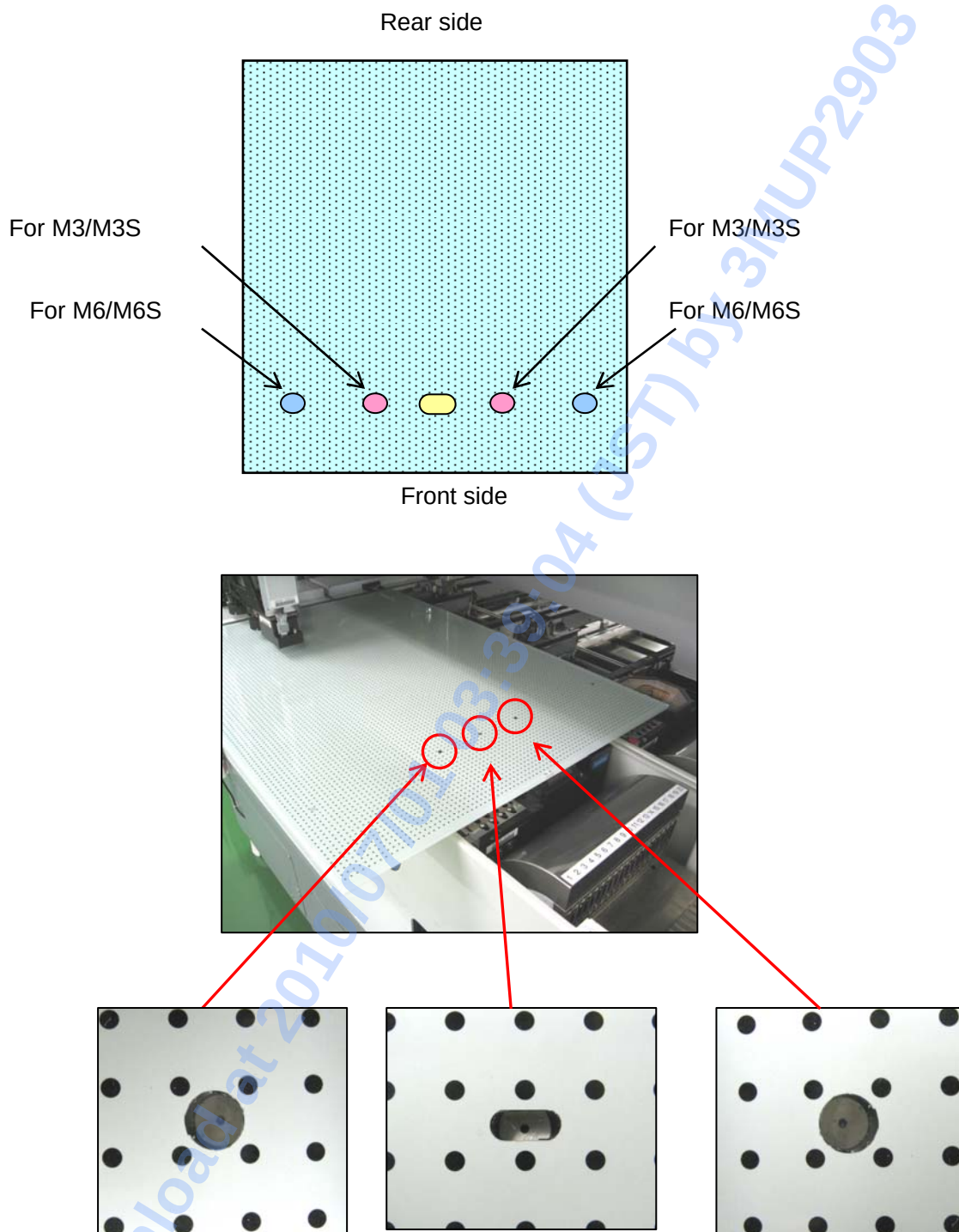
Caution: Ensure once more that the [EMERGENCY] button is pressed.

Caution: When carrying the matrix glass panel, pay attention not to damage the glass surface.



The serial number of the matrix glass panel comes to the module front side.

5. Adjust the matrix glass panel position so that the three fiducial marks on the conveyor come to the center of each three holes on the matrix glass panel.



6. Press the [START] button.

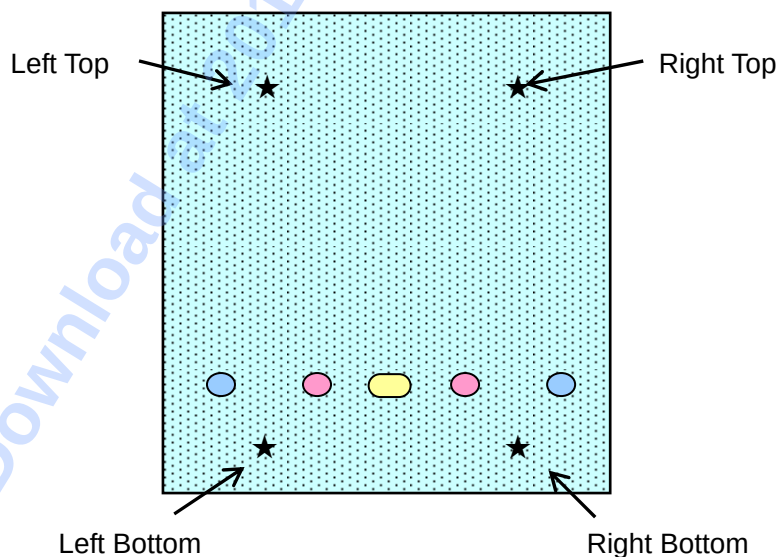


7. The buzzer beeps. (This is not the error alarm.)
8. The matrix measurement head begins to move and read three fiducial marks on the conveyor rail. Then, it moves to the Left Bottom position (See the picture and the drawing below).



Matrix Board Setting Gap

Left Top => (X um, Y um)
Right Top => (X um, Y um)
Left Bottom => (X um, Y um)
Right Bottom => (X um, Y um)

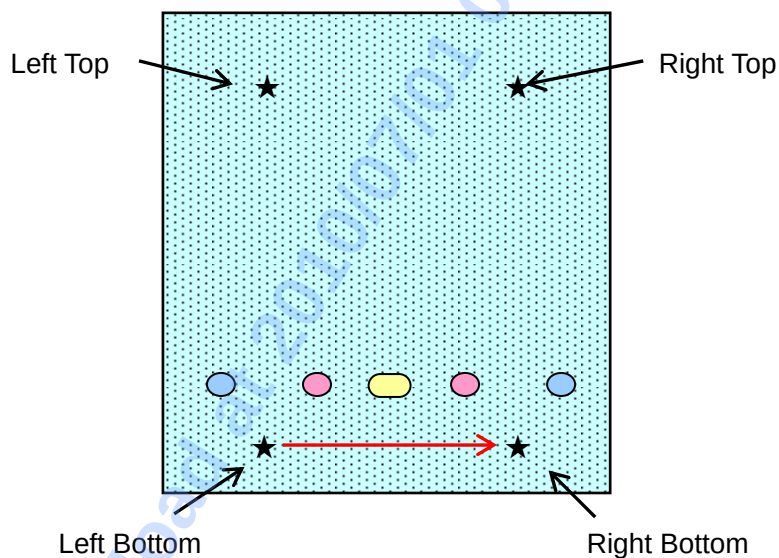


9. Press the [CYCLE STOP] button. The X and Y values of the Left Bottom position appear on the operation panel.



10. Press the [START] button. Then, the buzzer beeps. (This is not the error alarm.)

11. The matrix measurement head moves to the Right Bottom position.



12. Press the [CYCLE STOP] button. The X and Y values of the Right Bottom position appear on the operation panel.



13. Press the [START] button. Then, the buzzer beeps. (This is not the error alarm.)
14. The matrix measurement head moves to the Right Top position.
15. Press the [CYCLE STOP] button. The X and Y values of the Right Top position appear on the operation panel.



16. Press the [START] button. Then, the buzzer beeps. (This is not the error alarm.)
17. The matrix measurement head moves to the Left Top position.

18. Press the [CYCLE STOP] button. The X and Y values of the Left Top position appear on the operation panel.



19. Check that the X and Y values of all four positions are within the tolerance of +/- 1000 um.

Left Top => -1000 um < (X um, Y um) < +1000 um
Right Top => -1000 um < (X um, Y um) < +1000 um
Left Bottom => -1000 um < (X um, Y um) < +1000 um
Right Bottom => -1000 um < (X um, Y um) < +1000um

20. If the X and Y values of all four positions are out of the tolerance, adjust the matrix glass panel position so that all four positions are within the tolerance. To check that, repeat the steps 5 to 19.
21. When the X and Y values of all four positions meet the tolerance of +/- 1000 um, press the [START] button. Then, the buzzer beeps. (This is not the error alarm.)

22. Press the [CYCLE STOP] button to stop the matrix measurement.
23. Start the fine adjustment of the matrix glass panel position by following the steps below.
24. Press the [START] Button. Then, press the [MONITOR] button to see the H (horizontal) and V (vertical) values at the Left Bottom position.



Matrix Mark

H: 331.76

V: 262.32

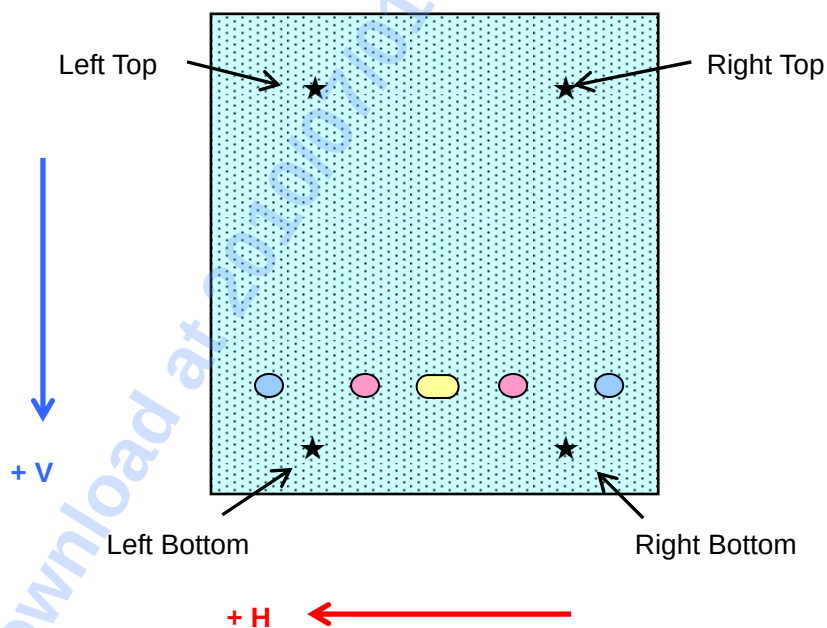
25. Ensure that the H and V values of the Left Bottom position are within the following tolerance.

Note: If the H and V values are close to 320 and 240 respectively, PAM results will be better.

Tolerance: $H = 320 \pm 10$ (pixel)

Tolerance: $V = 240 \pm 10$ (pixel)

26. Prepare a urethane hammer (if you don't have it, find a tool made by soft material such as rubber). Then, slowly and slightly pat the middle of the matrix glass panel sides so that the H and V values satisfy the tolerance.



27. Once the H and V values of the Left Bottom position meet the tolerance, press the [MONITOR] button and the [CYCLE STOP] button.
28. Press the [START] button and the buzzer beeps. (This is not the error alarm.) Then, the matrix measurement head moves to the Right Bottom position.
29. Press the [MONITOR] button and to see the H (horizontal) and V (vertical) values at the Right Bottom position.



30. Ensure that the H and V values of the Right Bottom position are within the following tolerance.

Note: If the H and V values are close to 320 and 240 respectively, the placement accuracy will be better.

Note: It is recommended that the V value difference between the Right Bottom position and the Left Bottom position be within one pixel. (For example, if V = 241.3 at the Right Bottom position and V = 241.9 at the Left Bottom position, they are within the tolerance.)

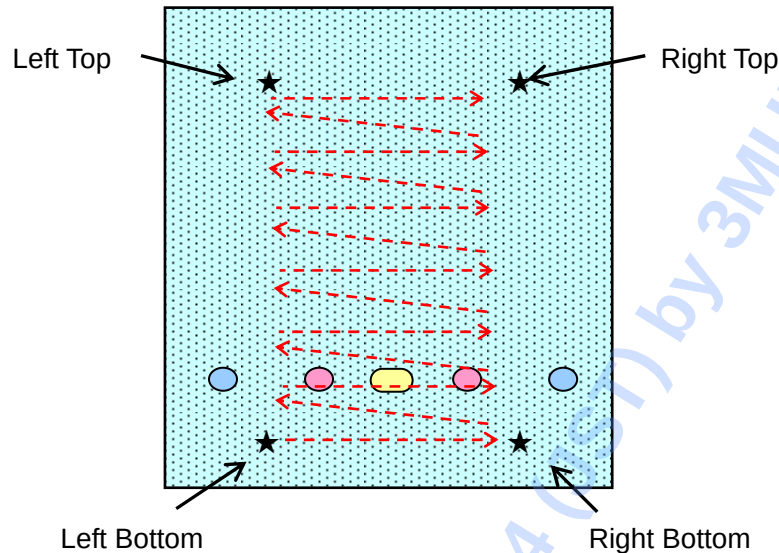
Tolerance: H = 320 +/- 10 (pixel)

Tolerance: V = 240 +/- 10 (pixel)

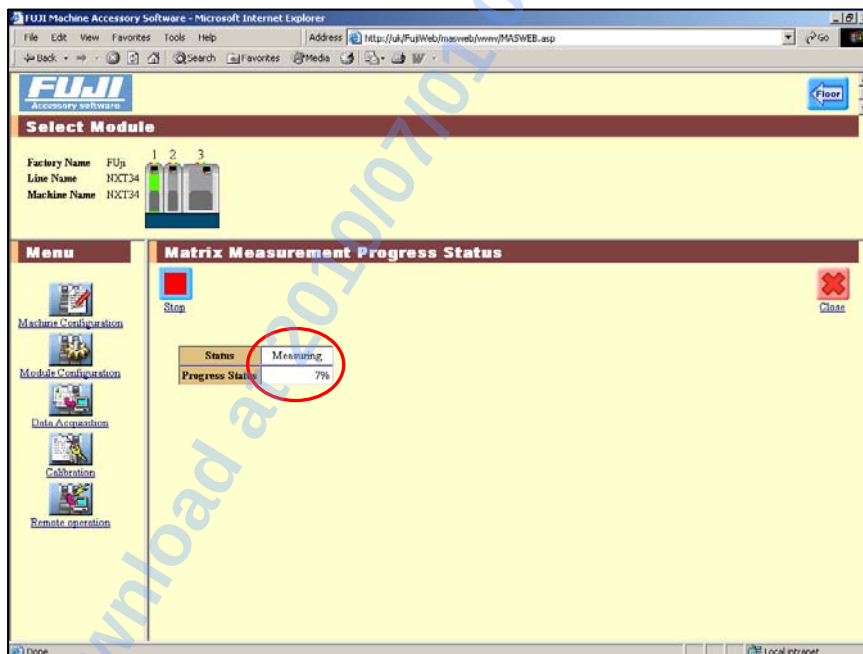
31. Prepare a urethane hammer (if you don't have it, find a tool made by rubber material). Then, slowly and slightly pat the middle of the matrix glass panel toward the direction to satisfy the tolerance.
32. Same as the Left Bottom position and the Right Bottom position, check the H and V values of the Right Top position and the Left Top position. If the H and V values are out of the tolerance, adjust the position of the matrix glass panel.
33. Once the H and V values of all four positions meet the tolerance, press the [START] button to start the matrix measurement. Then, the buzzer beeps. (This is not the error alarm.)

34. The matrix measurement head moves to the Left Bottom position and then follows the tracks shown in the drawing below. During this movement, two cameras (one is the mark camera, the other is the camera on the head) read the holes on the matrix glass panel.

Caution: Do not move the matrix glass panel during this movement.



35. The matrix measurement completes in approximately 30 minutes for M3/M3S modules and 20 minutes for M6/M6S modules.
36. The accessory software displays the percentage of the progress during the measurement.

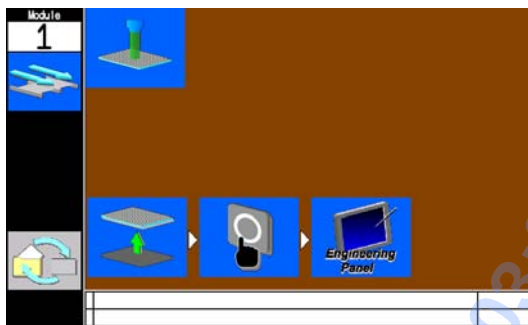


37. When the matrix measurement is complete, the matrix measurement head stops at the rear side of the module.
38. The pictogram to remove the matrix glass panel appears on the operation panel.

Caution: When this pictogram does not appear, the matrix measurement fails. When this failure occurs, one or several marks on the matrix glass panel are out of the tolerance. In this case, do not remove the matrix glass panel. Repeat the procedures from step 5.

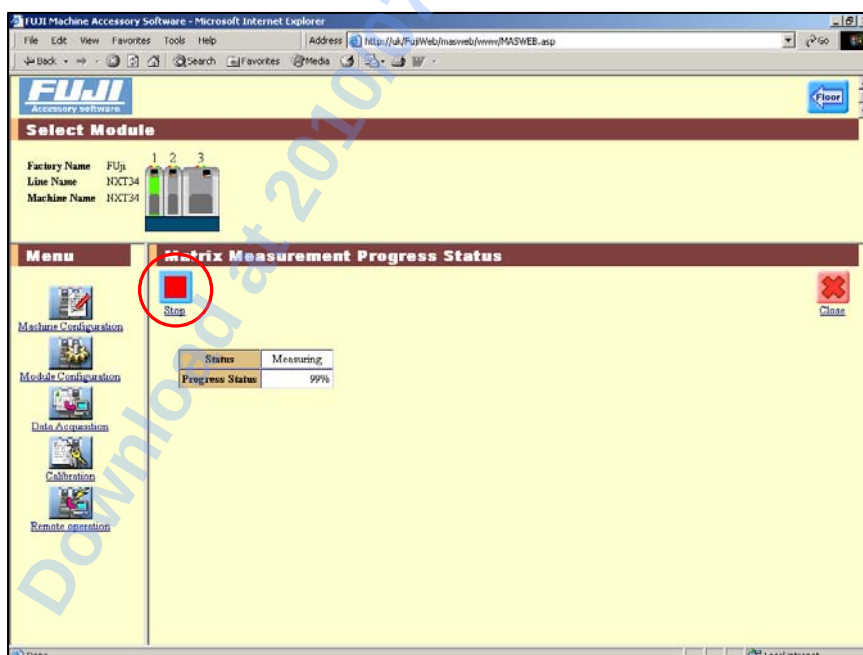
Caution: When the matrix measurement fails and the START button is pressed, the matrix measurement starts again and the progress status becomes 0 %.

Caution: Do not press OK button before removing the matrix glass panel.

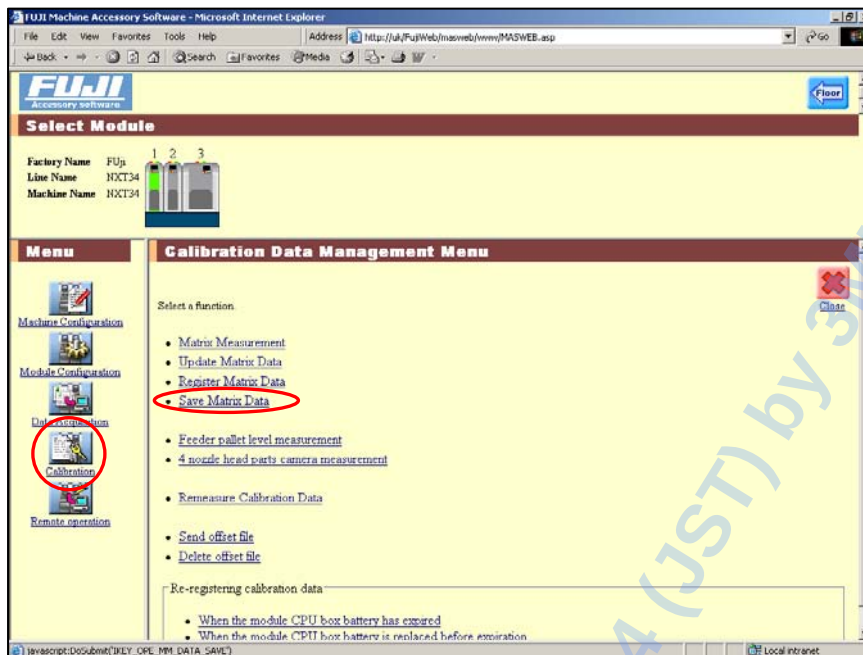


39. The progress status displays 99 %. Select [Stop].

Note: The percentage of the progress never shows 100 %.



40. Select [Calibration] – [Save Matrix Data] to save the matrix data to other media such as floppy disc or USB memory.

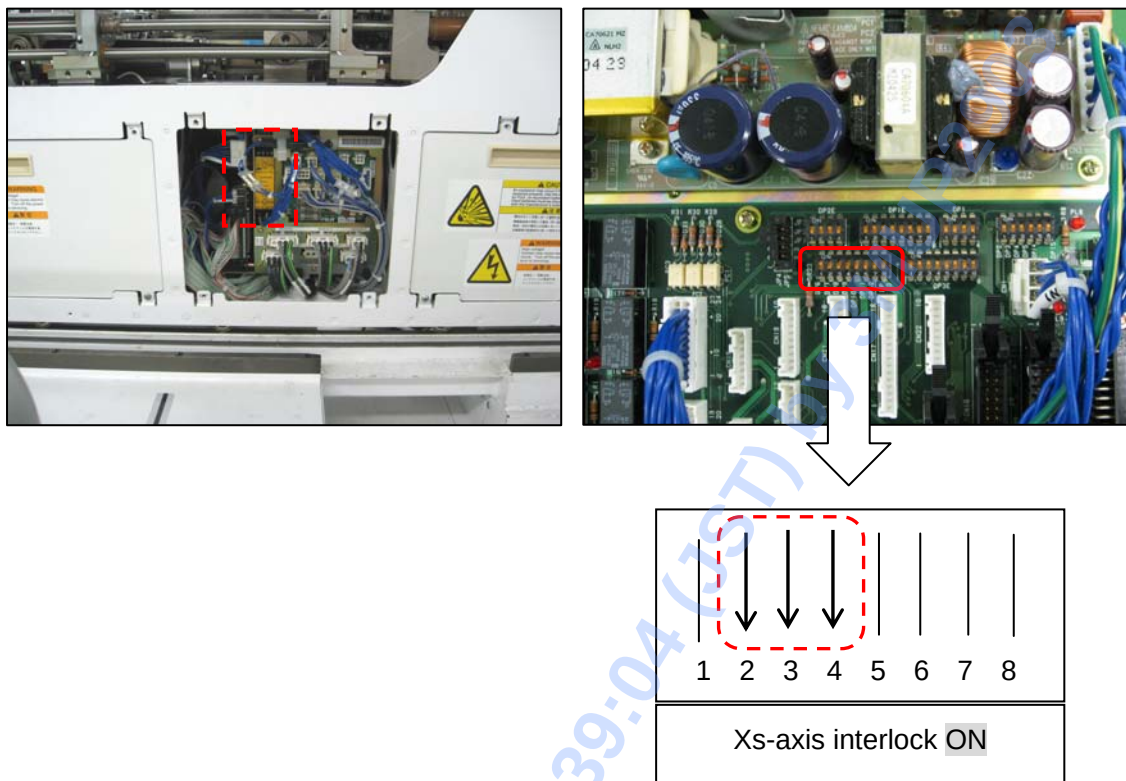


41. Press the [EMERGENCY] button and remove the matrix glass panel from the module.

Caution: Ensure that the matrix measurement head completely stopped and that the [EMERGENCY] button was pressed.

42. Turn OFF the base power.
43. Remove the matrix measurement head from the module. Then, reconnect the IEEE1394 cable removed from the mark camera unit.

44. To turn ON the Xs-axis interlock, change the dip switch settings of the control box back to the original condition.



45. When the modules next to the M3/M3S modules are removed for the matrix measurement, reinstall those modules.
46. Now the matrix measurement is complete.

6.4 Performing the Matrix Measurement (V4.20 or earlier)

Caution: The measurement procedure varies with the application software version. When the application software installed in the module is V4.20 or earlier, refer to the following procedures. When the application software version is V4.30 or later, refer to "6.3 Performing the Matrix Measurement (V4.30 or later)."

1. The operation panel displays the following screen and the [START] button flashes. Then, press the [START] button.
2. The matrix measurement head begins to move and read three fiducial marks on the conveyor. Then, the conveyor rail moves to the matrix measurement position.
3. Ensure that the matrix measurement head and conveyor stop. Then, press the [EMERGENCY] button.



4. Set the matrix glass panel (AA1RW00) on the module. Ensure that the serial number of the panel comes to the module front side.

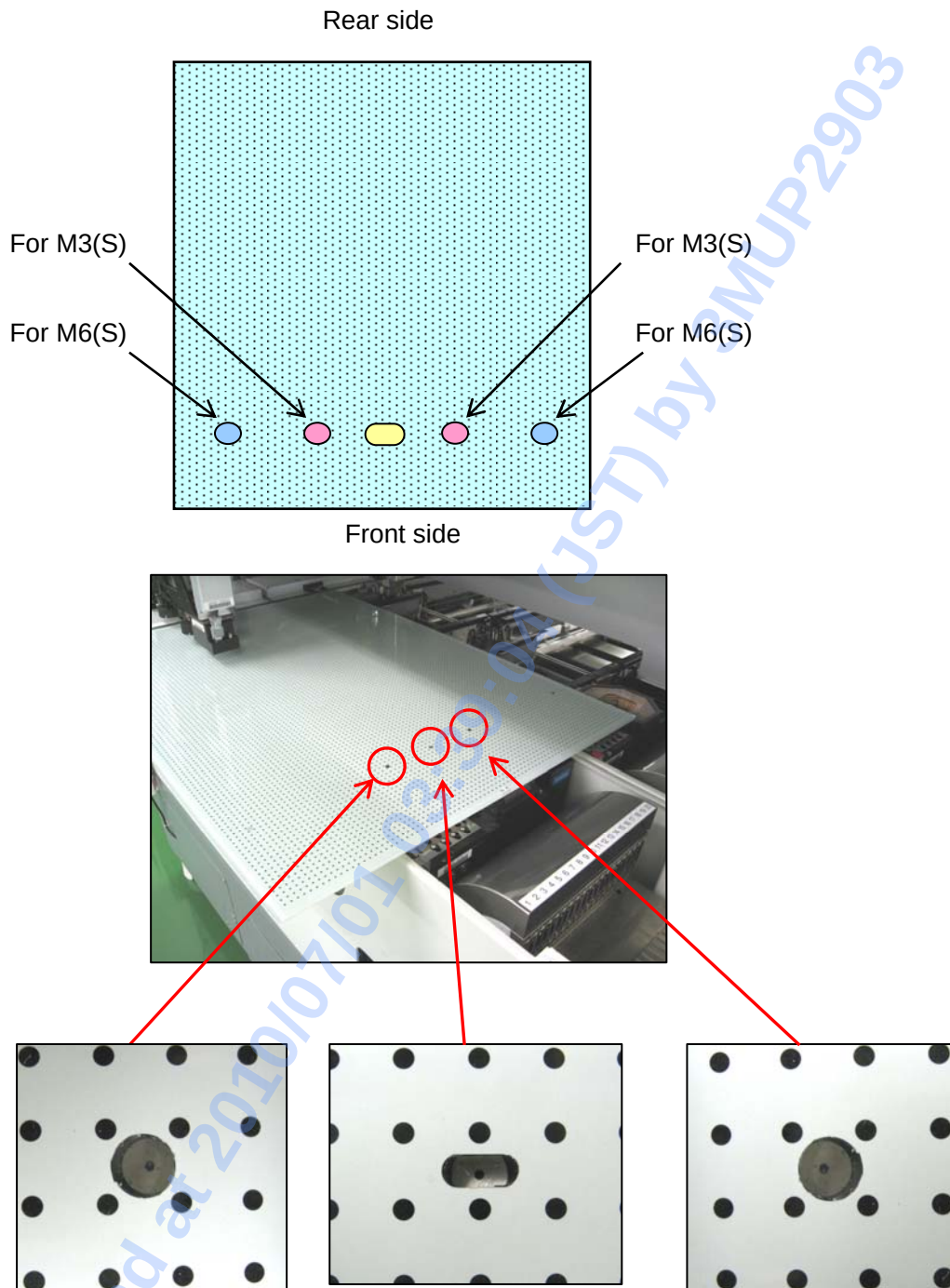
Caution: Ensure once more that the [EMERGENCY] button is pressed.

Caution: When carrying the matrix glass panel, pay attention not to damage the glass surface.



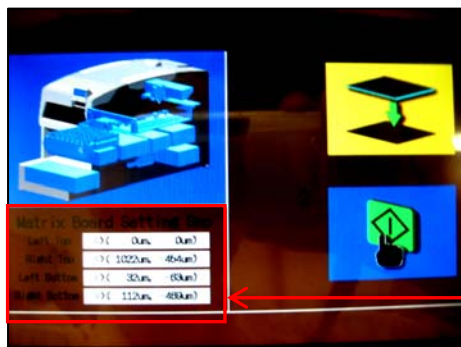
The serial number of the matrix glass panel comes to the module front side.

5. Adjust the matrix glass panel position so that the three fiducial marks on the conveyor come to the center of each three holes on the matrix glass panel.



6. Press the [START] button to start the matrix measurement.

- The matrix measurement head begins to move and read three fiducial marks on the conveyor rail. Then, it moves from the Left Bottom position to the Right Bottom position (See the picture and the drawing below).



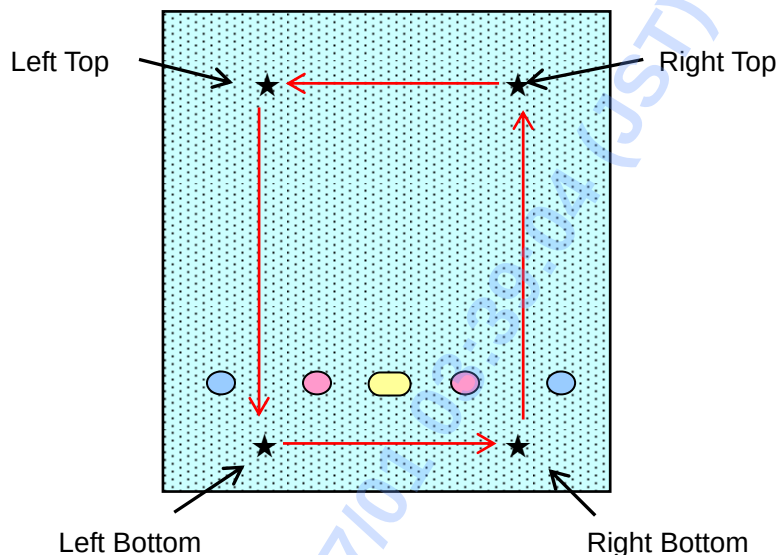
Matrix Board Setting Gap

Left Top => (X um, Y um)

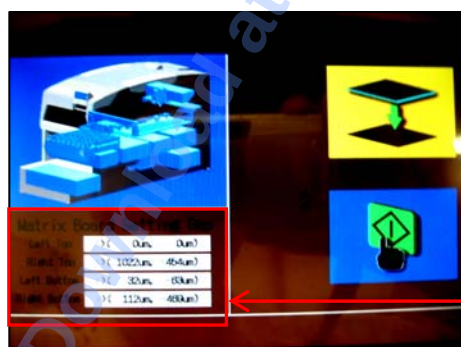
Right Top => (X um, Y um)

Left Bottom => (X um, Y um)

Right Bottom => (X um, Y um)



- Press the [CYCLE STOP] button. The X and Y values of all four positions appear on the operation panel.



Matrix Board Setting Gap

Left Top => (X um, Y um)

Right Top => (X um, Y um)

Left Bottom => (X um, Y um)

Right Bottom => (X um, Y um)

9. Check that the X and Y values of all four positions are within the tolerance of +/- 1000 um.

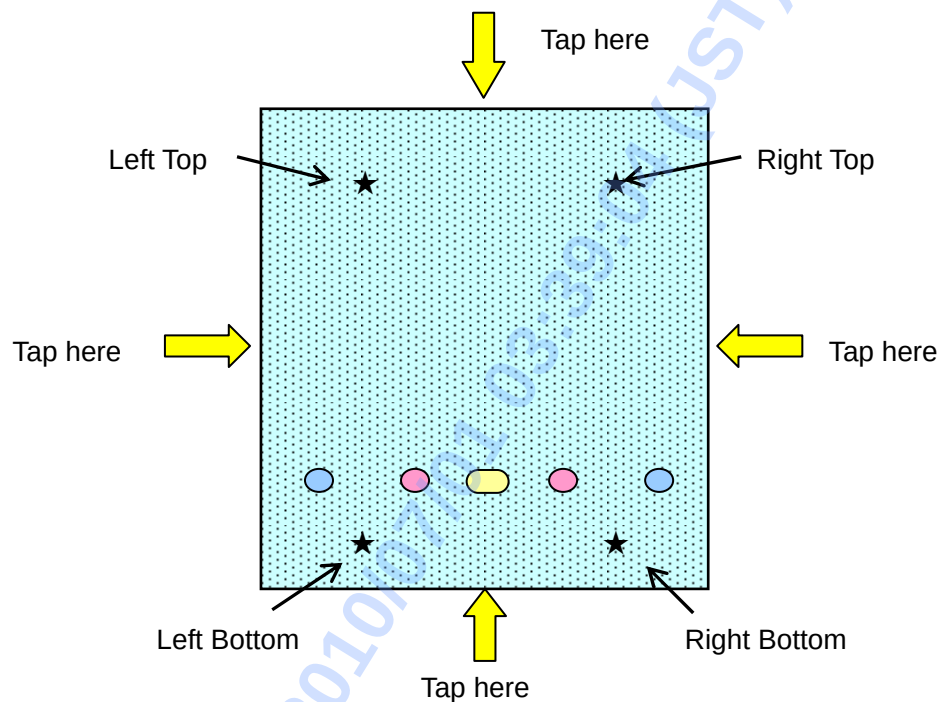
Left Top => -1000 um < (X um, Y um) < +1000 um

Right Top => -1000 um < (X um, Y um) < +1000 um

Left Bottom => -1000 um < (X um, Y um) < +1000 um

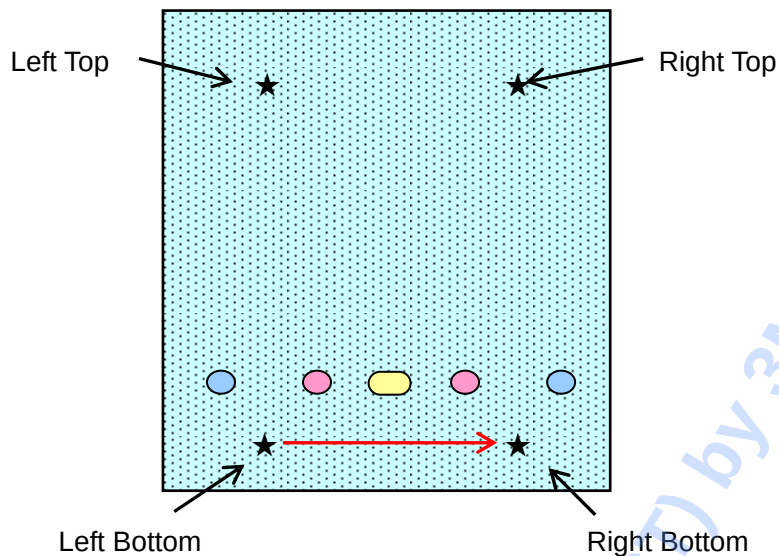
Right Bottom => -1000 um < (X um, Y um) < +1000um

10. If the X and Y values of all four positions are out of the tolerance, adjust the matrix glass panel position following the steps below.
11. Prepare a urethane hammer (if you don't have it, find a tool made by soft material such as rubber). Then, slowly and slightly tap the middle of the matrix glass panel sides so that all four positions are within the tolerance.



12. To check that all four positions are within the tolerance, repeat the steps 5 to 9.
13. Once the X and Y values of all four positions meet the tolerance of +/- 1000 um, press the [START] button.

14. The matrix measurement head moves from the Left Bottom position to the Right Bottom position.



15. Press the [MONITOR] button to see the H (horizontal) and V (vertical) values at the Left Bottom position.



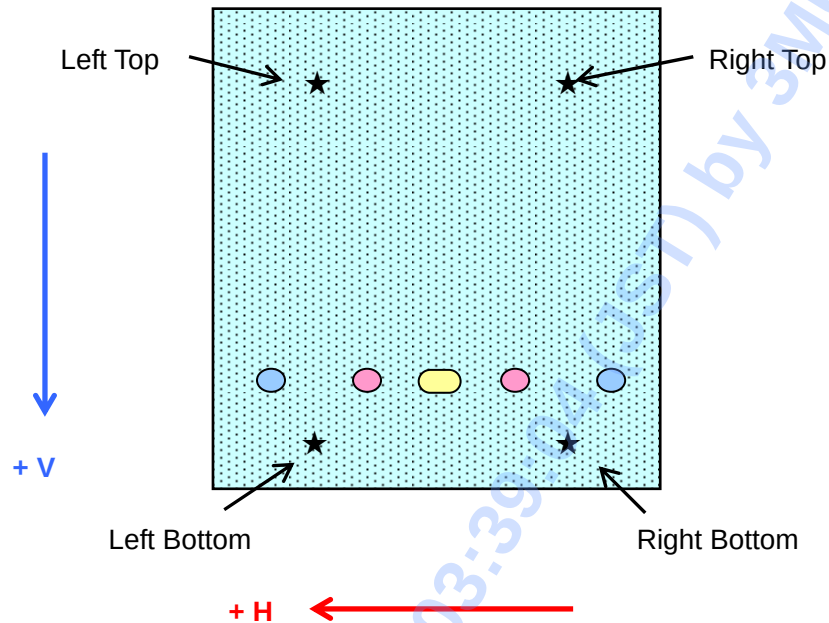
16. Ensure that the H and V values of the Left Bottom position and the Right Bottom position are within the following tolerance.

Note: If the H and V values are close to 320 and 240 respectively, the placement accuracy (PAM) will be better.

Tolerance: H = 320 +/- 10 (pixel)

Tolerance: V = 240 +/- 10 (pixel)

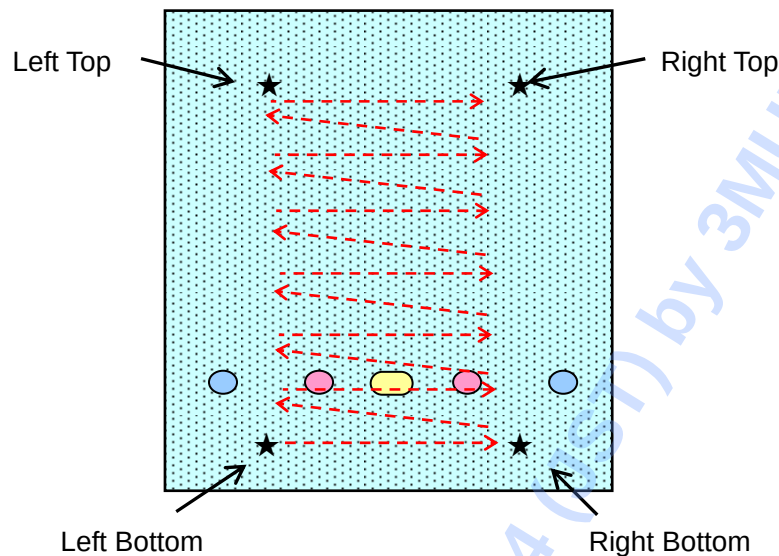
17. If the H and V values of the Left Bottom position and the Right Bottom position are within the tolerance, go to 20.
18. If the H and V values of the Left Bottom position and the Right Bottom position are out of the tolerance, go to the step 19.
19. Prepare a urethane hammer (if you don't have it, find a tool made by soft material such as rubber). Then, slowly and slightly pat the middle of the matrix glass panel sides so that the H and V values satisfy the tolerance.



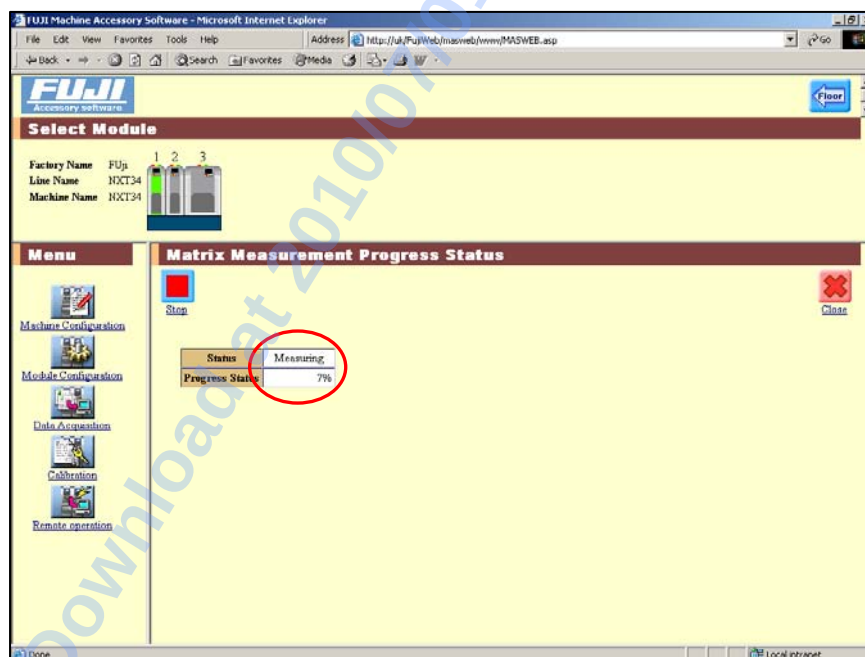
20. Once the H and V values of the Left Bottom and Right Bottom positions meet the tolerance, press the [CYCLE STOP] button. Then, press the [START] button to start the matrix measurement.

21. The matrix measurement head moves to the Left Bottom position and then follows the tracks shown in the drawing below. During this movement, two cameras (one is the mark camera, the other is the camera on the matrix measurement head) read the holes on the matrix glass panel.

Caution: Do not move the matrix glass panel during this movement.



22. The matrix measurement ends in approximately 40 minutes for M3/M3S module and 30 minutes for M6/M6S module.
23. The accessory software displays the percentage of the progress during the measurement.

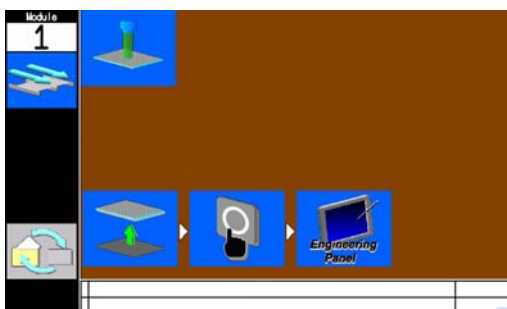


24. When the matrix measurement is complete, the matrix measurement head stops at the rear side of the module.
25. The pictogram to remove the matrix glass panel appears on the operation panel.

Caution: When this pictogram does not appear, the matrix measurement fails. When this failure occurs, one or several marks on the matrix glass panel are out of the tolerance. In this case, do not remove the matrix glass panel. Repeat the procedures from step 5.

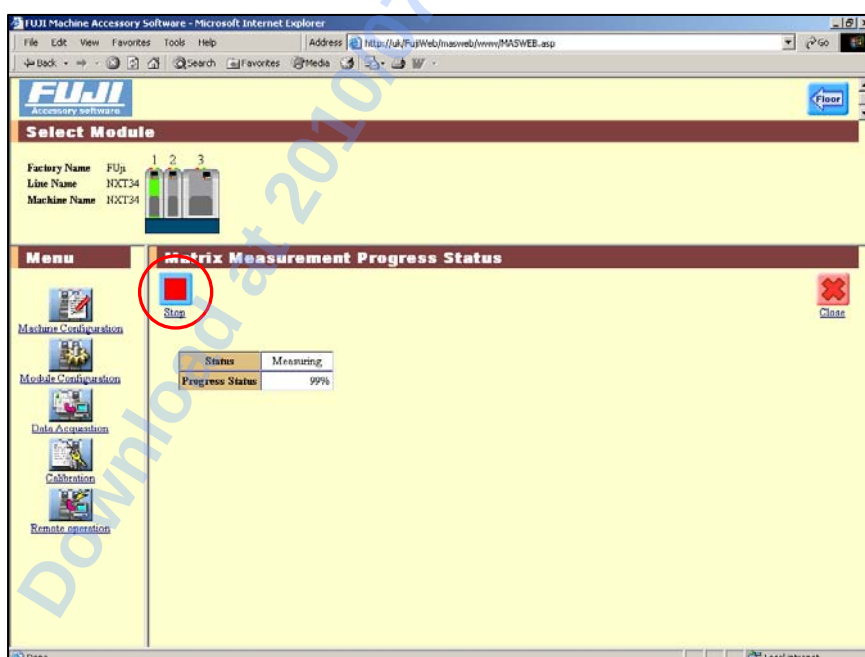
Caution: When the matrix measurement fails and the START button is pressed, the matrix measurement starts again and the progress status becomes 0 %.

Caution: Do not press OK button before removing the matrix glass panel.

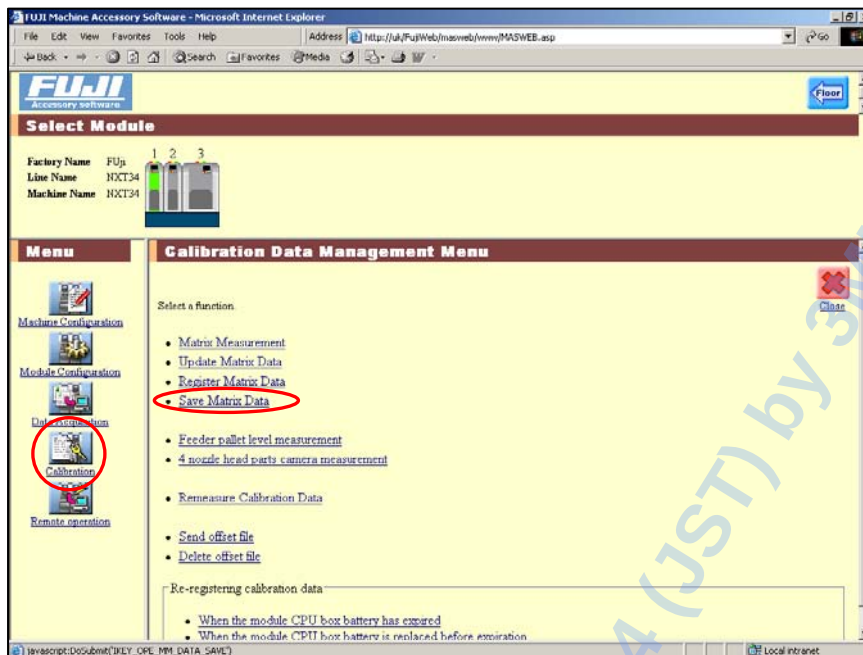


26. The progress status displays 99 %. Select [Stop].

Note: The percentage of the progress never shows 100 % even though the matrix measurement is complete.



27. Select [Calibration] – [Save Matrix Data] to save the matrix data to other media such as floppy disc or USB memory.



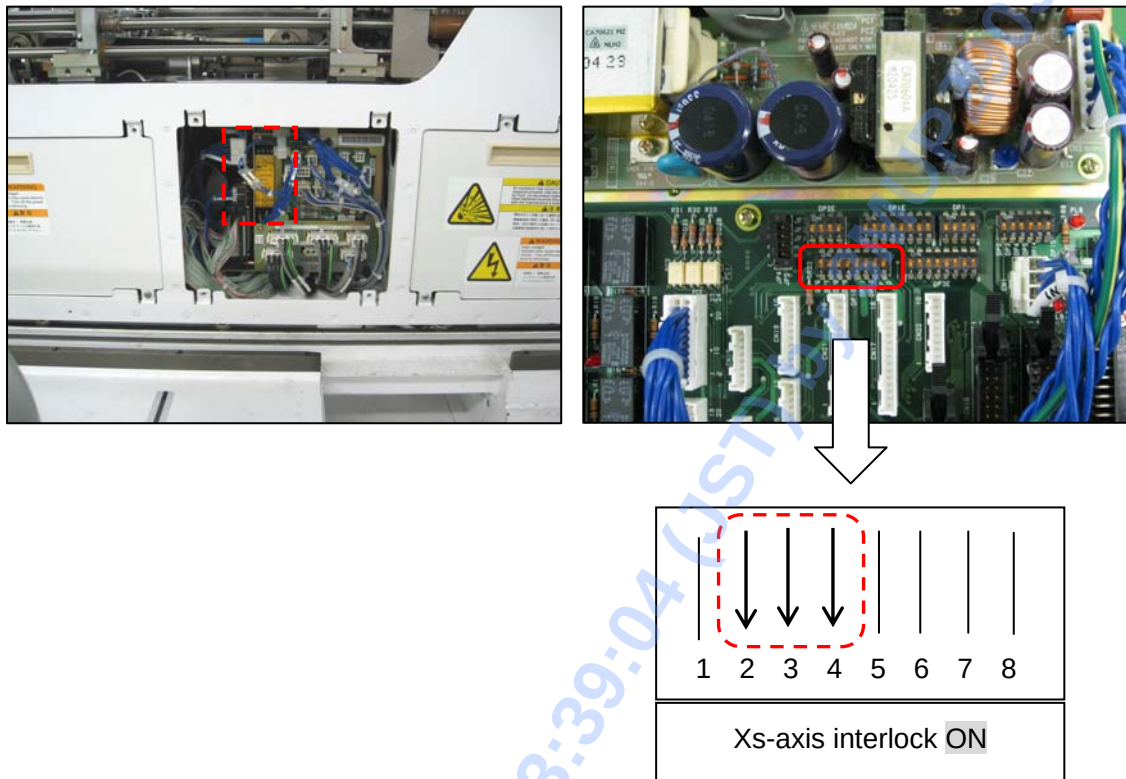
28. Press the [EMERGENCY] button and remove the matrix glass panel from the module.

Caution: Ensure that the matrix measurement head completely stopped and that the [EMERGENCY] button was pressed.

29. Turn OFF the base power. Then, reconnect the IEEE1394 cable removed from the mark camera unit.

30. To turn ON the Xs-axis interlock, change the dip switch settings of the control box to the original condition.

Note: It is not necessary to do this for M6/M6S modules.



31. When the modules next to the M3/M3S modules are removed for the matrix measurement, reinstall those modules.
32. Now the matrix measurement is complete.

Chapter 7

Matrix Measurement for NXTII

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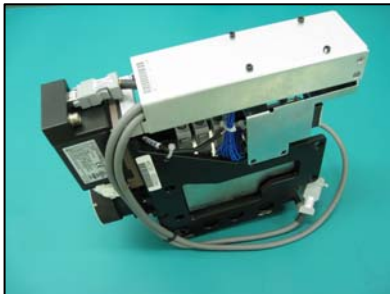
Download at 2010/07/01 03:39:04 (JST) by 3MUP2903

7. Matrix Measurement for NXTII

This chapter explains the matrix measurement procedure for NXTII. For the NXT procedure, refer to “Chapter 6 Matrix Measurement for NXT”.

7.1. Preparation for the Matrix Measurement

Required jigs:



Matrix Measuring Head (AA02S0*)



Matrix Measurement Glass Board (UM-GM)



IEEE1394 Cable Connector (RH39560)



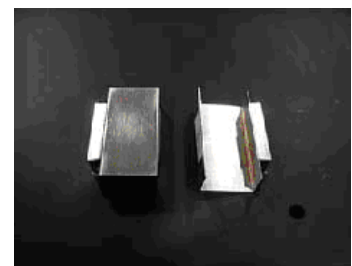
50 Pin Jumper Connector



M6II/IISP Adjustable Rail
Attachment Jig (PM0ACF0)



M3II Adjustable Rail
Attachment Jig (PM0ACE0)



Reference Rail Attachment
Jig (PB04DC0)

Caution:

Note: It is possible to use the matrix glass board (AA1RW00) used in M3/M3S for matrix measurement of M3II.

Note: The matrix glass board (AA1RW00) used in NXT cannot be used when performing matrix measurement for M6 II/M6 IISP because the fiducial mark position on a conveyor differs from M6/M6S conveyor.

Note: Turn ON the module (keeping-warm state) for 3 hours or more before performing Matrix Measurement.

Note: Be sure to push the emergency stop switch when you put in the body in a module when the module power supply is ON.

Note: Perform measurement where there is no module next to a measuring object module. (Check that there are no other peripheral equipment around the shuttle conveyor, double conveyor etc. next to a measuring object module)

Note: Matrix measurement cannot be performed on machines configured to use MTUL/LT. Attach a feeder pallet to perform matrix measurement on these machines.

Note: The matrix board will interfere with the auto backup pin stacker on machines equipped with this function. To avoid this, remove the bracket on which the 2D code is attached.

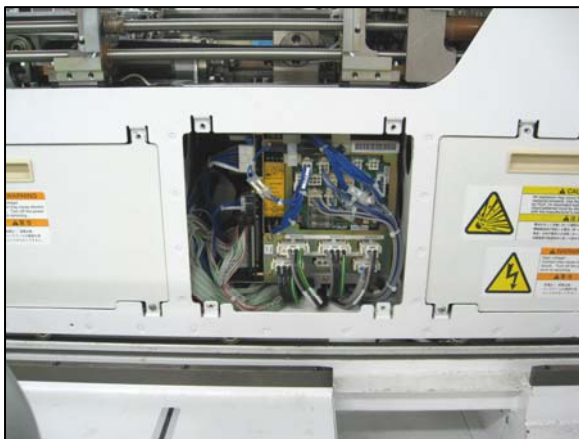
Note: Matrix measuring head (AA02S0) for NXT and NXTII are common.*

1. It is necessary to pull out the target module and the adjoining module when performing matrix measurement on an M3 II module. This is to prevent the matrix glass board interfering with the conveyor rail on the adjoining module and the Xs axis interfering with XY robot on the adjoining module.
2. It is no necessary to pull out both the target module and the adjoining module when performing matrix measurement on an M6S/II /SP module.
3. It is necessary to unlock the Xs axis when performing matrix measurement on an M3 II module. Be sure to turn off the base main power.

Note: Electric power is supplied to the control BOX even if the module power supply is OFF and the base power supply is ON. Check that the base main power is OFF.

Note: Xs axis is unlocked in order to make the movable region of Xs axis larger when performing matrix measurement on M3 II module.

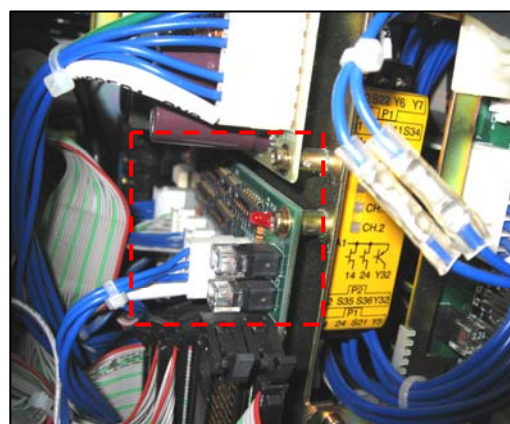
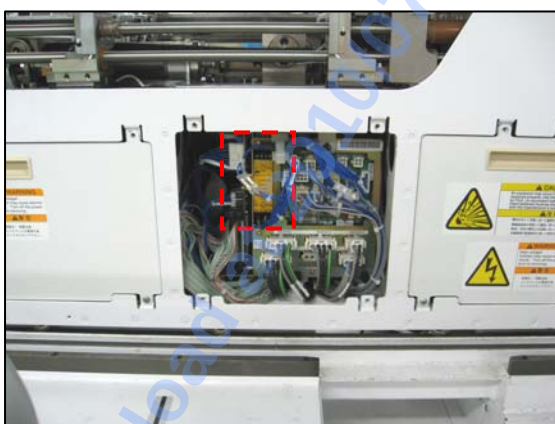
4. Remove the cover of M3 II module control BOX.



Note: The removal of the control BOX will be easy if using Module Change Unit (MCU) to pull out the module.



5. Locate the interface board on the side of control BOX.



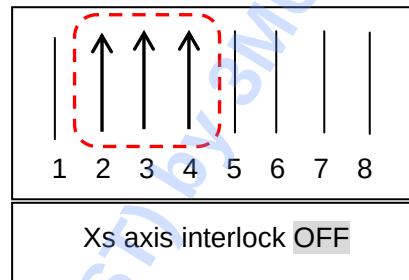
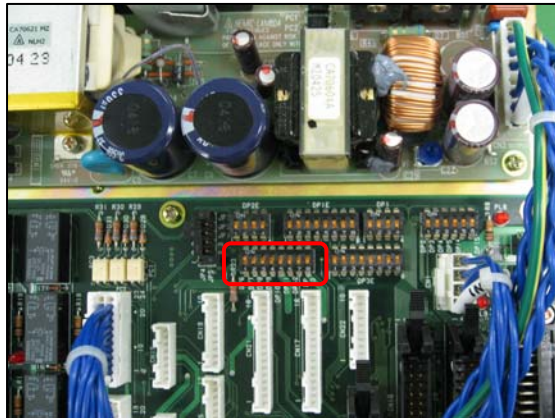
6. Record before changing the DIP switch settings.

7. Raise the DIP switch 2, 3, and 4 using tweezers or a small minus driver. (It is not necessary to change the DIP switch setting which has already raised)

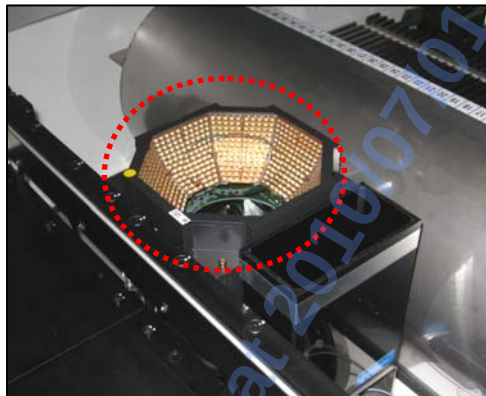
Note: Do not change the other DIP switch. Only change the DIP switch 2, 3, and 4.

Note: The original DIP switch setting differs according to module type and shipping periods.

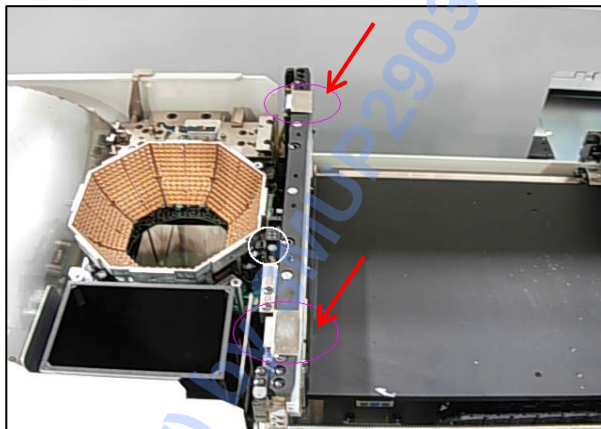
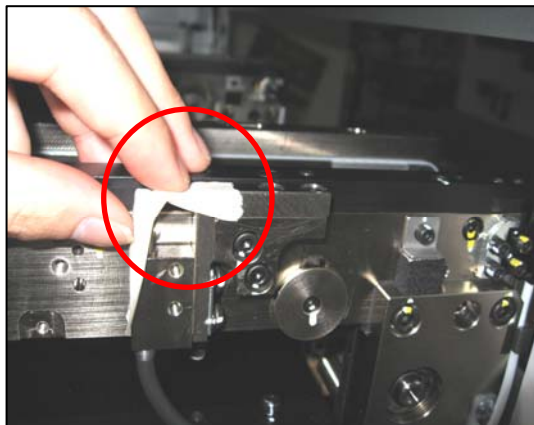
Note: It is not necessary to remove the control BOX from the module when performing the DIP switch settings.



8. Remove the nozzle station from the module.
9. Remove the parts camera cover. This is to prevent interference of the matrix glass board and the cover.



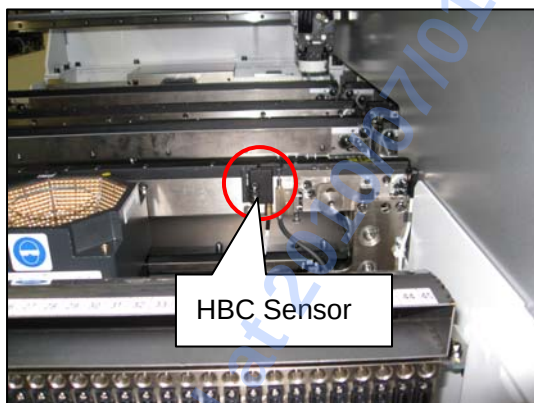
10. In order to prevent Z0 height check sensor interfering with a matrix glass board, stick a tape on the sensor before pushing in the sensor. Then attach the reference rail attachment jig (PB04DC0) to avoid interference of the HBC sensor bracket and the matrix board.



Warning: When there is no reference rail attachment jig (PB04DC0), it is necessary to remove the HBC (Hybrid Calibration) sensor bracket in order to avoid interference of the matrix glass board. Remove the two hexagon socket bolts (M04X10) which are fixing the bracket and temporarily place it where the conveyor and the other parts do not interfere.

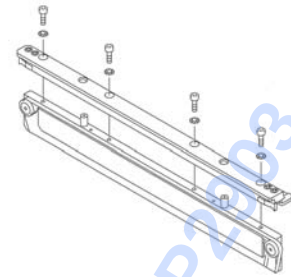
Note: It is not necessary to remove the HBC sensor from the air hose.

Note: It is necessary to measure the flatness when attaching the HBC sensor again.



11. When the automatic backup pin stocker is attached to the conveyor, remove the bracket where the 2D code is attached to avoid interference with the matrix board.

12. Remove the plate retainer and the rail of adjustable conveyor from the module.



M3 II Conveyor Adjustable Rail

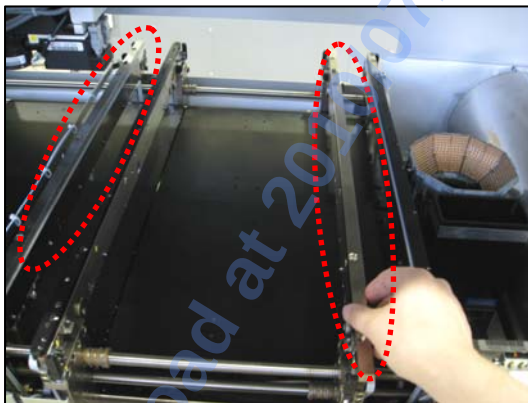
Module Type	Reference Rail	Adjustable Rail
M3 II	Bolt (x 6) + Collar (x 6)	Bolt (x 4) + 4 Washer
M6 II / II SP	Bolt (x 10) + Collar(x10)	Bolt (x 4) + 8 Washer

Note: When removing the rail, Z0 height check sensor attached to the conveyor reference rail side of the lane 1 cannot be removed.

Note: The reference mark positions between the lane 1 and the lane 2 differ.

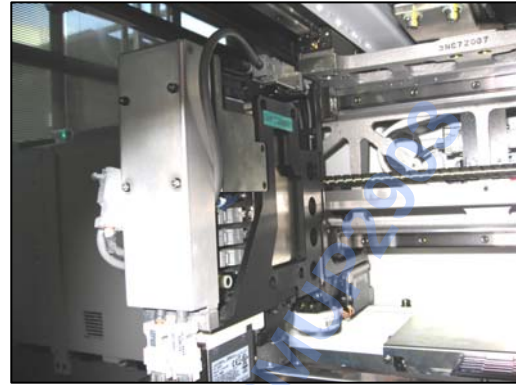
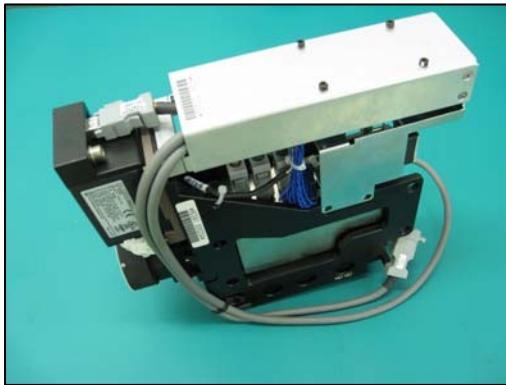
Note: Attach hexagon socket screw and washer which were removed back to the original place.

13. Attach the rail jig to the conveyor adjustable rail.



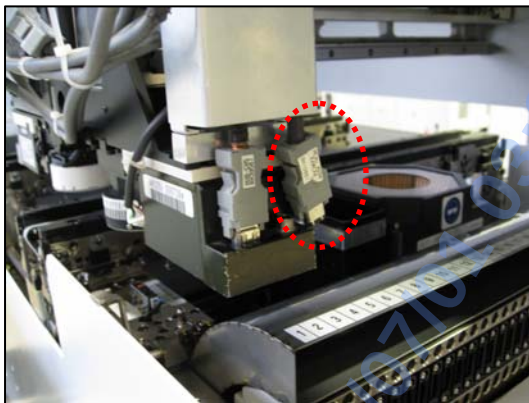
M3 II Use: PM0ACE0, M6 II / II SP Use: PM0ACF0

14. Attach the matrix measuring head (AA02S0*) to the module.



Note: Turns on the electricity for 2 hours and more (in the wake condition) after attaching the matrix measuring head. This is to raise the mark camera and the matrix measuring head camera internal component temperature to prevent shifts in the reading during measurement.

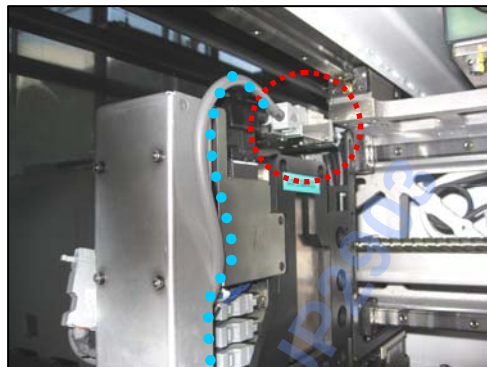
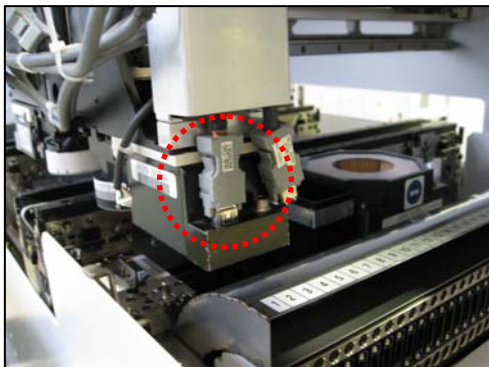
15. Remove the IEEE1394 cable which connected to the camera unit.



16. Connect the IEEE1394 jig cable connector to the matrix measuring head. Furthermore, insert an IEEE1394 cable on the matrix measuring head inner side cover to connect with XY robot connector.



IEEE1394 Jig cable connector (RH39560)

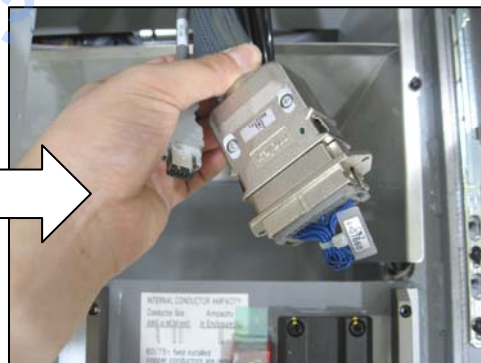
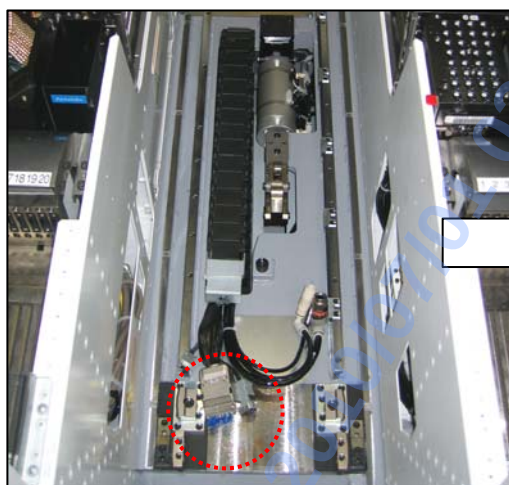


17. Perform step 18, 19, and 20 when performing matrix measurement on M3 II module.
Progress to step 23 when performing matrix measurement on M6 II/ISP module

18. Attach the 50-pin jumper connector to the 50-pin connector cable that was connected to the module removed in step 1.



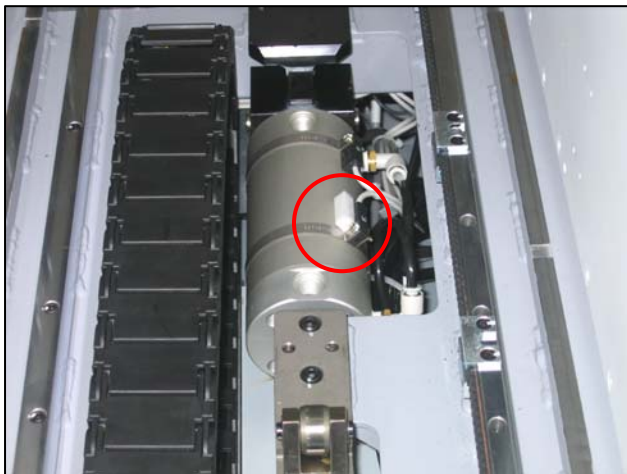
50 pin jumper connector



19. Set a scheme gauge, ruler, etc. to clamp position check sensor and docking check sensor and make it shade.



20. Attach a magnet to clamp check sensor and turn ON the sensor.

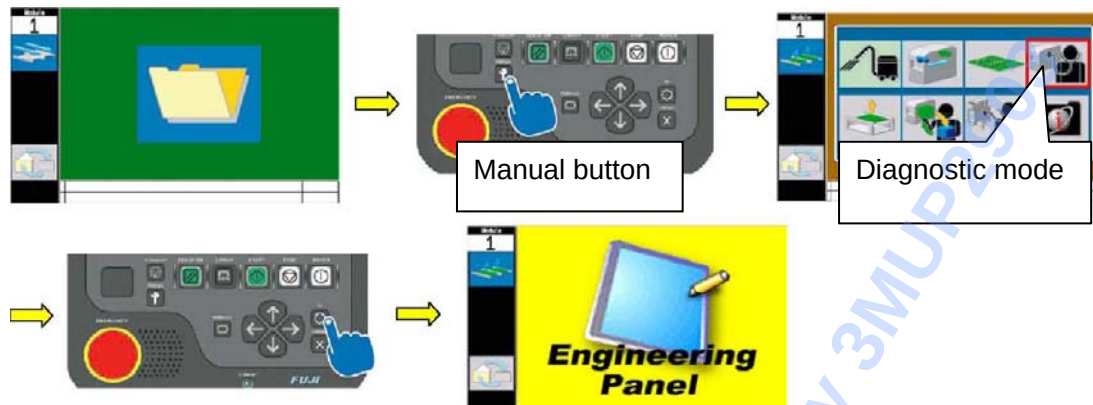


21. Turn ON the base main power and turn ON the module power.

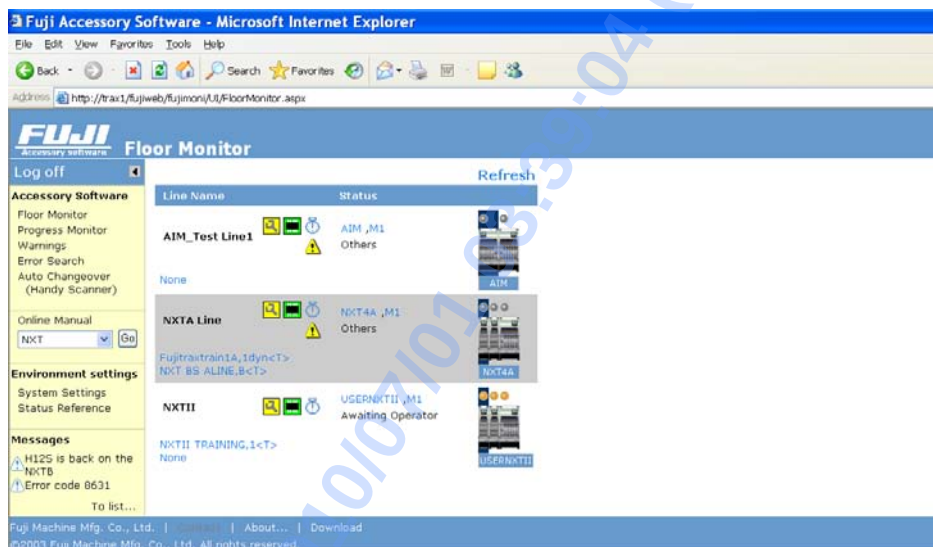
Note: After attaching the matrix measurement head, the power should be supplied in wake mode for at least two hours to prevent changes in temperature affecting the measurement results.

7.2. Performing the Directload of the Matrix Measurement

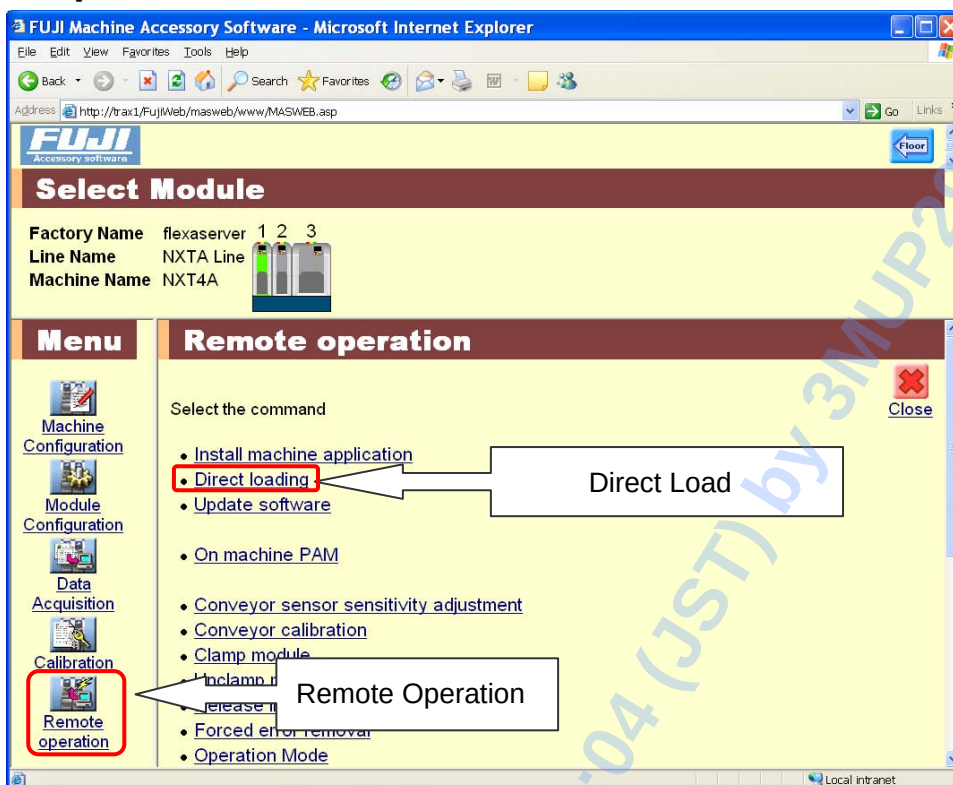
1. Change the module to the Diagnostic mode



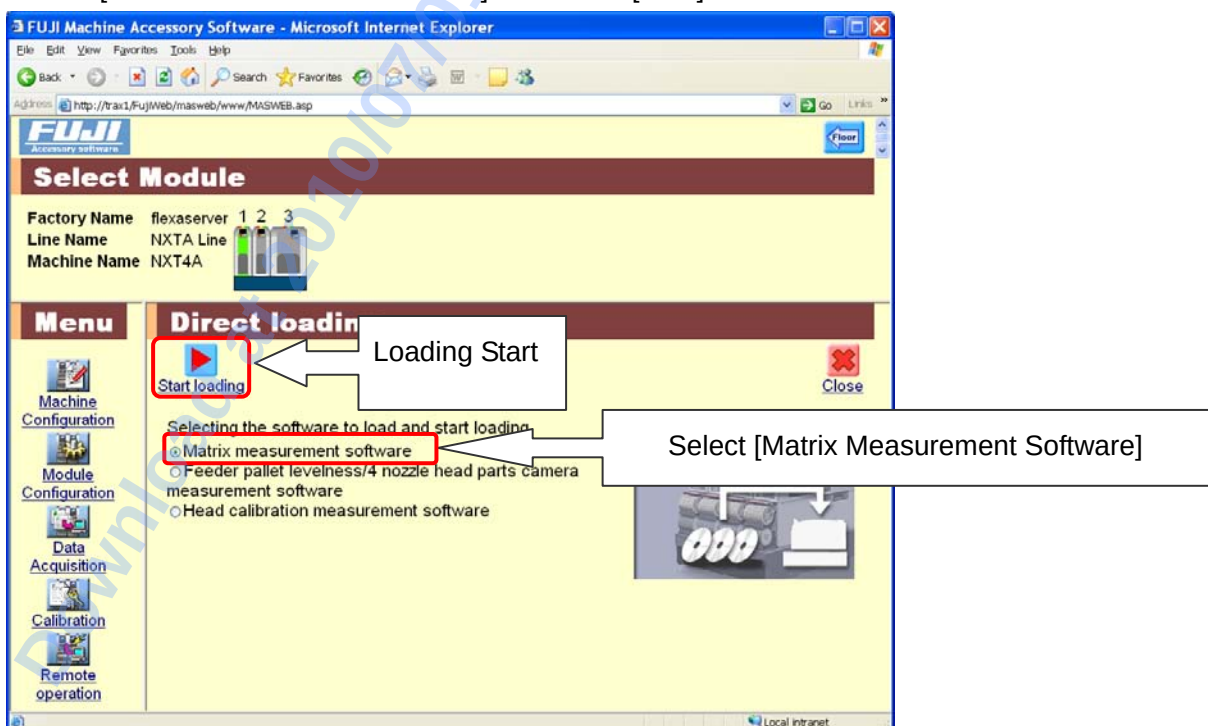
2. Start the NXT Accessory Software.



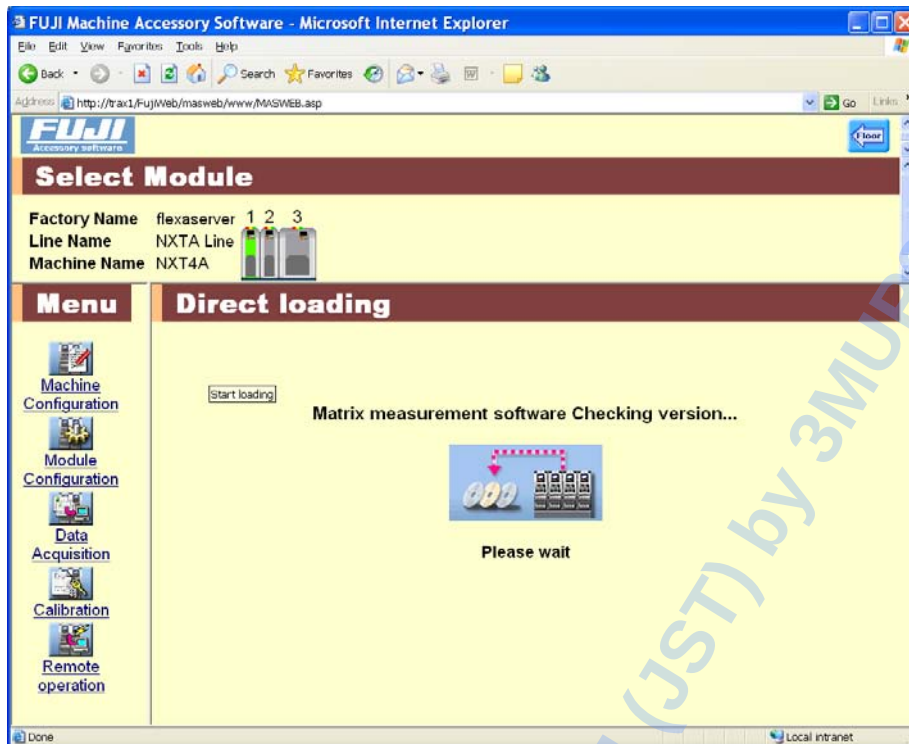
3. Select the module for matrix measurement, and select [Remote Operation] – [Direct Load].



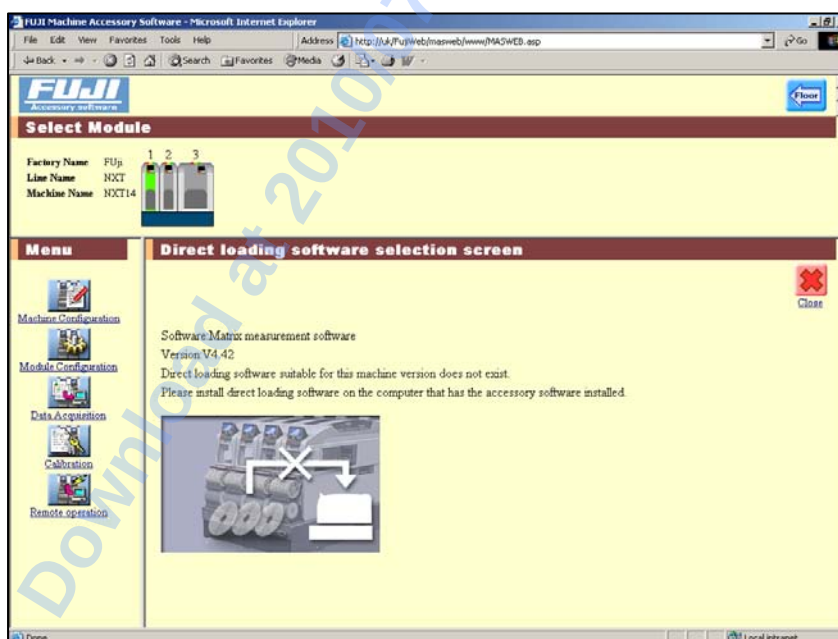
4. Select [Matrix Measurement Software] and select [Start].



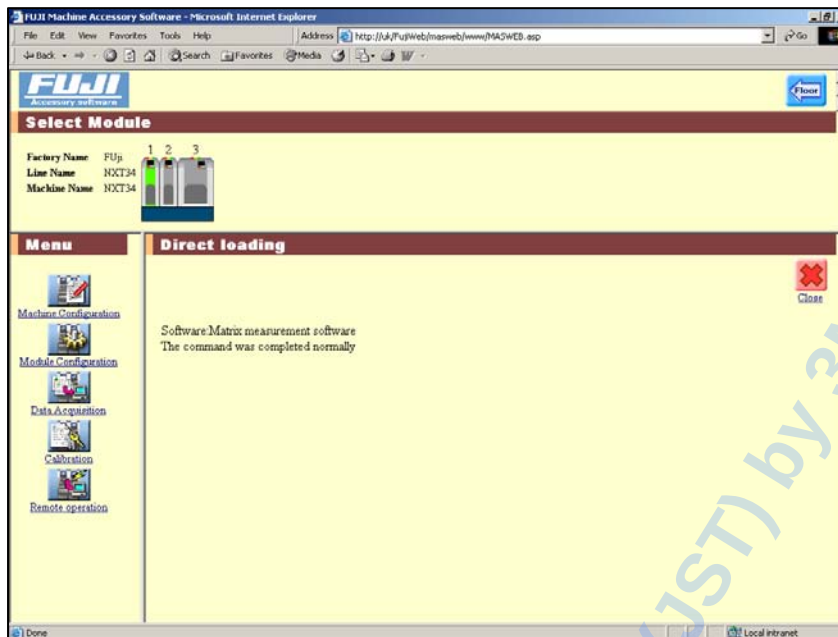
5. The following screen is displayed.



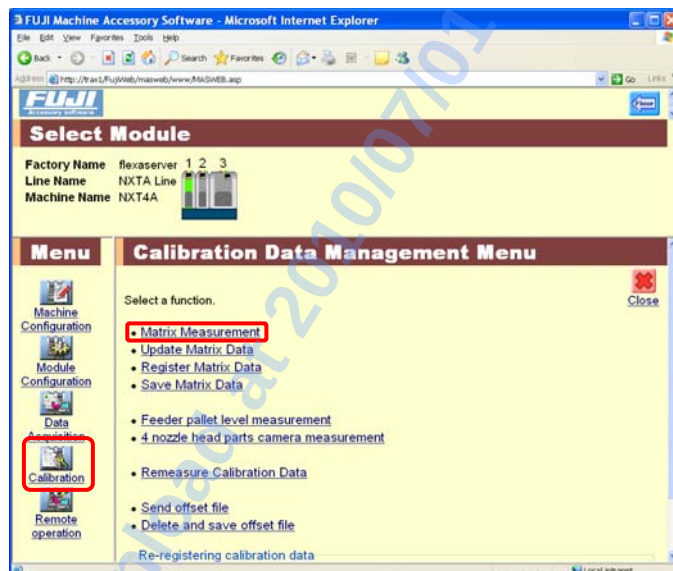
Note: When the following screen is displayed, the version of a module and matrix measurement software didn't matched. Check the module application version and download the matrix measurement software from NXT Software CD-ROM of the same version as the module version.



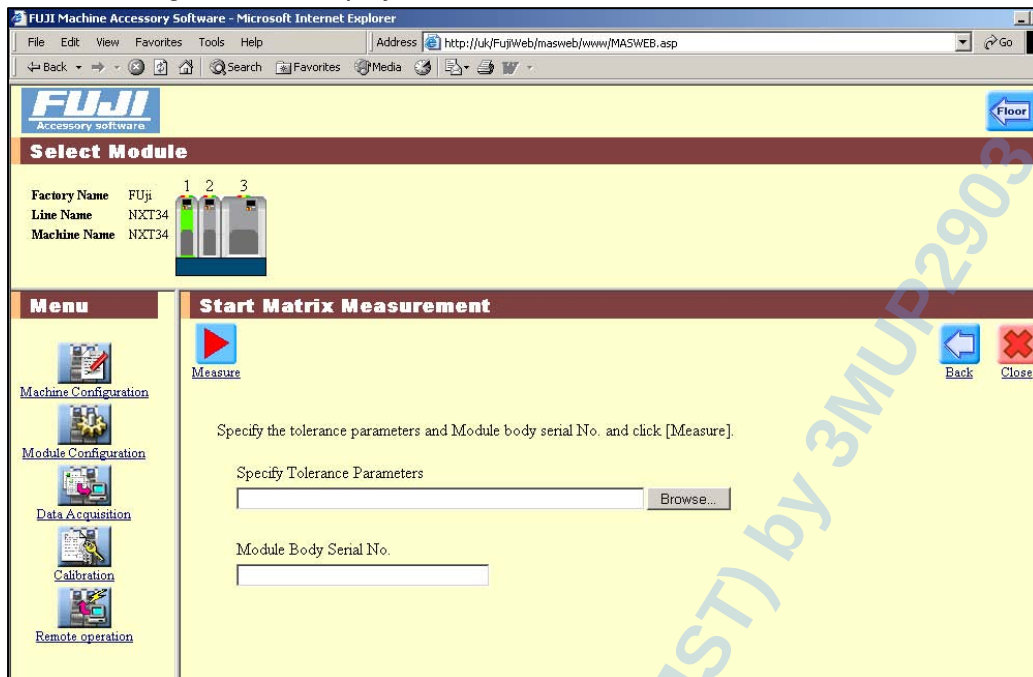
6. The following screen is displayed when the direct loading is completed.



7. The module will start reboot (reboot) if the direct load is completed. The module will be in a sleep condition (state). Then push the [Power] button to wake (start) the module.
8. Select [Calibration] – [Matrix Measurement] at Fuji Accessory Software.



9. The following screen is displayed.

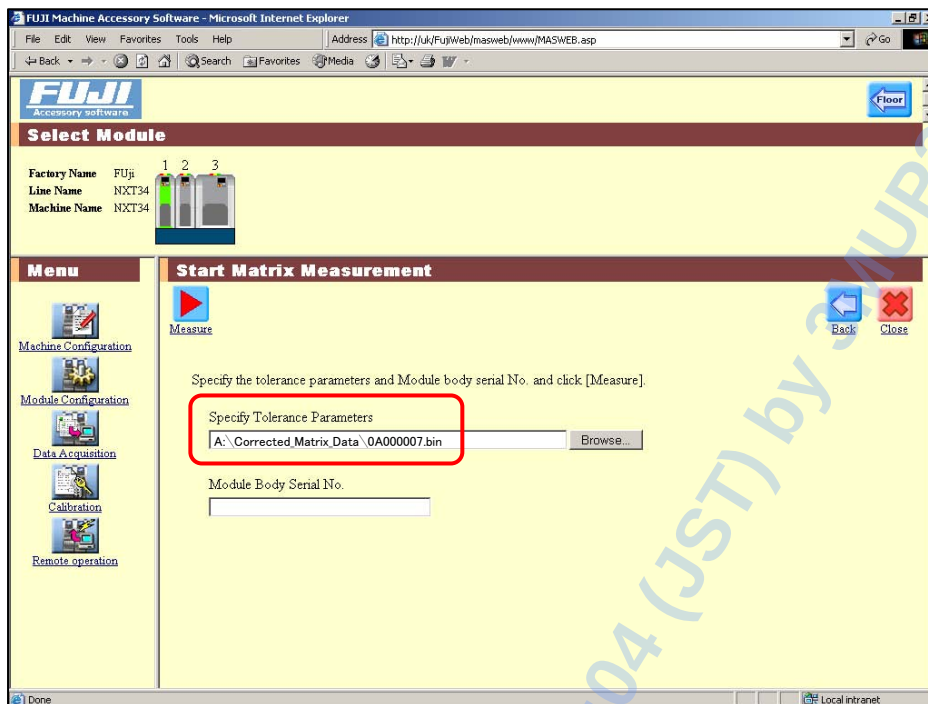


10. Set the floppy disk attached with the matrix glass board to the PC FD drive. This disk is a corrected matrix data file.

Note: Each matrix glass board has its own corrected data. Ensure that the serial number of the matrix board and the serial number of corrected matrix data are the same.

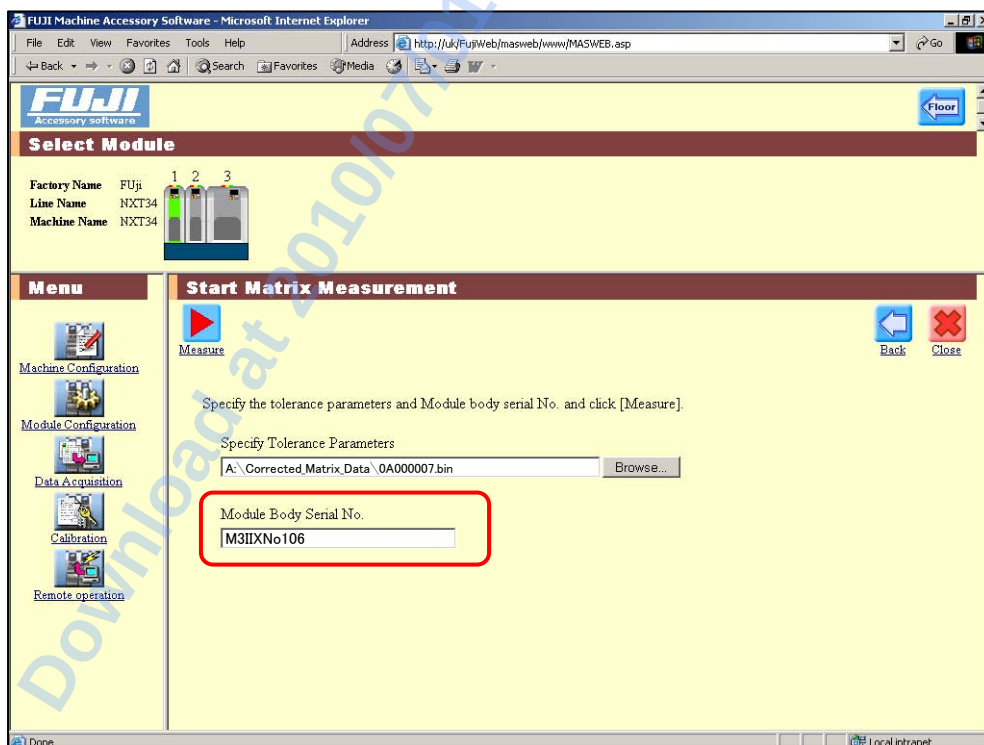


11. Select [Browse] and open the corrected matrix data file.

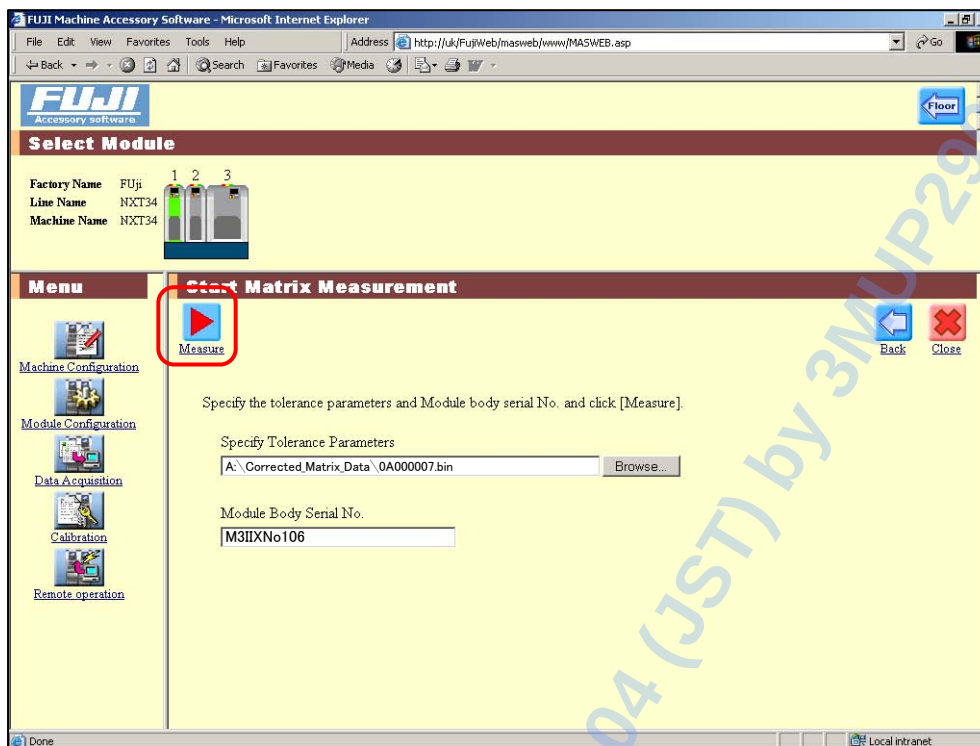


12. Input the module serial number. For example, M3 II module with serial number of 106 is inputted as M3 IIXNo106.

Note: Input M3 IIXNo or M6 IIXNo when module types are M3II or M6 II/SP.



13. Select [Start Measurement].



7.3. Performing Matrix Measurement

1. Operation panel is displayed as follows and the [START] button blinks. Press the [START] button.



2. The matrix measuring head begins to move, read the fiducial mark on a conveyor rail, and move to the width specified for matrix measurement. Then, it moves to the matrix measurement position and stops.



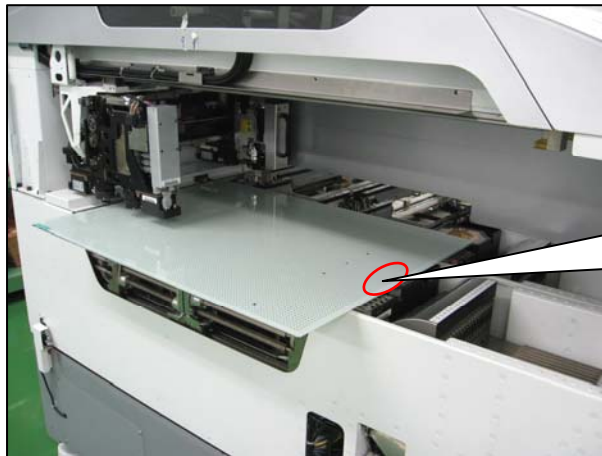
3. Check that the matrix measuring head and the conveyor have stopped and press the emergency stop button.



- Set the matrix glass board (AA02S0*) in the conveyor. At this time, confirm that the matrix glass board serial number has been set to the module front.

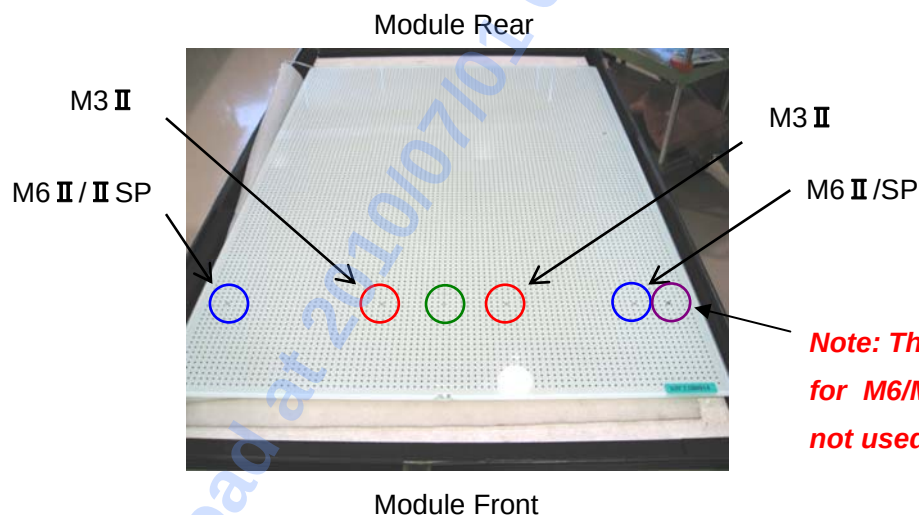
Warning: Recheck that the emergency stop button is pushed.

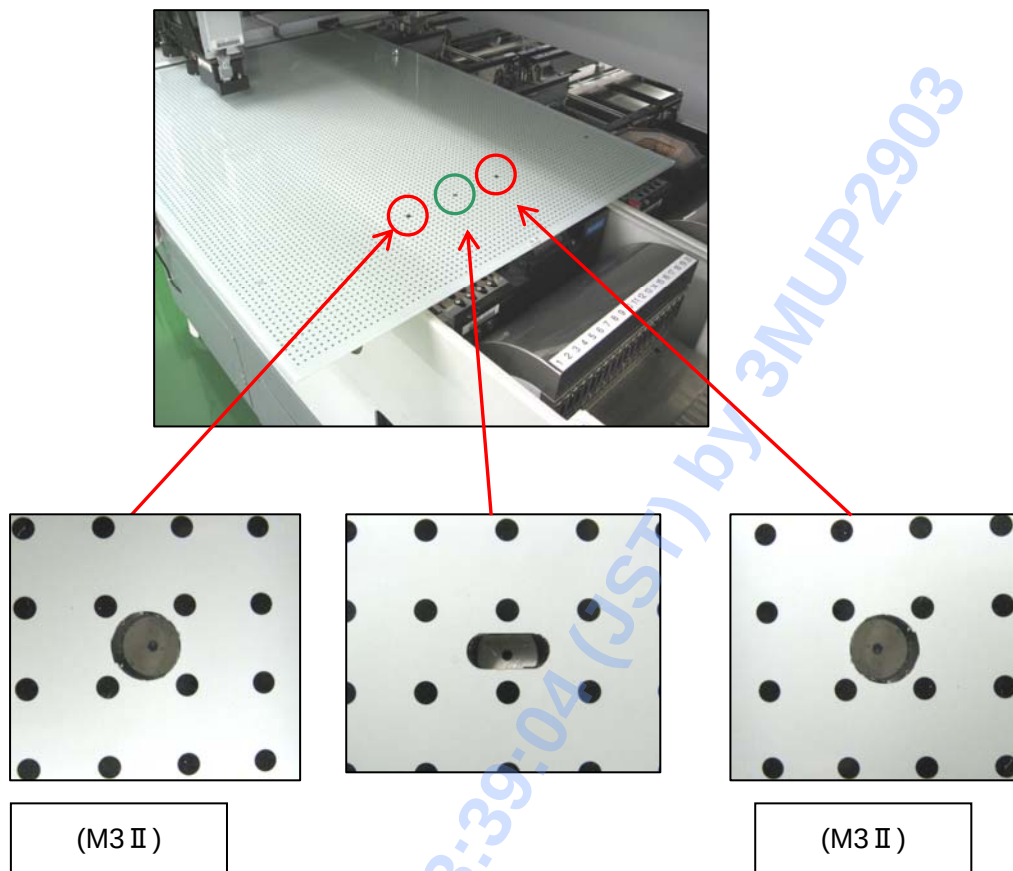
Warning: Be careful not to damage the glass surface when carrying the matrix glass board.



Confirm that the matrix glass board serial number has been set to the module front.

- Adjust the matrix glass board position and make so that the center of the fiducial mark on the conveyor come to the center of three holes on the matrix glass board. Note: This hole is used only for M6/M6S module. It is not used for M6 II/IISP.



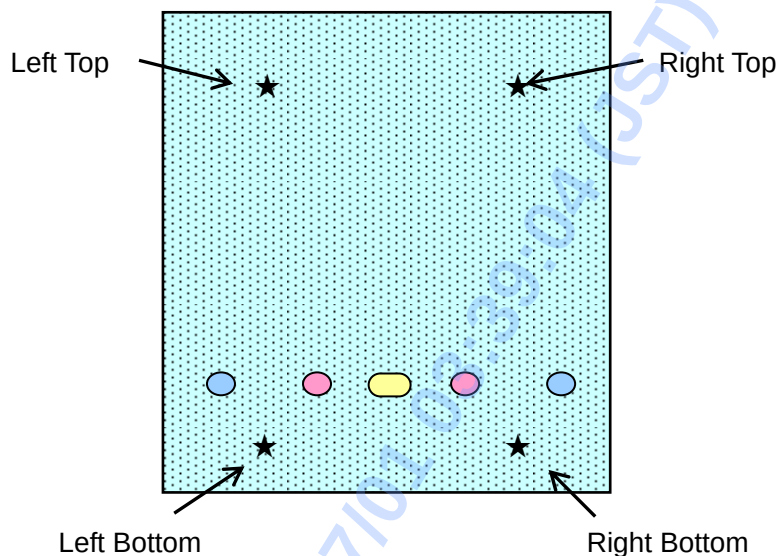
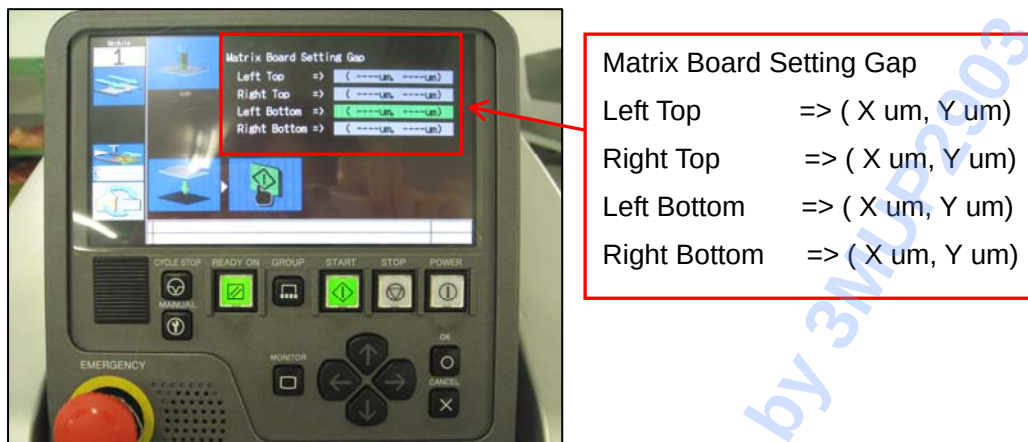


6. Press the [START] button.



7. The buzzer sounds. (This is not an error warning)

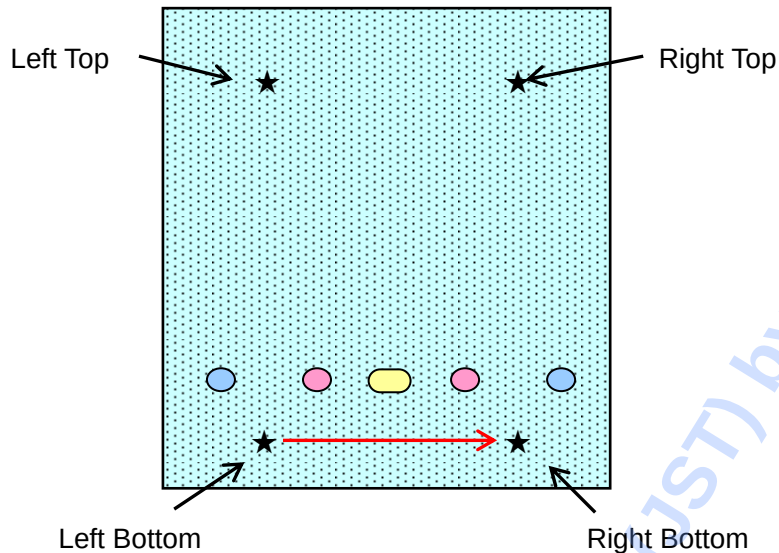
8. The matrix measuring head begins to move and read the fiducial mark on the conveyor rail. Then, the head moves to the Left Bottom position. (Refer to following graph)



9. Press [CYCLE STOP] button. The X and Y Coordinates of the Left Bottom Position are displayed on the operation panel.



10. Press the [START] button. The buzzer sounds afterwards. (This is not an error warning)
11. The matrix measuring head moves to the Right Bottom position.



12. Press [CYCLE STOP] button. The X and Y Coordinates of the Right Bottom Position are displayed on the operation panel.



13. Press the [START] button. The buzzer sounds afterwards. (This is not an error warning)
14. The matrix measuring head moves to the Right Top position.

15. Press [CYCLE STOP] button. The X and Y Coordinates of the Right Top Position are displayed on the operation panel.



16. Press the [START] button. The buzzer sounds afterwards. (This is not an error warning)
17. The matrix measuring head moves to the Left Top position.
18. Press [CYCLE STOP] button. The X and Y Coordinates of the Left Top Position are displayed on the operation panel.



19. Checks that the X and Y coordinate at all 4 points are 1000 um.

Left Top => -1000 um < (X um, Y um) < +1000 um

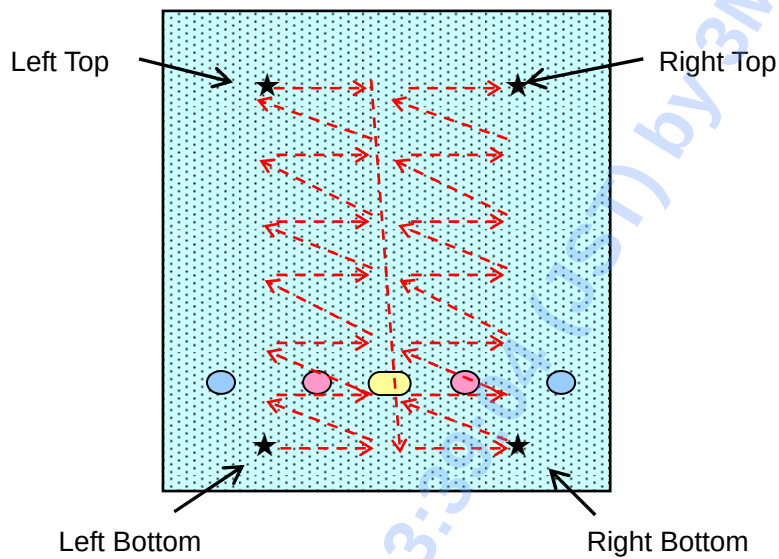
Right Top => -1000 um < (X um, Y um) < +1000 um

Left Bottom => -1000 um < (X um, Y um) < +1000 um

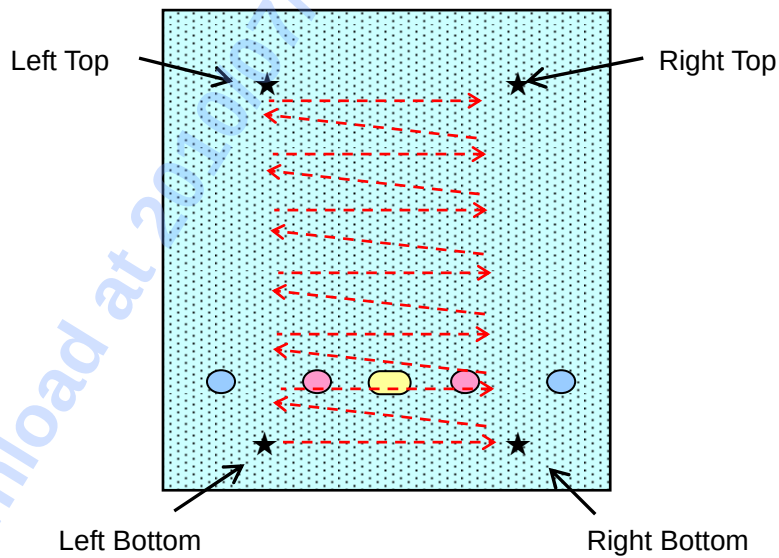
Right Bottom => -1000 um < (X um, Y um) < +1000um

20. Adjust the matrix glass board position so that the 4 points are within the tolerance.
Repeat step 5-19 to check whether all the 4 points are within the tolerance.
21. Push the [START] button if the X and Y coordinates of all 4 points are within +/- 1000 um tolerance. The head moves again and the buzzer sounds. (This is not an error warning)
22. Matrix measurement is started.

Note: The matrix measuring head on M3 II module moves as shown in the following figure.



Note: The matrix measuring head on M6 II/ II SP module moves as shown in the following figure.



23. Press the [CYCLE STOP] button to stop the matrix measurement.

24. Follow the next procedures to perform the last check on the matrix glass board position.
25. Press the [START] button. Then press the [MONITOR] button to display the H (horizontal) and V (vertical) value of the Left Bottom position.



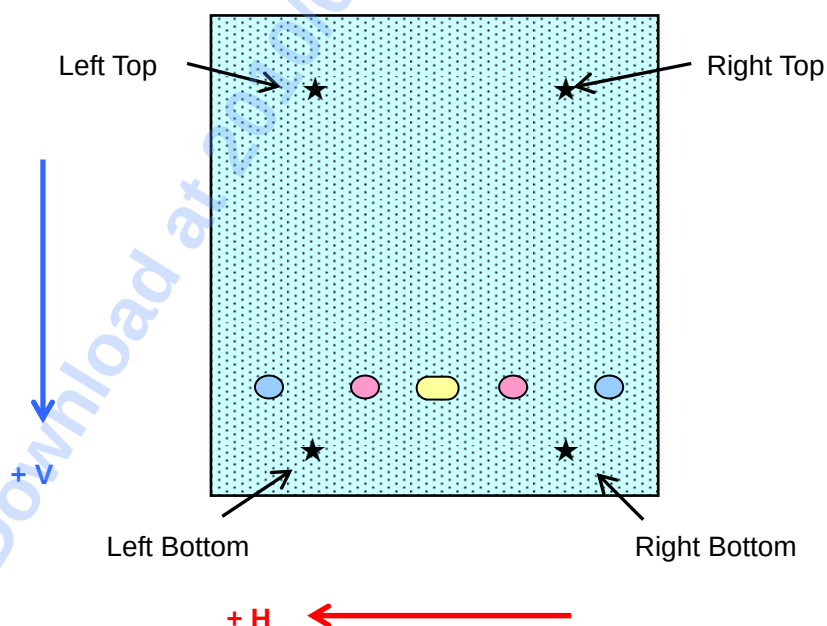
26. Check that the H and V values of the Left Bottom are within the following tolerance.

Tolerance: H = 320 +/- 10 (pixel)

Tolerance: V = 240 +/- 10 (pixel)

27. Prepare a rubber-made hammer. Strike the matrix glass board lightly so that the H and V values are within the tolerance.

Note: Strike the center of each side when striking the matrix glass board. The board will rotate if the corner of a glass board is striked and later the adjustment will become difficult.



28. Press the [MONITOR] button to return to the usual screen if the H and V values of the Left Bottom position are within the tolerance. Then press the [CYCLE STOP] button.
29. The buzzer sounds when the [START] button is pressed. (This is not an error warning) The matrix measuring head moves to the Right Bottom position.
30. Press the [MONITOR] button to check the H (horizontal) and V (vertical) values of the Right Bottom position.



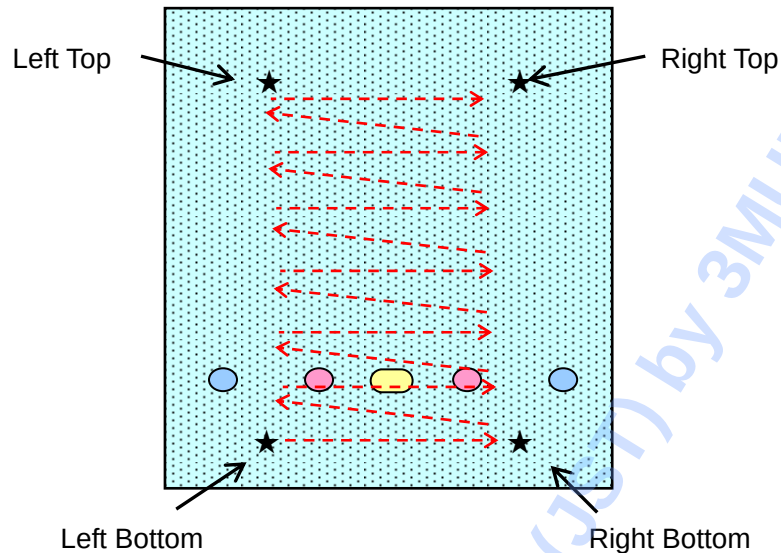
31. Check that the H and V values of the Right Bottom position are within the tolerance. Also check that the gap between the V value of the Right Bottom and Left Bottom are within 1 pixel. (For example, when the V value of the Right Bottom position = 241.3 (pixel) and the Left Bottom position = 241.9 (pixel), it is within the tolerance)

Tolerance: $H = 320 \pm 10$ (pixel)

Tolerance: $V = 240 \pm 10$ (pixel)

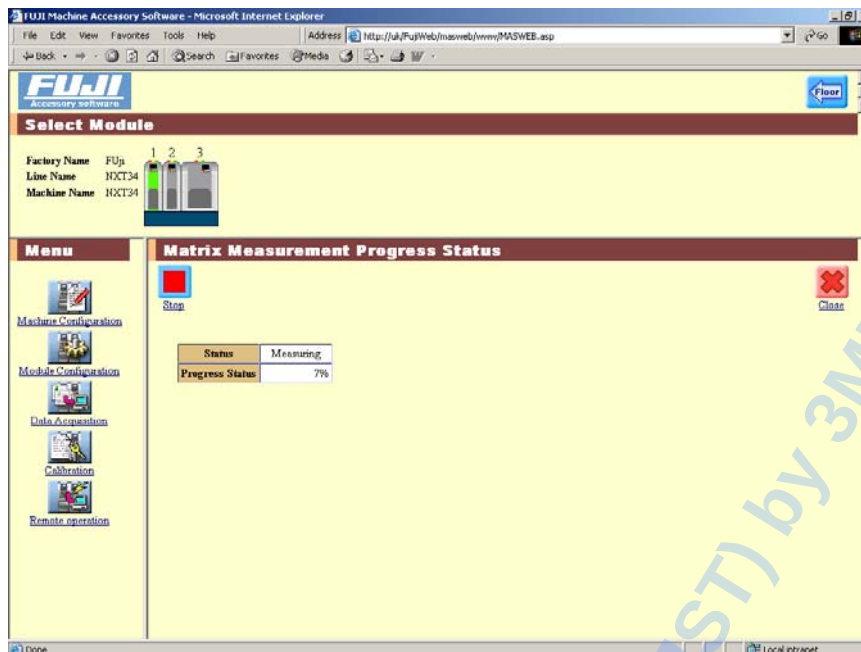
32. Prepare a rubber-made hammer. Strike lightly the side of matrix glass board to adjust the board position so that the H and V values are within the tolerance.
33. Perform adjustment of the Right Top and the Left Top positions same as the Left Bottom and the Right Bottom positions. Adjust the matrix glass board position when the value of H and V is outside the tolerance.
34. If the value of H and V of all 4 positions (Left Bottom, Right Bottom, Left Top, Right Top) are within the tolerance, press the [START] button to start the matrix measurement. The buzzer sounds afterwards. (This is not an error warning)

Note: Do not move the matrix glass board during measurement.



36. Perform matrix measurement for approximately 30 minutes. This is idling.
37. Press [CYCLE STOP] button to stop the matrix measurement once.
38. Press the [START] button to start the matrix measurement. This is the original measurement itself. Continue with the measurement if the value of H and V are within the tolerance. **The matrix measurement for M3II takes approximately about 90 to 120 minutes and approximately 50 minutes for M6 III/II SP respectively.**

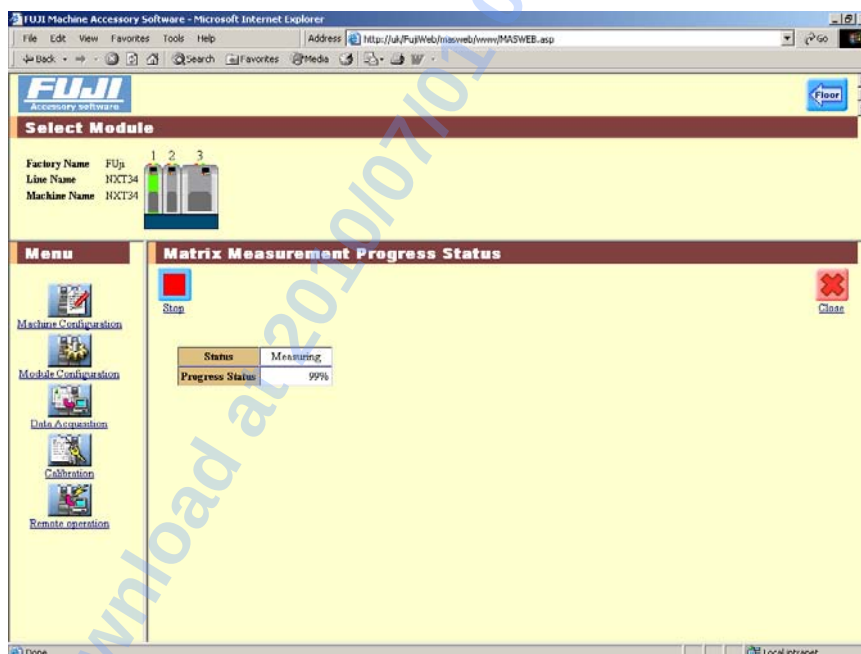
39. The progress status is displayed during the measurement at Accessory Software.



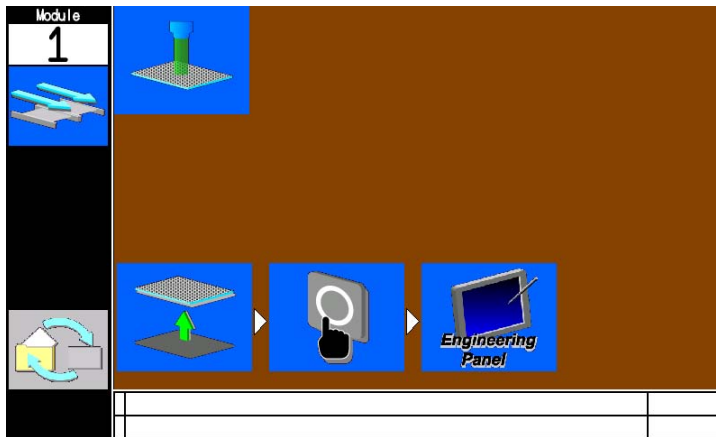
40. The matrix measuring head stops at the module rear position when the matrix measurement is finished.

41. The progress status becomes 99%. Select [Stop].

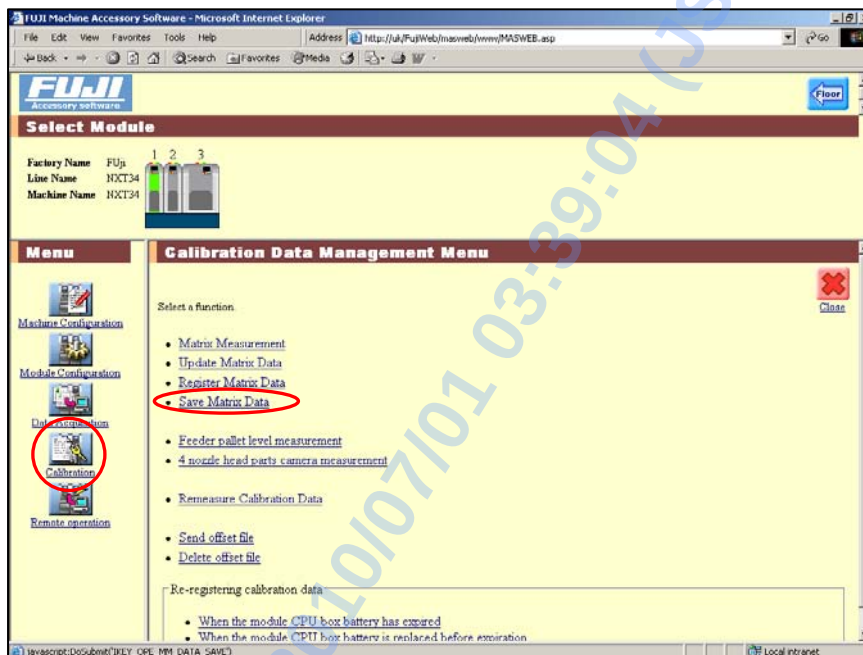
Note: The progress status will not become 100%.



42. At this time, the module operation screen displays as follows.



43. Select [Calibration Data Management] – [Save Matrix Data] to save the backup data on the other storage devices. This data will be used to re-register the matrix data when the module CPU box battery power runs out.



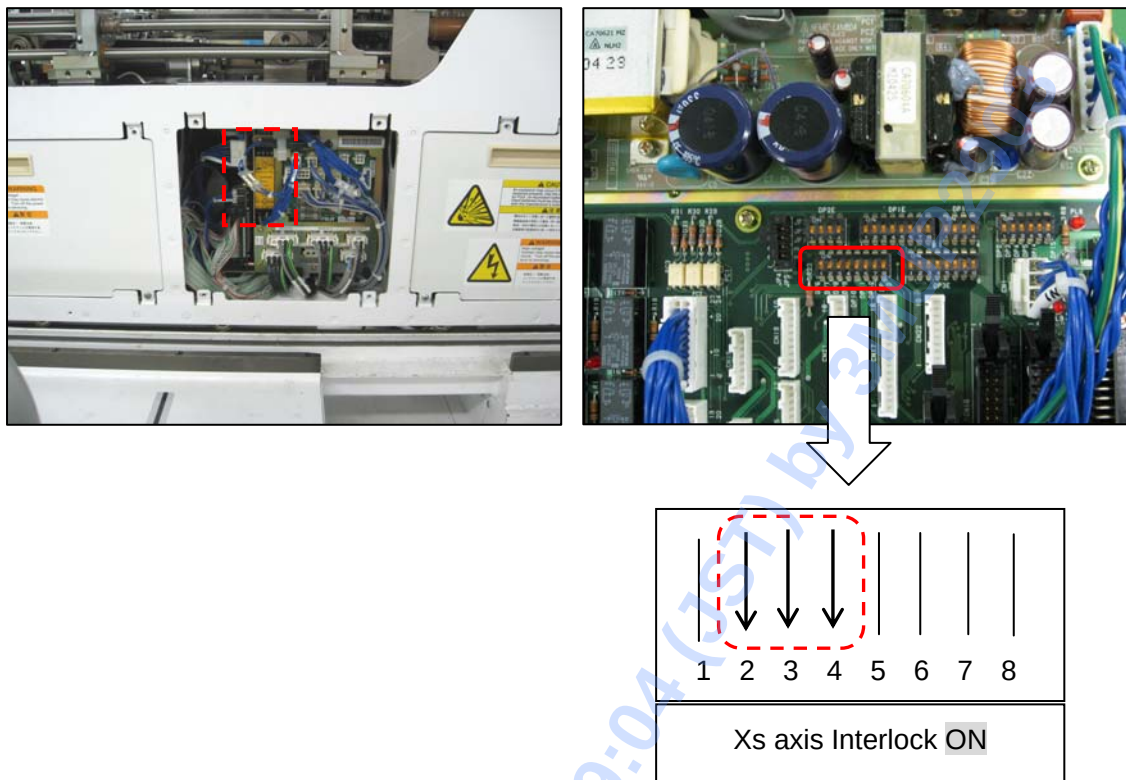
44. Press [EMERGENCY] button and remove the matrix glass board from the module.

Note: Be sure that the [EMERGENCY] button is pressed and the matrix measuring head stops completely.

45. Turn OFF the base main power.

46. Remove the matrix measuring head from the module. Connect the previously removed IEEE1394 cable again to the mark camera unit.

47. Turn ON the Xs axis interlock. Return the previously changed control box DIP switch settings to the original condition/state.



48. End the matrix measurement.

<MEMO>

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Chapter 8

Placing Quality Problems and Countermeasures

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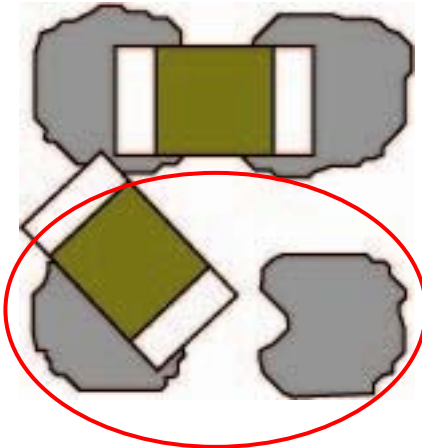
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8. Placing Quality Problems and Countermeasures

8.1 Examples of placing quality problems

The following are examples of placing quality problems

1. Shifted



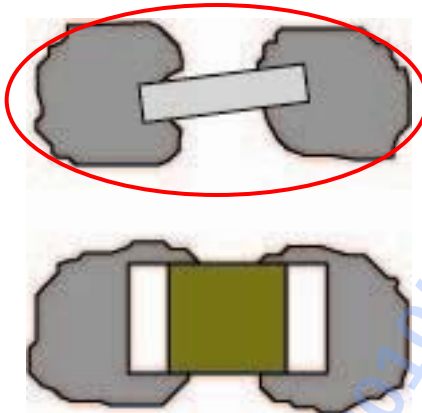
Parts are placed at a position shifted from the coordinates.

2. Missing



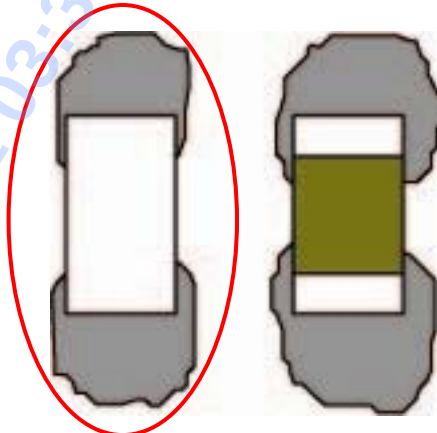
Parts are missing.

3. Side up



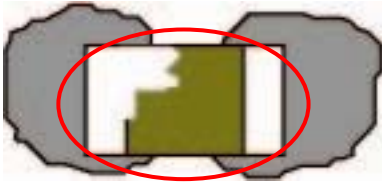
Parts are placed on their sides.

4. Upside down



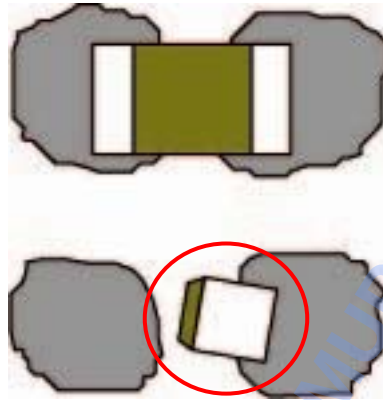
Parts are placed upside down.

5. Broken



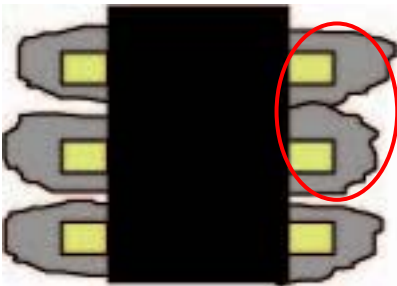
Parts are broken.

6. Tombstone



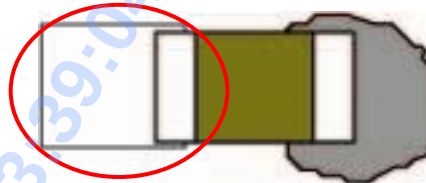
Parts are placed standing on their end.

7. Bridge



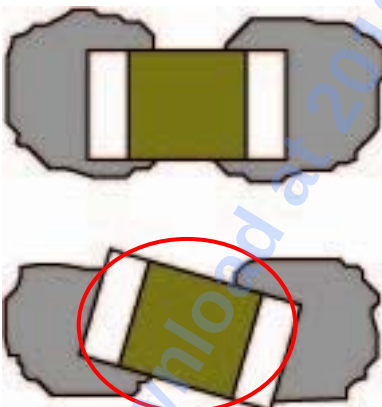
There is a solder bridge between pads

8. No Solder



Solder is missing from the pads

9. Skew

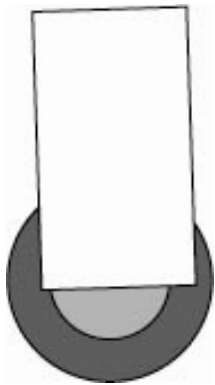


Parts are skewed

8.2 Causes of these problems

Almost all the problems described in the previous section are caused by unstable parts pickup.

Examples of unstable pickup:



Parts not picked up in the center



Parts picked up upside down



Parts picked up on their end

Placing quality problems occur when parts that are incorrectly picked up pass vision processing, or when the posture of parts changes after vision processing.

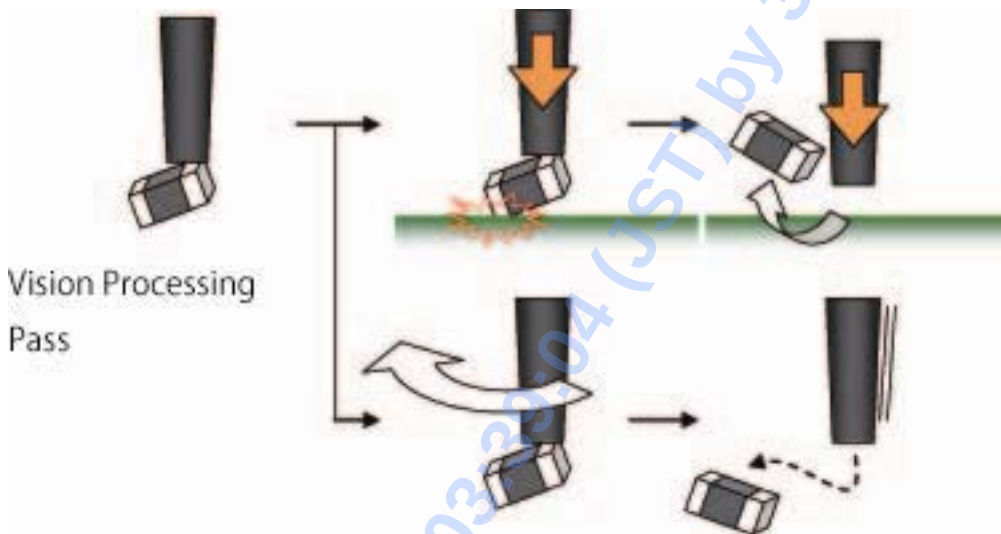
8.3 Mechanism of placing quality problems

This section explains how unstable pickup results in placing defects.

8.3.1 Missing/parts drop

Missing / Parts drop occurs mostly for rectangular chips such as 0603 (0201 in.), 1005 (0402 in.), and 1608 (0603 in.) The following two reasons are considered:

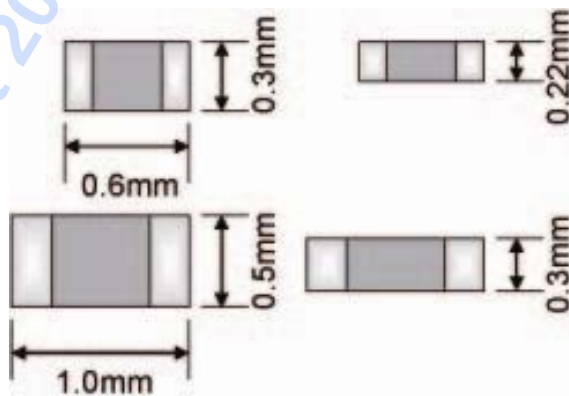
1. Parts picked up on their end, pass vision processing and are placed. As a result, parts hit the PCB surface, pop up and fall on to or off the PCB.
2. Parts picked up on their ends or edges pass vision processing and are placed. They are then dropped on to or off the PCB while being carried due to the excessive linear or rotational speed of the placing head.



* Dents can be found on PCB when the first case above occurs.

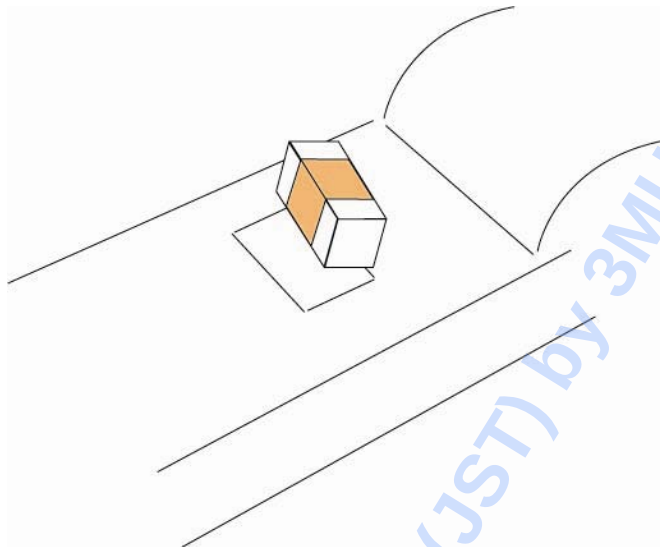
8.3.2 Pickup on the side

Parts most commonly picked up on their side are 0603R (0201R), 1005R (0402R), and tantalum capacitors. This is normally because part size is similar to part height and the setting for body size tolerance is too loose. In such cases the vision processing cannot detect parts picked up on their side.



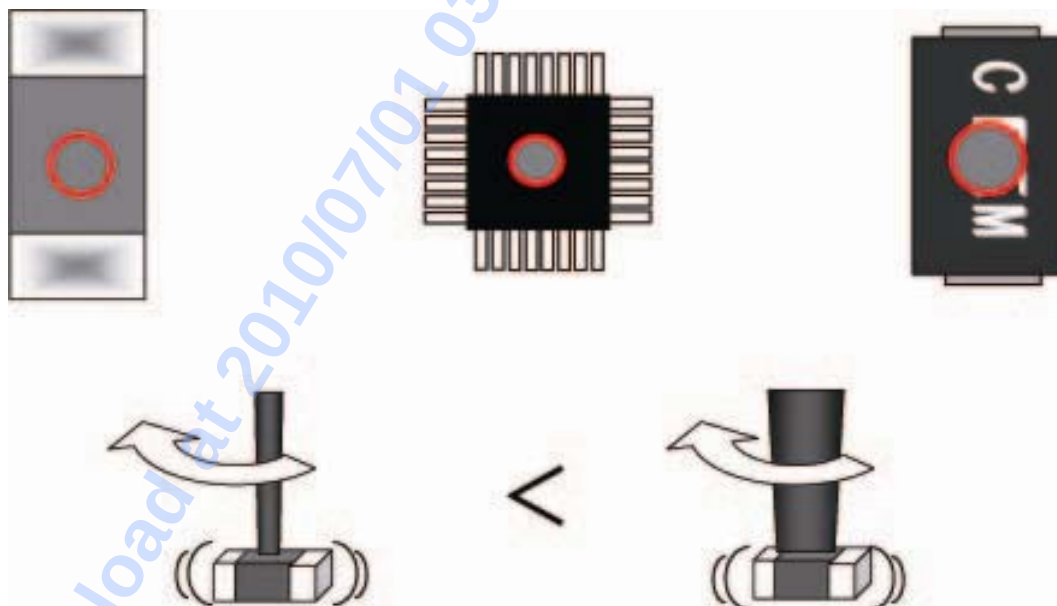
8.3.3 Upside down placement

Rectangular chips, diodes, and transistors are the parts most commonly placed upside down. When the top film is peeled from the carrier tape, the parts stick to the film or jump inside the cavity and flip over. The nozzle then picks them up as they are.

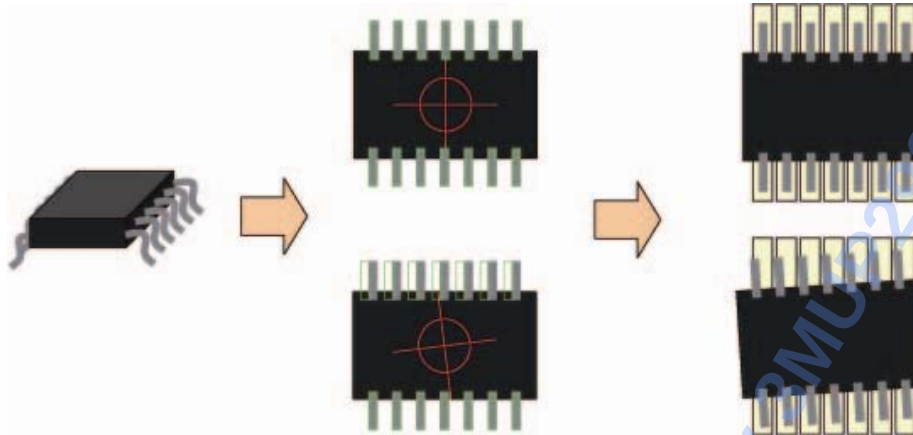


8.3.4 Skew

Skew occurs when the nozzle size or transport speed is inappropriate and the weight of the part exceeds the maximum transportable weight of the nozzle.



If the parts in question have leads and the amount of the skew is always stable, an incorrect setting in element data may be the cause of the problem.



* Particular care should be taken when element data is automatically generated with VPDplus.

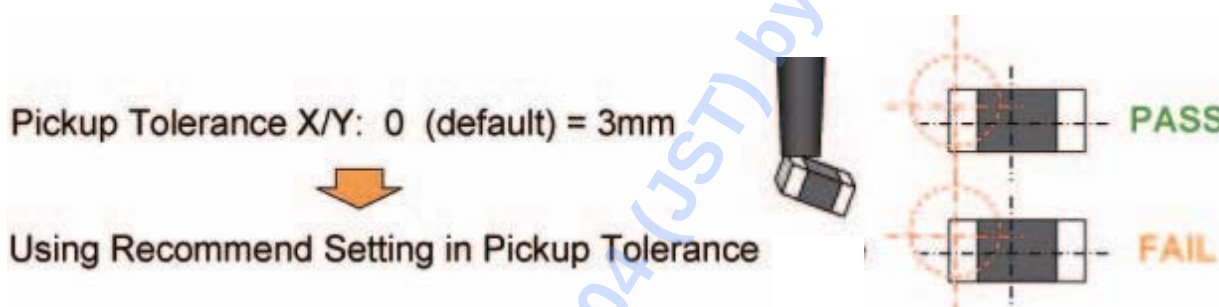
8.4 Countermeasures for placement quality problems

What measures should be taken in order to prevent the problems discussed in the previous section?

8.4.1 Review of pickup tolerance values

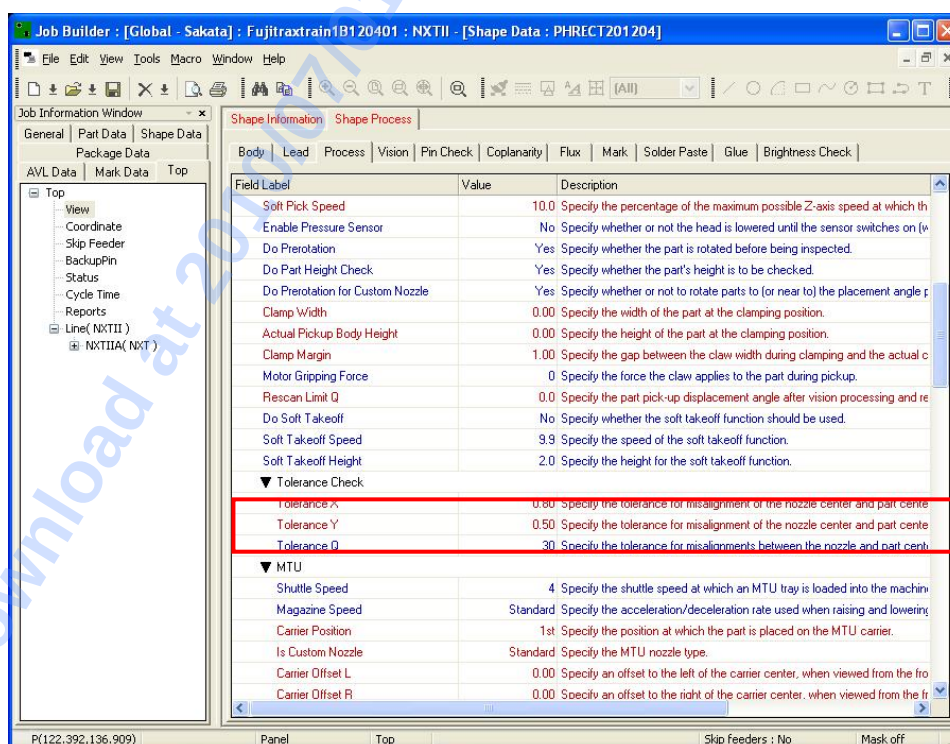
If the "Tolerance X" and "Tolerance Y" settings (pickup tolerance settings) in shape data are too large, vision processing may not be able to detect unstable parts pickup. The pickup tolerance checks the distance between the nozzle center position and the parts center position. If the pickup tolerance is too large, the vision processing cannot detect unstable pickup and the parts are placed as they are resulting in placement quality problems.

<Pickup Tolerance>



How to set the pickup tolerance:

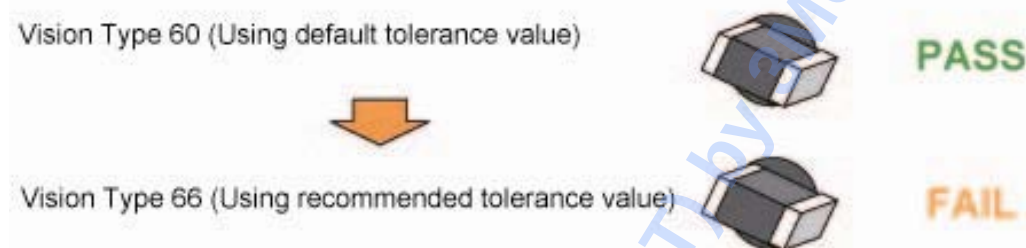
Start Job Builder, open the shape data for the parts in question and enter the recommended "Tolerance X" and "Tolerance Y" values in [Shape Process] – [Process] tab.



8.4.2 Usage of vision type 66 (applicable parts: 0402, 0603, 1005 and 1608)

Using vision type 66 makes it possible to detect parts picked up on their end which cannot be detected with vision type 60. Recommended tolerance values must be entered in "Body tolerance X/Y" and "Pickup Tolerance X/Y" when using vision type 66.

Parts pickup check



<Notes>

- Vision errors may occur more frequently for vision type 66 due to its sensitivity to foreign objects on the nozzle such as dust.
- The vision error rate is higher when using vision type 66 as all parts for which pickup errors occur are rejected.

8.4.3 Recommended body/pickup tolerance values for chip parts

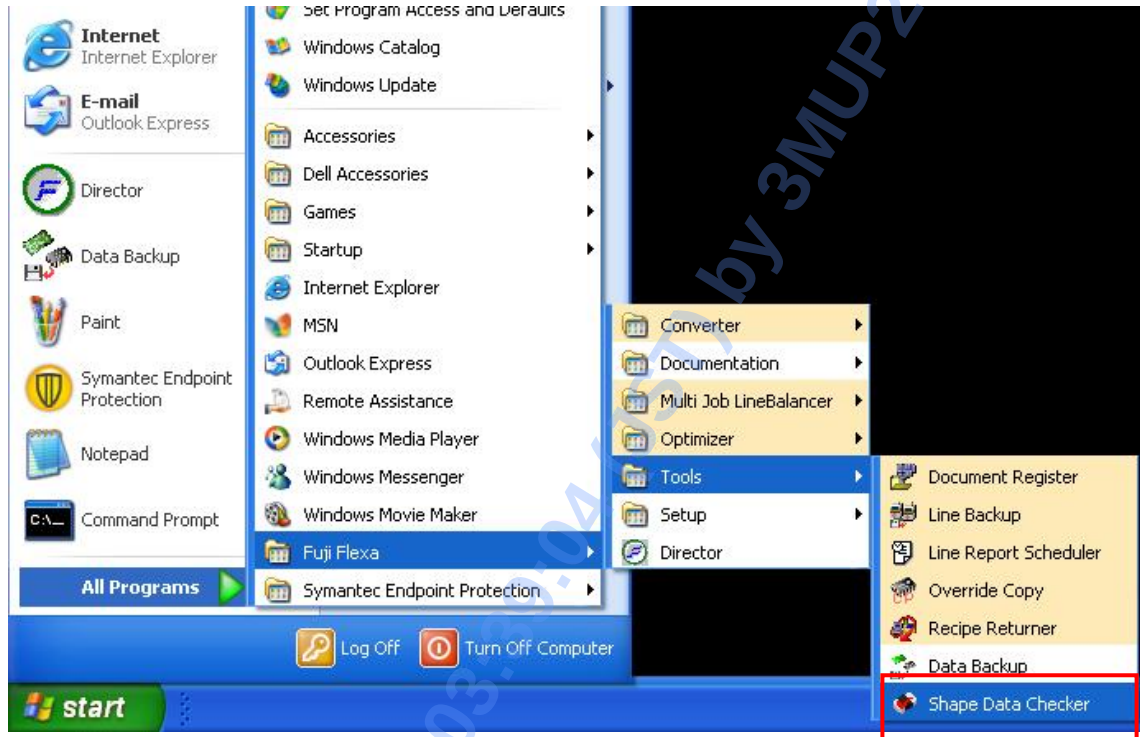
Refer to "Fuji Tech Report (Appendix) #10010A" for details of the recommended tolerance values (including IPS) for rectangular chips.

8.4.4 Using the Fuji Flexa shape data checker

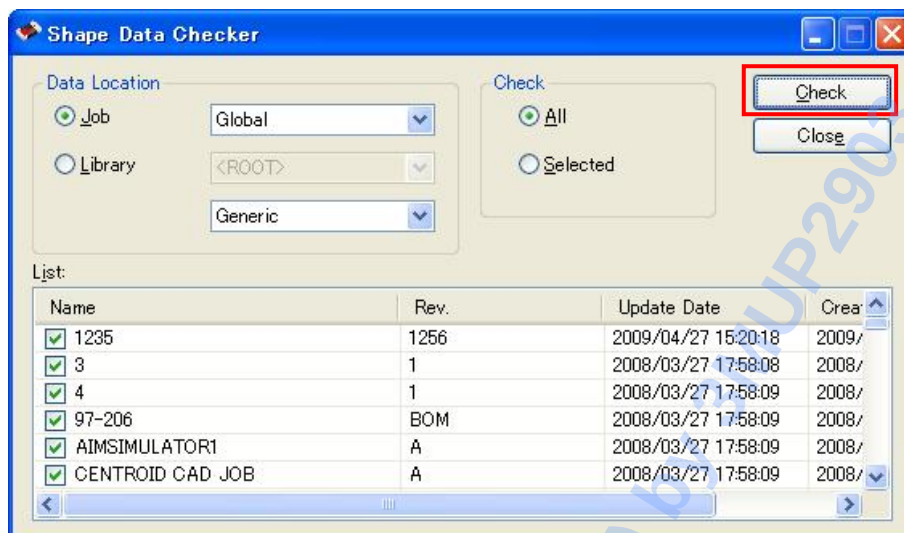
Shape Data Checker is available in Fuji Flexa version V5.0.0 and higher. Shape Data Checker can be used to compare Body Size and Pickup Tolerance values with those recommended by Fuji.

<How to use Shape Data Checker>

1. Select [Start] – [All Programs] – [Fuji Flexa] – [Tools] – [Shape Data Checker] .



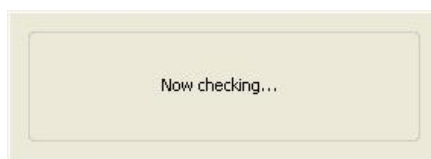
2. Select the desired settings in the [Shape Data Checker] window and then click [OK].



<Settings for Shape Data Checker>

Section	Outline	Details
Data Location	Specify the location of the Shape Data to be checked	Job: shape data in jobs are checked. Library: Shape data in Director's Shape Library is checked. Use the dropdown list to specify the location of the shape data. Global / Local: Select the job directory in which the job is stored when setting to "Job" for "Data Location". <ROOT> / *****: Specify the group where the job is stored when the settings are "Job", "Selected" for "Data Location", "Check", respectively. Generic/NXT/XPF...: Specify whether items in "Generic" or those for individual machine models (NXT, etc.) in "Override" are checked.
Check	Specify whether all shape data or only the specified shape data should be checked.	All: All shape data in the location specified in "Data Location" is checked. Selected: individual shape data in the location specified in "Data Location" is checked. Select the checkbox for shape data you wish to check.
List	The list of shape data to be checked.	Selected shape data will be checked.

<"Now checking" window>



3. The following result window is displayed after the check is finished.

Result					
Shape Name	Rev.	Field Name	Value	Recom...	Result
1005C(97-206)	BOM	Body Size Tolerance X	0.000	0.250	Default
1005C(97-206)	BOM	Pickup Tolerance X	0.000	0.300	Default
1005C(97-206)	BOM	Body Size Tolerance Y	0.000	0.125	Default
1005C(97-206)	BOM	Pickup Tolerance Y	0.000	0.175	Default
1608(97-206)	BOM	Body Size Tolerance X	0.160	0.320	Too Tight
1608(97-206)	BOM	Pickup Tolerance X	0.000	0.480	Default
1608(97-206)	BOM	Body Size Tolerance Y	0.080	0.200	Too Tight
1608(97-206)	BOM	Pickup Tolerance Y	0.000	0.280	Default
2125(97-206)	BOM	Body Size Tolerance X	0.200	0.400	Too Tight
2125(97-206)	BOM	Pickup Tolerance X	0.000	0.600	Default
2125(97-206)	BOM	Body Size Tolerance Y	0.120	0.313	Too Tight
2125(97-206)	BOM	Pickup Tolerance Y	0.000	0.375	Default
3216(97-206)	BOM	Body Size Tolerance X	0.300	0.640	Too Tight
3216(97-206)	BOM	Pickup Tolerance X	0.000	0.960	Default
3216(97-206)	BOM	Body Size Tolerance Y	0.160	0.320	Too Tight
3216(97-206)	BOM	Pickup Tolerance Y	0.000	0.480	Default
RECT3216160(AIMSI...	A	Body Size Tolerance X	0.60	0.640	Too Tight
RECT3216160(AIMSI...	A	Pickup Tolerance X	1.00	0.960	Too Loose
RECT3216160(AIMSI...	A	Body Size Tolerance Y	0.40	0.320	Too Loose
RECT3216160(AIMSI...	A	Pickup Tolerance Y	0.70	0.480	Too Loose
RECT3216160(CENTR...	A	Body Size Tolerance X	0.60	0.640	Too Tight
RECT3216160(CENTR...	A	Pickup Tolerance X	1.00	0.960	Too Loose
RECT3216160(CENTR...	A	Body Size Tolerance Y	0.40	0.320	Too Loose
RECT3216160(CENTR...	A	Pickup Tolerance Y	0.70	0.480	Too Loose
RECT2012040(CP643...	A	Body Size Tolerance X	0.200	0.400	Too Tight

Output CSV File... Cancel

<Explanation of the Items in each column of the "Result" window>

Column Name	Details
Shape Name	Shape Data file name
Rev.	Revision of the job
Field Name	Shape data item name
Value	Values set in the item
Recommendation	Fuji recommended value
Result	Judged result after comparing the set value with the recommended value OK (letters in black): Specified value is the same as the recommended value. Default (letters in green): There is no entry for these items. Too strict (letters in blue): Specified values are smaller than the recommended values. Too loose (letters in red): Specified values are larger than the recommended values.

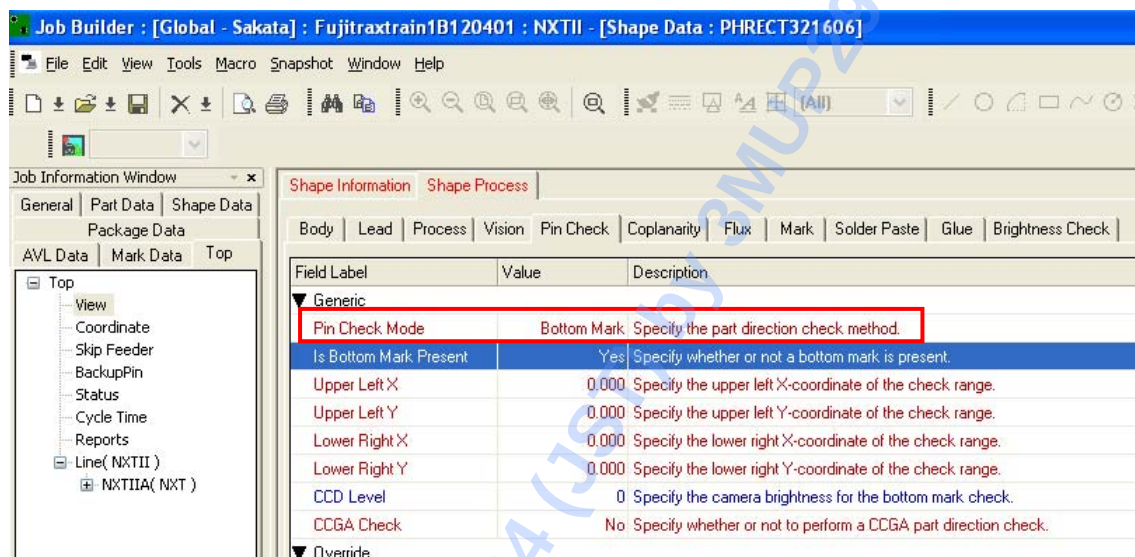
Note: clicking [Output CSV File ...] will output the results in a CSV file.

8.4.5 Detecting upside-down parts function

There are two ways to detect parts placed upside-down.

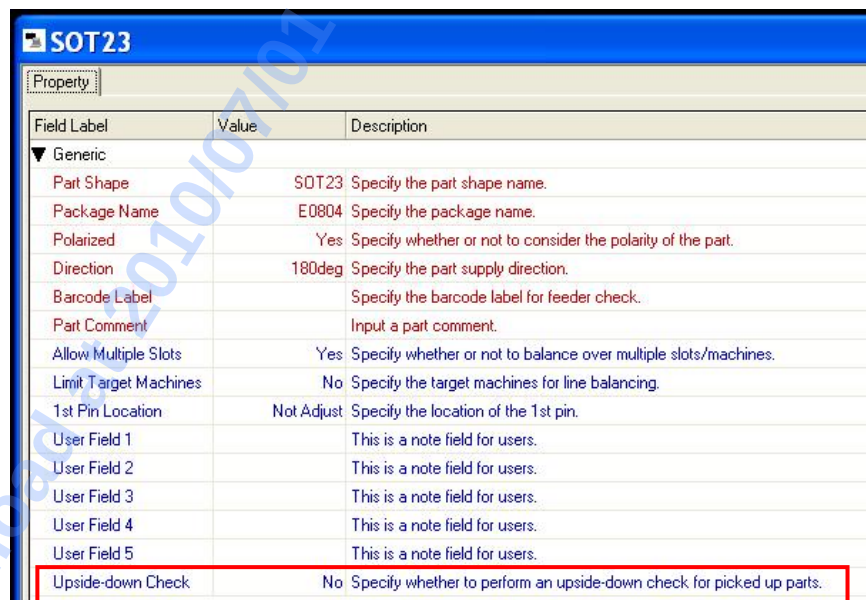
- Bottom Mark Check (Rectangular Chip Parts)

Select the [Pin Check] tab in [Shape Process], and then set the [Pin Check Mode] to [Bottom Mark]. Other items should be left at the default value.



- Using an IPS head (Leaded Parts)

Leaded parts placed upside-down can be detected using an IPS head. Open the shape data and select "Yes" for "Upside-down Check".



* There are some limitations on the part types for which upside-down check can be performed on the IPS head.

8.4.6 Review of shape data settings (nozzle size, transport speed, etc.)

Shape data settings may need to be reviewed to solve parts shifting or skewing problems. Inappropriate nozzle size or transport speed settings may cause these problems as a result of the part weight exceeding the maximum transportable weight for the nozzle.

Items which need to be reviewed

- Nozzle settings: Minimum Diameter, Maximum Diameter, Name

Shape Information Shape Process		
Body Lead Process Vision Pin Check Coplanarity Flux Mark Solder Paste Glue Brightness Check		
Field Label	Value	Description
▼ Generic		
Machine Process Combination		Specify the transport speed and nozzle name combination for each machine configuration.
Process Options	0	Specify an appropriate code when performing special processing.
▼ Nozzle		
Minimum Diameter	1.0	Specify the minimum nozzle diameter which can be used to pick up the part with this assem.
Maximum Diameter	1.3	Specify the maximum nozzle diameter which can be used to pick up the part with this assem.
Name		Specify the nozzle name when using a custom nozzle.
Load Check	No	Specify whether to check the nozzle load at constant intervals during production.
Type Number	0	Specify the code for the nozzle used for pick-up and placement of parts when in PAM mod.

- Parts transport speed: Transport Speed

Job Builder : [Global - Sakata] : Fujitraxtrain1B120401 : NXTII - [Shape Data : PHRECT321606]		
File Edit View Tools Macro Snapshot Window Help		
Shape Information Shape Process		
Body Lead Process Vision Pin Check Coplanarity Flux Mark Solder Paste Glue Brightness Check		
Field Label	Value	Description
▼ Place		
Do Place Check	No	Specify whether or not a parts confirmation check is performed.
Offset X	0.00	Specify an offset value for part placement in the X-direction.
Offset Y	0.00	Specify an offset value for part placement in the Y-direction.
Offset Z	0.00	Specify an offset value for part placement height.
Pressure	0	Specify the amount of pressure to be used for part placement.
Transport Speed	100	Specify the head acceleration rate from pick-up to placement.
Transport Speed Z	100	Specify the head acceleration rate from pick-up to placement(Z axis).
Transport Speed Q	100	Specify the head acceleration rate from pick-up to placement(nozzle rotation).
Soft Place Speed	0.0	Specify the Z-axis deceleration/acceleration.
Place Motion Selection	0	Specify whether to apply placing pressure.
Placing Thrust Pressure	50.000	Specify the applicable pressure when using placing pressure.
Placement Wait Time	0.000	Specifies the amount of time the Z-axis is stopped at the low position.
Stack Placing Down Speed	0.3	Specifies the lowering speed of the Z-axis during Package-on-Package placement.
Stack Placing Up Speed	0.3	Specifies the raising speed of the Z-axis during Package-on-Package placement.
Do Place Precision Priority	No	Specify whether or not to perform placement motion with high placement accuracy.
Table Speed	High	Specify the table speed.
Slow Place Speed	0	Specify the Z-axis transport speed.
Mode	Standard	Specify the placing speed.

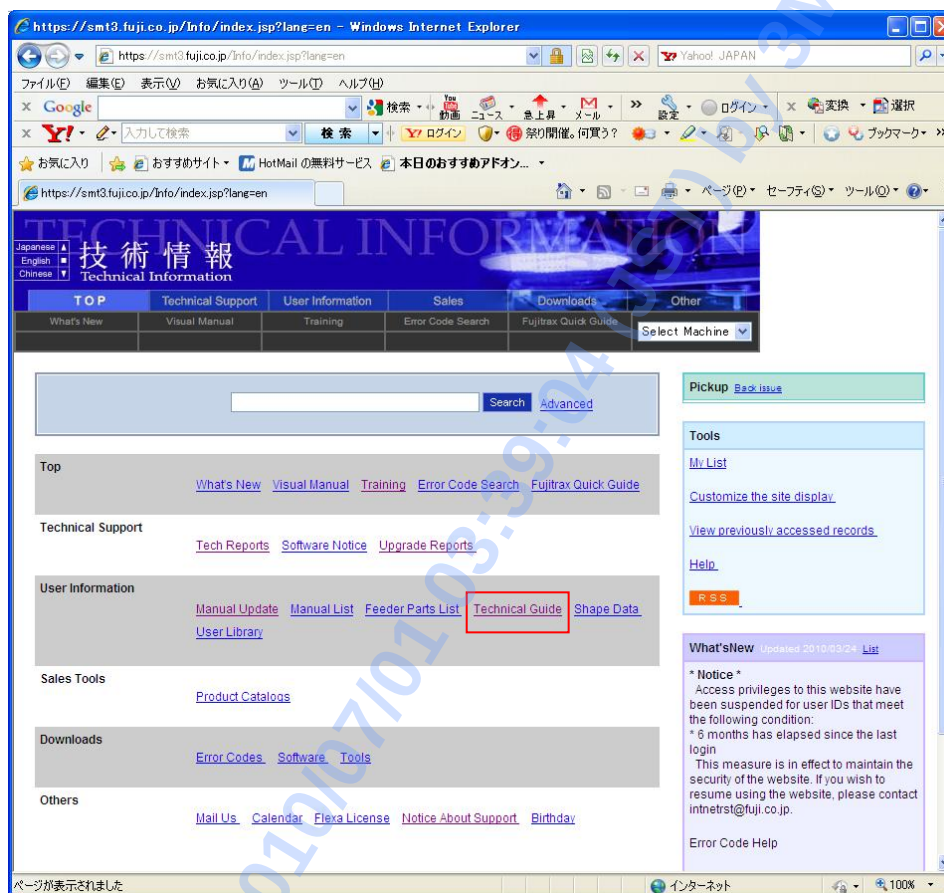
Notes: Refer to the following documents when making these settings.

- Technical Guide: Guideline for selecting nozzles for each part. Available for download from the Fuji Website.
- Fuji Tech Report (Appendix) No. 08059AR “Transportable weight of NXT-2 nozzles”

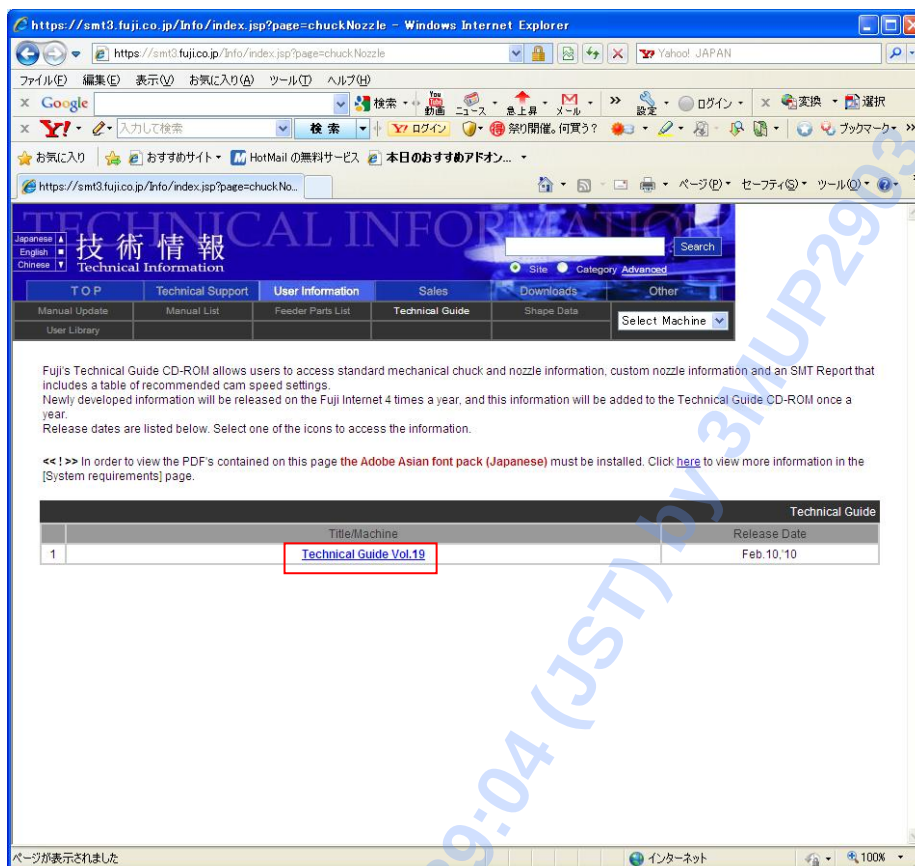
* The documentations listed above can be downloaded from the Fuji Technical Information page <http://smt3.fuji.co.jp/Info/>

<How to access the “Technical Guide”>

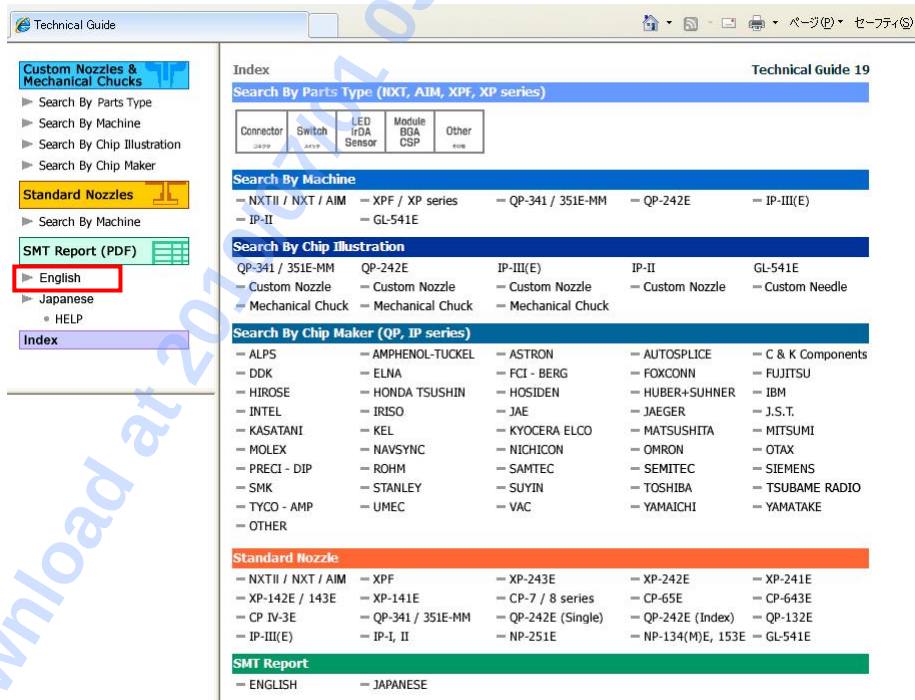
1. Click [Technical Guide] on the “Technical Information” page.



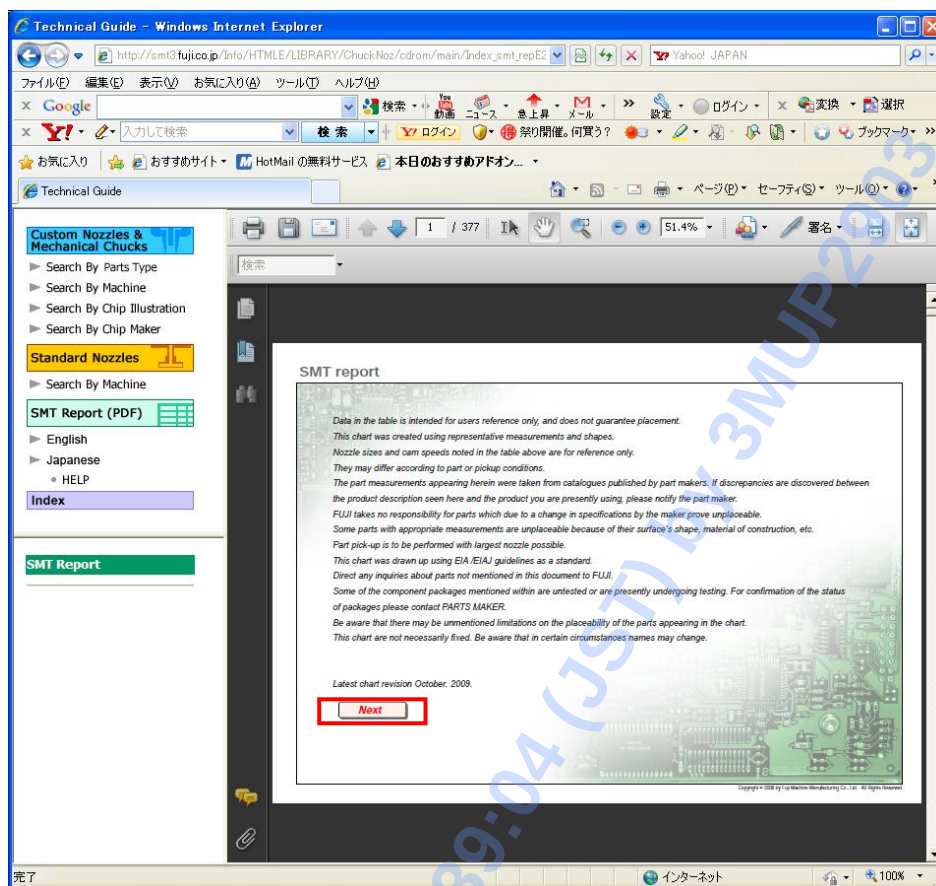
- Click [Technical Guide Vol. ****].



- Click [English] under [SMT Report].



4. Click [Next].



5. Click the machine name in the target part type cell.

Rectangular chips		Melf-type		Mini-mold		Tantalum electrolytic		Chip coil	
Fixed resistors		Fixed resistors		transistors,		capacitors			
Ceramic capacitors		Ceramic capacitors		diodes, LEDs					
NXT II	CP-7, 8 series	NXT II	CP-7, 8 series	NXT II	CP-7, 8 series	NXT II	CP-7, 8 series	NXT II	CP-7, 8 series
NXT, AIM	CP-65	NXT, AIM	CP-65	NXT, AIM	CP-65	NXT, AIM	CP-65	NXT, AIM	CP-65
XPF	CP-6 series	XPF	CP-6 series	XPF	CP-6 series	XPF	CP-6 series	XPF	CP-6 series
XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series
XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II
XP-141E	NP-1 series	XP-141E	NP-1 series	XP-141E	NP-1 series	XP-141E	NP-1 series	XP-141E	NP-1 series
XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E
QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)	
Trimmed resistors		Trimmed capacitor		Aluminium electrolytic		Surface acoustic		ZNR surge absorber	
				capacitor		wave filter			
NXT II	CP-7, 8 series	NXT II	CP-7, 8 series	NXT II	CP-7, 8 series	NXT II	CP-7, 8 series	NXT II	CP-7, 8 series
NXT, AIM	CP-65	NXT, AIM	CP-65	NXT, AIM	CP-65	NXT, AIM	CP-65	NXT, AIM	CP-65
XPF	CP-6 series	XPF	CP-6 series	XPF	CP-6 series	XPF	CP-6 series	XPF	CP-6 series
XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series	XP-142E, 143E	CP IV series
XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II	XP-242E, 243E	IP-1, II
XP-141E	NP-1 series	XP-141E	NP-1 series	XP-141E	NP-1 series	XP-141E	NP-1 series	XP-141E	NP-1 series
XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E	XP-241E, 341E	QP-132E
QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)		QP-2, 3, IP-III(E)	

6. The recommended setting table for nozzle size and "Transport Speed" is displayed.

SMT report
INDEX

Machine Type
NXT II

Remarks

The "Q" suffix on the end of the nozzle size indicates nozzles fitted with a rubber pad.
 The "M" suffix on the end of the nozzle size indicates nozzles designed for MELF parts.

Symbols:
 ○ = Placement possible.
 △ = Possible with an H12MS head, H12S head with 6 nozzle, H08 head with 4 nozzles, Q04 with camera image center control, or an H04 head with 4 nozzles, but throughput reduced.
 X = Not possible.

■ Placement is possible. Numbers in the table indicate transport speed.
 % indicates the Shape data Place TransportSpeed, TransportSpeedZ, and TransportSpeedQ.
 100%=H, 70%=Mid, 30%=Low
 (% is based on the head with symbol ○.)

NXT II

NXT, AIM

XP

XP-143E, 142E

XP-242E, 243E

XP-141E

XP-241E, 341E

QP-2, 3, IP-III(E)

CP-7, 8 series

CP-65

CP-5 series

CP IV series

IP-I, II

NP-1 series

QP-132E

Rectangular chips
 Fixed resistors, Ceramic capacitors

Part name	0402	0803	1006	1808	2012	3218	3225	4632
Tape width x feed pitch	8 x 2	8 x 2	8 x 2	8 x 4	8 x 4	8 x 4	8 x 4	12 x 8
Dimensions(mm)	L	0.4	0.6	1.0	1.6	2.0	3.2	4.5
	W	0.2	0.3	0.5	0.8	1.2	1.6	2.5
	T	0.12 ~ 0.2	0.22 ~ 0.3	0.3 ~ 0.5	0.45 ~ 0.8	0.5 ~ 1.25	0.5 ~ 1.25	0.8 ~ 2.0
Weight (g)	~ 0.0001	~ 0.0003	~ 0.0013	~ 0.0055	~ 0.0076	~ 0.0223	~ 0.0490	~ 0.1390
H12MS Head	○	○	○	○	○	○	○	○
H12S Head	○	○	○	○	○	○	○	○
H08 Head	○	○	○	○	○	○	○	○
Q04 Head	○	○	○	○	○	○	○	○
H04 Head	X	X	X	○	○	○	○	○
H02 Head	X	X	X	○	○	○	○	○
H01 Head	X	X	X	○	○	○	○	○
QF Head	X	X	X	○	○	○	○	○
φ0.3	100%							
φ0.4		100%						
φ0.7			100%					
φ1.0				100%				
φ1.3					100%			
φ1.5M						100%		
φ1.8							100%	
φ1.8M								100%
φ2.5, φ2.5G							100%	70%
φ2.5M								
φ3.7, φ3.7G								
φ5.0, φ5.0G								
φ7.0, φ7.0G								
φ10.0, φ10.0G								
φ15.0, φ15.0G								
φ20.0, φ20.0G								

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8-17

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8.4.7 Using the IPS head

Use of IPS (Intelligent Part Sensor) head reduces placing quality problems. The following functions are available:

- Nozzle crack check
- Parts picked up on end check
- Parts return check
- Parts picked up upside-down check (for certain leaded parts)



(H12HSQ)

Setting for IPS head

Setting for parts presence check (parts return check) function:

[Shape Data] – [Shape Process] – [Place] – [Do Place Check]: Yes

Shape Data : SOT23		
Shape Information Shape Process		
Body Lead Process Vision Pin Check Coplanarity Flux Mark Solder Paste Glue Brightness Check		
Field Label	Value	Description
Is Custom Nozzle	Standard	Specify the MTU nozzle type.
Carrier Offset L	0.00	Specify an offset to the left of the carrier center, when viewed from the front of the machine.
Carrier Offset R	0.00	Specify an offset to the right of the carrier center, when viewed from the front of the machine.
▼ Place		
Do Place Check	No	Specify whether or not a parts confirmation check is performed.
Offset X	0.00	Specify an offset value for part placement in the X-direction.
Offset Y	0.00	Specify an offset value for part placement in the Y-direction.
Offset Z	0.00	Specify an offset value for part placement height.
Pressure	0	Specify the amount of pressure to be used for part placement.
Transport Speed	100	Specify the head acceleration rate from pick-up to placement.
Transport Speed Z	100	Specify the head acceleration rate from pick-up to placement(Z axis).
Transport Speed Q	100	Specify the head acceleration rate from pick-up to placement(nozzle rotation).
Soft Place Speed	0.0	Specify the Z-axis deceleration/acceleration.
Place Motion Selection	0	Specify whether to apply placing pressure.
Placing Thrust Pressure	50.000	Specify the applicable pressure when using placing pressure.
Placement Wait Time	0.000	Specifies the amount of time the Z-axis is stopped at the low position.
Stack Placing Down Speed	0.3	Specifies the lowering speed of the Z-axis during Package-on-Package placement.
Stack Placing Up Speed	0.3	Specifies the raising speed of the Z-axis during Package-on-Package placement.
Do Place Precision Priority	No	Specify whether or not to perform placement motion with high placement accuracy.
Table Speed	Ultra High	Specify the table speed.
Slow Place Speed	0	Specify the Z-axis transport speed.
Mode	Standard	Specify the placing speed.
Panel Stopping Position Correction	No	Specify whether the part requires panel stopping position correction.

Settings for the parts height detection (parts picked up on end check) function:

[Shape Data] – [Shape Process] – [Process] – [Pick] – [Do Part Height Check]:
 Yes

Field Label	Value	Description
▼ Pick		
Do Auto Offset	Yes	Specify whether automatic compensation is performed for pick-up offset.
Pick Check Mode	No Check	Specify the method to check if a nozzle has picked up a part during pickup.
Offset X	0.00	Specify an offset value from the center of the feeder cavity to the pick-up point in the X-direction.
Offset Y	0.00	Specify an offset value from the center of the feeder cavity to the pick-up point in the Y-direction.
Offset Q	0.00	Specify the nozzle rotation angle when picking up a part using a long nozzles, mechanical chucks.
Offset Z	0.00	Specify an up or down offset for the pickup point of the part in a feeder cavity.
Tray Pick Offset Z	0.0	Specify an up or down offset for the pickup point of the part in a tray cavity.
Soft Pick Speed	10.0	Specify the percentage of the maximum possible Z-axis speed at which the part is carried after pickup.
Enable Pressure Sensor	No	Specify whether or not the head is lowered until the sensor switches on (when the nozzle is moving to pick up a part).
Do Perotation	No	Specify whether the part is rotated before being inspected.
Do Part Height Check	Yes	Specify whether the part's height is to be checked.
Do Perotation for Custom Nozzle	Yes	Specify whether or not to rotate parts to (or near to) the placement angle prior to vision processing when using custom nozzles.
Clamp Width	0.00	Specify the width of the part at the clamping position.
Actual Pickup Body Height	0.00	Specify the height of the part at the clamping position.
Clamp Margin	1.00	Specify the gap between the claw width during clamping and the actual claw width.
Motor Gripping Force	0	Specify the force the claw applies to the part during pickup.
Rescan Limit Q	0.0	Specify the part pick-up displacement angle after vision processing and reperform vision processing after compensation.
Do Soft Takeoff	No	Specify whether the soft takeoff function should be used.
Soft Takeoff Speed	9.9	Specify the speed of the soft takeoff function.
Soft Takeoff Height	2.0	Specify the height for the soft takeoff function.
▼ Tolerance Check		

Part height tolerance setting:

[Shape Data] – [Shape Information] – [Body] – [Height Tolerance]:

Enter the value for parts height tolerance.

Field Label	Value	Description
▼ Generic		
Length(X)	1.500	Specify the body size in the X-direction of the part or die.
Width(Y)	2.900	Specify the body size in the X-direction of the part or die.
Length Tolerance(X)	0.000	Specify the part's X-direction length tolerance.
Width Tolerance(Y)	0.000	Specify the part's Y-direction length tolerance.
Height	1.10	Specify the height of the part or die.
Height Tolerance	0	Specify the part's height tolerance(%).
Color	Black	Specify whether the part's body is black or white.
Stand Off	0.00	Specify the size of the gap between the part body and the panel surface.
▼ Override		

Setting for parts picked up upside down check for leaded parts:

[Part Data] – [Upside-down Check]: Yes

SOT23

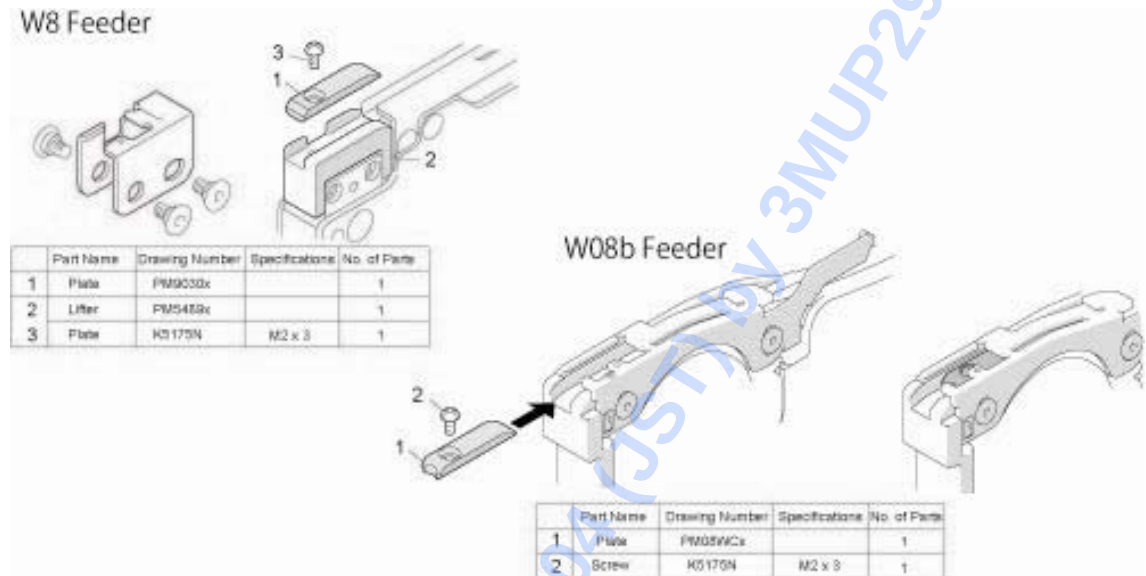
Property

Field Label	Value	Description
▼ Generic		
Part Shape	SOT23	Specify the part shape name.
Package Name	E0804	Specify the package name.
Polarized	Yes	Specify whether or not to consider the polarity of the part.
Direction	180deg	Specify the part supply direction.
Barcode Label		Specify the barcode label for feeder check.
Part Comment		Input a part comment.
Allow Multiple Slots	Yes	Specify whether or not to balance over multiple slots/machines.
Limit Target Machines	No	Specify the target machines for line balancing.
1st Pin Location	Not Adjust	Specify the location of the 1st pin.
User Field 1		This is a note field for users.
User Field 2		This is a note field for users.
User Field 3		This is a note field for users.
User Field 4		This is a note field for users.
User Field 5		This is a note field for users.
Upside-down Check	No	Specify whether to perform an upside-down check for picked up parts.

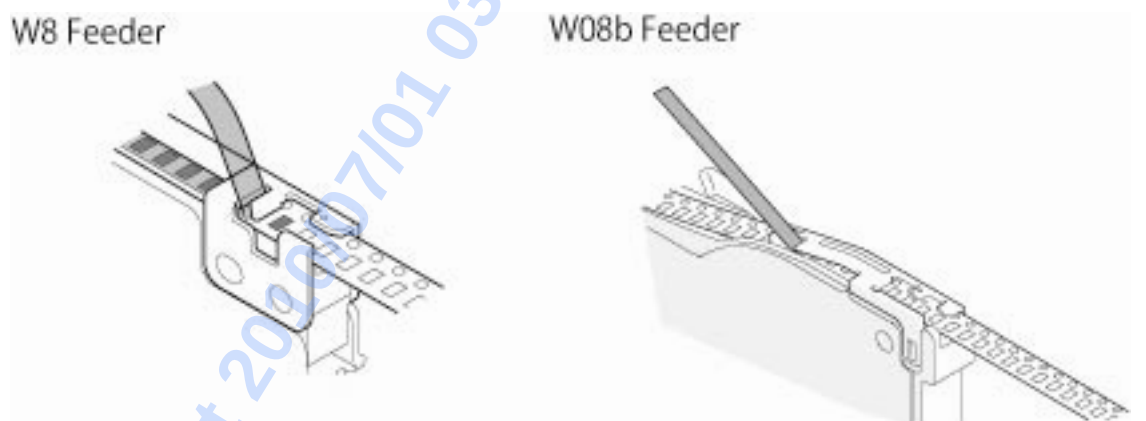
8.4.8 Feeder ESD protection

Dislodgement of parts due to electrification from peeling is mainly caused by using cover tape with no ESD protection or weak ESD protection. In such cases, parts may be picked up on their ends or upside down. Fuji recommends attaching the ESD plate to the feeder to prevent the adverse effects of ESD.

1. Attaching ESD plate



2. Slit fold back position



Stable pickup can be achieved by folding back the top film at the slit to prevent parts being dislodged by the electrified top film.

8.4.9 Nozzle cleaning

Unstable parts pickup may be caused by dirty nozzles. The following optional units are available in order to perform efficient nozzle cleaning.

- Brushing Unit Feeder Type



- Auto Nozzle Cleaner



8.4.10 Changing nozzles due to statistical warnings

This function is used to skip nozzles which exceed the specified error rate. The skipped nozzle is replaced by another nozzle and production is continued. The machine stops with an error if there is no alternative nozzle.

<Setting / Usage>

1. Perform the following setting in machine accessory software.

[Module Configuration] – [Statistical Nozzle error settings]

The screenshot shows the 'FUJI Machine Accessory Software - Microsoft Internet Explorer' window. The address bar displays 'http://accserver/FujiWeb/masweb/www/MASWEB.asp'. The page has a 'Select Module' section with fields for 'Factory Name' (User Training), 'Line Name' (NXTII B Line), and 'Machine Name' (NXTIIB). Below this is a 'Menu' on the left with icons for 'Machine Configuration', 'Module Configuration' (highlighted with a red box), 'Data Acquisition', 'Calibration', and 'Remote operation'. The main area is titled 'Module Configuration' and contains a 'Category' dropdown menu set to 'Statistical Nozzle error settings' (highlighted with a red box). To the right of the dropdown is a 'Send to Machine' button. Below these is a table with settings:

Settings	Current Value	Set Value	Unit
Change nozzle when an error occurs	OFF	☒ ON ☐ OFF	
Error count to decide statistical nozzle errors	4	4	

The 'Set Value' for the error count is also highlighted with a red box. A 'Close' button is located in the top right corner of the configuration area.

- “Change nozzle when an error occurs”: ON
 - “Error count to decide statistical nozzle errors”: (Specify the number of times an error should occur for a statistical skip to be implemented).
2. Click “Send to Machine”.
 3. Set spare nozzles in the nozzle station and start production.

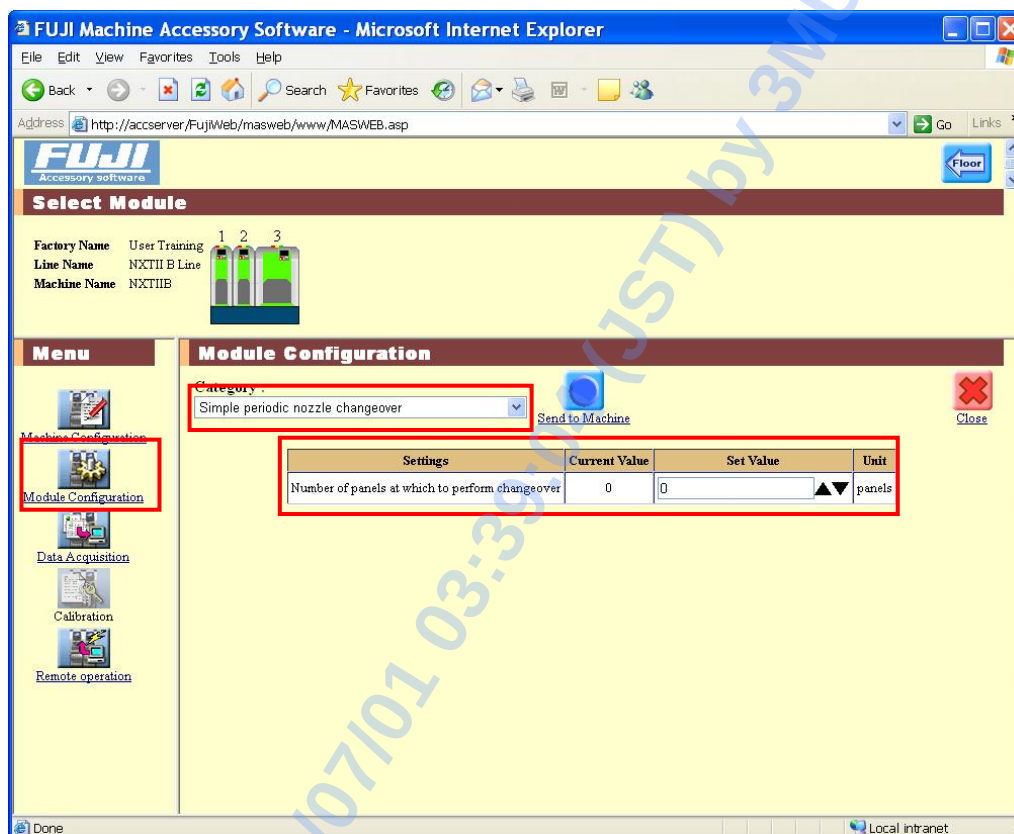
8.4.11 Simple periodic nozzle changeover function

This function automatically changes the nozzles after the number of produced panels reaches the quantity specified at the start of production. The nozzles are automatically exchanged during production with the same type of nozzles that were set in the nozzle station prior to starting production.

Setting/usage

1. Perform the following settings in machine accessory software.

[Module Configuration] – [Simple periodic nozzle changeover]



- “Number of panels at which to perform changeover”: (Specify the number of panels at which to perform nozzle changeover. The possible range is from “0” to “99,999”.)
2. Click “Send to Machine”.
 3. Load spare nozzles in the nozzle station and start production.

Chapter 9

Conveyor operation during unit breakdown

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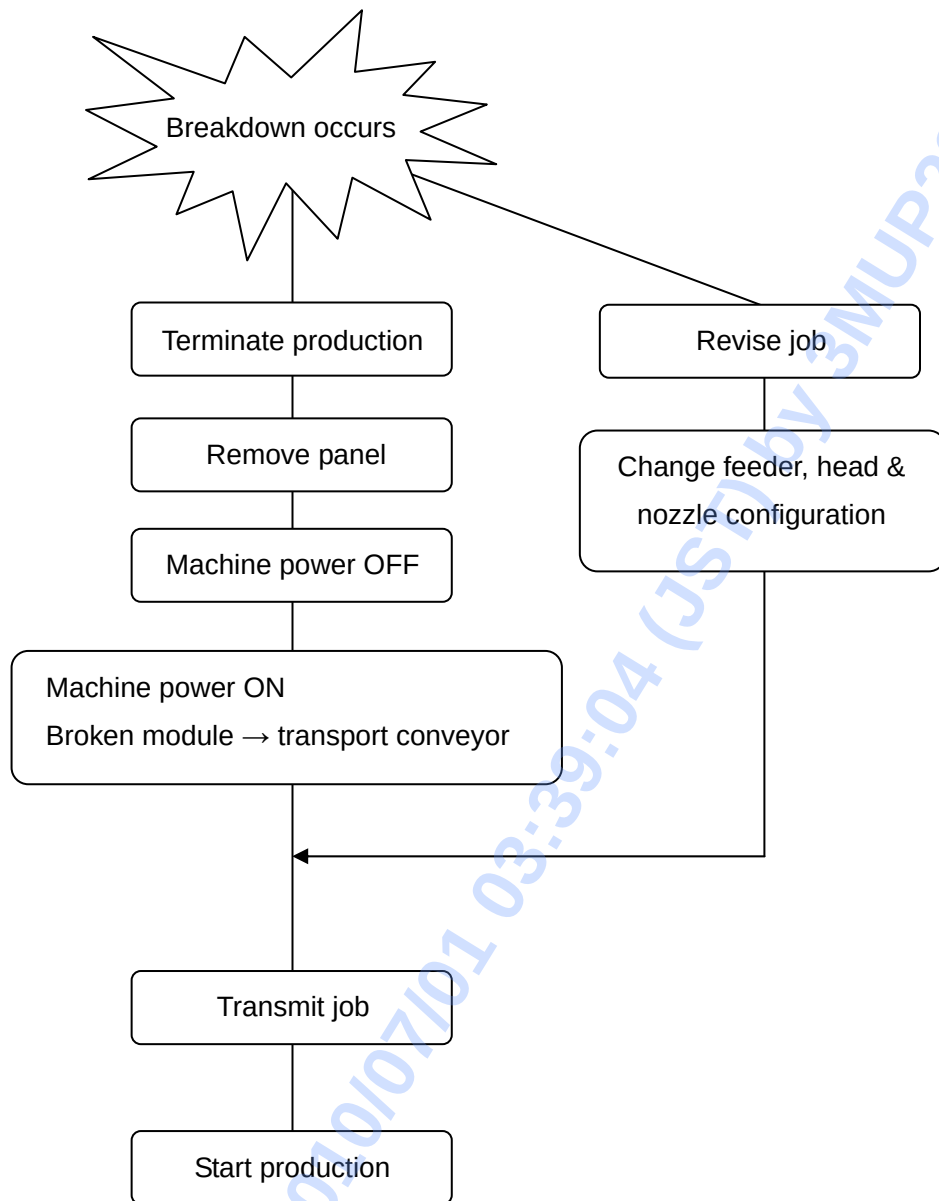
9. Conveyor operation during unit breakdown

If one module is disabled due to a problem such as a servo or camera breakdown, production cannot be performed on the entire line. In such cases, this function can be used to continue production by operating the disabled module as a conveyor only. However, parts assigned to the disabled module must be reassigned to other modules, so this function can only be used when there are a sufficient number of empty feeder slots in the line. Furthermore, this function cannot be used when certain parts, such as the CPU board, have broken-down. Please refer to the table below for details.

9.1 Conveyor Operation Function Availability

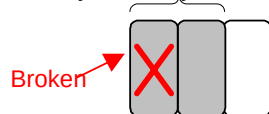
Breakdown where operation is permitted	Breakdown where operation is not permitted
• Servo amplifier • Servo motor • Encoder • XY-robot • Head • Robot cable • Waste tape processing unit • Nozzle changer • Nozzle station • Mark camera • Parts camera	• CPU board • Servo board • Vision processing board • Direct I/O • Remote I/O • Serial communication (RS232C) • IEE1394 • Operation panel • Device unit • Safety door • Conveyor • Module control box • Air pressure

9.2 Procedure flow



Note: in the case of pair module production, both modules must be used as transport conveyors. (Neither module can be used for production.)

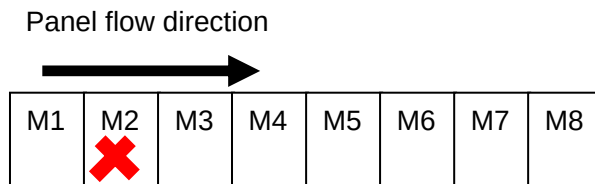
When performing paired production, set either normal operation or panel conveyance only for both module.



9.3 Work Procedure

A. Breakdown occurs

A breakdown occurs at module 2 (M2). Production cannot be continued in the current format.



B. Stopping production

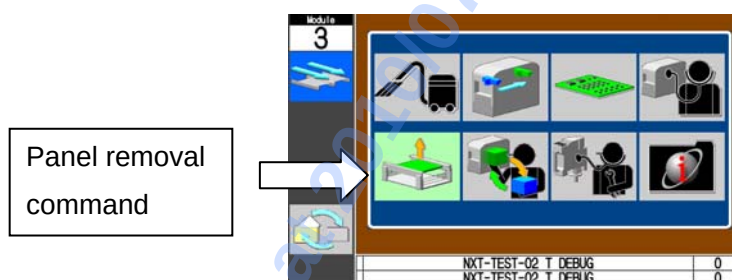
Press the [STOP] switch on the module after the disabled module to finish production and unload the panels from the line. In the example above, the [STOP] switch should be pressed on the M3 module to finish production and unload panels from modules M3 to M8.

C. Removing panels

Remove all panels from the disabled module and any preceding modules. In the example above this would be modules M1 and M2.

Use the following procedure if the manual command is available

1. Select the [Panel Removal] command in the manual screen, and then press the START switch.



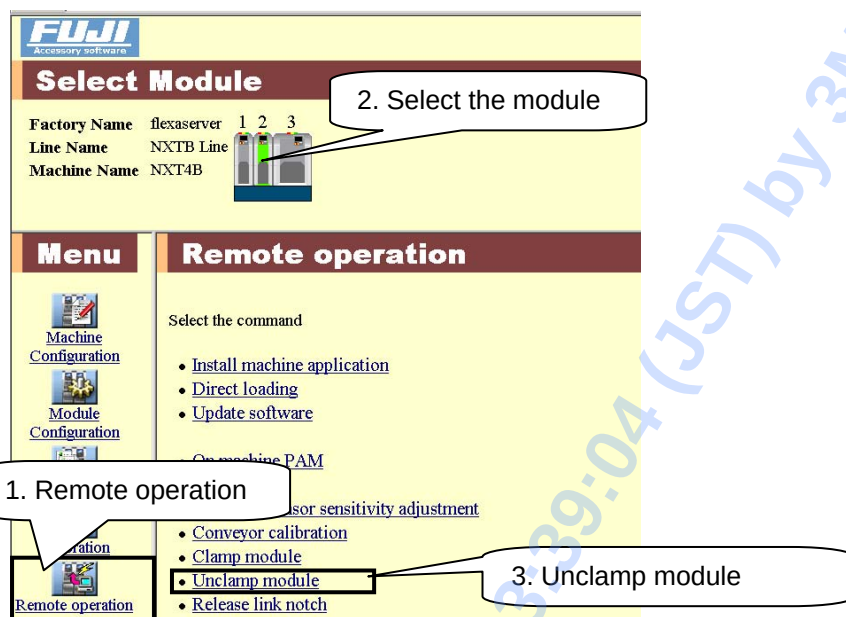
2. Use the manual command to unclamp the module, and then remove the panel.
3. Press the Emergency SW and the unclamp the board by hand.

Caution: production cannot be continued for the removed panels.

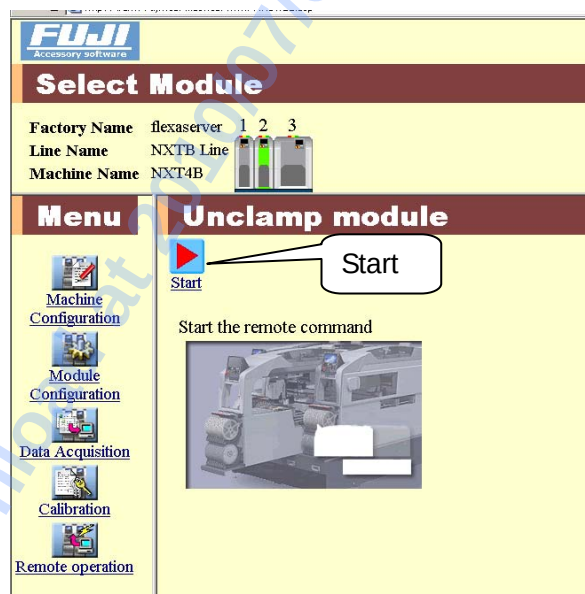
Use the following procedure if the manual command is unavailable.

If the manual command is unavailable due to the breakdown, use the NXT Accessory Software remote command to unclamp the module and remove the panel.

1. Select the broken-down module, and then select [Remote Operation] – [Unclamp Module].



2. Click [Start] to unclamp the module.



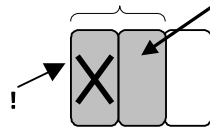
Caution: production cannot be continued for the removed panels.

D. Machine power OFF

In order to find the cause of the module breakdown, pull the module forward and turn the main power OFF. If the head is damaged for example, replace the head

E. Changing the job

Use Fuji Flexa to create a job that is compatible with the disabled module. Allocate the feeders and devices from the disabled module to another module(s). In the case of pair module production, production should not be performed on the pair module either.



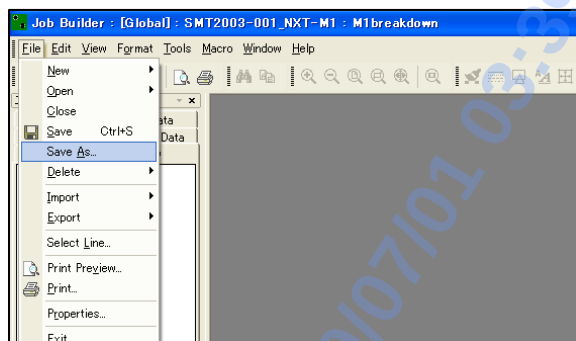
Do not perform production on
this module either.

Breakdown

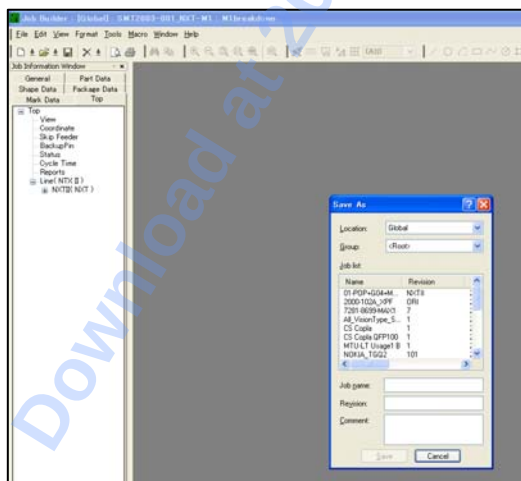
The procedure is explained below.

1. Open the job.
2. Save the job under a different name.

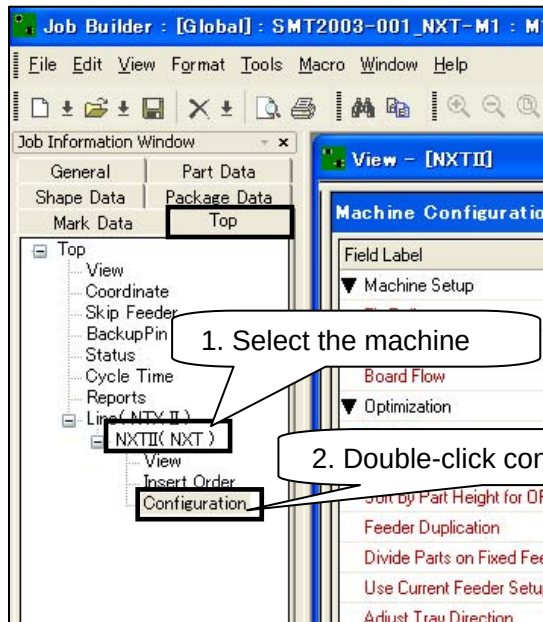
Select [File] – [Save As...]



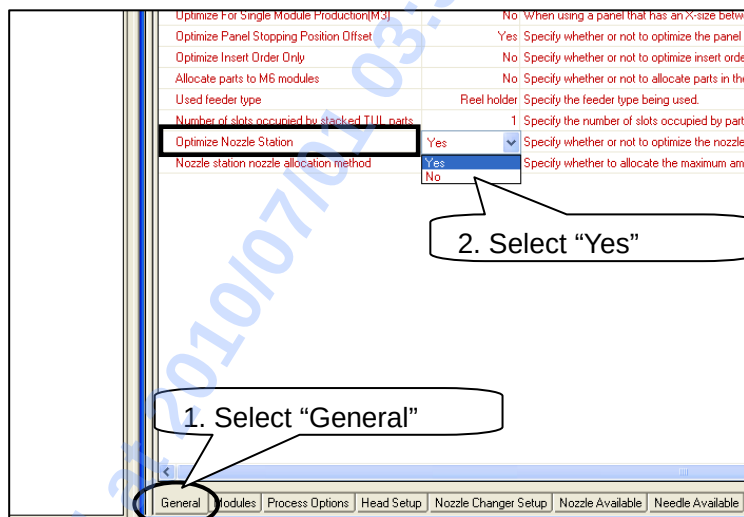
3. Enter the Job name and revision, and then click [Save].



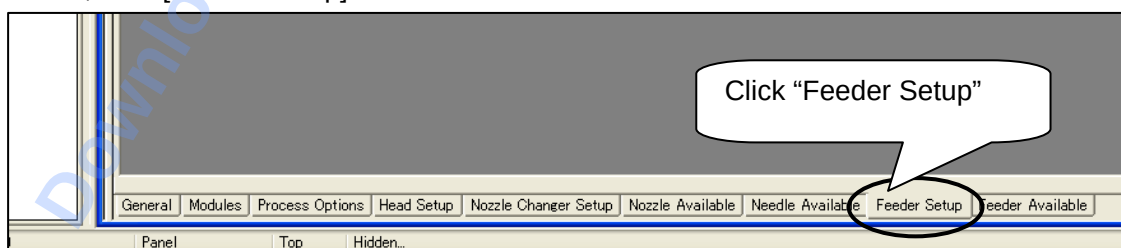
4. Select the disabled machine in the Job Information Window and double-click [Configuration].



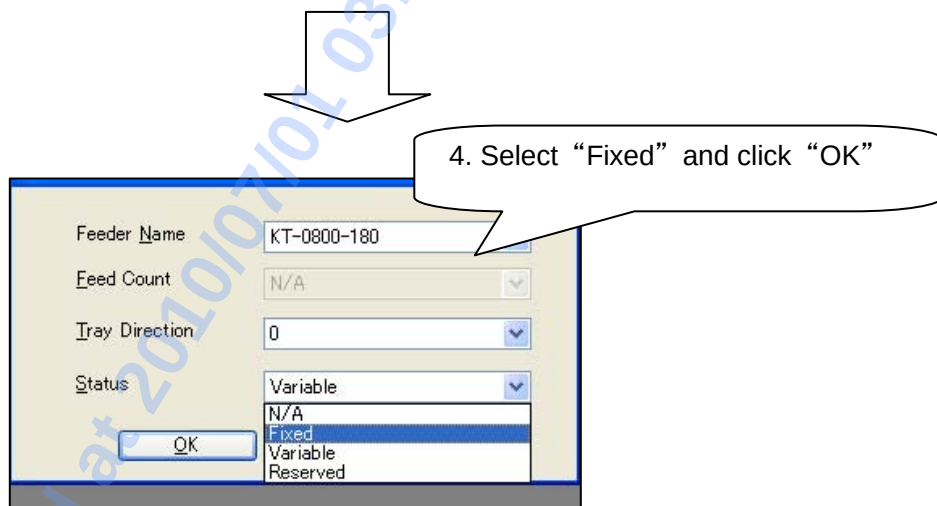
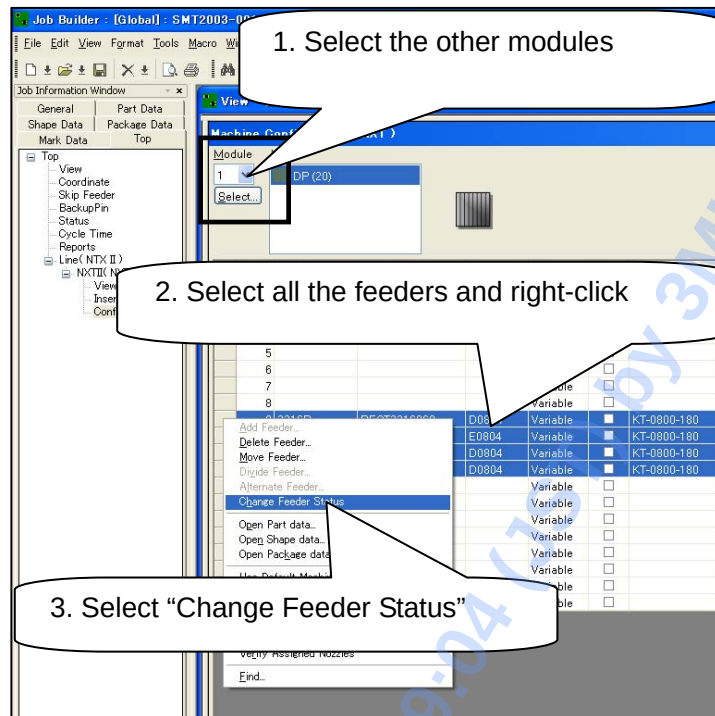
5. Select the [General] tab - [Optimization] – [Optimize Nozzle Station] – [Yes] to activate optimization for the nozzle changer.



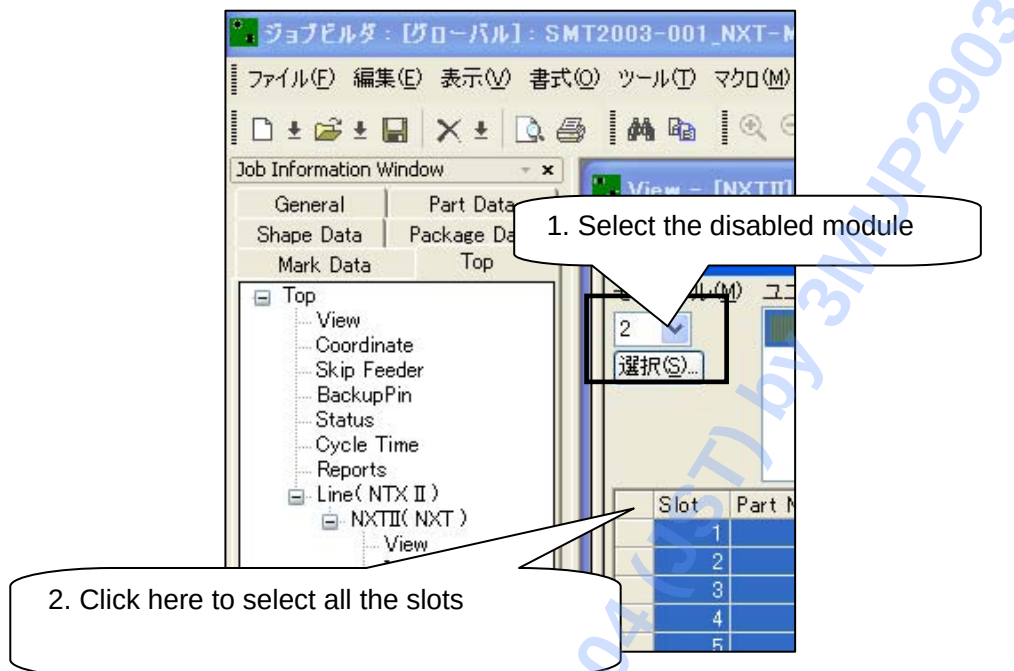
6. Next, click [Feeder Setup].



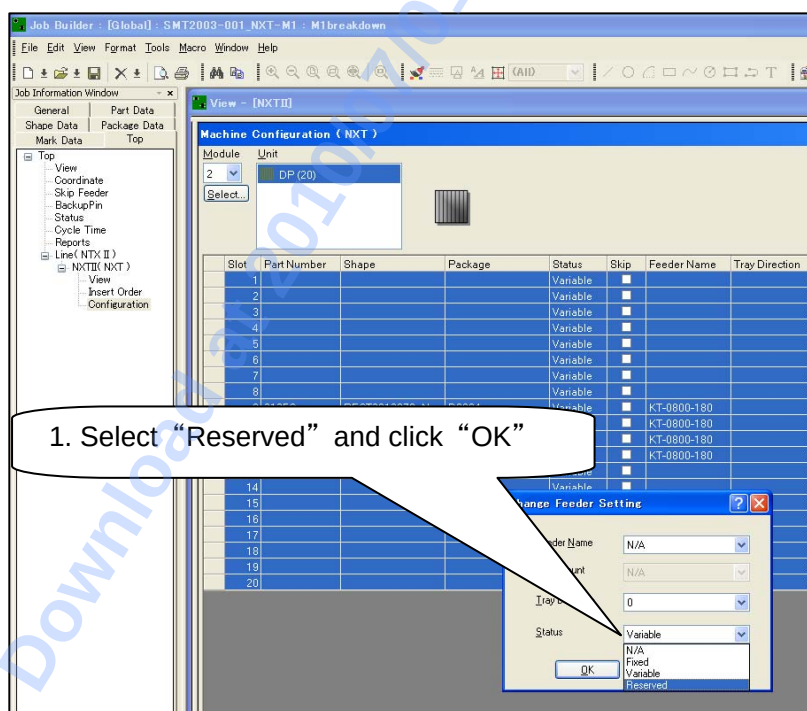
7. Set the feeder status to fixed on all modules except the disabled module. In this example, M2 is disabled, so feeder status should be set to fixed on modules M1 and M3 to M8.



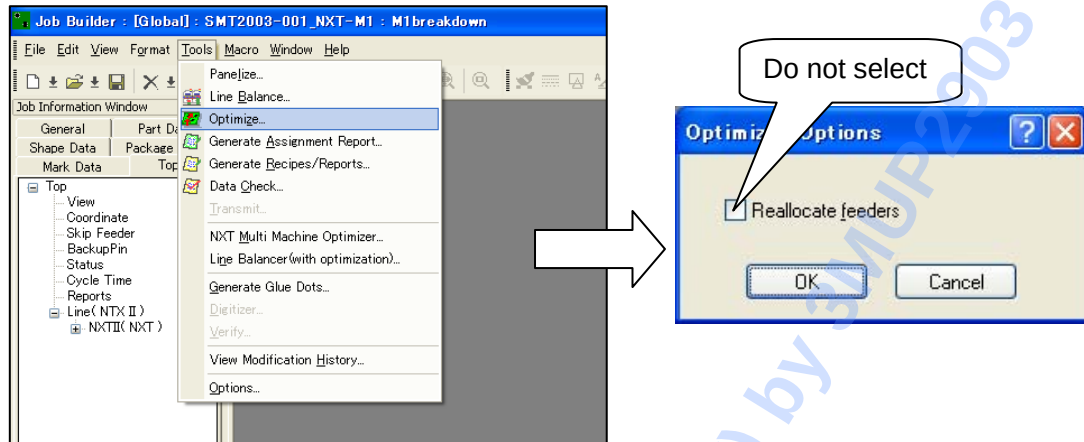
8. Select the disabled module (and the pair module in the case of pair production), and select all the feeder slots.



9. Right-click, select [Change Feeder status], and then change the feeder status to “Reserved”. The feeder information display is cleared when “Reserved” is selected.



10. Select [Tools] – [Optimize...] to open the optimizer. Do not select the “Reallocate feeders” checkbox.

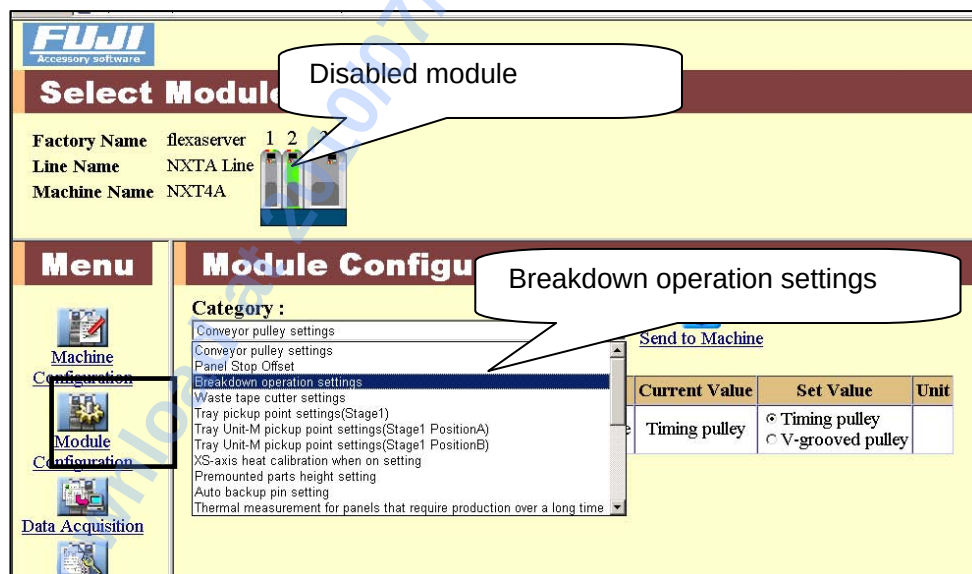


11. Check that the feeders assigned to the disabled module have been reassigned to other modules.

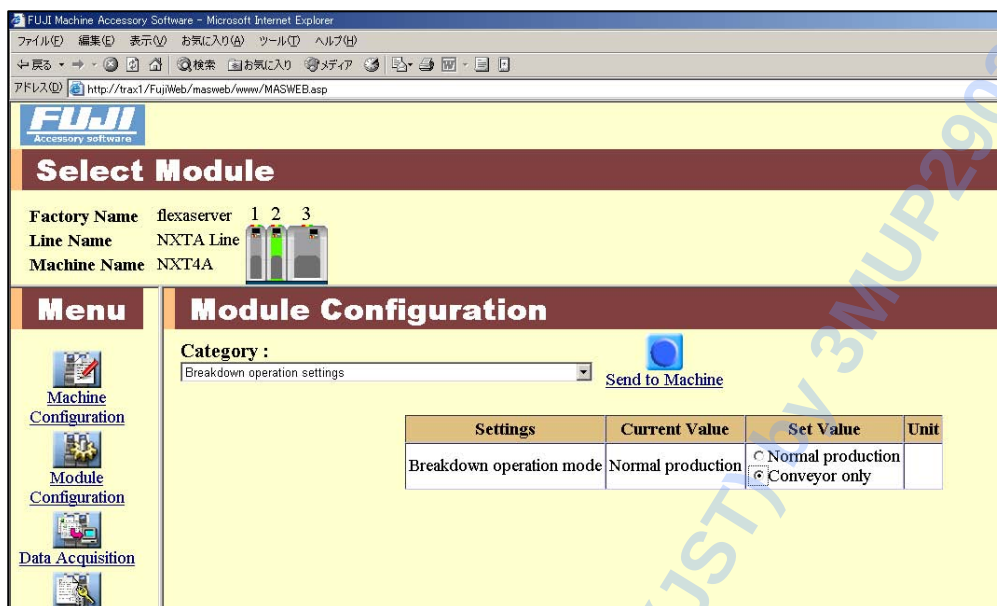
12. Editing the job is now complete.

F. Turning the power ON

1. Turn the NXT main switch ON and wait until the module enters sleep mode.
2. Select the disabled module in NXT Accessory Software, and then select [Module Configuration] – [Breakdown operation settings].



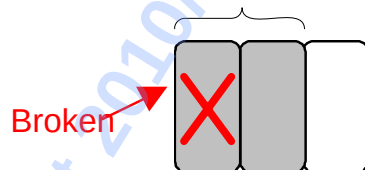
3. Select "Conveyor only" and click "Send to Machine".



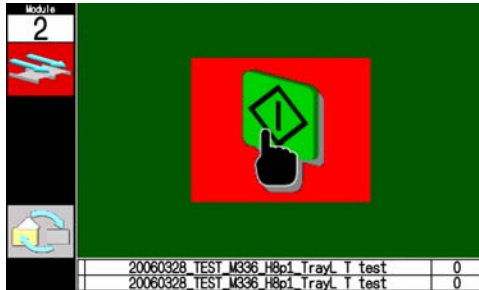
Caution

Note: in the case of pair production, [breakdown operation settings] should be selected for both modules.

When performing paired production, set [Conveyor only] for both modules.



- After making the settings, press [Power On] on the operation panel to wake the module.
The settings are complete once the following screen displays.



G. Transmitting the job

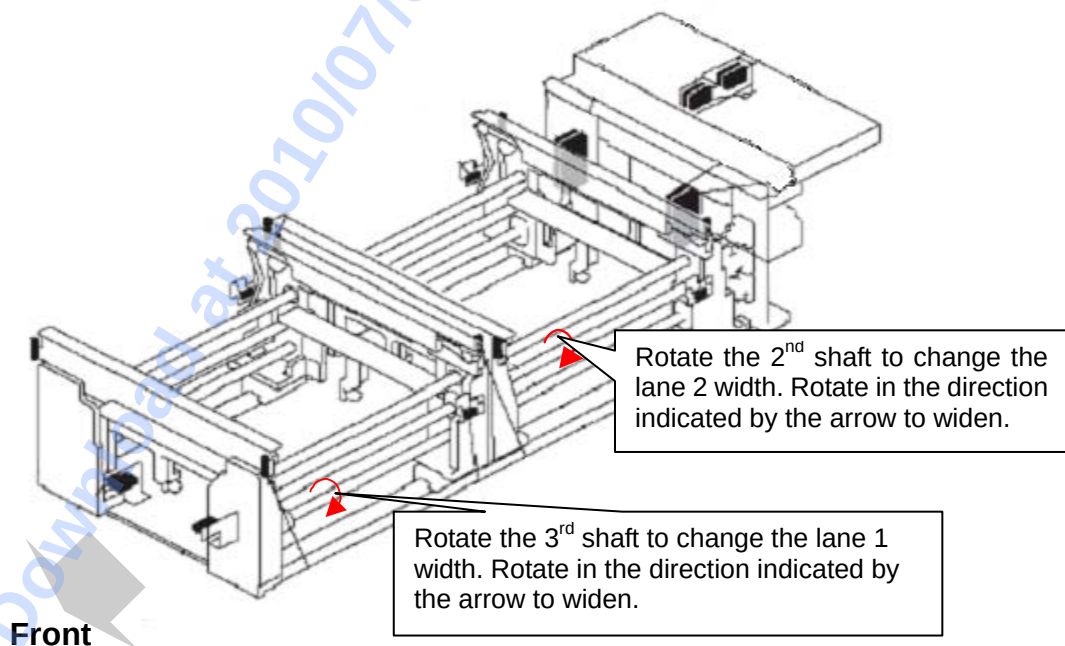
Transmit the job edited in step “E”.

*Note: when transmitting the job a panel removal message may display regardless of whether there is a panel in the module or not. This is because the machine cannot check for panels immediately after manual command operation. Follow the instructions and then **retransmit the job** to exclude the disabled module from production.*

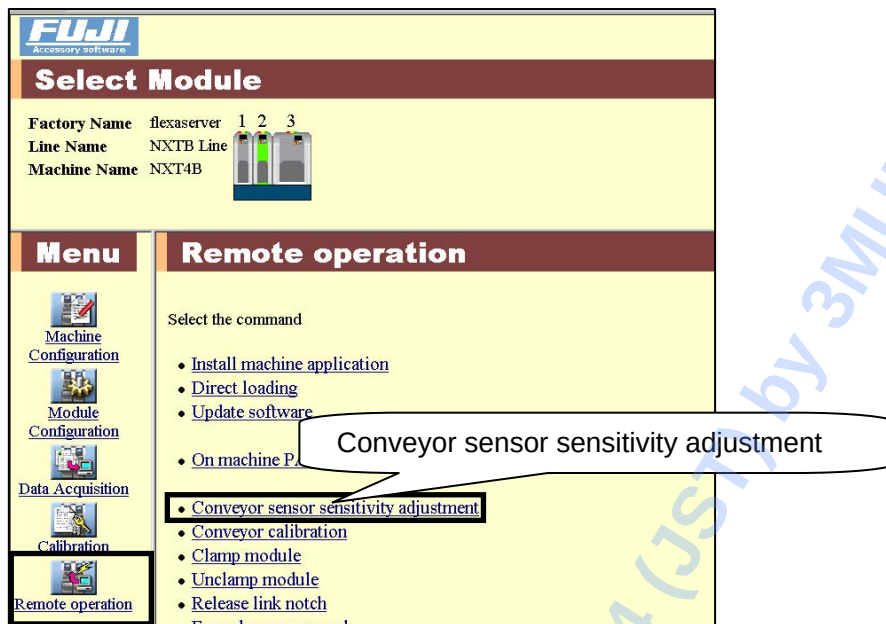
H. Starting production

Automatic conveyor width change is disabled, so the conveyor width change must be performed manually when required. The manual conveyor width change procedure is explained below.

- Pull out the module and then press the Emergency Stop switch.
- Follow the procedure shown in the diagram below to change the conveyor width.



3. Select the module in Accessory Software, and then select [Remote Operation] – [Conveyor sensor sensitivity adjustment] – [Send] to adjust the conveyor sensitivity.



4. Perform test production to check that all the parts are properly mounted, and then start production.

Chapter 10

Acquiring Trace Data for Different Types of Error

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10. Acquiring Trace Data for Different Types of Error

Trace data is acquired in order to investigate errors and other problems on the machine. However, as there are various methods of acquiring trace data, we would like to use this chapter to explain how to choose the right method for the particular problem you are experiencing.

10.1 Overview of the various trace data acquisition methods

As shown below, there are 4 methods of acquiring trace data. Please refer to “2.3 Trace data acquisition procedures” for the actual trace data acquisition procedures.

1. Acquiring trace data from Accessory Software

This method should be used when an error screen displays after detection of an error, or when the module has been stopped with the Cycle Stop button (or Emergency Stop Switch) following unusual operation.

2. Acquiring trace data that was saved automatically following an OS or Application error

Use this method to acquire trace data when an OS error or application error (Penguin Symbol) has occurred.



3. Forced acquisition of trace data with the main switch ON

Use this method when operation is impossible due to the screen hanging up, and a communication error prevents you using Accessory Software to acquire the trace data.

4. Temporarily saving trace data on the machine

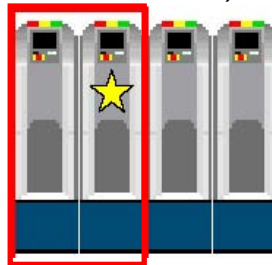
As the machine has a limited amount of memory, trace data must be acquired soon after a problem occurs. If trace data is not saved in time, information required to analyze the problem may be lost. This function makes it easy to save trace data whenever a problem occurs.

10.2 Selecting the target module(s) for trace data acquisition

The target module(s) for trace data acquisition vary according to the type of problem and operating environment. Follow the example below to select the target module(s) for acquiring trace data.

1. The error occurs during pair module production

In this case, trace data should be acquired from the target module (☆) and its pair module (the two modules within the red frame).



2. The error occurs during panel conveyance (Error code: **** 07**)

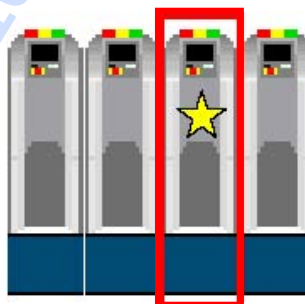
3. The error code indicates a communication error between adjoining modules (Error codes such as 8CB30000 which instruct you to save trace data from the adjoining modules.)

Acquire trace data from the target module (☆) and the two adjoining modules (the three modules within the red frame).



4. For other problems

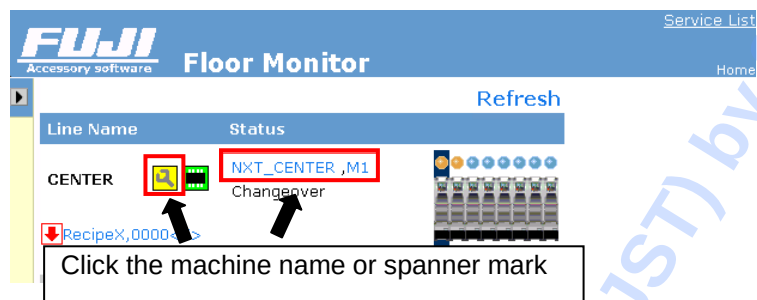
Acquire trace data from the target module (☆) only.



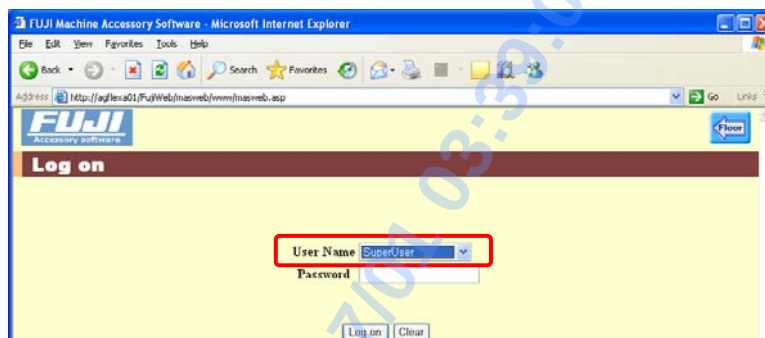
10.3 Trace data acquisition procedures

10.3.1 Acquiring trace data with Accessory Software

1. Press CYCLE STOP or EMERGENCY STOP on the target module (and adjoining modules if necessary). Leave the module(s) as they are. Do not perform any operation until the trace data has been acquired.
2. Start Accessory Software, and then click the machine name or spanner mark in Floor Monitor.



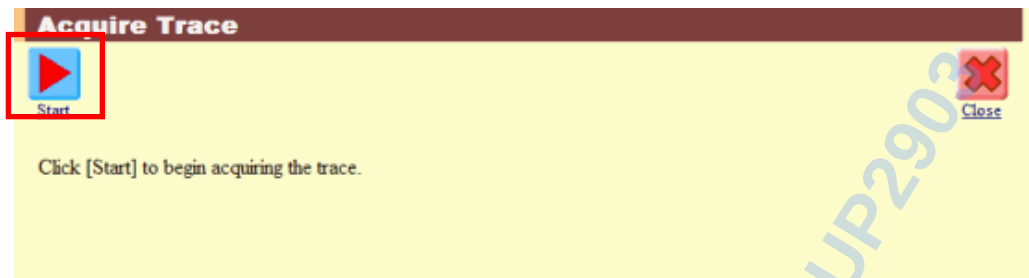
3. Enter "superuser" as the user name in the login screen.



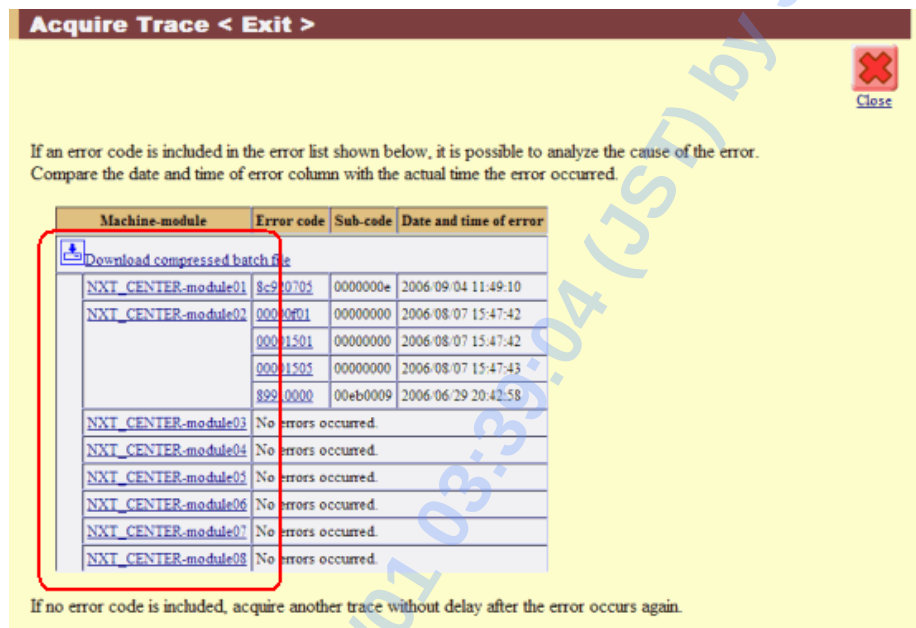
4. Select the target module, and then click [Data Acquisition] – [Acquire Trace].



5. Click [Start].



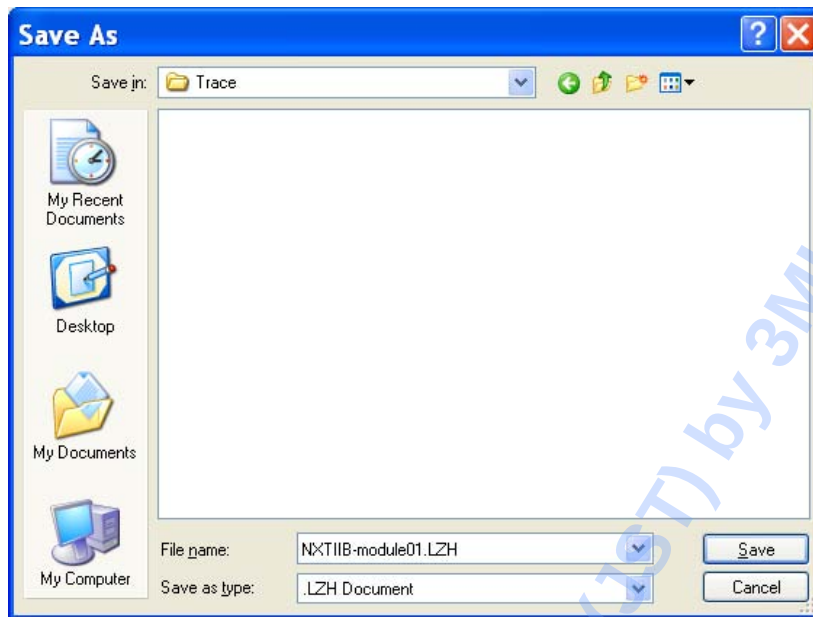
6. Click the trace data you wish to download.



7. Follow the instructions in the File Download dialog to download the trace.

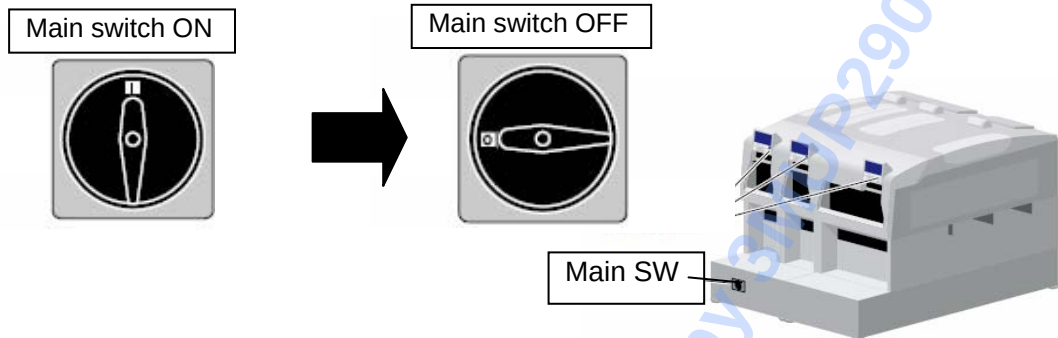


8. Select the save destination and save the file.

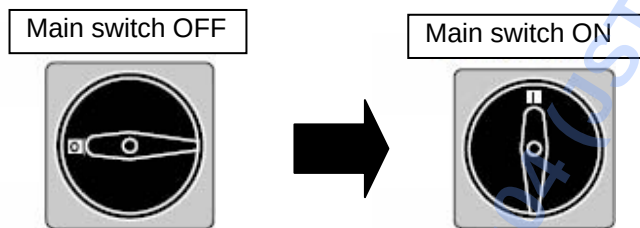


10.3.2 Acquiring trace data that was automatically saved following an OS error

1. Turn the main switch OFF on the base on which the OS error occurred.



2. Wait 30 seconds and turn the base main switch ON.



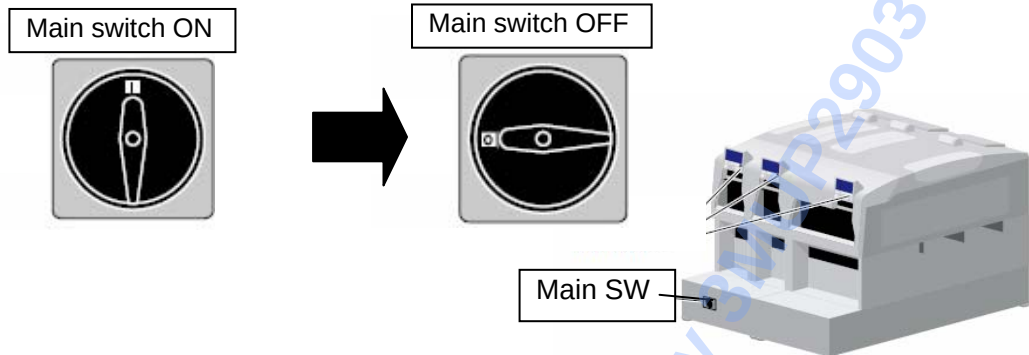
3. Boot up the machine and wait for the modules to enter sleep mode. (The machine takes longer to boot up following an OS or Application error).



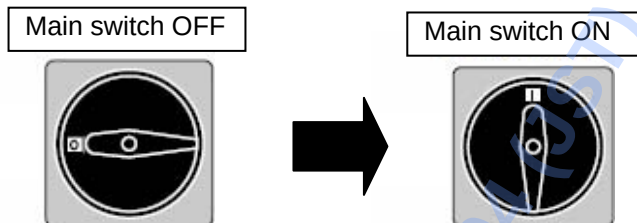
4. Once the module has entered sleep mode, acquire the trace data using the procedure described in "10.3.1 Acquiring trace data with Accessory Software".
5. Send the trace data to your Fuji representative. Also, include as much information as possible about the conditions at the time the error occurred.

10.3.3 Forced acquisition of trace data with the breaker ON

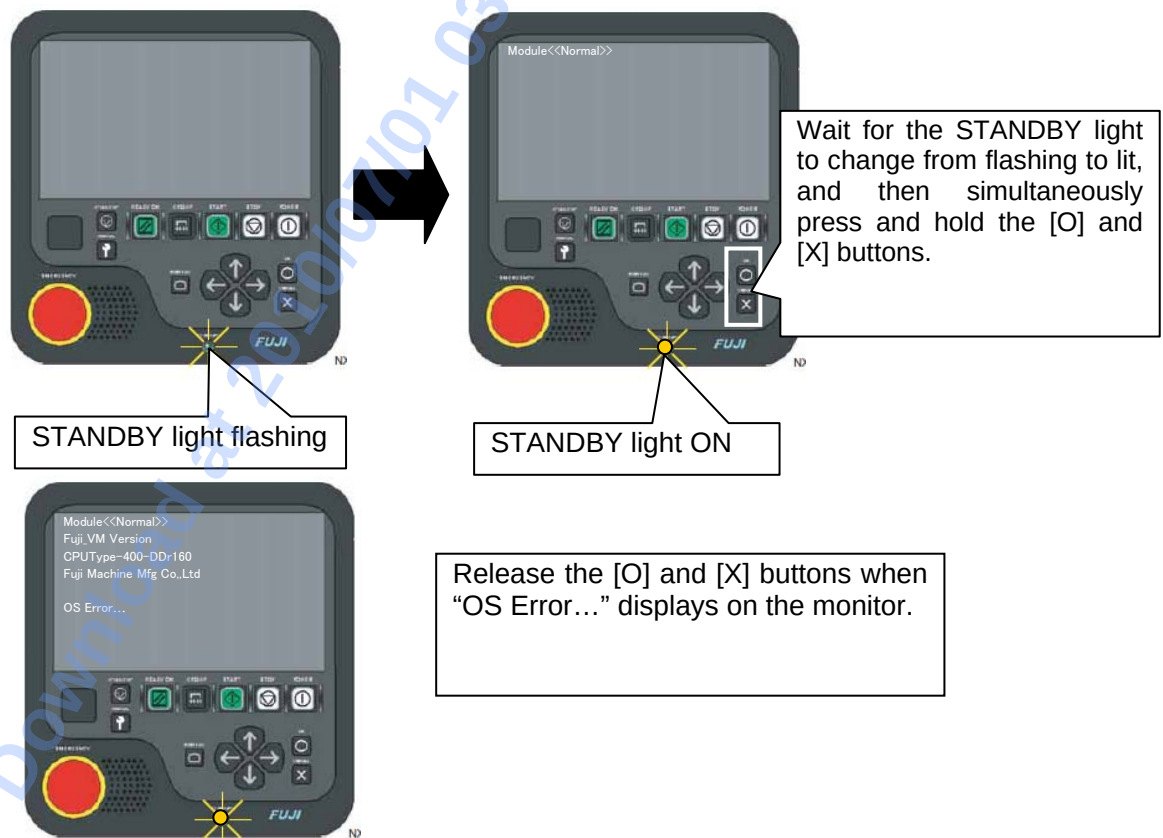
1. Turn the main switch OFF on the machine that has hung up.



2. Wait 30 seconds and then turn the base main switch ON.



3. As the module boots up, wait for the STANDBY light on the operation panel to change from flashing to lit, and then press and hold the [O] and [X] buttons simultaneously when the text starts appearing on the screen. Release the [O] and [X] buttons when "OS Error..." displays on the screen.



4. Wait for the module to enter sleep mode. (It takes longer than normal for the module to enter sleep mode).



5. Once the module has entered sleep mode, acquire the trace data using the procedure described in "10.3.1 Acquiring trace data with Accessory Software".

10.3.4 Temporarily saving trace data on the machine

1. Set the module to one of the statuses shown below, and then press the STOP button while holding down the CYCLE STOP button.
 - Waiting for the START button
 - Waiting for changeover
 - Waiting for cancellation of error
 - Waiting for manual mode command



2. The busy (hour glass) icon displays on the monitor.



3. Once the busy (hour glass) icon disappears, acquire the trace data using the procedure described in "10.3.1 Acquiring trace data with Accessory Software".

10.4 Trace data acquisition methods for various problems

Choose the appropriate trace data acquisition method for the problem you are experiencing. Send the trace data to your Fuji representative and include as much information about the error as possible.

Using the appropriate method to acquire trace data immediately after a problem occurs will help us to solve the problem as soon as possible. Thank you for your cooperation.

10.4.1 When production stops with an error, or when the Emergency Stop button or Cycle Stop button was pressed due to abnormal operation.

1. Refer to “10.2 Selecting the target module(s) for trace data acquisition” to select the target module(s).
2. If you need to restart production quickly, perform the procedure in “10.3.4 Temporarily saving trace data on the machine”. If you can wait some time before starting production again, perform the procedure in “10.3.1 Acquiring trace data with Accessory software”.

10.4.2 When an OS or Application error (Penguin Symbol) occurs

Follow the procedure in “10.3.2 Acquiring trace data that was automatically saved following an OS error”.



10.4.3 When the machine has hung up, and a communication error prevents you using Accessory Software to acquire the trace

Follow the procedure in “10.3.3 Forced acquisition of trace data with the breaker ON”.

Chapter 11

Countermeasures for Various Error Types

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11. Countermeasures for Various Error Types

11.1. The standby lamp does not switch from flashing to lit during startup

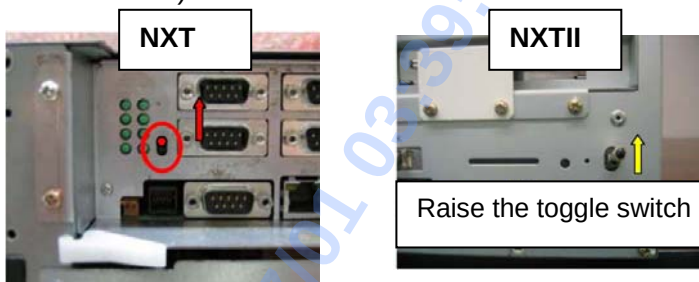


Details

The standby lamp does not switch from flashing to lit during startup and the operation panel is blank.

Countermeasure

Check the status of the standby lamp when the CPU BOX on the target module is booted up in safe mode. (Raise the toggle switch on the module CPU box and reboot the module).



The standby lamp keeps flashing

There may be a hardware problem. In this case, check the following.

Possible causes	Countermeasures
Communication error between the module CPU Box and operation panel "PANEL LINK"	● Check the connector connection at the rear of the operation panel ● Check the "PANEL LINK" connection on the module CPU box ● Replace the operation panel ● Replace the "PANEL LINK" harness connecting the operation panel to the CPU BOX
Module CPU BOX hardware defect	● For NXT: replace the module CPU BOX ● For NXTII: replace the module CPU board

上

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

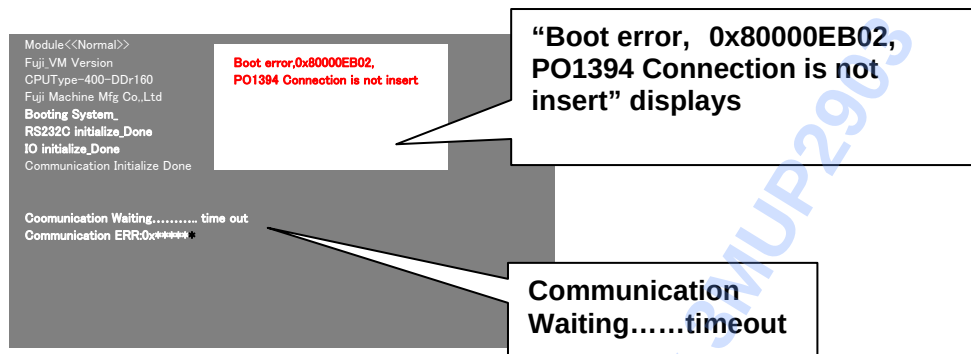
The standby lamp changes from flashing to lit

The problem may be caused by the module CPU box application software not operating properly. In this case, check the following.

Possible causes	Countermeasures
Module CPU BOX defect	● Perform Emergency Install for the module CPU box
Module CPU BOX internal problem	● NXT: replace the module CPU BOX ● NXTII: replace the module CPU board

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

11.2 “Communication Waiting.....timeout” error during startup → “PO 1394 Connection is not insert” displays



Details

Communication error between the base and module.

Countermeasure

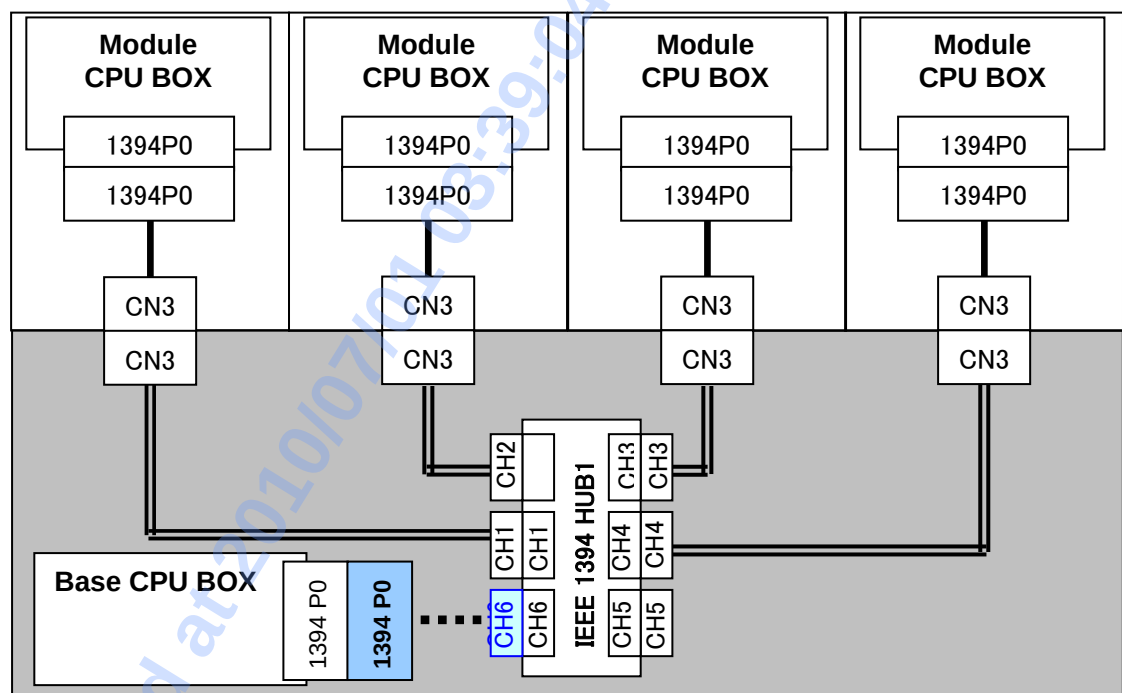


Fig.1 Block Diagram

A. If the error occurs on all modules on the base

There may be a communication problem between the base CPU BOX and 1394 HUB.
(see thepath in the Fig.1 block diagram).

Caution: be sure to turn the main switch OFF before performing the procedures below.

Possible causes	Countermeasures
Poor contact or disconnection between the base CPU BOX and IEEE1394HUB. (see the path in the Fig.1 block diagram)	● Check the base CPU BOX "1394PO" connector connections ● Check the IEEE1394HUB CH6 connector connections ● Replace the harness
IEEE1394HUB failure	Replace the IEEE1934HUB
Base CPU BOX (CPU board failure)	NXT: replace the base CPU BOX NXTII: replace the base CPU board

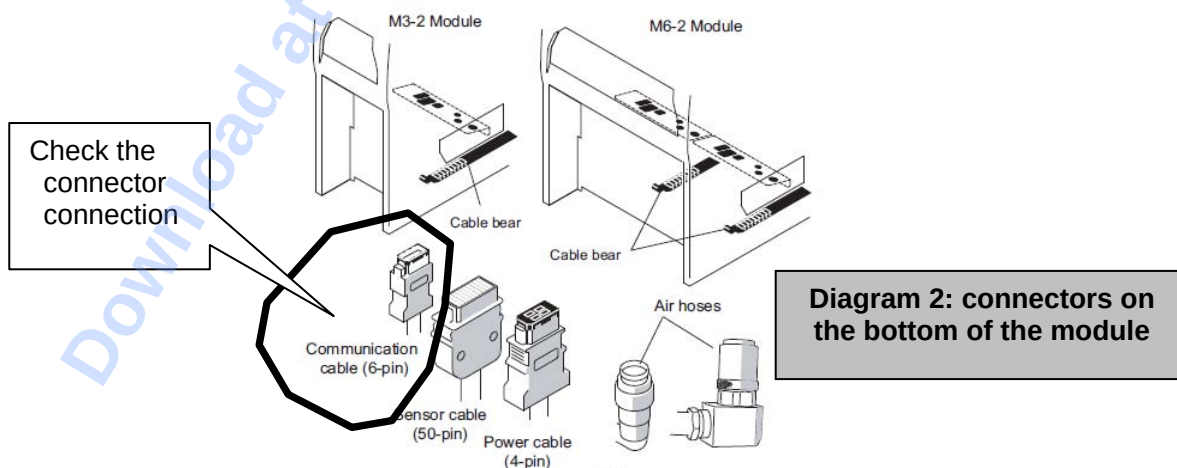
Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

B. The problem is limited to a specific module

- There may be a communication problem between the base IEEE1394HUB and the connector on the bottom of the module.
(see the ===== path in the Fig.1 block diagram)

Caution: be sure to perform the procedure below with the main switch OFF.

Possible causes	Countermeasures
Poor contact or disconnection between the base IEEE1394HUB and connector on the bottom of the module. (see ===== path in Fig.1 block diagram)	● Check the connection of the IEEE1394HUB CH6 connector ● Check the connection of the CN3 cable connector on the bottom of the module (see Fig.1/2 diagrams). ● Replace the harness inside the base cable bear
Failure of the IEEE1394HUB	Replace the IEEE1934HUB



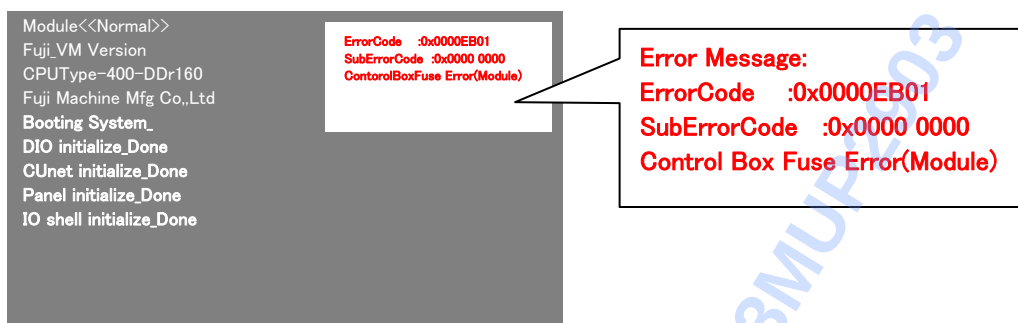
- There may be a communication problem between the module CPU box and the connectors on the bottom of the module.
(See the  path in the Fig.1 block diagram).

Possible causes	Countermeasures
Poor contact or disconnection between the module CPU box and the connectors on the bottom of the module. (See the  path in the Fig.1 block diagram).	● Check the module CPU BOX "1394PO" connector. ● Replace the harness.
Module CPU BOX (CPU box) failure	NXT: replace the module CPU BOX. NXTII: replace the module CPU board.

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

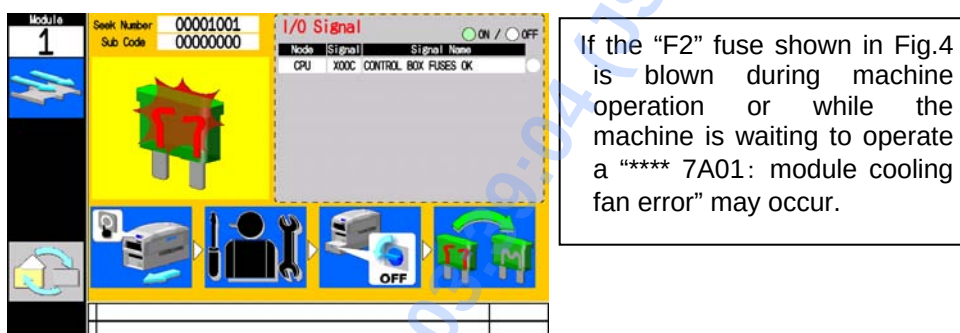
11.3 Module control box fuse error (Error code: **** 1001)

The following error occurs when a blown fuse is detected during module startup.



The following error occurs when a blown fuse is detected during machine operation or while the machine is waiting to operate.

Error code:**** 1001



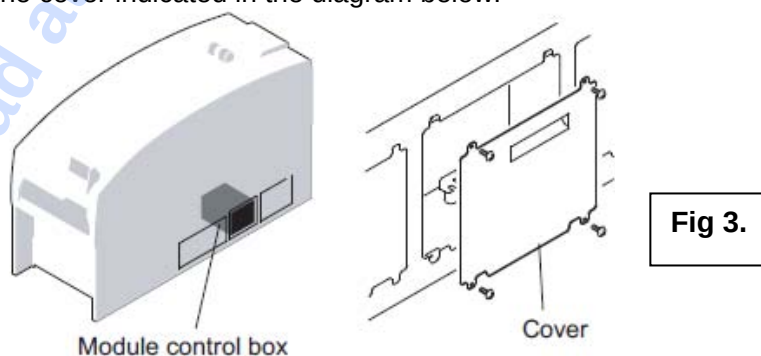
Details

Blown fuse in the module control box.

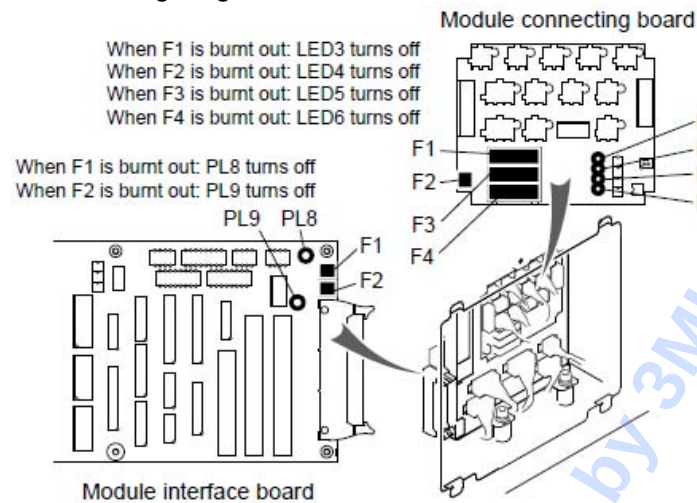
Countermeasure

Replace the fuse in the module control box.

1. Pull out the module to the right of the module on which the error occurred. If the error occurred on the module furthest to the right, pull out that module and remove the side cover.
2. Remove the cover indicated in the diagram below.



3. Attach an anti-static wrist strap to your wrist.
4. Refer to the following diagram and find the burnt out fuse from the LED status.

**Fig 4.**

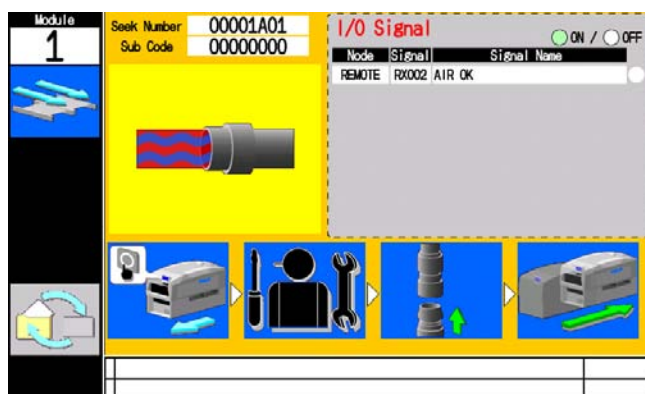
5. Turn the main switch OFF.

Caution: be sure to turn the power OFF at the main switch before replacing the fuse

6. Replace the burnt out fuse in the module control box.

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

11.4 Module air pressure error



Details

The module air pressure is too low.

Countermeasure

A. When the error occurs on all modules on the base

There may be a problem with the air supply to the base. Check the following items and take action as required.

- Check that there are no problems with the air supply from the factory compressor.
- Check that the air hose is connected to the air inlet valve inside the base.

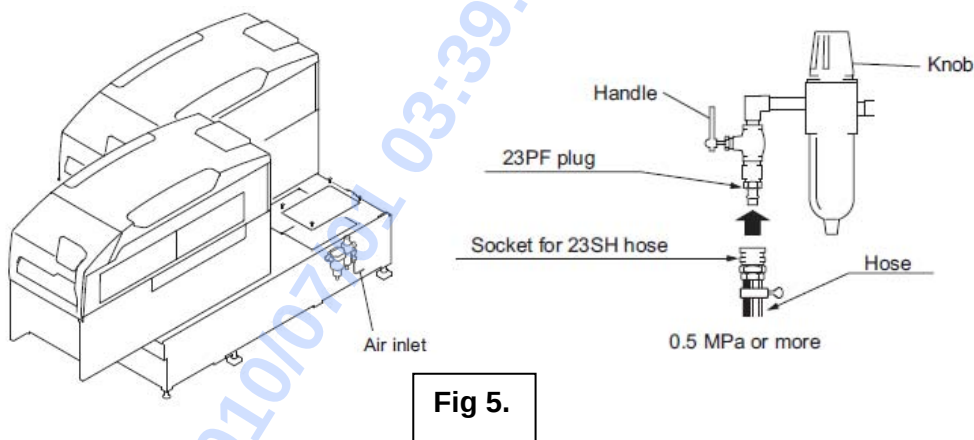


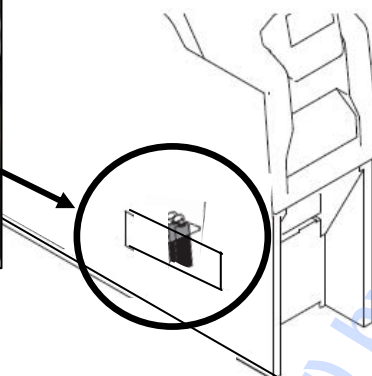
Fig 5.

- Check that there are no problems with the connection between the factory air supply outlet and the NXT base air inlet.
- Clean the filter regulator. This should be cleaned yearly (every 8000 hours). (Refer to the procedure described in "Cleaning the air regulator" in the NXT II Mechanical Reference.)

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

B. If the error is limited to a particular module

The module pressure sensor is located on the left side of the module. (In the case of the first module, the side-cover needs to be removed to access the sensor). Set the MODE switch to "ME", and then check the pressure displayed on the sensor indicated by the arrow.

**Fig 6.****If the air pressure is less than 0.4 MPa**

Pull out the module, turn the knob to open the cover on the bottom of the module, and then check the air hose connection inside (refer to the diagram below).

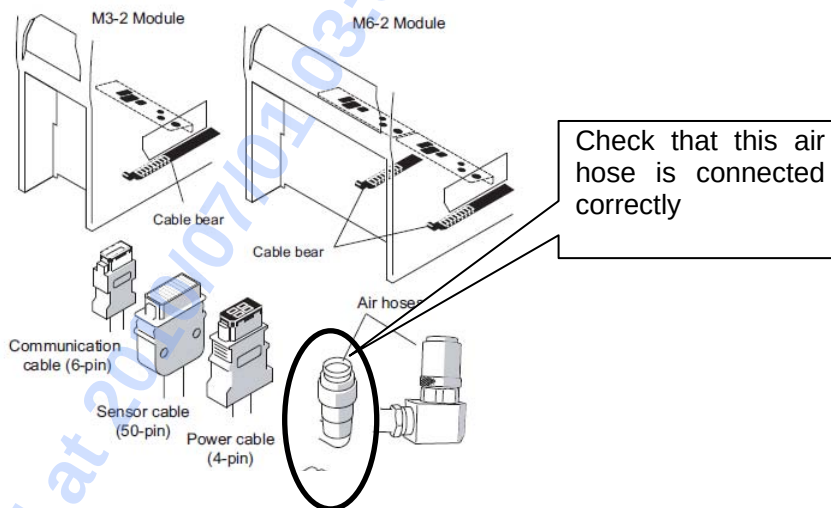
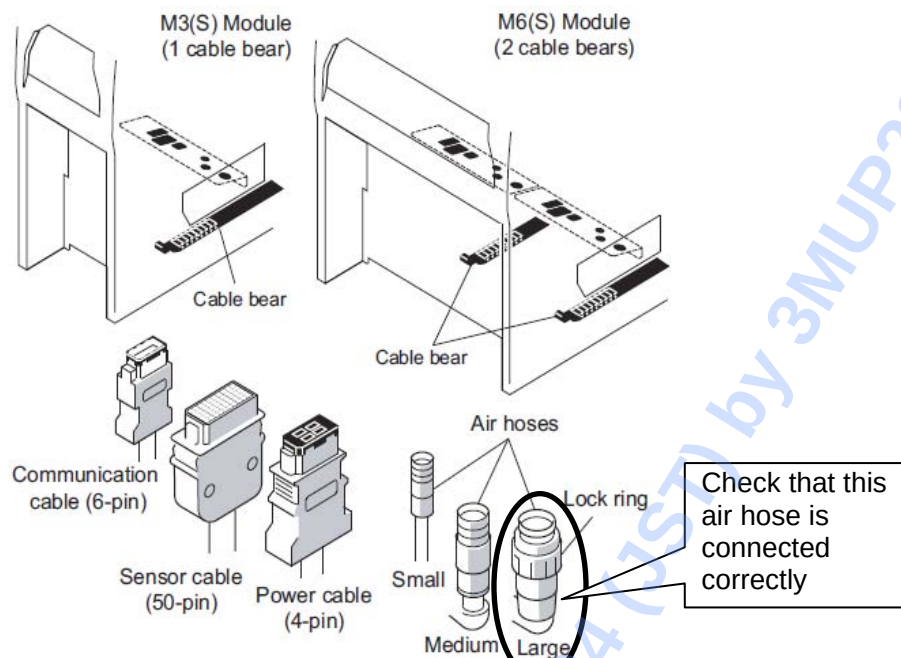
Fig 7. For NXTII

Fig8. For NXT

If the air hose is connected correctly, check that there is no air leakage inside the module.

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in appendix 2).

If the air pressure is above 0.4 MPa, but there is still an error

Check the sensitivity of the module pressure sensor. (Refer to the module pressure sensor adjustment procedure in the NXT II Mechanical Reference).

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in appendix 2).

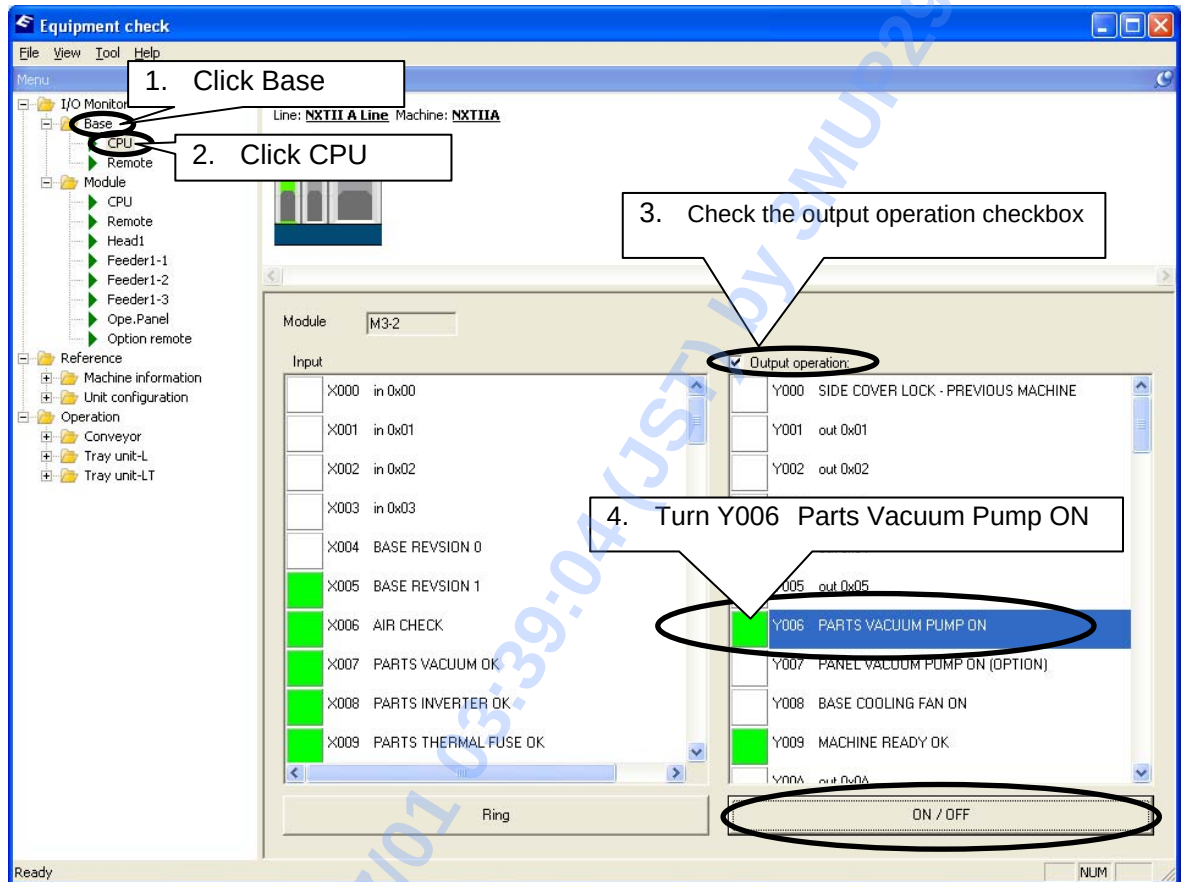
11.5 Vacuum pressure error(Error code: **** 6501)

Details

Vacuum pressure is too low

Countermeasure

1. Use the IO command to turn the vacuum pump ON.



2. Check the vacuum pressure displayed on the front of the base.
(The vacuum sensor is the right-hand sensor).



If the vacuum pressure does not reach -59.0 kPa (-58.9~0)

1. Check that there is no air leakage inside the base.
2. If there is no leakage, clean the vacuum pump filter.
(Refer to "Cleaning the vacuum pump filter" in the NXT II Mechanical Reference for the procedure).
3. If the running time on the base exceeds 10000 hours, replace the vacuum pump's inlet valves, outlet valves and gaskets. (The replacement procedure for the different types of base is detailed in the "Replacing Consumable Parts" section of the NXT II Mechanical Reference. Follow the procedure indicated for your type of base).

8M, 4M, 2M Base (NXT)

NXT Mechanical Reference – Replacing Consumable Parts - Replacing the Vacuum Pump's Inlet Valves/Outlet Valves & Gaskets.

2M-2 Base (NXTII)

NXTII Mechanical Reference – Replacing Consumable Parts – Replacing the Vacuum Pump's Inlet Valves/Outlet Valves & Gaskets.

4M-2 Base (NXTII)

NXTII Mechanical Reference – Replacing Consumable Parts – Replacing the Vacuum Pump's Inlet Valves/Outlet Valves & Gaskets (for the 4M-2 pump).

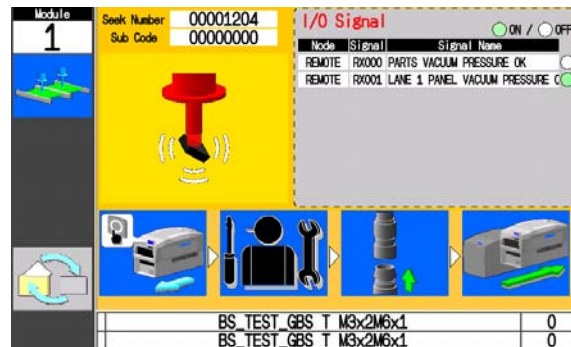
Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in appendix 2).

If the air pressure reaches -59.0kPa (-99.9~-59.0) but there is still an error

The base vacuum sensor sensitivity adjustment may be wrong so please check it. (Refer to the base vacuum sensor sensitivity adjustment in the NXT (II) Mechanical reference for the procedure).

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in appendix 2).

11.6 Module vacuum error (Error code: **** 1203, ****1204)

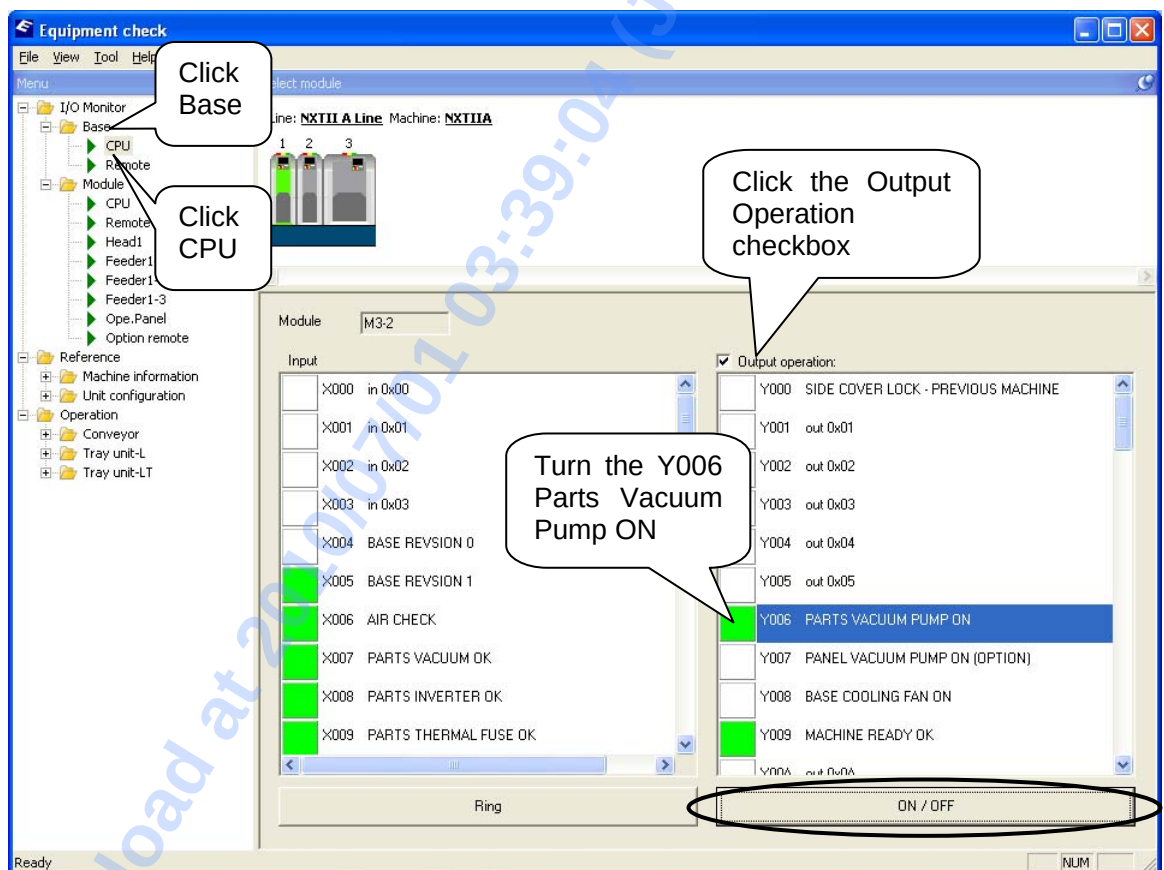


Details

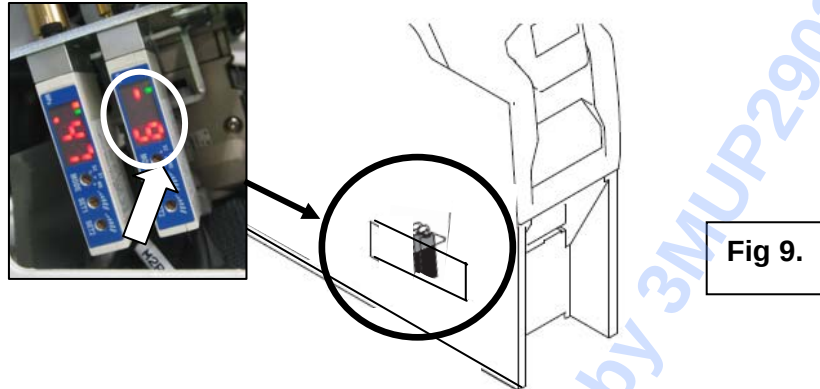
The module vacuum pressure has decreased.

Countermeasure

1. Use the IO command to turn the parts vacuum pump ON.



2. Pull out the module, and locate the module vacuum pressure sensor on the left side of the module. (If the module is the first one, it is necessary to remove the side cover). Set the MODE switch to "ME" and check the reading on the vacuum sensor indicated by the arrow in the diagram below.



If the vacuum pressure does not reach -59.0 kPa (-58.9~0)

1. Check that there are no air leaks within the module.
2. Pull out the module, turn the knob to open the cover on the bottom of the module, and then check the air hose connection inside (refer to the diagram below).

Fig.10 For NXT II

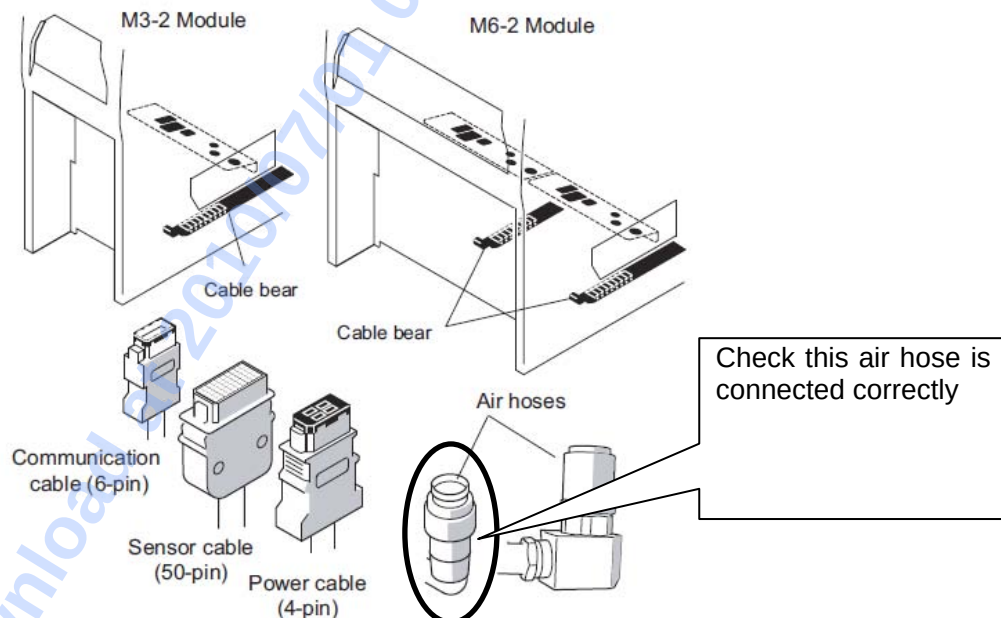
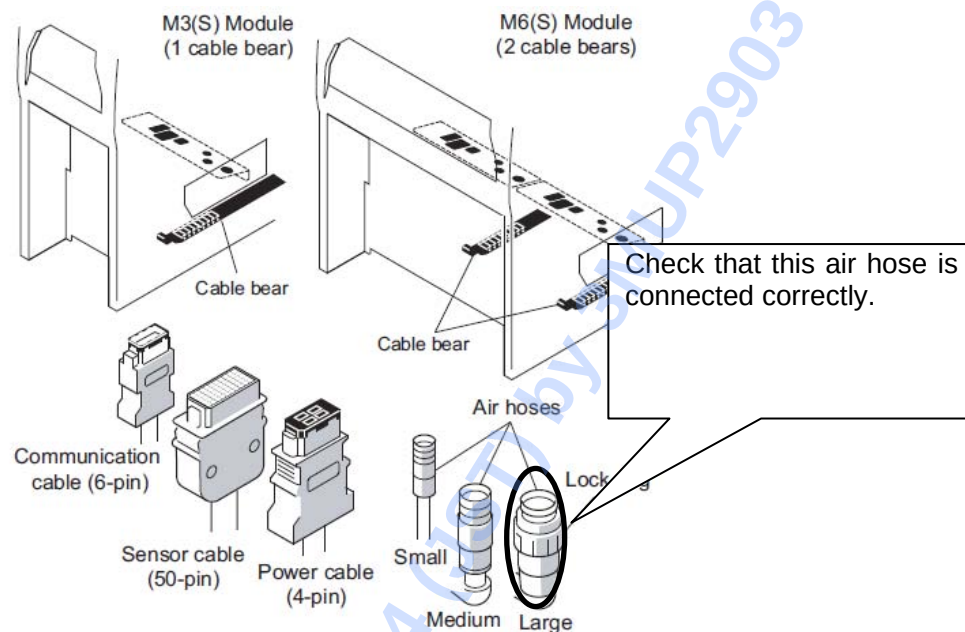
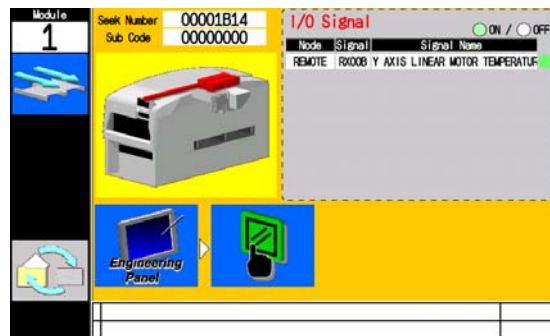


Fig11. For NXT

3. Replace the head to see if it resolves the problem.

Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in appendix 2).

11.7 Y-axis linear motor temperature error (Error code: ****1B14)



Details

A Y-axis linear motor temperature error has been detected.

Countermeasure

Caution:

Be careful not to touch the linear motor shaft or coil directly after module operation. You may be burned by the high temperature. Be sure to wait until the shaft has cooled down before attaching the protective cover and performing the procedure.

Caution:

The module linear motor exerts a strong magnetic force. People with heart pacemakers should keep at least 400 mm away from the Y-axis linear motor. Failure to do so could result in the pacemaker malfunctioning.

Check that the linear motor cooling fan is operational.

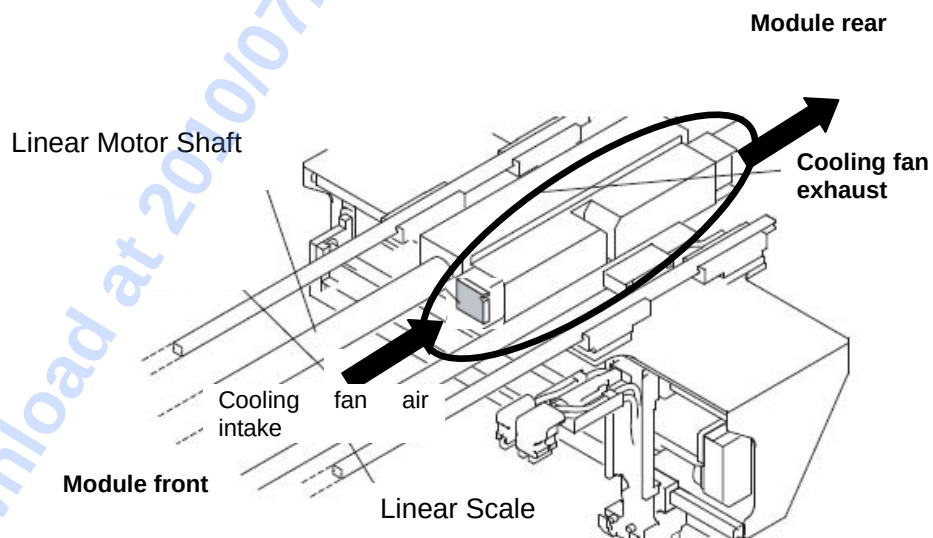
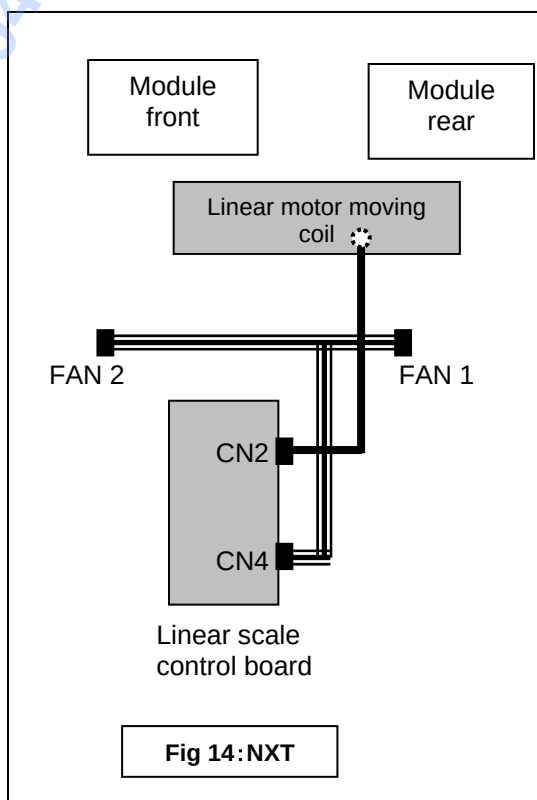
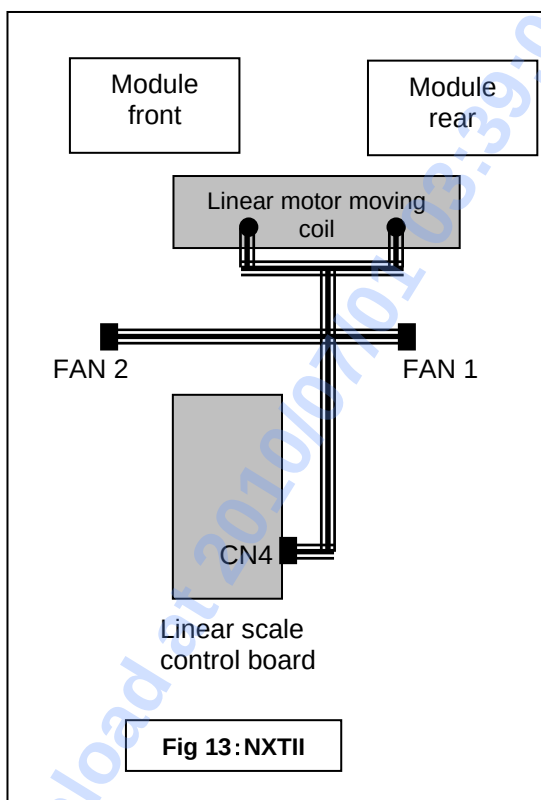


Fig.12

A. Follow the procedures below if the cooling fan is not operational

Possible causes	Countermeasures
Cooling fan failure	Replace the cooling fan
Linear scale control board failure	Replace the linear scale control board
Poor contact or disconnection between the linear scale control board and cooling fan	- Check the CN4 connection on the linear scale control board - Check the FAN1 and FAN2 connector connections - Check the conduction between the linear scale control board CN4 and FAN 1(refer to diagrams 13 and 14) - Check the conduction between the linear scale control board CN4 and FAN 2 (refer to diagrams 13 and 14). - Replace the cooling fan/thermistor harness (NXTII: figure 13  harness) (NXT: figure 14  harness)

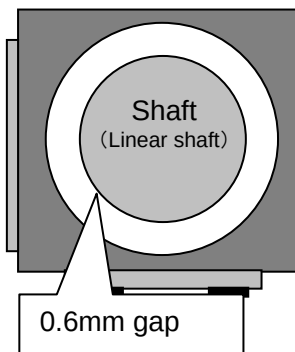


Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

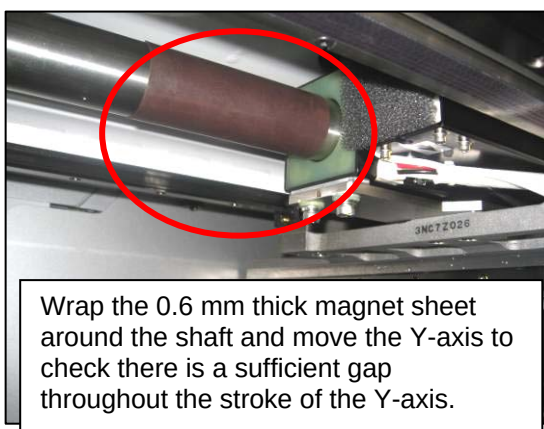
B. Follow the procedures below if the cooling fan is operating

Possible causes	Countermeasures
Cooling fan filter blockage	Clean the cooling fan filter (Refer to "Cleaning the linear motor cooling fan filter" in the NXT (II) mechanical reference).
Linear scale control board failure	Replace the linear scale control board
Servo amplifier error	Replace the servo amplifier (servo amp upgrade is required following replacement)
Thermistor cable defect	Replace the cooling fan/thermistor cable (NXTII: Fig 13  harness) (NXT: Fig 14  harness) <u>Caution: The thermistor cable cannot be replaced separately on the first batch of NXTs produced. For these machines, the whole linear motor should be replaced.</u>
Overheating problem caused by friction between the linear shaft and moving coil.	Use a magnetic sheet to confirm that the gap between the linear shaft and moving coil is 0.6 mm or more along the whole length of the shaft. (See Fig.15). If the gap is less than 0.6 mm, it should be adjusted by an engineer who has attended the linear motor replacement training.
Y-axis linear motor operation defect	Replace the linear motor. This procedure should only be performed by someone who has attended linear motor replacement training.

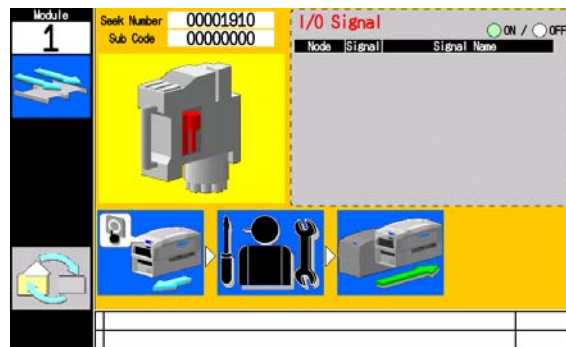
Fig.15



Magnet sheet
Part No. //A TM134-2



Contact the Call Center if you cannot solve the problem after trying the countermeasures above. (Use the call center enquiry form in Supplement 5).

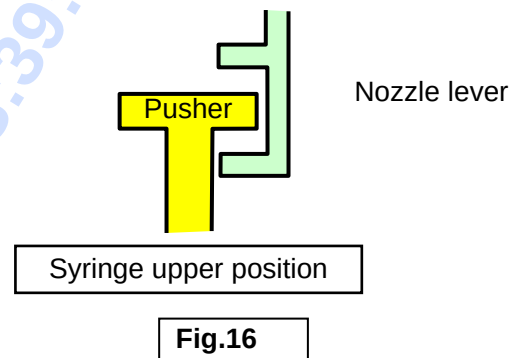
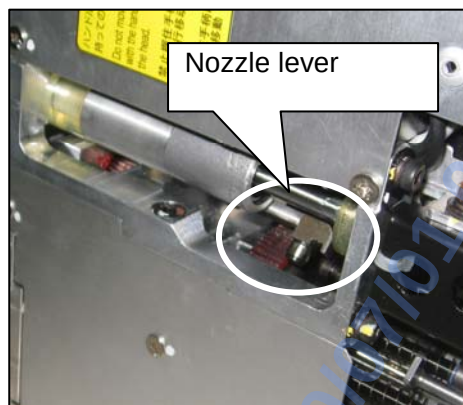
11.8 Nozzle lower lever error (Error code: ** 1910)****Details**

Nozzle lower lever error (H12 series, H08, H04 head)

Countermeasure

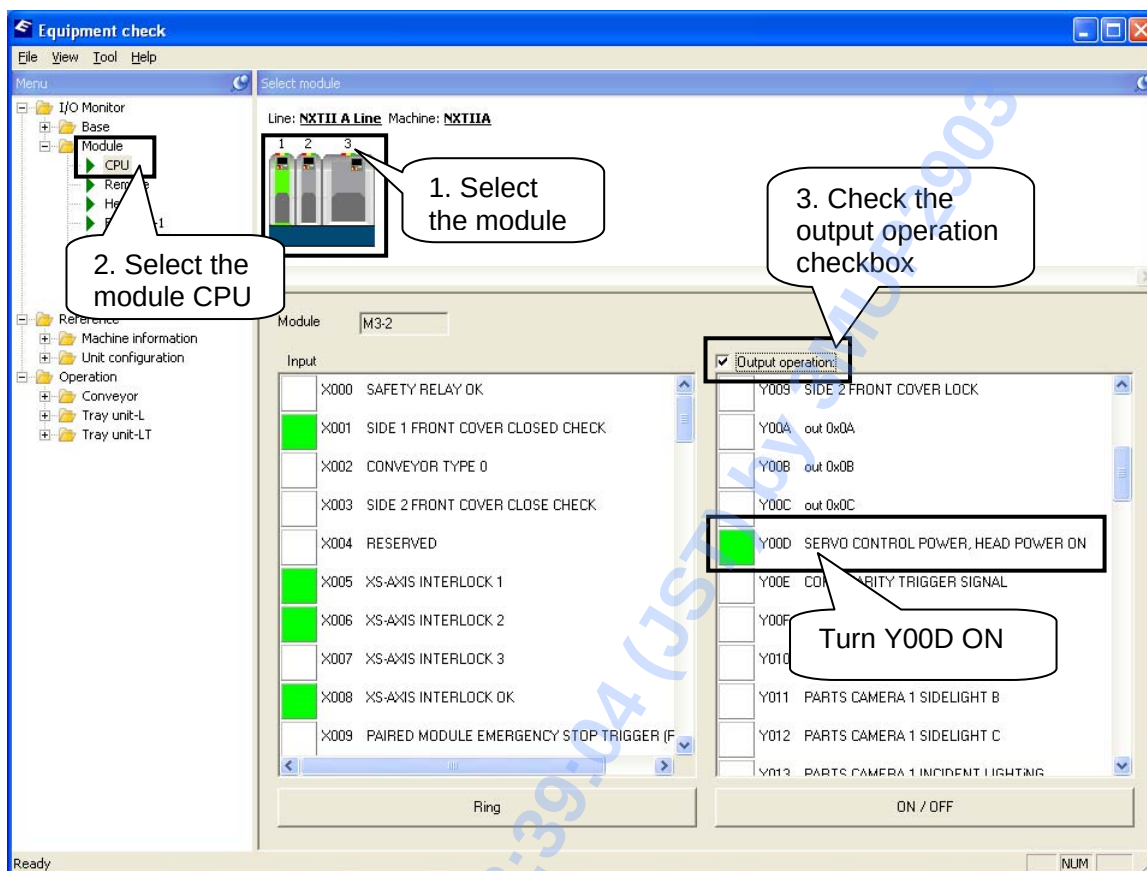
The nozzle lever is designed to escape to the side if it interferes with the syringe pusher during R-axis rotation. The above error occurs when this happens.

1. Pull out the module, and check that all the syringes are at the upper position. With the lever still pushed to the side, manually raise the Z-axis to the upper end position. Return the nozzle lever to its normal position, and then lower the Z-axis so that the syringe pusher is between the claws on the lever as shown in diagram 1 below.

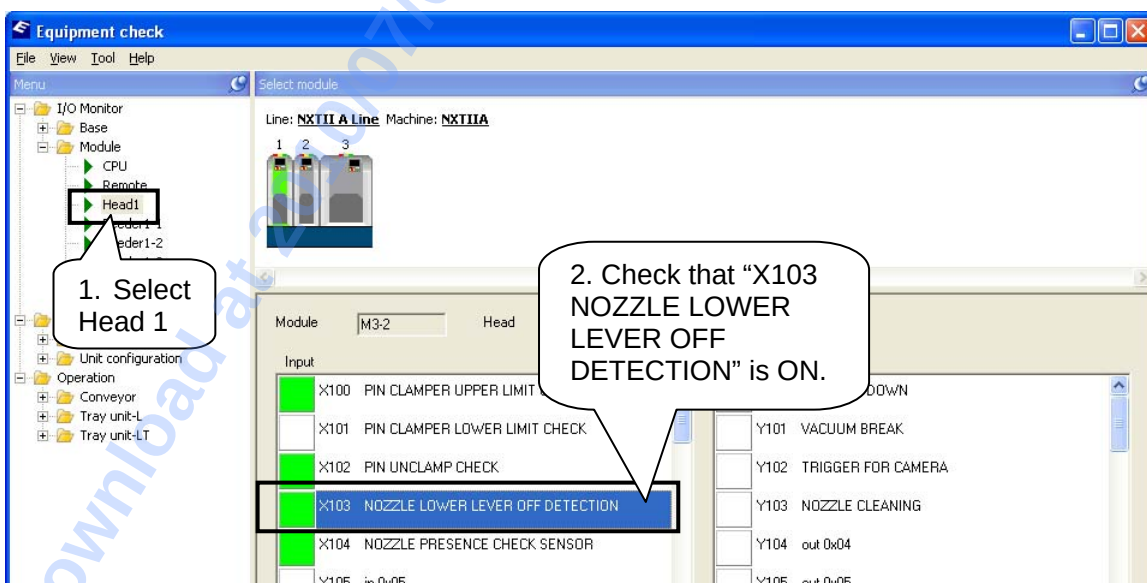


2. Rotate the R-axis by hand to confirm that the nozzle lever does not interfere with the syringe pushers.

3. Check the nozzle lower lever signal. Use the IO command to turn “Y00D SERVO CONTROL POWER, HEAD POWER ON”.



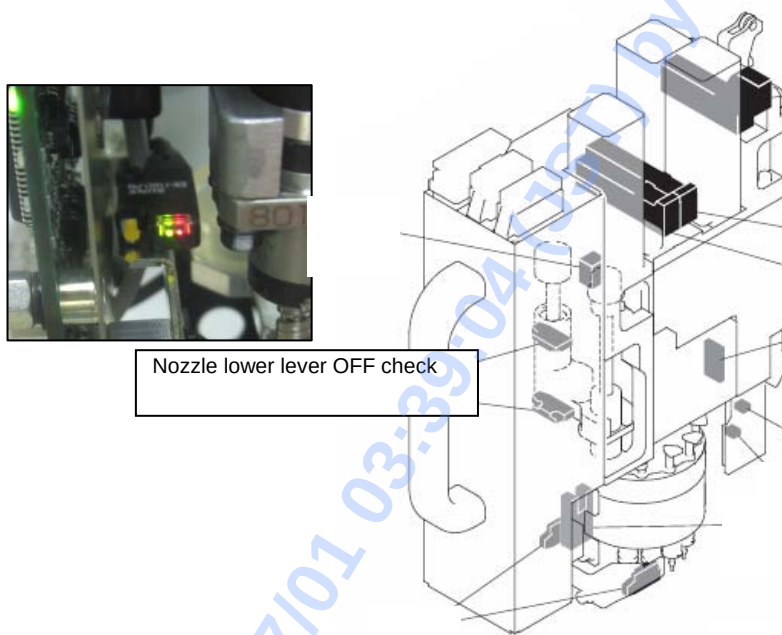
4. Check that “X103 NOZZLE LOWER LEVER OFF DETECTION” is ON. (If “Y00D SERVO CONTROL POWER, HEAD POWER ON” is not turned ON in the previous step, the I/O detection will not function).



5. If the I/O is OFF, check that the nozzle lower lever is at the normal position (not at the escape position).
6. If the nozzle lower lever is at the normal position, check that the sensor is functioning correctly. Check that the sensor LED and I/O signal change with the position of the nozzle lower lever as indicated below.

Nozzle lower lever is at the normal position (the syringe pusher is at the correct height).
The sensor's green and red LEDs are ON.
I/O input signal "X103 NOZZLE LOWER LEVER OFF DETECTION" is ON.

Nozzle lower lever is at the escape position
The sensor's red LED is OFF and the green LED is ON.
I/O input signal "X103 NOZZLE LOWER LEVER OFF DETECTION" is OFF.



If the LEDs and IO signals do not function as described above, please contact the Call center. (Use the call center enquiry form in Supplement 5).

11.9 Application error screen



Normally, this error does not occur.

Follow the procedure described in “10. Acquiring trace data for various error types” – “Trace data acquisition methods for various problems” – “10.4.2 When an OS or Application error (Penguin Symbol) occurs” to acquire the trace data.

Select the appropriate trace acquisition method for the error you are experiencing, and then send the trace data to the Call Center.
(Use the call center enquiry form in Supplement 5).

Supplement

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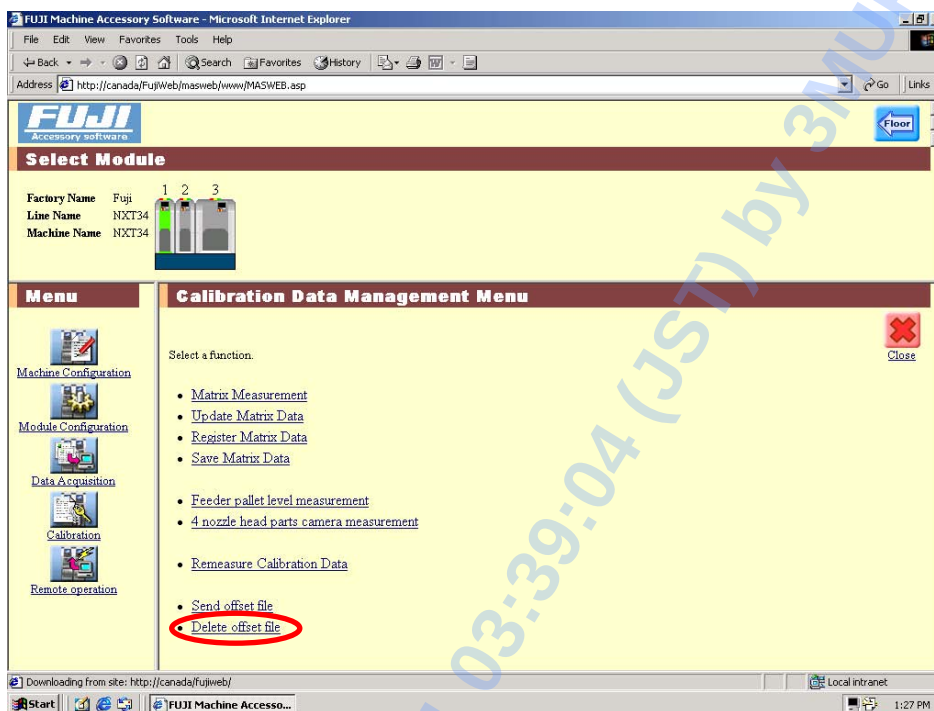
CONFIDENTIAL

Download at 2010/07/01 03:39:04 (JST) by 3MUP2903

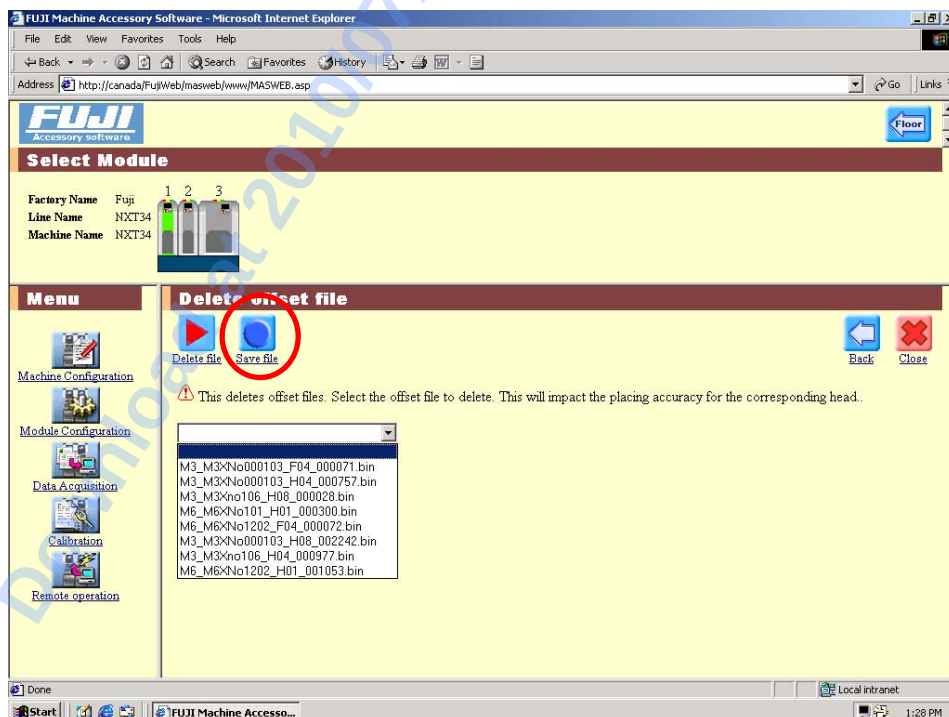
Supplement 1. Backing up the Base CPU BOX Data

<Backing up the Final offset file>

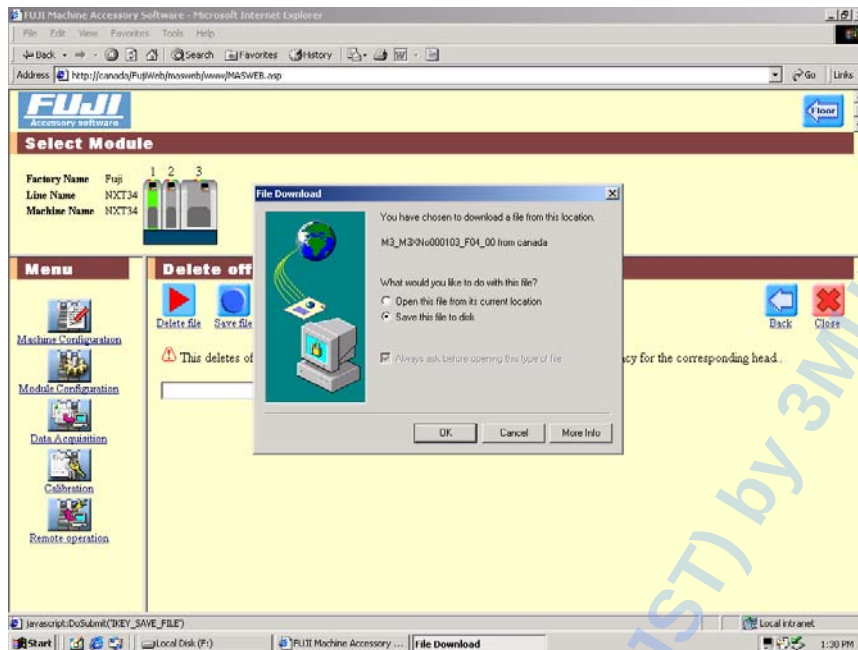
1. Open the Accessory Software.
2. Select the module which to backup the file and then click [Calibration].
3. Click the [Delete offset file].



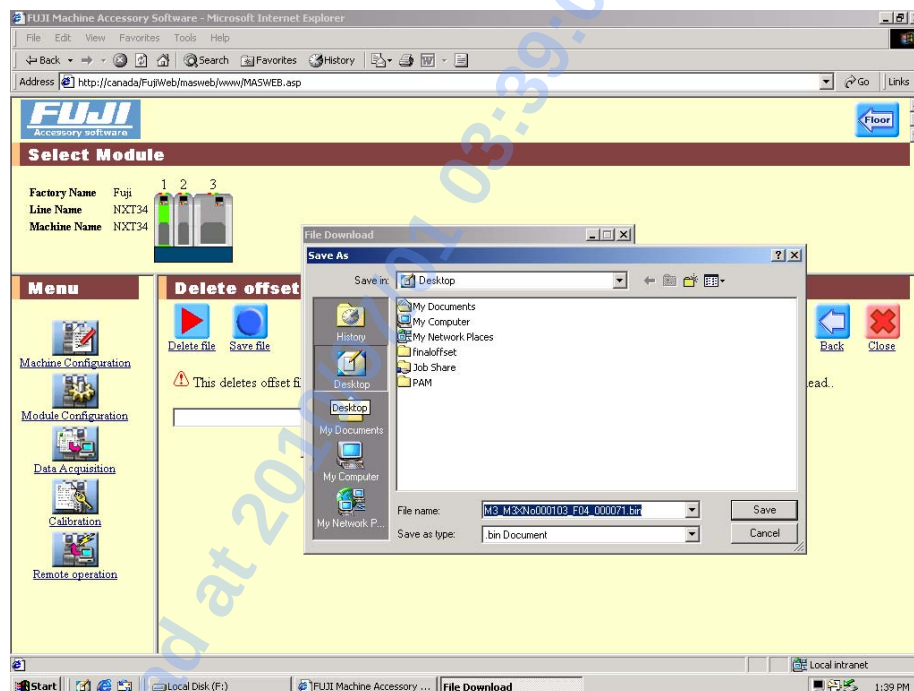
4. Select the Final offset file which to save and press the [Save file].



5. Select the “Save this file to disk” and press “OK”.

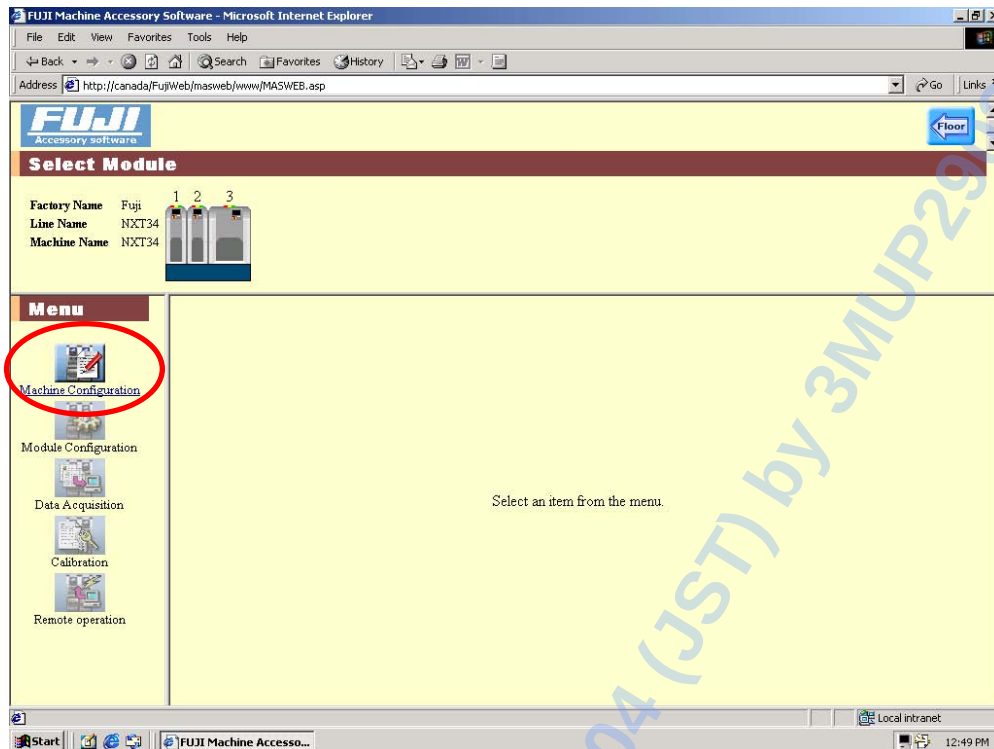


6. Select the location and press “Save”.

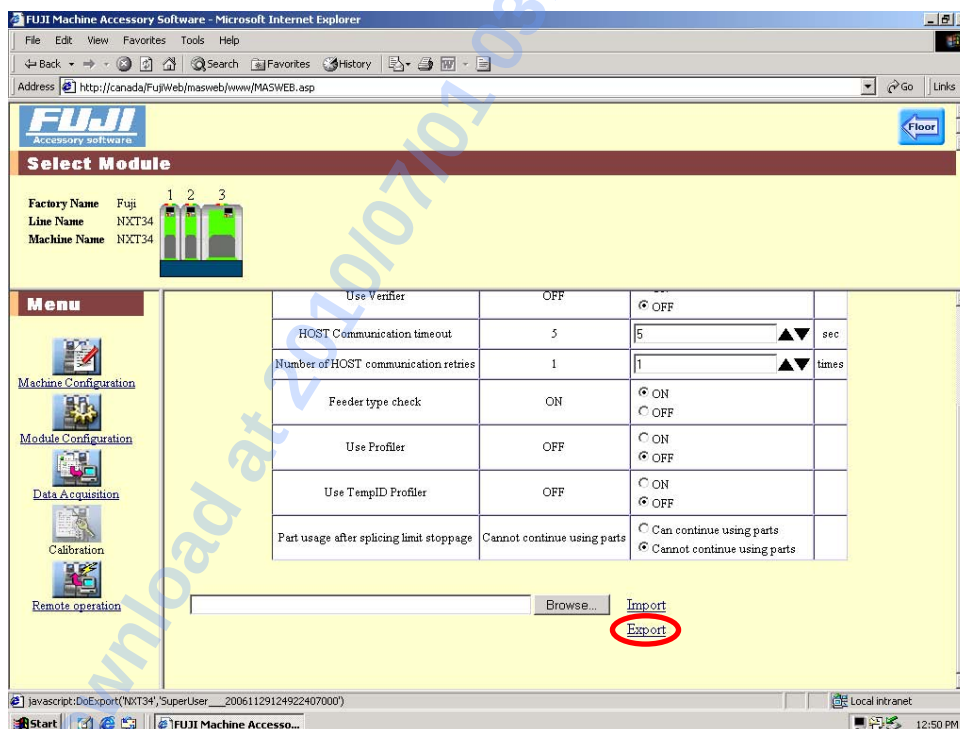


<Backing up the machine configuration>

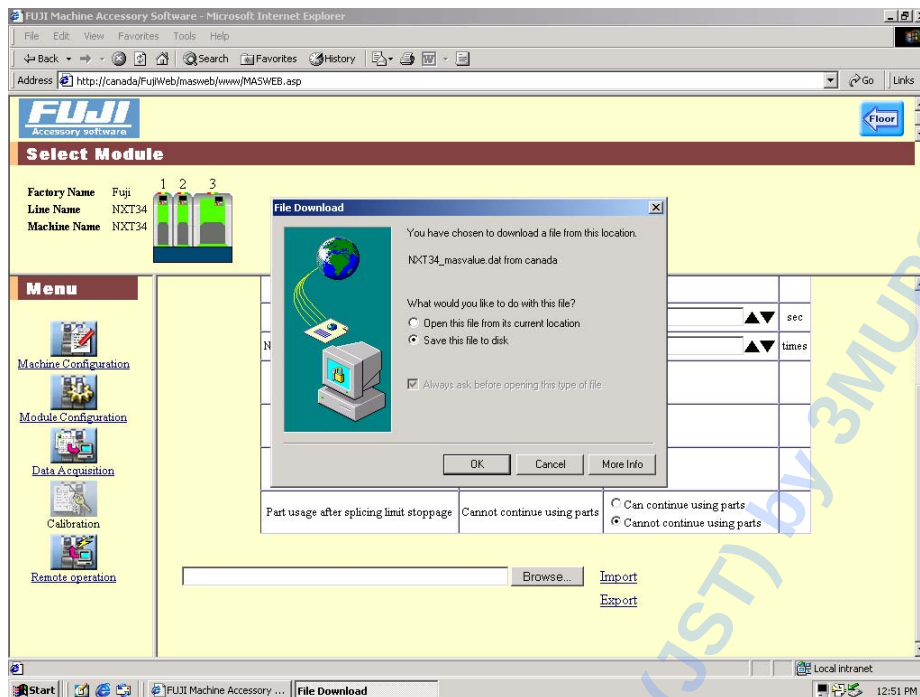
1. Open the Accessory Software and click the [Machine Configuration]



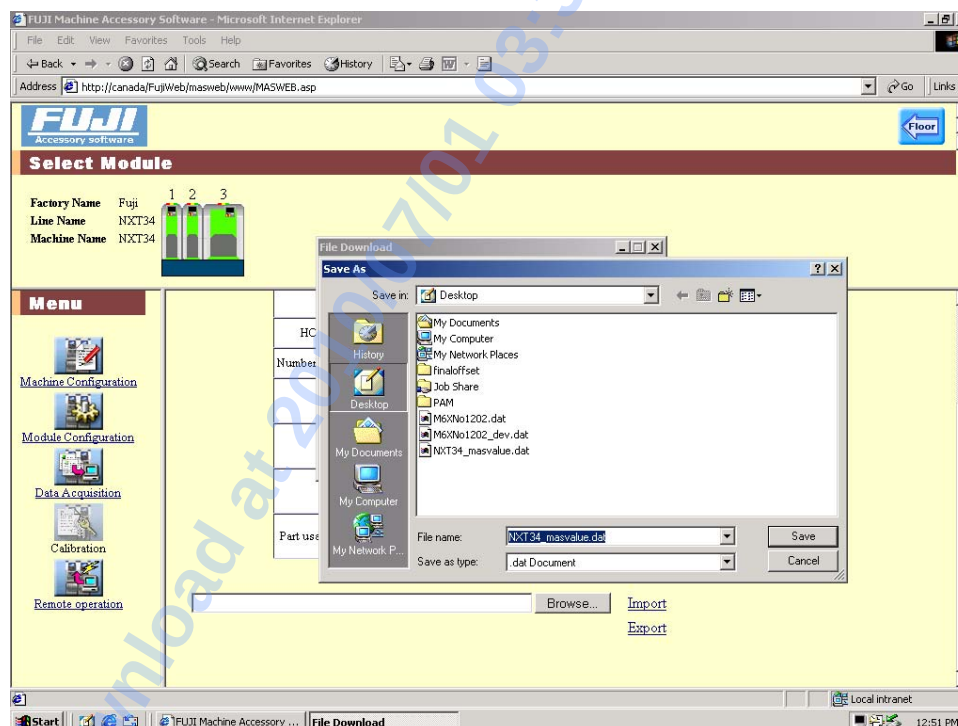
2. Click the [Export].



3. Select "Save this file to disk" and press [OK].



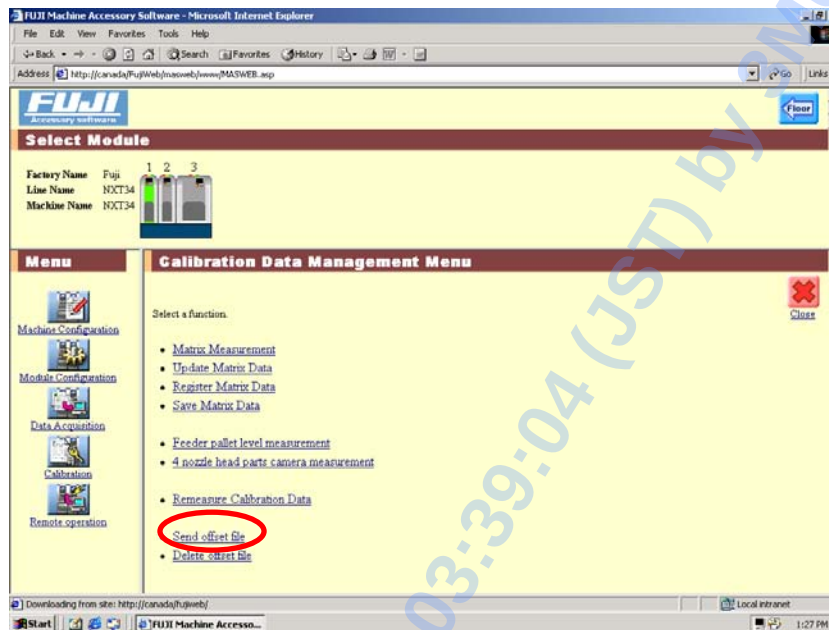
4. Select the save destination and input the file name and then press [Save].



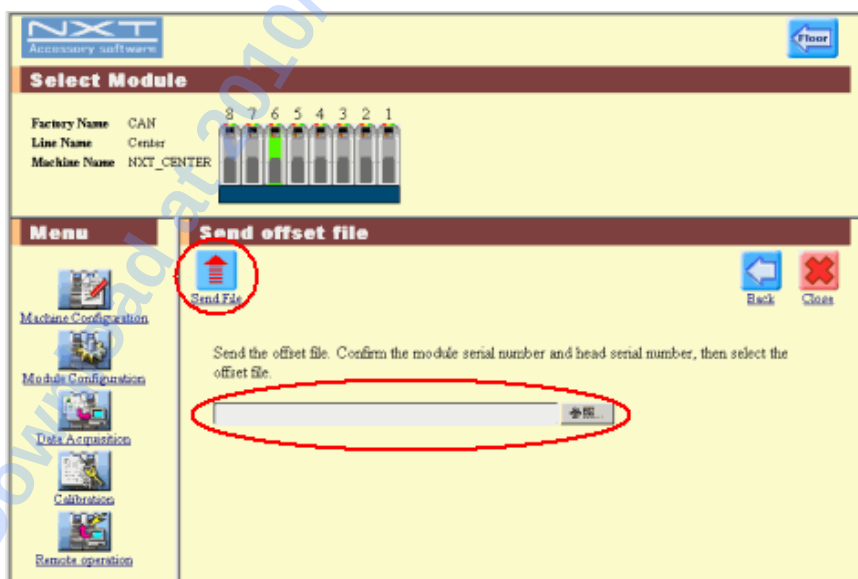
Supplement 2 Registering the Base CPU BOX Data

<Registering Final Offset data to the CPU>

1. Open the Accessory Software and send the backup offset data to the machine.
2. Select the module to which to send the file and then click [Calibration].
3. Click [Send offset file].

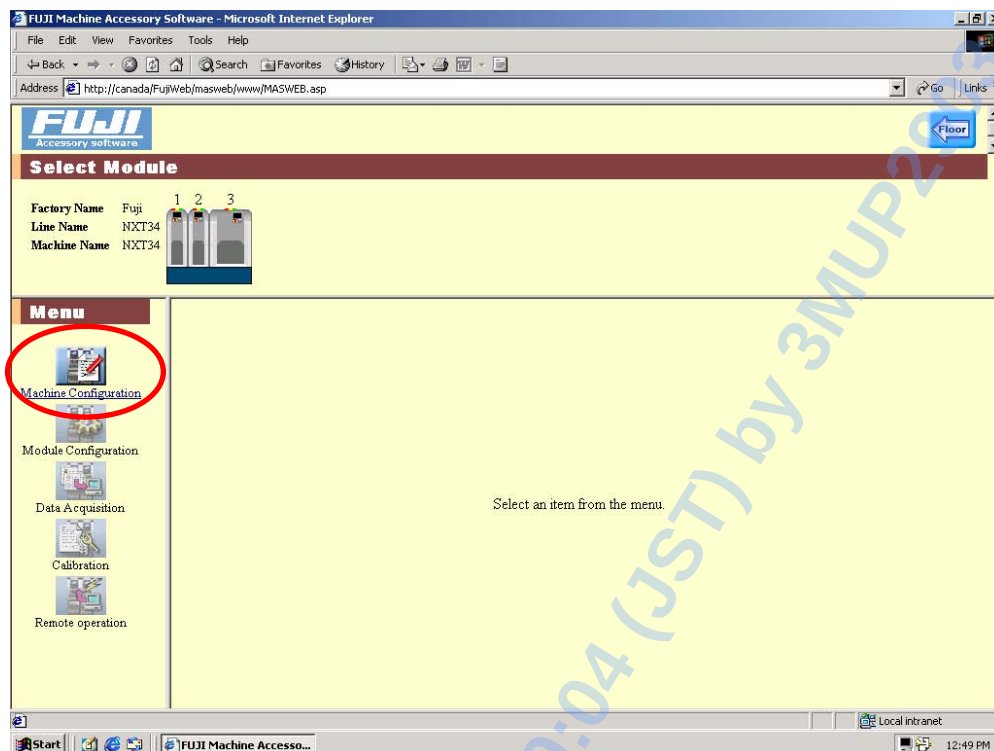


4. Select the file to send to the module and then click [Send File] to send the selected file.

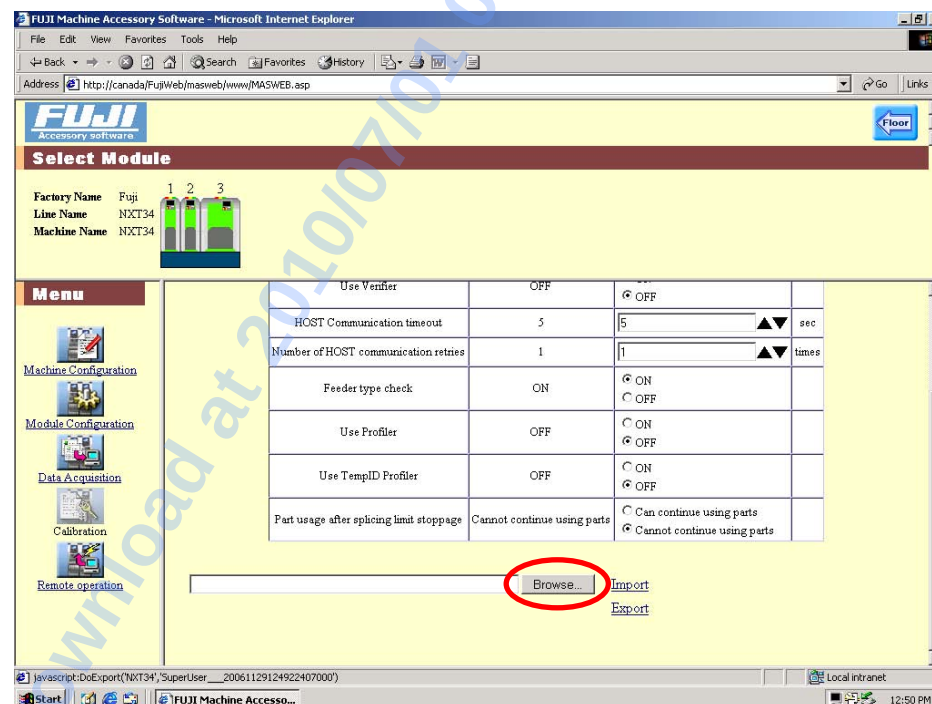


<Registering Machine configuration to the machine>

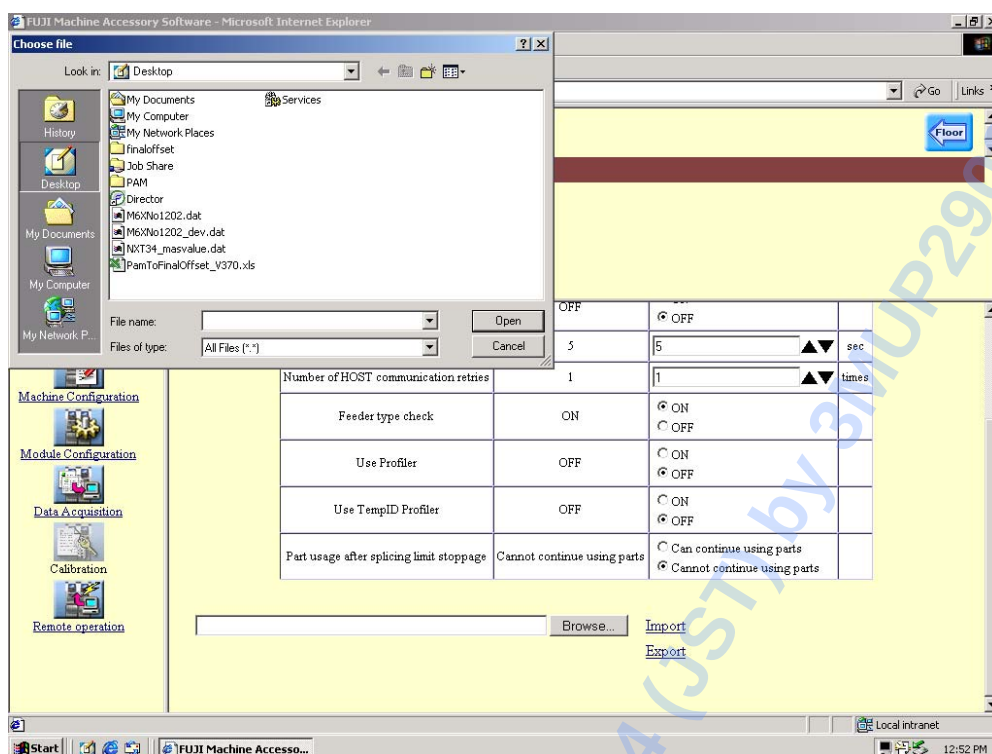
1. Open the Accessory Software and click the [Machine Configuration]



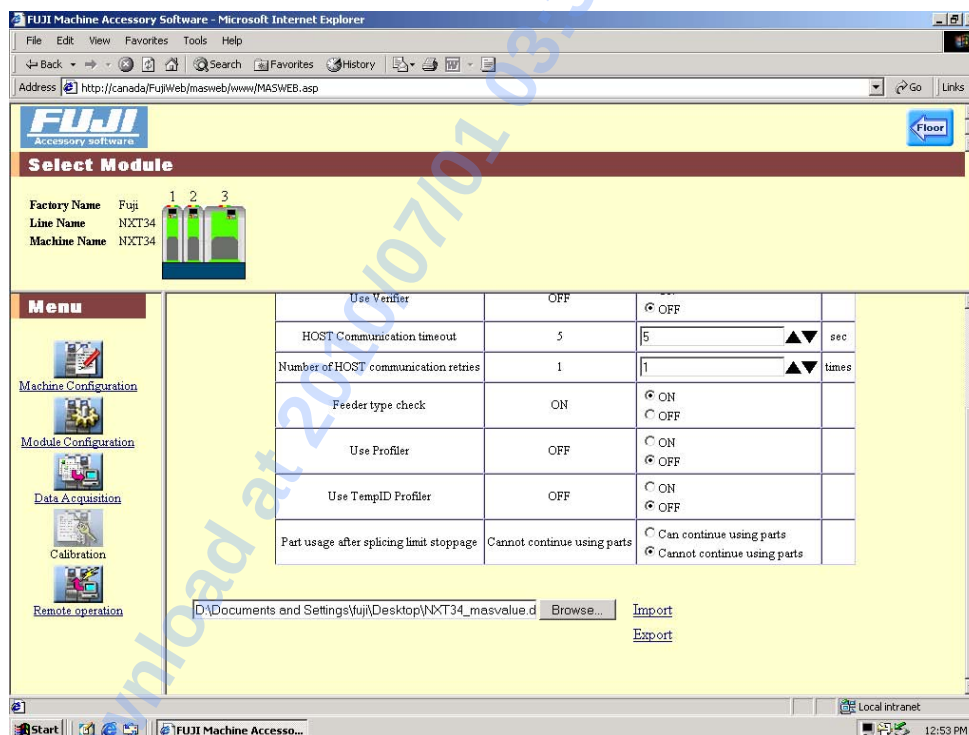
2. Press the [Browse].



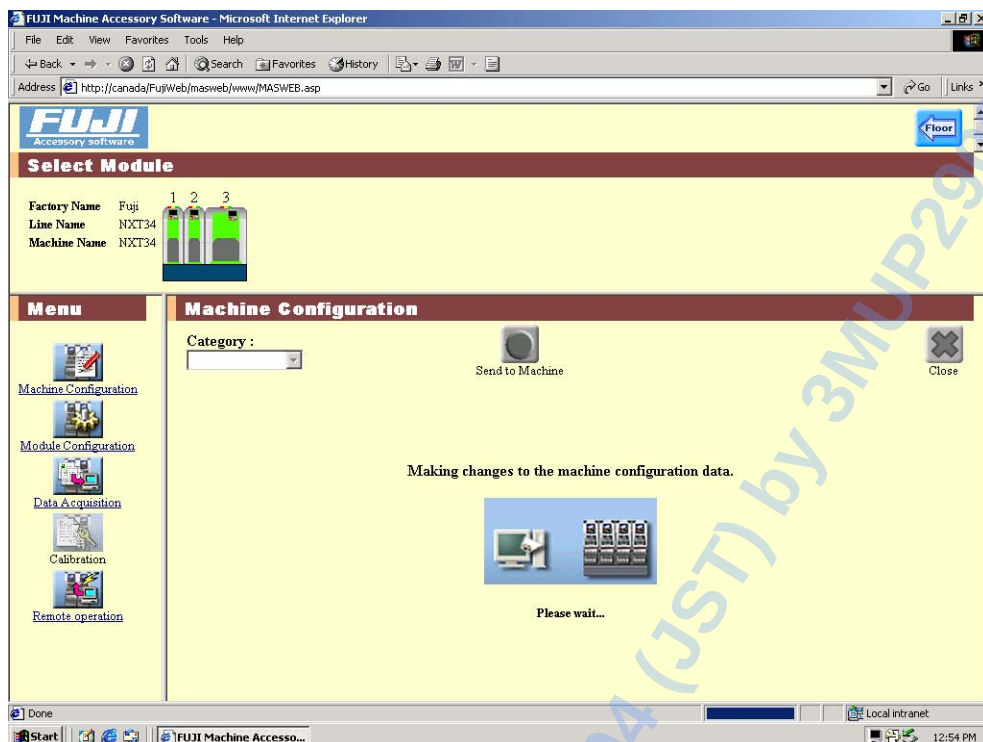
3. Select the file backed up before and press [Open].



4. Click the [Import].



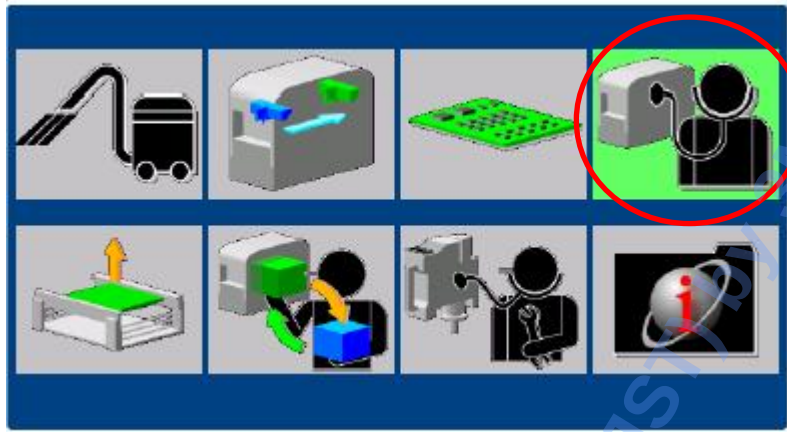
5. Machine configuration should now be registered.



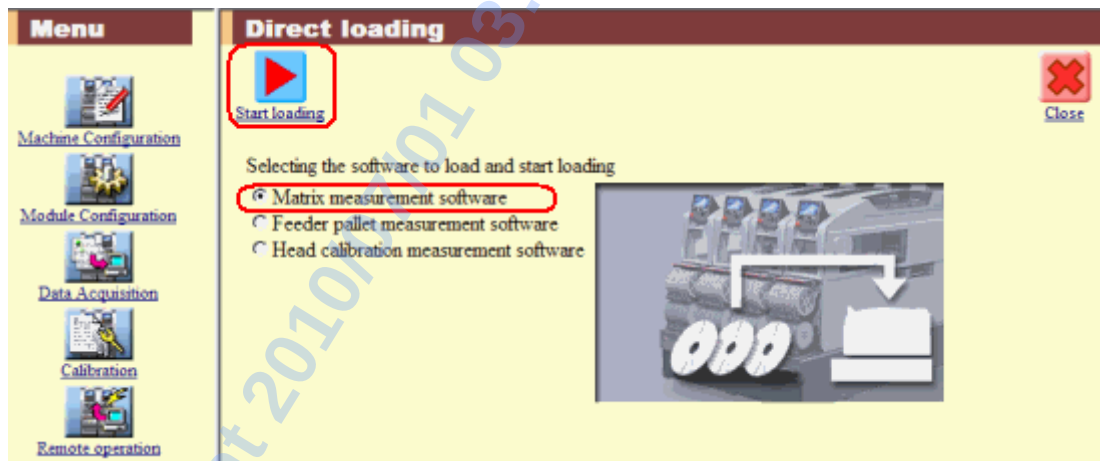
Supplement 3 Backing up the Module CPU BOX Data

<Backing up the Matrix measurement data >

1. Select [Manual command] – [Diagnosis command] at the operation panel.



2. Select the module for direct loading and click [Remote Operation] - [Direct load].
(If the Matrix software is not installed in the PC that NXT Accessory Software is installed, install the Matrix measurement software to this PC beforehand.)
3. Select the [Matrix measurement software] and click [Start loading].

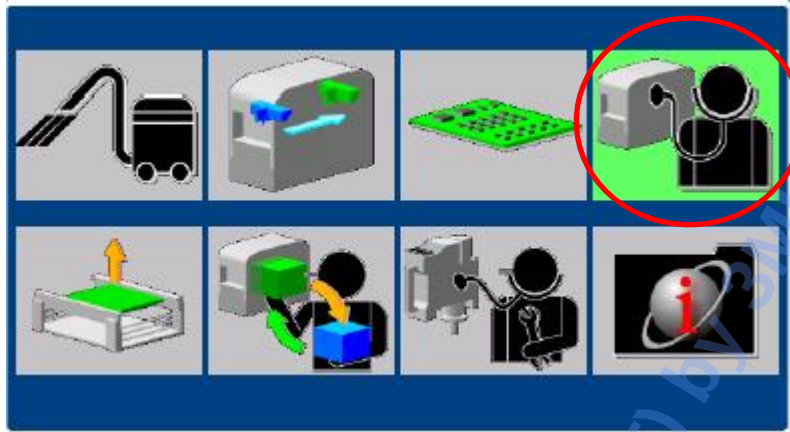


4. The machine automatically reboots after direct loading is finished. When the module becomes a waiting for the next operation status, which means "ready" status after loading the software, press [Manual] switch on the operation panel.
5. Select the module for which to saving a measurement data, and then click [Calibration] – [Save Matrix Data]. ([Calibration] is available after one module is selected.)
6. Select the matrix measurement data file.

7. Click [Save]. Restart the machine and escape from Matrix measurement mode.

<Backing up feeder pallet level measurement data>

1. Select [Manual command] – [Diagnosis command] at the operation panel.

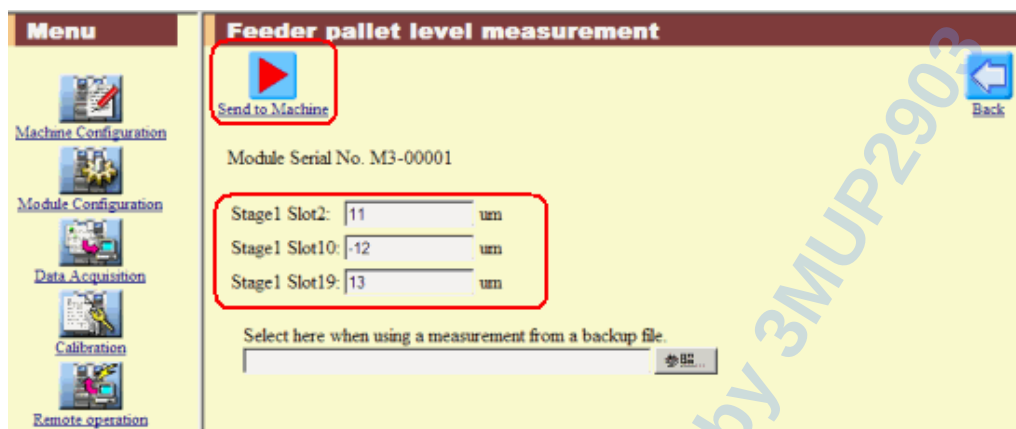


2. Install the Matrix measurement software to a PC that NXT Accessory Software is installed. Select the module for direct loading and click [Remote Operation] - [Direct load].
3. Select the [Feeder pallet measurement software] and click [Start loading].



4. The machine automatically reboots after direct loading is finished. When the module becomes a waiting for the next operation status, which means “ready” status after loading the software, press [Manual] switch on the operation panel.
5. Select the module for which to saving a measurement data, and then click [Calibration] – [Feeder Pallet Level Measurement]. ([Calibration] is available after one module is selected.)

6. Check the 3 points that measurement data is entered. Click [Send to Machine]. Once registered the backup file downloaded dialog box appears. Download the data to make a backup.



7. Restart the machine and escape from feeder pallet level measurement mode.

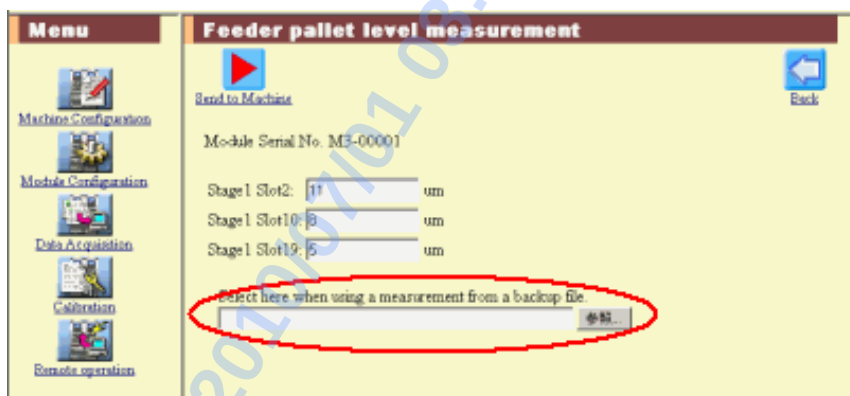
Supplement 4 Registering the Module CPU BOX Data

<Registering Matrix measurement data to CPU>

1. Select [Manual command] – [Diagnosis command] at the operation panel.
2. Install the Matrix measurement software to a PC that NXT Accessory Software is installed. Select the module for direct loading and click [Remote Operation] - [Direct load].
3. Select the [Matrix measurement software] and click [Start loading].
4. The machine automatically reboots after direct loading is finished. When the module becomes a waiting for the next operation status, which means “ready” status after loading the software, press [Manual] switch on the operation panel.
5. Select the module for which to saving a measurement data, and then click [Calibration] – [Register Matrix Data]. ([Calibration] is available after one module is selected.)
6. Select the matrix measurement data file.
7. Click [Register].
8. Restart the machine and escape from Matrix mode.

<Registering “Feeder pallet levelness measurement data” to CPU>

1. Select [Manual command] – [Diagnosis command] at the operation panel.
2. Select the module for direct loading and click [Remote Operation] - [Direct load].
(If the Measure tool is not installed in the PC that NXT Accessory Software is installed, install the Measure tool software to this PC beforehand.)
3. Select the [Feeder pallet measurement software] and click [Start loading].
4. The machine automatically reboots after direct loading is finished. When the module becomes a waiting for the next operation status, which means “ready” status after loading the software, press [Manual] switch on the operation panel.
5. Select the module for which to saving a measurement data, and then click [Calibration] – [Feeder Pallet Level Measurement]. ([Calibration] is available after one module is selected.)
6. Follow the procedure below to register the feeder pallet level measurement results from a backup file.
7. Select the backup file from the [Feeder pallet level measurement] page. The backed up results display.



8. Press the [Send to Machine].

If the following file is selected, an error occurs.

- The file is not a feeder pallet level measurement file.
- The results in the file are for a different module.

Supplement 5. Service inquiry form

Use the following form when sending an email enquiry to the Call Center.

Subject: User/Machine Type/Phenomenon	

Agent	:
Reporter	:
Customer	:
Machine	:
S/N	: (Xno.)*For NXT, trouble module S/N.
W/T (Working Time) :	
Machine Ver.	:
Flexa/Trax Ver.	:

Subject: User/Machine Type/Phenomenon	
Content	
Details of the phenomena	
Duplication procedure & Occurrence condition	
Trouble frequency	
User environment: (Operating computer specification and OS. System configuration)	
Attach data (Trace, job, image log files and etc.)	
*Please compress the files when sending.	
Best regards,	
Your name	
Agent name	
E-mail address	